**AIR FORCE MANUAL**

**No 355–2**

# MANUAL OF AIR DEFENCE

Information given in this manual is not to be communicated either directly or indirectly to any person not authorized to receive it.

## AIR HEADQUARTERS

## 29 JANUARY, 2021

AIR FORCE MANUAL　　　　　　　　　　　　　　　　　　AIR HEADQUARTERS

No 355-2　　　　　　　　　　　　　　　　　　　　　　29 January, 2021

**AIR DEFENCE**

# MANUAL OF AIR DEFENCE

1. Air Defence System of a country provides early warning of airborne objects and enables the defending forces to commit all available resources for neutralizing such impending threats. The purpose of this Air Force Manual is to lay down broad guidelines for all personnel employed in Pakistan's Air Defence System to ensure effective and efficient management of Air Defence resources. This manual also provides guidance for other agencies, both civil and military, who liaise, coordinate or share responsibility of Air Defence with the PAF.

2. The information contained herein covers all aspects of day-to-day operations, training and evaluation of all Air Defence elements. Specific techniques, procedures, which are of a higher security grade are issued separately from time to time.

3. All local orders and instructions issued for the conduct of the 'Air Defence Mission' are to be supplementary to this Manual. No variation or departure from the laid down instructions is permissible except when issued as authorized amendments to this Air Force Manual.

BY ORDER OF CHIEF OF THE AIR STAFF, PAKISTAN AIR FORCE:

**JAVAD SAEED**
Air Marshal
Deputy Chief of the Air Staff (Air Defence)

OFFICIAL:

**(SYEDA ASIFA KAMRAN)**
Wing Commander
Deputy Director Regulations

This supersedes AFM No 355-2, dated 12 March, 2004 along with its all changes.

**OPI**: Air Def (Trg & Eva)　　　　　　　　　　　　　{AHQ/79002/AD (Trg)}

# CONTENTS

## CONTENTS (Contd)

## CONTENTS (Contd)





Revised on 29 Jan 21

## CONTENTS (Contd)

## CONTENTS (Contd)

# CONTENTS (Contd)

## CONTENTS (Contd)





Revised on 29 Jan 21

# RECORD OF CHANGES

| CHANGE | | | CHANGED BY | | |
|---|---|---|---|---|---|
| S No | Date | Rank | Name | Sig | Date |
| | | | | | |

# CHAPTER – 1

# AIR DEFENCE GENERAL

## Introduction

1. The advent of stand-off weapons with precision engagement capability, lethal and extended kill envelops of interceptors and GBADS and large force application has added new dimension to aerial warfare. Tactics and techniques need a constant reappraisal alongside improvement in Air Defence system to counter emerging threat scenarios and ensure sovereignty of airspace. To obviate possibility of a pre-emptive strike, highest state of readiness and proficiency of Air Defence forces is essential. Needless to emphasize that a fully alert Air Defence organization is vital for safeguarding the national aerial frontiers and thwart enemy's efforts to cause critical damage to national war potential.

2. Pakistan Air Force is primarily responsible for the Air Defence of Pakistan. Pak Army and Navy assist PAF in this important task by providing point or limited area defence. Civil Defence Organizations; although not a direct part of the military defence forces; contribute to the Air Defence effort by "Minimizing the Effects of Hostile Air Action".

## Definition

3. The Air Defence may be defined as "Combination of measures designed to nullify the attacking hostile airborne objects or reduce the effectiveness of such attacks after they have been launched. It also includes any of the possible measures taken against violations of national or any other airspace of defined dimensions beyond the national airspace, considered essential for national defence."

4. The explanation of important terms used in the definition is as under:

(a) **Combination of Measures**. It means that Air Defence is not a singular action but a combination of activities that are undertaken by various elements of Air Defence system.

(b) **Nullify**. It is interpreted as either destroying the hostile airborne object or forcing it to divert it in such a manner that it becomes unable to inflict any damage.

(c) **Reduce the effectiveness**. It means to take such measures which minimize the effects of attack by hostile air objects. The prime objective of Air Defence always remains to nullify the attacking airborne objects. However, there is always a possibility of such objects to penetrate the defence. Therefore, an effective Air Defence is also required to reduce / minimize their effects.

(d) **After they have been launched**. It signifies the nature of Air Defence operations as ‘reactionary’. It involves measures taken to negate offensive actions by enemy or any other deliberate airspace violations.

(e) **Violation of National Airspace**. It means unauthorized entry of an airborne object in national airspace or deviation from authorized air route Corridor / area without prior clearance.

(f) **Airspace of defined dimensions beyond the National Airspace**. It means a region in which the country has right of prior information to ensure airspace sovereignty.

**Broad Categories of Air Defence**

5. Air Defence can be classified into the following two categories:

(a) **Active Air Defence**. Active Air Defence is the direct concern of fighting forces and encompasses all measures adopted to prevent the enemy from attacking Vulnerable Areas and Points (VAs / VPs). All efforts must be directed towards destroying or diverting greatest number of attacking airborne objects as far away as possible from the VP / VA.

(b) **Passive Air Defence**. All the measures adopted to minimize the effects of hostile air action fall in the category of passive Air Defence. They include such measures as deception and dispersal.

**Objectives of Air Defence**

6. Basic objective of Air Defence is to discourage the enemy from attacking the VAs / VPs. In war the primary objectives of Air Defence are:

(a) To prevent hostile air forces from inflicting such critical damage that the nation's survival is jeopardized.

(b) To preserve the military means by conducting sustained defensive and offensive operations by air to defeat the enemy.

(c) To aid in sustaining the nation's will to resist the enemy.

**Air Defence Strategy**

7. The primary purpose of an Air Defence system is to deter a potential aggressor from launching air strikes. If deterrence is not successful, the Air Defence system should be capable of inflicting an unacceptable loss to the enemy air power. The potential aggressor must be convinced about the credibility and effectiveness of an Air Defence system and unacceptable attrition, he is likely to suffer in case he chooses to attack. Fear of unacceptable losses is likely to impose certain restrictive conditions upon the enemy and, to a degree, limit the courses of action that his

forces can employ against the defended territory. Moreover, an effective Air Defence system greatly reduces the enemy's chances of achieving an initial advantage through a pre-emptive strike and may cause many complications in his operations, resulting in major failures of his plans.

**Air Defence Doctrine**

8. The accepted doctrine of Air Defence is **to attack the enemy as far away from the VP / VA as is operationally feasible**. The doctrine also envisages capability of increasing resistance against enemy as he approaches his objective, be it from any direction, by employing an ever-increasing number of weapons.

**Inter-dependence of Defensive and Offensive Forces**

9. An alert Air Defence system possesses the means to destroy attacking Air Forces or to reduce the effectiveness of their attacks. However, purely defensive measures have limitations which make them incapable of ensuring immUnity from Air attacks. In the event of war the Air Defence measures against hostile air attack are to be supplemented immediately by strong counter air blows against the homeland sources of enemy's strength. One of the primary objectives of these operations is to destroy the enemy's offensive and defensive capabilities at the source, before their full strength can be brought to bear directly against Pakistan in sustained operations. Therefore, offensive forces are to be available concurrently with the Air Defence at the outset of hostilities.

10. The relative importance of the Air Defence and strike forces is such that neither force can be increased to the point where its requirements would have the effect of denying essential strength to the other. Both are mutually supporting and operate with a singleness of purpose.

**Functions of Air Defence**

11. Air Defence Forces exploit their capabilities to the utmost to provide security from enemy air attack. Air Defence recognizes that maximum security requires the elimination of the enemy's total power to attack by air and that, it is never complete until hostile air forces, wherever they may be, cease to exist as an organized, effective fighting force. Air Defence performs its missions by destroying the invading aircraft, air / surface to surface missiles by imposing tactical restrictions upon their employment and when attacks are delivered, minimizing their effects. The primary tasks of Air Defence are:

(a) **To Gain Command of the Air**. The ultimate security of Pakistan demands the ability to assume the command of the airspace in the areas of conflict. Command of the air is an indispensable condition of victory. In all military operations security from air attack is a primary requirement. Command of air is gained and maintained by exploiting the capabilities of total air power. The Air Defence system, in performing its functions, achieves effects which contribute in gaining command before and during a conflict.

Prior to war, its capability to minimize surprise in air attack and to start immediate operations to destroy invading aircraft and missiles contribute greatly to the desired position of air superiority. During conflict, command of air is attained by the destruction of the enemy's air power. After command of the air is established, Air Defence forces may participate in other operations to complete the defeat of the enemy.

(b) **To Defend Areas Essential to the Prosecution of the War**. The Nation's strength to sustain protected warfare is contained largely in defence installations, regional concentrations of plants, supplies, transportation and communication system, power facilities and skilled manpower. Certain among these regions contain resources of such critical importance that they are indispensable to the nation's capacity and will to face hostilities, and therefore, must be preserved. Air Defence forces are organized deployed and conducted, to provide the highest degree of security to these vital elements.

(c) **To Deny Freedom of Action to Hostile Aircraft**. The attacker holds the initiative in warfare. He may exercise this advantage by planning his attack to achieve surprise. Air Defence should be capable of restricting or denying this freedom of action to the enemy. Firstly Air Defence is to constitute such a strong threat of destruction that an enemy would be constrained to forego attacks on the most strongly defended areas, or sacrifice surprise through delay or indecisive tactics. Secondly, Air Defence should be capable of denying freedom of action in the air to hostile aircraft by confronting them with such concentrated and flexible opposition that they are unable to carry out their attack as planned, and consequently, loose freedom of action at the most critical time of their operations.

(d) **To Destroy Attacking Aircraft and Missiles**. Successful Air Defence is achieved when hostile aircraft or missiles are destroyed before they reach their targets. However, because of inherent limitations in all Air Defence systems, complete destruction of enemy air attacks cannot be assured. In the performance of this function, the mobility and flexibility of Air Defence Forces are exploited to the fullest to exercise the greatest kill capability in providing protection to the critical areas.

(e) **To Divert Attacking Aircraft**. Because there is no method which assures complete Air Defence, hence security is also sought through different measures, short of destruction. The diversion of hostile air power from attacks on vital resources is a relatively limited function. When an activity of the Air Defence system causes enemy aircraft to turn away from its objective and retire without attacking, then the function of diversion achieves successful Air Defence within the circumstances of the attack concerned.

(f) **To Deceive Attacking Aircraft**. The purpose of this function is to mislead or confuse hostile aircraft seeking to attack. When this function is performed successfully, it prevents identification of vital area to the enemy and causes him to waste his offensive efforts through gross misjudgement in his operations.

(g) **To Minimize the Effects of Enemy Air Attacks**. This function is concerned with the non-combative activities and precautions that can be performed for the purpose of minimizing damage, casualties and the disruption of normal activities before and after the attack occurs. The successful performance of this function calls for thorough organization of civil defence, air-raid warning system, First aid, fire fighting services etc and avoidance of unnecessary and costly air raid alarms and work stoppages.

(h) **To Participate in other Actions on Achievement of Command of the Air**. When an acceptable degree of security has been obtained through command of the air, appropriate forces of the Air Defence system may be employed elsewhere to further hasten the conclusion of the war by engaging in other operations against the enemy.

**Principal Operations of Air Defence**

12. Air Defence reaches its planned form in the readiness of its forces to conduct coordinated operations against enemy air attacks. The cumulative result of the provisions and plan of Air Defence is the ability to conduct Air Defence operations where and when necessary. Maximum effectiveness of the Air Defence forces depends upon two indispensable conditions. The first of these is the possession of timely and accurate intelligence information indicating the enemy's capabilities, possible courses of action and significant movement of his air force. The second condition is the ability of the Air Defence system to take prompt action based on the information that is available from intelligence sources and from the facilities of the system itself.

13. The principal operations of Air Defence are:

(a) Combative Operations

(b) Deceptive Operations

(c) Preventive Operations

14. **Combative Operations**. Combative operations are the coordinated measures directly employed to neutralize hostile aircraft or missiles. Precise co-ordination of many activities is necessary in these operations. Pre-attack intelligence information on the enemy's movements is vitally needed. Early warning and concurrent intelligence is also necessary. A sensitive communication system is to be provided to move information rapidly to the points where it will be needed most urgently. The system is to be organized for immediate decisions and rapid action after the decisions have been made. The proper disposition of available forces is of dominant consideration so that maximum forces may be brought to bear against the attacker.

Normally, combative operations are performed in four successive steps; Detection, Identification, Interception and Destruction:

(a) **Detection**. Detection is conducted to supply information on which to base decisions to conduct timely actions against any invading aircraft or missile, to alert the terminal and other defence weapons and the execution of other measures. The information obtained through detection is to be displayed, simultaneously for the appropriate Air Defence commanders. Whenever possible, means of detection should provide overlapping coverage. Detective operations are put in motion on the basis of information supplied by intelligence sources, other external means of alerting, and the Air Defence system itself. The essential components of detective operations are early warning and GCI stations capable of continuous surveillance, the mobile ground observers and an integrated communication and display system. Detective operations combine the capabilities of all available means, which are given below:

(i) **Radar**. It is the basic mean of detection which is conducted from fixed or mobile locations and from airborne platform. Airborne detective operations increase the flexibility of system.

(ii) **Passive Detection**. It is the detection of enemy's electronic and thermal emissions.

(iii) **Visual Detection**. It is the employment of ground observers for obtaining the detection coverage when such is not available from other means, or for complementing the available coverage.

(b) **Identification**. This is concerned with identification, by various methods, of all airborne objects penetrating or using National airspace. Identification process involves one or more of the three basic means given below:

(i) **Automatic, Identification Devices, IFF and SIF**. These devices identify aircraft in flight, simultaneously with detection. They eliminate the complicated manual procedures, involved in other means of identification.

(ii) **Correlation**. This is matching of the position and manner of operation of detected aircraft with available flight plan data.

(iii) **Visual Inspection**. This means the identification of aircraft by employing visual means. It could be accomplished through interception or visual recognition from the ground.

(c) **Interception**. Normally, interception is conducted in a planned sequence after detection and identification. This involves the control and guidance of the intercepting weapon to a point from which it can complete an attack on the intercepted aircraft or missile. Decisions to conduct interception are made at the SMCC or ADOC where the knowledge of air situation in the area concerned is the most pertinent, accurate and readily available. Interception is accomplished by directing the interceptor aircraft or surface to air weapons for action through operational communication means, providing necessary guidance up to the desired point by accepted methods and monitoring their engagements. Recovery of the friendly interceptor aircraft after the success or failure of the interception is also part of this operation.

(d) **Destruction**. Destruction of attacking enemy aircraft is the prime object of the Air Defence combative operations and is accomplished with the weapons available to the system. Following the determination of hostility, the decision to destroy, activates a sequence of events which thereafter takes place so rapidly as to be practically automatic and concurrent. Within the 360 degrees Air Defence concept, weapons of destruction are employed in a manner appropriate to each attack. These weapons fall into following two general categories:

(i) **Long Range Weapons**. Long range weapons are employed in the protection of any defended objective. They attack and destroy hostile aircraft which approach defended areas from any direction and as far as possible from all potential objectives of the enemy. They are characterized by their flexibility and mobility. Examples of such weapons are manned or unmanned interceptor aircraft and missiles.

(ii) **Short Range Weapons**. Short range weapons are employed in point defence 'and are generally restricted to the destruction of enemy aircraft approaching specific targets. These weapons are limited in range and are normally unrecoverable. Generally, mass application is the characteristic of these weapons. These weapons are mainly anti-aircraft guns or rocket propelled-ground-to-air ordnance. Their effectiveness is limited by the extent to which the enemy enters their range of fire and their capability for fast and accurate target acquisition.

15. **Deceptive Operations**. These operations supplement combative operations. By employing various techniques and devices, in co-ordination with combative operations, they have the capabilities to confuse attacking forces and to cause them to make gross errors. Examples of deceptive operations are given below:

(a) Camouflage

(b) Decoys

(c) Smoke screens

(d) Controlled lighting or black-out

(e) Radar and radio jamming

(f) Control of electronic emissions to deny their use to enemy

16. **Preventive Operations**. Tasks designed specifically to reduce the susceptibility of commUnities and facilities for damage from air attacks constitute preventive operations. These operations also facilitate early recovery of such area, if attacked. When performed successfully, they minimize the effects of enemy air attack. Examples of preventive operations are given below:

(a) Air raid warning system

(b) Use of shelters

(c) Dispersal of commUnities and facilities

(d) Control of transportation and evacuation

(e) Fire-fighting services

(f) Organization and control of medical aid facilities

(g) Disposal of unexploded bombs and missiles

(h) Decontamination involving chemical, biological or nuclear conditions

17. These operations combine the co-coordinated efforts of military and civilian agencies, in preparation of techniques to reduce the susceptibility of areas and resources to damage from air attack and also to facilitate rapid recovery from attack. Inter-related communications, continuous co-ordination, and a control organization responsive to both civilian and military agencies are required.

**Components of Air Defence**

18. The main components of a highly organized and fully developed Air Defence system are as follows:

(a) A Control and Reporting (C&R) System

(b) Interceptor Aircraft

(c) Surface-to-Air Guided Weapons (SAGWs)

(d) Anti-aircraft Guns

(e) Balloon Barrage System

(f) Concealment and Deception

(g) Civil Defence

19. **Control and Reporting System (C&R)**. The task of the Control and Reporting System is to provide warning of the enemy aircraft and missiles, continue tracking them thereafter, and to co-ordinate and direct defensive measures. The major elements of the C&R are as follows:

(a) **Air Surveillance or Raid Reporting Organization**. Air surveillance is the systematic observation of air space by electronic, visual or other means, primarily for the purpose of detecting, identifying and determining the movement of all aircraft and missiles, friendly or hostile, within the airspace under observation. Information about all aircraft movements is obtained from a carefully coordinated network of radar stations of various types including fixed, mobile or airborne and other observer Units so as to provide complete coverage of all approaches to the defended area. Under certain circumstances naval vessels, specially earmarked for the purpose of reporting air movements, can supplement the informational to land stations. All information about movement of aircraft etc is collected and displayed by air surveillance organization for use by controls organization.

(b) **Control Organization**. The information obtained by the surveillance or reporting organization is used mainly for the control and direction of the air battle, though all other active, passive and civil defence cells work from the same display. The control organization is responsible for expeditiously assessing the air situation, identifying all moving objects in the airspace, and if necessary, ordering counter action, in sufficient time to affect a 'Kill' before a hostile aircraft can achieve its objective. This is done by either scrambling or diverting the interceptor aircraft or activating the anti-aircraft weapons, as the case may be. When the aircraft are used to intercept the hostile raid, they are handed over to the most conveniently placed Ground Controlled Interception (GCI) Station or AEWACs (as per situation). A control team, consisting of intercept controllers and their assistants, then directs the interceptor aircraft with the help of radar or other means to a position where the interceptor pilot can visually or electronically take up the shooting down of the hostile aircraft. Interceptors may be handed over to other GCI stations in the sector or to the adjacent operational sectors, if necessary, for a successful and timely interception.

(c) **Interceptor Recovery Service**. To increase control capacity, and assist in the recovery of interceptors after combat, an interceptor recovery service is included in the organization. This service is available through its own radar display Dicky Fix (D/F) stations. It is responsible for safely guiding back to base or, if necessary, to a diversionary airfield, all interceptors on completion of assigned missions. The responsibility of the recovery service ends when the interceptors change over to the local radio control of the airfield.

(d) **Intelligence and Radio Warning Organization**. The task of this organization is to deceive, divert and hinder enemy air operations by denying the enemy aircraft or missiles the use of his radio and radar aids, or to aid the defence by exploiting the enemy's radio and radar transmissions. This section collects and interprets all information derived from all possible sources and advises the concerned agencies about enemy tactics and weak points.

(e) **Inter-Communication System**. An efficient, rapid and secure system of intercommunication is essential to link many components of an Air Defence system. No Air Defence System, however, simple or complex, can function with any degree of efficiency without proper means of passing information and instructions with the minimum delay between the various elements of the organization. In addition to the R/T or landlines network of communications, the system should be able to carry the air situation picture instantaneously to all those who require it.

20. **Interceptor Aircraft**. The task of interceptors is to destroy enemy raiders before they reach the point of weapon release, and as far away as possible from the point or area being defended. With the ever-changing tactics of fighter bombers, the problem of interception has increased manifold. The interceptors should carry radio navigation and radar equipment, so that with the assistance of GCI controllers, they can complete the mission of intercepting assigned enemy 'raids' under all conditions of light or weather. The predominant features of an effective interceptor fighter are its rate of climb, ceiling, speed, manoeuvrability, lethality, all aspect weaponry, Airborne Intercept radar (Al), Avionics and operational radius.

21. **Surface-to-Air-Guided Weapons (SAGWs)**. The task of SAGW system is essentially similar to that of interceptors. The deployment of SAGW depends on the performance of the system and the nature of the target or targets to be defended. To make the most efficient use of SAGWs they are integrated with interceptors in a common Air Defence System and thus come under the control of the Air Defence Commander.

22. **Anti-Aircraft Guns**. Anti-aircraft guns complement interceptors and SAMs in their mission. Anti-aircraft guns, sited in the target area, attempt to engage and destroy as many hostile aircraft or missiles as are not at that moment engaged by interceptors or SAMs. Simultaneous engagement by interceptors, SAMs and the guns may be undertaken in certain instances. Anti-aircraft guns are assigned to the Pakistan Army. In Air Defence operations, however, the PAF Air Defence organization exercises operational control over them.

23. **Balloon Barrage**. The balloon barrage is deployed on airfields and vulnerable points as an effective Air Defence measure to prevent hostile aircraft from delivering weapons while remaining at low levels. The main objectives of balloon barrage are:

(a) To force the raider to climb for attack thus exposing itself to the terminal defences.

(b) To make it ineffective in weapon delivery by creating obstructions in its path.

(c) To cause physical damage to the attacking aircraft, should it choose to ignore the presence of balloons.

24. **Concealment and Deception**. Both visual and radar methods of concealment and deception may be employed. Visual methods include blackout, decoys and camouflage. Radio counter measures can also be employed to disrupt electronic navigation systems and to divert aircraft from their targets by interference with radar bombing systems. Dispersal is also a sound means of concealment; it has the possible advantage of reducing the effects of air attack and may, if carried out widely and deliberately, make a target unsuitable or unprofitable for attack.

25. **Civil Defence**. Civil defence is the most important aspect of passive Air Defence as its aim is to minimize the results of enemy air action. The tasks of civil defence are to reduce the scale of dislocation of services, causalities and destruction. The civil defence organization does not come under the control of the Air Defence commander but close liaison between the civil defence organization and the Air Defence system is always maintained. In this manner, the up-to-date information obtained from the control and reporting system is available to the civil defence authorities whenever an air attack is anticipated and precautions are taken well in time.

**Essential Characteristics of an Air Defence System**

26. The following are the characteristics of an effective Air Defence system:

(a) **Flexibility**. The capability to react rapidly, counter every method of air attack, and support any type of ground or air operations is called flexibility.

(b) **Mobility**. The capability to shift the Air Defence forces rapidly and is able to deploy and operate in any environment. Mobility only applies to mobile radars and other Air Defence assets.

(c) **Responsiveness**. Can react quickly and selectively to crises in all area. Air Defence Weapons should possess a variety of configurations that can be tailored deployed and employed on short notice to meet mission requirements.

(d) **Versatility**. Possess a variety of capabilities which can be applied in varying levels of intensity. Capable of operation throughout a broad spectrum of attack profiles, tactics, speeds, altitudes and manoeuvres.

(e) **Economy**. Modular capability permits a controlled response. The acceptance of 'acceptable' performance instead of 'maximum' performance reduces initial and follow-on or follow-up costs.

(f) **Standardization**. Standardization reduces maintenance, training and support costs. However, it should not degrade other essential characterization.

(g) **Reliability**. It means performance without interruption by component failure or environmental conditions must function properly at the right place and right time.

(h) **Maintainability**. This is the capability to restore a specific condition within a given time period by available technicians.

(j) **Survivability**. It is the preservation of the capability to perform all assigned mission. Mobility and redundancy are major factors in survivability. A capability to operate under adverse conditions will enhance the chances of survival.

**Essential Characteristics of Air Defence Forces**

27. The recent technological and tactical advancements to counter new developments in the air offensive require a constant 0update for the Air Defence System. The essential characteristics of Air Defence forces are:

(a) **A Constantly High Degree of Readiness**. Air Defence forces are to be prepared to function under unfavourable and difficult conditions imposed by the enemy in exercising the initiative in attack. They are to have the capability to maintain, over a long period of time, fluctuating operations ranging from standby conditions to full alert.

(b) **Speed of Action**. These forces are to have the capability to alert and engage them swiftly. When command decisions have been made, the forces are to perform their functions and tasks with exceptional speed, because only a limited time will be available once the forces which the enemy has committed to attack have been detected.

(c) **Endurance**. Air Defence forces must be capable of sustaining their operations for all periods of time during which the enemy is capable of conducting air attacks.

(d) **All Weather Operational Capability.** Since the enemy has the option of attacking during day or night and in all weather conditions, the Air Defence forces are also to have the capability to operate during all these conditions.

# CHAPTER – 2

# COMMAND AND ORGANIZATION OF AIR DEFENCE UNITS

## Relationship of Command, Control and Coordination

1. Air Defence of Pakistan is one of the primary responsibly of Pakistan Air Force. PAF, while safeguarding aerial frontiers, controls the efforts of all forces engaged in Air Defence mission. In order to operate Air Defence as a single cohesive body, a single commander is designated to discharge Air Defence functions. Chief of the Air Staff (CAS) is the commander of Air Defence System. He is assisted in his task by Deputy Chief of the Air Staff (Air Def), Deputy Chief of the Air Staff (Ops) and Air Officer Commanding, Air Defence Command.

2. **Full Command and Control**. On behalf of CAS, Air Officer Commanding Air Defence Command, exercises full operational command and control over all forces, which are allocated and primarily assigned to the Air Defence system.

3. **Operational Control**. Operational Control comprises those functions of command involving the assignment of Air Defence tasks and authoritative directions, including deployment, re-deployment and employment of Air Defence Weapon systems necessary to accomplish the Air Defence mission. Specifically, operational control provides the AOC Air Defence Command with authority for determining the rules of engagement of all weapons utilized in the Air Defence system, irrespective of the force to which they belong to or are assigned, in accordance with the policy issued by Air Headquarters.

4. **Control and Coordination of Civilian Agencies**. During all Air Defence operations, the Air Defence Command exercises control over the efforts of all civilian agencies whose activities augment the Air Defence system. The efforts of all such civilian groups are merged into the system through coordination between the appropriate civilian authorities and the Air Defence Commander.

5. **Coordination with Friendly Countries**. The Air Defence functions will be coordinated when required with the functions of friendly systems adjacent to or supporting Pakistan Air Defence System in accordance with the Government policies.

6. The organogram of Air Defence setup at Air Headquarters is placed at **Appendix 'A & B'** to this AFM.

## Organization of Air Defence Command

7. The Air Defence Organization is centred around the highest operational controlling formation i.e Headquarters Air Defence Command which is responsible for supervising the implementation of Air Defence policies issued by Air Headquarters and for effective control of Air Defence activities of PAF, both during

8. peace time and in war. The Air Defence Command is commanded by an Air Marshal who is directly responsible to the Air Defence Commander (CAS) and through him to the Supreme Commander of Armed Forces (President of Pakistan) for the country's defence.

9. The AOC Air Defence Command exercises operational control over all elements of Air Defence namely:

(a) **Air Defence Operations Centre (ADOC)**. ADOC is the highest echelon of PAF AD network, where the country-wide air situation is displayed and command and control function is executed. The Recognized Air-Maritime Picture (RAMP) is received at ADOC through all under command SMCCs. The organogram of Headquarters Air Defence Command (ADC) and layout of ADOC are shown at **Appendix 'C'** to this AFM.

(b) **Sector Mission Control Centre (SMCC)**. SMCCs are the operational centres of control, over subordinate Units in their areas of responsibility. There are a number of Sector Headquarters. Sector Headquarters has an integrated operations room, called Sector Mission Control Centre (SMCC) from where Sector Air Defence battle is fought. The organogram of a Sector Headquarters and layout of SMCC are shown at **Appendix 'D'** respectively to this AFM.

(c) **Sub-Control Centres (SCCs)**. These are established within a sector for receiving, filtering and upward telling of aircraft reports received from various Air Defence Units under their control.

(d) **Generic Mission Control Centre (GMCC)**. GMCC Units in static or mobile versions are the control and reporting centres of AD Network. The GMCC receives data from multiple heterogeneous sensors and after filtration generates a composite air picture which is up told to SMCC. A GMCC is capable of conducting interceptions in all height regimes. Each GMCC has low level radars as its integral part and deployed as per requirement. The organogram of a GMCC Headquarters and layout of GMCC are shown at **Appendix 'E'** respectively to this AFM.

(e) **High Level Radar Squadron**. This includes various High-level radars of PAF which up tell their data to GMCC. The organogram of a typical High-level Radar squadron is shown at **Appendix 'F'** to this chapter.

(f) **Airborne Early Warning Command and Control System (AEWACS)**. AEWACS platforms are essential in executing Air Defence mission as they offset the limitations of ground-based sensors. When declared on Air Defence Alert (ADA), they can be given scramble as per the laid down procedures for the purpose of floating Air Defence environment, enhanced trans-frontier visibility and various utilizations in EW operations. The organogram of AEWACS squadron and layout of AEWACS crew are shown at **Appendix 'G'** respectively to this AFM.

(g) **Fighter Interceptors on Air Defence Alert (ADA)**. Fighter interceptors on ADA are main instruments of tactical actions with diverse type of aircraft.

(h) **Surface-to-Air Missiles (SAMs)**. These include various types of SAM elements of the PAF. The organogram of a CSS is shown at **Appendix 'H'** to this AFM.

(j) **Mobile Observer Units (MOUs)**. These are composed of base and field elements for visual observation and reporting of an aircraft movement within their area of responsibility. The organogram of MOU is shown at **Appendix 'J'** to this AFM.

(k) **Balloon Barrage Units (BBUs)**. These are deployed at all PAF airfields and other important VPs of national interest.

(l) **Army and Naval Anti-Aircraft Artillery / SAM Units**. These Units are specifically allocated for Air Defence purposes.

(m) **Passive Defence Elements**. These include such measures as Civil Defence, black out, control of sirens, radio and TV transmissions etc.

(n) **Civil Aviation**. Only control of civil aircraft movement during war / emergency.

10. **Sector Mission Control Centres**. Pakistan's airspace is divided into a number of operational sectors; each one is responsible for the Air Defence of its designated region. It serves as the nerve center of Air Defence operations. Each sector is under the operational control of a Sector Commander. The Sector Commander, on behalf of the AOC Air Defence Command, exercises operational and functional control over all the Units allocated for Air Defence Operations within his Sector. Close co-ordination is also maintained with other armed forces and civilian agencies involved in the Air Defence. Following are the major functions performed by SMCC:

(a) **Air Situation Monitoring**. This is the basic function of SMCC which is performed by the following actions:

(i) Receipt and display of track data input from the subordinate operation centres.

(ii) Track correlation from multiple sites.

(iii) Assignment of track numbers.

(iv) D/R and extrapolation of tracks.

(v) Maintaining status boards.

(b) **Generation and Sharing of Recognized Air Picture (RAP)**. SMCC receives a composite air picture from single/multiple GMCCs and AEWACS in the form of tracks and generates a RAP through the following actions:

(i) Receipt of all flight plans from various agencies / sources.

(ii) Flight plans entry, updating, correlation, and display.

(iii) Identification of tracks is carried out through the following means:

(A) Parameters / position correlations.

(B) Electronic Method (IFF / SIF).

(C) Visual Identification (VID) by fighter interceptor / MOUs.

(iv) RAP is shared in the following manner:

(A) Cross-tell of identified tracks to adjacent SMCCs and AEWACS (Back told + own AEWACS tracks)

(B) Downward tell to GMCC.

(C) Forward tell to ADOC.

(c) **Threat Evaluation and Generation of AD Response**. Threat evaluation is carried out by continuously monitoring the air situation and carrying out identification of each track being reported to SMCC. A track not correlating with any flight plan information available with SMCC is declared as Unknown (Assumed Hostile / Friend) or Hostile, and tactical action is taken as per existing procedures.

(d) **Routine Tasks**

(i) Receipt of all flight plans from various agencies / sources.

(ii) Maintaining active link operations with AEWACS.

(iii) Allocating training missions to under command GCI Units.

(iv) Maintaining GBADS on required alert status and issuing of control orders.

(v) Coordinating interceptor aircraft, Ack Ack and SAM defences.

(vi) Keeping a D/R of all the raider tracks, downward telling of raiders' information to appropriate low looking radars.

(vii) Assisting aircraft in distress.

(viii) Coordinating air rescue missions.

(ix) Accepting requests for FFS, if possible.

(x) Reporting flight plan irregularities and air space violations.

(xi) Providing flight plans for radar performance control.

(xii) Keeping a watch on the operational performance of the MOUs deployed in the field and scheduling period of operations for them.

(xiii) Keeping a watch on the operational capabilities and limitations of each radar Unit within the area of responsibility and taking remedial actions, if required.

(xiv) Scheduling period of operations and maintenance of each radar Unit within the area of responsibility.

(xv) Maintenance of daily log of operations.

(xvi) Security of radio & radar transmissions.

(xvii) Recording / replaying of operational data whenever required.

(xviii) Digital Simulation.

(e) **Enhanced Role during Hostilities**. During hostilities / escalations, different elements of AD will be enhanced in numbers and alert status. In such scenarios, in addition to the above-mentioned tasks, each SMCC will be entrusted with the responsibility of executing its **'DCA plan'**. Hence the role and task of SMCC during hostilities is increased but not limited to the followings:

(i) Enhanced requirement of airspace management.

(ii) Management of CAPs in AOR.

(iii) Management of enhanced ADA.

(iv) Management of AEWACS anchor and its protection through Barrier CAP (BARCAP).

(v) Passing air raid warnings to civil defence setups through CDLO.

(vi) Enhanced coordination with Air Space Coordination Cell (ASCC) when established.

# CHAPTER – 3

# AIR DEFENCE SETUP AT AIR HEADQUARTERS

## Introduction

1. The Chief of the Air Staff is the Air Defence Commander and responsible to the Supreme Commander (President of Pakistan) for all Air Defence matters. The Air Defence Commander is assisted by the Deputy Chief of the Air Staff (Operations) and Deputy Chief of the Air Staff (Air Defence) at Air Headquarters. Duties of PSOs and personnel at Directorates of Air Defence are given in the succeeding paragraphs.

## Duties and Responsibilities Deputy Chief of the Air Staff (Ops) Secretariat

2. **Deputy Chief of the Air Staff (Operations)**. As a Principal Staff Officer he is responsible to Chief of the Air Staff (CAS) for following matters pertaining to Air Defence:

(a) Control and direction of all AD Units of the Pakistan Air Force.

(b) Assist the CAS on all operational matters related to sovereignty of Pak Airspace.

(c) Conduct of entire AD Ops during wartime (OCAO & DCAO).

(d) Formulation of employment strategy with embedded AD concept.

(e) Issuance of ROEs for employment of AD assets.

(f) Formulation of AD Force Goals in coordination with DCAS (AD).

(g) Define operational capability for induction of new AD equipment.

(h) Development review and updating of Air Defence Operational plans in coordination with AOC, ADC.

(j) Operational Training of Fighter Controllers (Saffron Bandits, Ex Hawk Eyes, DACT Camps etc).

(k) Jointly represent PAF on AD matters at Tri-services forums along with DCAS (AD).

3. **Assistant Chief of the Air Staff (Air Defence)- ACAS (AD)**. He is responsible to the DCAS (Ops) for the following:

(a) Advise DCAS (Ops) on matters related to Air Defence Operations, GBADS Ops and Ops Plans.

(b) Coordinate and liaison on all operational matters relating to the AD with the following:

(i) Joint Staff Headquarters.

(ii) Sister services.

(iii) Operations and Plans Sub-Branches at Air Headquarters.

(iv) Air Defence Command and Regional Air Commands.

(v) Civil Aviation Authority.

(vi) Civil Defence Authority.

(c) Representing Air Headquarters at Services Headquarters meetings pertaining to AD Operational matters.

(d) Formulation and monitoring implementation of AD Ops policies and ROEs.

(e) Policy matters pertaining to AD War plans in coordination with ACAS (Plans).

(f) Formulation of deployment plan of Air Defence sensors and weapons.

(g) Determine Air Defence Force Goal for overall Ops preparedness of AD in co-ordination with ACAS (Plans).

(h) Coordinate with DCAS (AD) Secretariat for provisioning of functional support in all matters related to AD Operations.

(j) Coordinate with DCAS (AD) Secretariat to process AD budgetary requirements related to DAD (Ops) and DAD (Plans).

(k) Assist DCAS (Ops) at Tri-services level on all matters related to AD Ops, GBADS Ops and Plans.

4. **Director of Air Defence (Operations) – Dir AD (Ops)**. He is responsible to the ACAS (Air Def) for the following:

(a) All matters pertaining to operations and functioning of Air Defence Units as well as on any suitable changes or improvements in present tactical concepts in the Air Defence Systems.

(b) Scrutinizing post deployment reports of mobile Air Defence elements and suggesting measures to enhance their operational effectiveness.

(c) Analysing and evaluating the Air Defence exercises conducted by the three Regional Air Commands and the Air Defence Command, with a view to make them as realistic as possible.

(d) Complete up-gradation of equipment (hardware / software).

(e) Coordination of Passive Air Defence measures with concerned agencies.

(f) Maintaining liaison with Civil Defence Authorities for effective planning and exercising of the Civil Defence System.

(g) Liaison with Dte of Ops / ATS / CAA on matters directly affecting Air Defence operations and issuance of policies on such activities.

(h) Planning and subsequent evaluation of probe missions.

(j) Recording and submitting all air violations of Pakistan air space and hijacking incidents for perusal of the Air Staff.

(k) Coordination with other Directorates / Services for conduct of exercises (PAF, Army, Navy and International exercises).

(l) Ensuring availability of AD assets for tests and trials and coordinating their conduct.

(m) Preparation and record keeping of all briefs / presentations on Ops matters to the Air Staff.

(n) Maintaining and forwarding of Coalition flying data to the Air Staff.

(p) Coordinate all AD operational matters related to Force Multipliers.

(q) All other tasks as directed by ACAS (AD).

5. **Deputy Director of Air Defence (Operations) – DDAD (Ops)**. He is responsible to Dir Air Def (Ops) for:

(a) All matters pertaining to Air Defence operations.

(b) Progress / follow up on the CAS directives and IGAF visits reports after visit to Air Defence Bases / Units.

(c) Recce, movement, deployment and calibration of Air Defence assets.

(d) Planning / monitoring and analysis of probe missions.

(e) Airspace violations and flight plan irregularities.

(f) Complete up-gradation of equipment (hardware / software).

(g) Coordination with Ops Dte for providing flying effort for Air Defence Operations.

(h) All matters relating to MOUs and AAA.

(j) Liaison with CAA through Dte of Ops / ATS.

(k) Record of establishment / allocation / transfer of equipment of GMCCs/ MMCCs and other radar squadrons, including generators, radio relays and connectivity equipment.

(l) Maintaining operational data for quick availability of information.

(m) Coordination with other Directorates / Services for conduct of exercises (PAF, Army, Navy and International exercises).

(n) Ensuring availability of AD assets for tests and trials and coordinating their conduct.

(p) Preparation and record keeping of all briefs / presentations on Ops matters to the Air Staff.

(q) All other tasks as directed by Dir AD Ops.

6. **Assistant Director Air Defence Operations (Force Multipliers) – ADAD Ops (Force Multipliers)**. He is responsible to DDAD (Ops) for the following:

(a) Deal with all AD operational matters related to Force Multipliers (AEW&C, AAR, ISR platforms, EW and PDS).

(b) Monitoring of day to day Air Defence activities.

(c) Coordination for movement / deployment of Air Defence Units.

(d) Coordination with HQ ADC for conduct and monitoring of probe missions.

(e) Follow up action related to air violations and flight plan irregularities.

(f) All matters relating to MOUs and AAA deployments.

(g) Updating of RCI of radars and VPDs of SAM Units.

(h) Coordinating with Directorates of HPR and LLR regarding maintenance activities of radar Units.

(j) Preparation and record keeping of all briefs / presentations on Ops matters to the Air Staff.

(k) Maintaining and forwarding of Coalition flying data to the Air Staff.

(l) All other tasks as directed by Deputy Dir AD Ops.

7. **Deputy Director Air Defence Operations (GBADS) – DDAD Ops (GBADS)**. He is responsible to Dir Air Def (Ops) for the following:

(a) Training and operational activities of SAM Units.

(b) Monitoring the operational availability of SAMs specialist equipment and coordinating with Dte of Missiles and Dte of MUnitions on matters relating to serviceability of equipment.

(c) Scrutinizing SAM Units' ops deployment plans.

(d) Issue / revision and amendment of the relevant publications concerning SAMs.

(e) Evaluation of new SAM equipment planned for induction in the PAF and induction of newly acquired SAMs.

(f) Maintaining operational data for quick availability of information.

(g) Progress / Follow up of CAS / Air Staff directives, IGAF visits on issues related to PAF GBADS.

(h) All other tasks as directed by DAD Ops.

8. **Director Air Defence (Plans) – Dir AD (Plans)**. He is responsible to the ACAS (Air Def) for:

(a) Initiate and implement all actions related to identified Force goals of AD Branch in coordination with Dir AD (Policy).

(b) Formulate and revise AD War Plans and ensure safe custody of Top Secret Registry.

(c) Coordinate with HQ ADC on progress made with respect to preparatory actions for execution of respective War Plans.

(d) Coordinate and liaison with sister services and other organizations on matters pertaining to smooth execution of AD War Plans.

(e) Coordinate and liaison with Plans Sub Branch and other Directorates to fulfil the requirements of AD War Plans.

(f) Review & identify war plans related infrastructure requirements of AD Units, if required, and coordinate processing of land acquisition cases of War sites with Dir AD (SP&D).

(g) Ensure availability of AD war sites for quick deployment.

(h) Plan and coordinate swift deployment of AD assets at random intervals during peace time for ensuring war time preparation.

(j) Forecast Mob Reserve & other wartime requirements for AD Units and coordinate their provision with Dir AD (SP&D).

(k) Identify and initiate cases for Project Development Indigenization Board (PDIB) approvals / scaling cases pertaining to AD War Plans and coordinate their provisioning with Dir AD (SP&D).

(l) Identify and initiate cases for cases for AD budgetary requirements related to AD Ops / AD Plans in coordination with Dir AD (SP&D).

(m) Process all matters pertaining to Balloon Barrage Units.

9. **Deputy Director Air Defence (Plans) - DDAD (Plans)**. DDAD (Plans) is responsible to Dir AD (Plans) for the following:

(a) Custody and handling of Top-Secret Documents.

(b) Process and handle all matters pertaining to AD War Plans.

(c) Undertake scrutiny of WP observations and process incorporation of necessary changes in the relevant portions of WP.

(d) Coordinate with Directorates of Air Defence (SP&D / Policy) for allocation of resources (Mob Reserve, SMT / HMT, Security) required for smooth movement, deployment and Operations during any contingency / war.

(e) Coordinate with other Directorates for activation of all AD Units on C&M status and relocation of war sites, if required, as per Air Staff directives.

(f) Coordinate to practice swift deployments of AD Units for war time preparedness and addressing the observations.

(g) Evaluate AD war sites through arrangement of periodic visits by AD Units with respect to presence of screening / interference sources.

(h) Ensure coordination with sister services HQs / LEAs and other Directorates for safe and expeditious movement / deployment of AD assets during war times.

(j) Undertake any other task assigned by Dir AD (Plans).

10. **Deputy Director Air Defence (Operations Development) - DDAD (Ops Dev)**. He is responsible to Dir AD (Plans) for the following:

(a) Represent Dte of AD at other Directorates on matters pertaining to AD Ops development.

(b) Prepare ASR / URDs (pertaining to AD Ops requirements) in coordination with other stake holders.

(c) Prepare ROEs & SROEs in coordination with other agencies for proficient employment of AD assets.

(d) Work out war related infrastructure requirements of AD Units and if required, coordinate with Dte of AD (SP&D), land acquisition / infrastructure development cases pertaining to AD war sites.

(e) Process routine / additional funds requirements (MTG, Contingency etc) pertaining to ACAS (AD) Sectt.

(f) Identify and process cases for PDIB approvals pertaining to Dte of AD (Plans) in coordination with Dte of AD (SP&D).

(g) Coordinate with Dte of AD (SP&D) for processing of the cases for Standing Equipment and Scale Committee Meetings (TO&E) pertaining to war time resources

(h) Coordinate with Dte of AD (SP&D) to process AD budgetary requirements related to AD Plans.

(j) Undertake all matters pertaining to Balloon Barrage Units.

(k) Ensure Security & Administration of the ACAS (AD) Sectt.

(l) Undertake any other task assigned by Dir AD (Plans).

**Duties and Responsibilities Deputy Chief of the Air Staff (AD) Secretariat**

11. **Deputy Chief of the Air Staff (Air Defence)**. As a Principal Staff Officer he is responsible to Chief of the Air Staff (CAS) for the following:

(a) Assist the CAS on matters pertaining to Air Defence.

(b) Support / assist DCAS (Ops) in all matters related to Air Defence.

(c) Pursuance of Air Defence developmental strategy and Force Goals; identified by Ops Branch.

(d) Operational preparedness and readiness of Air Defence Units.

(e) Development, review and pursuance of Air Defence support plans, in coordination with Ops Branch.

(f) Provision of resources and capacity building of Air Defence Branch encompassing human resource, infrastructure and assets.

(g) All matters relating to maintenance, repair indigenous development, spare support and procurement of Air Defence radars, $C^2$ / automation systems and their associated equipment.

(h) Career progression of Officers of Air Defence branch in consultation with DCAS (Ops) and AOC, ADC.

(j) Evaluation, selection and induction of new Air Defence equipment and up-gradation of held equipment.

(k) Formulation of Air Defence budgetary requirement in coordination with DCAS (Ops).

(l) Jointly represent PAF on Air Defence matters at Tri-services forums along with DCAS (Ops).

(m) Evaluation of Air Defence Units for war preparedness.

(n) Basic training, system training and CTP of Fighter Controllers.

(p) Conduct of Operational Category Boards of Air Defence Officers.

(q) Airmen training of Air Defence Branch.

12. **Assistance Chief of the Air Staff (Resource Management) – ACAS (RM)**. The ACAS (RM) is responsible to the DCAS (AD) for the following:

(a) Assist DCAS (AD) on all matters related to Air Defence.

(b) Co-ordination and liaison with following agencies for Air Defence matters:

(i) JSHQ, Pak Army & Pak Navy

(ii) Operations, Plans, Support, Admin and Engineering Sub Branches at Air Headquarters to ensure smooth execution of peace / war time role of Air Defence Units.

(iii) Air Defence Command and Regional Air Commands.

(c) Liaison with Pak Army and Navy for requisite support to PAF AD deployments within Army and Navy setups respectively.

(d) Assisting DCAS (AD) on matters related to Air Defence Committee meeting and all other tri-services Air Defence aspects.

(e) Keeping DCAS (AD) Sectt abreast on Air Defence operational matters.

(f) Short-listing of Air Defence systems in pursuance of PAF Force-Goals

(g) Monitor selection, evaluation and induction process of new Air Defence equipment along with upgrade / modernization projects related to Air Defence for force development and acquisition in line with the Air Staff vision.

(h) To supervise all organizational matters relating to Air Defence Branch. Additionally, ensuring processing of Admin / Organizational cases including review of establishment(s), whenever change in organizational structure, role & task or assets of Air Defence Units is deemed necessary.

(j) Keeping close liaison with Personnel Branch and HQ ADC regarding matters related to career progression of AD officers.

(k) To supervise development, review and pursuance of Air Defence support plans in coordination with concerned Branches at AHQ.

(l) Scrutinizing and finalizing the resource requirements of Air Defence Units encompassing human resource, infrastructure and equipment / assets.

(m) Ascertaining requirements of Ops works / infrastructure pertaining to AD Units through HQ ADC and its pursuance with Dte of Works.

(n) Planning yearly budgetary requirements for new procurements as well as for maintenance of Air Defence related equipment in coordination with ADC and concerned secretariats at AHQ.

(p) Ensure judicious utilization of allocated funds in line with AHQ policies.

(q) To coordinate with ACAS (NW) and ACAS (AD) for provisioning of communication facilities during peace and war.

(r) Plan availability of requisite security troops and security equipment for Air Defence sites, particularly those located off-base, in coordination with Security Sub-Branch.

(s) Ensure implementation of remedial measures to address the observations / deficiencies related to AD Branch, highlighted during the visits of CAS, PSOs and IG Branch to Air Defence Units.

(t) Formulation of staff level talks / agenda points for the Air Staff and pursuance of agreed / finalized points of different forums

(u) Formulation of criteria regarding selection of Air Defence officers for foreign assignments and mutual exchange visits to allied countries.

13. **Director Air Defence (Policy and Resource Management) – Dir AD (Pol & RM)**. He is responsible to the ACAS (RM) for:

(a) Shortlisting / selection of Air Defence systems in pursuance of PAF Force Goals and in accordance with their viability for PAF.

(b) Processing and coordinating with concerned Dtes and HQ ADC during entire process of evaluation / induction of new equipment and any indigenous upgrade in Air Defence.

(c) Providing regular updates on Air Defence acquisition / development projects to the Air Staff.

(d) Processing cases related to Air Defence policy and organizational matters including relocation of Air Defence Units and issuance of Admin instructions.

(e) Processing and finalization of new / review of Establishment cases pertaining to AD Units in coordination with concerned directorates and HQ ADC.

(f) Preparing and following progresses on CAS / PSOs directives and IG Branch inspection points on matters related to Air Defence Branch.

(g) Coordination and correspondence with JSHQ and Services HQs on Air Defence matters related to Dte of AD (Policy & RM).

(h) Formulation of Staff level talks / agenda points for the Air Staff visits and pursuance of agreed / finalized points at different concerned forums.

(j) Processing and managing conduct of bilateral visits to Air Defence setups by Allied and PAF delegations.

(k) Assisting ACAS (RM) on career progression of Air Defence officers in coordination with Personnel Branch.

(l) Representing Air Defence Branch as a specialist member in selection panel of SSC / BLPC AD officers or wherever Branch opinion is required.

(m) Any additional task as specified by ACAS (RM).

14. **Deputy Director Air Defence (Projects) – DDAD (Proj)**. He is responsible to Dir AD (Policy & RM) for:

(a) Providing input to concerned directorates for formulation of user Requirement Document (URD) and Air Staff Requirement (ASR) pertaining to new Air Defence equipment.

(b) Analyzing, scrutiny and academic evaluation of technical proposals for Air Defence new systems / upgrades in light of approved ASR and their suitability for PAF.

(c) Conduct of evaluation trials during selection and subsequent induction of Air Defence systems in PAF as per defined Force Goals.

(d) Coordinating with OR&D Sub-Branch and other concerned directorates during various stages of induction process besides formulation of Factory and Site Acceptance Procedures.

(e) Maintaining liaison with concerned agencies at AHQs and HQ ADC to pursue and track conduct of Ops evaluation / trials of Air Defence new / upgrade equipment.

(f) Following up Air Defence projects and systematically maintain their updated record for regular appraisal to the Air Staff.

15. **Deputy Director Air Defence (Policy and Organization) – DDAD (Policy & Org)**. He is responsible to Dir AD (Policy & RM) for:

(a) Processing and coordination with concerned directorates at AHQ for new / review of establishment cases pertaining to Air Defence setups and subsequent issuance of necessary Admin Instructions.

(b) Processing relocation cases of Air Defence Units in coordination with concerned directorates at AHQ and HQ ADC.

(c) Coordination with Security Branch at AHQ for availability of requisite security troops for Air Defence Units.

(d) Compilation of progresses / follow-up of CAS / Air Staff directives and IG Branch inspection points on matters related to Air Defence setups.

(e) Maintaining record of resource establishment (HR, Ops Eqpt, IT Eqpt etc) in respect of Air Defence Units.

(f) Looking after administrative and discipline matters of the Directorate.

16. **Deputy Director Air Defence (Coordination) – DDAD (Coord)**. He is responsible to Dir AD (Policy & RM) for:

(a) Coordination and correspondence with JSHQ and Services HQs on Air Defence matters related to Dte of AD (Pol & RM).

(b) Coordination with concerned directorates of AHQ related to Tri-services Air Defence matters including Air Defence Committee meeting points.

(c) Compilation of agenda / talking points of Air Staff visits and Staff-level forums.

(d) Progress / follow-up of all agenda points related to Working Groups in coordination with Directorate of Infrastructure and respective DAs.

(e) Initiation and processing of part cases pertaining to visits of Allied / Own Air Defence delegations.

(f) Coordination with concerned sections at MoD and MoFA for issuance of Government letters pertaining to visits of Allied delegations in Air Defence domain.

(g) Coordination with FA (AF) / Dte of Accts and MoFA for issuance of Government letters pertaining to visits of PAF / foreign delegations in Air Defence Branch.

(h) Liaison with concerned specialist Dtes and MoD for issuance of AFIs / Corrigendum related to Air Defence Branch.

(j) Compilation of Air Defence returns / updates for perusal of the Air Staff.

17. **Assistant Director Air Defence (Projects) – ADAD (Proj)**. He is responsible to DD AD (Proj) for:

(a) Liaise with concerned Dtes and HQ ADC pertaining to evaluation / trials of new Air Defence inductions / upgradations.

(b) Preparing appraisals on Ops related matters (weekly IAF update, airspace violations, monthly briefs, Air Staff Presentations) for the perusal of the Air Staff.

(c) Assisting in coordination with concerned directorates at AHQs for formulation of URDs & ASRs.

(d) Assisting in evaluation of technical proposals shared by OEM regarding Air Defence systems.

(e) Tracking movement and deployment of AD assets for Air Staff appraisal.

18. **Assistant Director Air Defence (Policy and Organization) - AD AD (Pol & Org)**. He is responsible to DD AD (Policy & Org) for:

(a) Coordination with concerned agencies / directorates at AHQ and ADC for finalization of progresses on Air Staff directives.

(b) Processing of Org Memos and Admin Instructions of Air Defence Units.

(c) Assisting in maintaining updated record of resource establishment (HR, Ops Eqpt, IT Eqpt etc) for Air Defence Units.

(d) Maintaining establishment data of Air Defence Units for quick and accurate availability of information.

(e) Assisting DDAD (Policy & Org) in smooth functioning of administrative matters and discipline issues of the Directorate.

19. **Director Air Defence (Support Plans & Development) – Dir AD (SP&D)**. He is responsible to ACAS (RM) for:

(a) Maintaining close liaison with DAD (Plans) and HQ ADC in periodic review of AD Ops Plans.

(b) Formulation of AD Support Plans in coordination with HQ ADC, based on the approved AD Deployment Plans, and inclusion of the AD resource requirements in WP Support Plans.

(c) Pursuance of support plans with respective Air Headquarters Directorates so as to ensure timely allocation / provisioning of resources to AD Units in line with War Plans requirements.

(d) Reviewing existing authorization / holding of various resources in coordination with HQ ADC so as to timely address deficiencies affecting War Plan execution by AD Units.

(e) Reviewing existing scaling of AD Units in co-ordination with HQ ADC. Process new scaling cases with concerning AHQ Directorates for subsequent issuance of Govt letters accordingly.

(f) Analyzing yearly Budgetary requirements for AD related projects and requisite War Preparedness of AD Units. Subsequently, processing the same for Air Staff approval and projection/ approval during BPC Meetings.

(g) Maintaining close coordination with AHQ (Dte of Budget) for allocation of approved AD related funds and with concerned Specialist Directorates at AHQs to ensure timely indenting, procurement and disbursement of funds under respective Heads.

(h) Maintaining close liaison with AHQ (Dte of CM), DP (Air) and respective Specialist Directorates at AHQ throughout the procurement process for timely disbursement of allocated funds as per project timelines.

(j) As Specialist Dte for AD Mob Reserve Equipment, reviewing the Mob Reserve authorization of various AD Units, vetting BOO conducted by AD Units for disposal of Mob Reserve items, and undertaking entire process of procurement of deficient Mob Reserve after necessary budget allocation.

(k) Maintaining close liaison with HQ ADC for initiation and scrutinization of all AD Ops works and subsequently coordinate with AHQ (Dte of Works) for their timely processing and allocation of funds.

(l) Coordinate with AHQ (Dte of Infra) for inclusion of AD developmental plans in line with future PAF plans.

(m) Undertaking coordination with GHQ and NHQ for support requirements to AD field deployments located within Army / Navy setups.

20. **Deputy Director Air Defence (Support Plans) – DDAD (Supp Plan)**. He is responsible to Dir AD (SP&D) for the following:

(a) Maintaining close liaison with AHQ (Dte of Plans), DAD (Plans) and HQ ADC in periodic review of AD Ops Plans.

(b) Coordinating with HQ ADC for timely completion of AD Support Plans and ascertaining their applicability with ops deployments and specific field requirements.

(c) Ensuring inclusion of the AD resource requirements in WP Support Plans. Subsequently, coordinating with concerning AHQ Directorates for inclusion of AD requirements in their respective support plans for subsequent provisioning of resources (whenever required).

(d) Reviewing Mob Reserve authorization of various AD Units, vetting BOO conducted by AD Units for disposal of Mob Reserve items, and undertaking entire process of procurement of deficient Mob Reserve after necessary budget allocation.

(e) Ascertaining existing authorization and scaling of various resources in coordination with AHQ (Dte of LAS) and HQ ADC and if required, initiate new scaling cases in support of field deployments.

(f) Maintaining close liaison with concerning Air Headquarters Directorates for timely processing and provisioning of resources including Ration, POL, Clothing etc for sustained field deployments by AD Units.

(g) Coordinating with HQ ADC for arrangement of critically required resources to undertake peacetime field deployments and if required, initiate cases for immediate funds requirements for procurement of required resources.

(h) Maintain close liaison with AHQ [Dte of AD (Plans)] and HQ ADC for periodical review of AD Support Plans and AFL 4-2 for their accuracy and efficacy.

(j) Any other task, as specified by ACAS (RM) and DAD (SP&D).

21. **Deputy Director Air Defence (Infra Development) – DDAD (Infra Dev)**. He is responsible to Dir AD (SP&D) for the following:

(a) Coordinating with Air Defence Command for timely initiation, scrutinization and rationalization of Ops works for subsequent infrastructure development of various AD projects.

(b) Initiating and prioritizing cases for seeking Air Staff approval through AHQ (Dte of Infra) & (Dte of Works) considering ops urgency of the requirements.

(c) Coordinating with AHQ (Dte of Works) and HQ ADC for initiating BOOs as per the Ops requirements.

(d) Maintaining close coordination with RACs and HQ ADC for timely completion and submission of BOOs as per the AHQ guidelines.

(e) Carrying out scrutiny / endorsement of infrastructure related BOOs by DAD (SP&D) and same are to be forwarded to AHQ (Dte of Works) for subsequent processing / allocation of funds.

(f) Maintaining close liaison with AHQ (Dte of Works) for timely resolving any anomalies / objections as highlighted during BOO processing.

(g) Maintaining close liaison with HQ ADC for periodic updates / progress of Ops Works for perusal of the Air Staff.

(h) Ensuring timely submission of progress reports on CAS Directives, Air Board Decisions, PSOs meetings and PSOs / IGs visits reports in coordination with specialist Directorates / Regional Commands / Bases.

(j) Administration of Dte of Air Def (SP&D) setup.

22. **Assistant Director Air Defence (Budget) – ADAD (Bgt)**. He is responsible to Dir AD (SP&D) for the following:

(a) Analyzing yearly Budgetary requirements for AD related projects and requisite War Preparedness of AD Units.

(b) Coordinating with various AD sponsoring / OPI Directorates for timely sharing of their associated CAS / PDIB approved AD related cases for subsequent inclusion in overall AD budgetary requirements.

(c) Preparing comprehensive AD financial estimates for seeking Air Staff approval sequel to which AD requirements and related funds are to be projected under respective heads for projection/ approval during BPC Meetings.

(d) Coordinating with AHQ (Dte of Bgt) for allocation of funds as approved under various heads during BPC to specialist Directorates / AD field Units.

(e) Maintaining close liaison with various AHQ agencies including Specialist Directorates (for timely raising of indents), Dte of Bgt (for funds availability), Dte of CM (for processing of indents) and DP (Air) (for contractual formalities, procurement / delivery of equipment and timely resolving payment modalities).

(f) Coordinating with Dte of CM, Dte of LAS, DP (Air) and HQ ADC for timely disbursement of current and last financial year carryover funds.

(g) Maintaining close liaison with Dte of Bgt, Dte of LAS, Dte of CM and DP (Air) for updating of funds consumption state and prepare periodic funds utilization cases for subsequent perusal of Air Staff through FUR meetings.

(h) Processing scaling cases pertaining to AD setups in AHQs Scale Committee Meetings (SCMs) at Dte of LAS for subsequent issuance of Govt Letters.

23. **Assistance Chief of the Air Staff (Training & Evaluation) – ACAS (Trg & Eval)**. The ACAS (Trg & Eval) is responsible to the DCAS (AD) for the following:

(a) To formulate and review Cyclic Training Programme (CTP) of Fighter Controllers, and monitor its effective implementation.

(b) To formulate, review and implement Annual Training Programme (ATP) for Fighter Controllers, Radar Operators and Allied / Foreign Trainees.

(c) To evaluate Ops preparedness of Air Defence Units and their adherence to laid down policies / procedures.

(d) To monitor categorization status of Fighter Controllers and organize their Ops Categorization accordingly.

(e) To review categorization, evaluation and training policies for Air Defence Units

(f) Allocate GCI task to Air Defence Units / Training institutes and ensure task completion in coordination with ACAS (AD) and Headquarters Air Defence Command.

(g) To coordinate with ACAS (AD), ACAS (RM) and Personnel Branch for detailing of Fighter Controllers for in-land, foreign and professional courses.

(h) To plan and organize GCI and GBADS competitions in coordination with Headquarters Air Defence Command.

(j) To review and monitor BS Aviation programme of Air Defence Trainees and fulfil their training requirements as per HEC policy.

(k) To plan and evaluate Air Defence exercises in-coordination with ACAS (AD) and Senior Air Staff Officer (SASO), Air Defence Command, PAF.

(l) To plan and conduct seminars on topics related to Air Defence systems / latest developments.

(m) To review Air Defence publications and disseminate data on Tactics, Techniques and Procedures (TTPs) in coordination with Tactics Development School, Air Power Centre of Excellence (TDS, ACE).

24. **Director Air Defence (Training & Evaluation) – Dir AD (Trg & Eval)**. He is responsible to ACAS (Trg & Eval) for the following:

(a) To prepare, review and implement Cyclic Training Programme (CTP) of Fighter Controllers at Air Defence Units.

(b) To prepare, review and implement Annual Training Programme (ATP) for Fighter Controllers and Radar Operators, in coordination with Headquarters Air Defence Command.

(c) To formulate, review and implement ATP for Allied / Foreign trainees in coordination with Air Defence training institutes.

(d) To plan and conduct audit to evaluate the training standards and operational health of Air Defence Units.

(e) To plan, prepare, review and conduct Ops Categorization of Fighter Controllers.

(f) To allocate GCI task as per CTP and requirements of Air Defence training institutes and ensure task completion, in coordination with Dte of AD (Ops) and Air Defence Command, PAF

(g) To monitor BS Aviation program of Air Defence trainees and fulfil training requirements in accordance with HEC policy.

(h) To plan and conduct Inter-Squadron Competitions (GCI / Spada System) for enhancing system proficiency and ameliorate professional skills of Air Defence officers.

(j) To monitor visit / inspection reports of CAS, DCAS (AD) and IG Branch and coordinate for progress on matters pertaining to Air Defence training.

(k) To prepare, review and publish Air Defence publications, in coordination with ACE, PAF.

(l) To prepare and publish Air Defence Magazine / Journal and provide relevant reading material to Air Defence Units.

(m) To continually evaluate threat vis-à-vis changes in employment concepts and incorporate changes in training of Fighter Controllers (Tactics, Techniques and Procedures), in coordination with ACE, PAF.

25. **Deputy Director Air Defence Training & Evaluation (GCI) – DDAD {Trg & Eval (GCI)}**. He is responsible to Dir AD (Trg & Eval) for the following:

(a) To ensure training of Fighter Controllers as per Cyclic Training Programme (CTP) / laid down policies.

(b) To ensure completion of requisite GCI task of Fighter Controllers, in coordination with DDAD (Ops) and Headquarters Air Defence Command.

(c) To coordinate training of Air Defence personnel (Air Defence officers & Radar Operators) on existing and newly inducted Air Defence systems.

(i) To plan system courses as per inland ATP and detail AD personnel in coordination with Personnel Branch and Headquarters Air Defence Command.

(ii) To plan and detail Fighter Controllers for courses at ABMS and maintain record for systematic training commensurate to individual's experience / professional milestone.

(d) To assist in planning and conduct of categorization and training evaluation.

(e) To assist in scheduling, planning and conduct of GCI competition while, maintaining a close coordination with Air Defence Command.

(f) To deal with correspondence regarding training activities of Air Defence Units.

(g) To track and update progress of visit / inspection reports of CAS, DCAS (AD) and IG Branch.

(h) To assist in detailing Fighter Controllers for exposure on AEWACS / Fighter Squadrons in coordination with Directorate of Ops and Headquarters Air Defence Command.

(j) To monitor implementation of FAM / Induction training of Radar Operators at SMCC / GCI Units.

(k) To continually evaluate threat in respect of new inductions / developments vis-à-vis employment concepts and suggest changes in Fighter Controllers training, in coordination with ACE, PAF.

(l) To evaluate the efficacy of Vision Based Simulator system (VCC / COMTAS) to impart meaningful training as per CTP and suggest changes / up-gradation in the simulator system.

26. **Deputy Director Air Defence Training & Evaluation (GBADS) – DDAD {Trg & Eval (GBADS)}**. He is responsible to Dir AD (Trg & Eval) for the following:

(a) To ensure training of Fighter Controllers as per Cyclic Training Programme (CTP) / laid down policies.

(b) To coordinate training of SAM operators (Air Defence officers & Radar Operators) on existing and newly inducted GBAD Systems:

(i) To plan and conduct GBADS training courses as per inland ATP and detail AD personnel in coordination with Personnel Branch and Headquarters Air Defence Command.

(ii) To assist in detailing Air Defence officers for SPADA training Module at CCS and GBADS courses at No 455 & GBADS School, in coordination with training institutes.

(c) To conduct evaluation and audit of SAM Units and monitor progress on audit reports.

(d) To assist in scheduling, planning and conduct of SPADA competition along with preparing engagement profiles and system quiz, in consultation with HQs ADC.

(e) To deal with correspondence regarding training activities of SAM Units and ensure availability of requisite study material in SAM Units.

(f) To monitor implementation of familiarization / induction training programme of Radar Operators at SAM Units.

(g) To continually evaluate threat in respect of new inductions / developments vis-à-vis employment concepts and suggest changes in SAM operators training, in coordination with ACE, PAF.

27. **Assistance Chief of the Air Staff (Radar Engineering) – ACAS (RE)**. The ACAS (AD) is responsible to the DCAS (AD) for the following:

(a) Policy matter relating to maintenance of Air Defence ground radar systems, CRCs, Air Defence Data Automation, aircraft simulators and associated data handling equipment, power generators and Air Defence centre's environmental control system.

(b) Provide specialist support for evaluation, selection and induction of new Air Defence ground radar systems, Air Defence automation systems, aircraft simulators and associated equipment.

(c) Coordination and liaison with Dte of Air Defence and Air Defence Command for maintenance issues of Air Defence assets.

(d) Policy matters regarding up-gradation / modifications in High Level / Low Level radars, Air Defence Automation System, aircraft simulators and other ancillary equipment as suggested by manufacturers.

(e) Tasking of development / special tasks to NO 108 AED and indigenous development projects at No 118 SED, PAF.

(f) Formulation of policy for training on Air Defence System and support equipment.

(g) Formulation of maintenance support plans for Air Defence Sensors, aircraft simulators and related Systems.

(h) Coordination with ACAS (Log) and supply depots for effective spares support and procurement actions.

(j) Development plans and budget forecast estimates in respect of Air Defence, High Power / Low Level Radars, Air Defence Automation System, aircraft simulators and ancillary / support equipment.

(k) Ensuring actions on the CAS's directives, staff visits and inspectorate reports on matters relating to Air Defence equipment.

(l) To ensure effective depot level maintenance support to all Air Defence Units.

(m) Co-ordination and liaison with PAC Kamra, KARF, PEC, KRL, NDC, AWC or any other agency on matters relating to Air Defence Systems repairs, indigenous development etc.

28. **Director Air Defence Automation Systems – Dir ADAS**. He is responsible to the ACAS (RE) for:

(a) Organizing, controlling, and directing the maintenance activities for Vision Systems and their ancillary equipment.

(b) Formulating policies, AFOs and AFLs for the maintenance support of Vision Systems.

(c) Planning and forecasting, necessary upgrades to existing Vision Systems and induction of new systems to support Air Defence operations.

(d) Analysis of critical defects which affect system performance and developing their solutions.

(e) Ensure actions on CAS directives.

(f) Ensuring implementation of Air Staff directives, IGAF and staff visit reports.

(g) Coordinating with other services / specialist directorates for support and resources required for smooth operation of Vision Systems.

(h) Directing and controlling tasks required for modification, technical investigation and emergency servicing through No 108 AED / 118 SED.

(j) Reviewing and definitizing the requirement of spares for providing defined Air Defence Operations.

(k) Planning and forecasting budget estimates.

(l) Ensure PALMS utilization by all concerned Units for supply and maintenance management.

(m) Management of technical orders, SOPs and maintenance instructions.

29. **Deputy Director Vision Systems (DD VS)**. He is responsible to Dir ADAS for:

(a) Coordinating with other depots / Units for support and resources required for smooth operations / upgrade of Vision Systems.

(b) Tracking of installed hardware serviceability and availability.

(c) Monitoring manning level of automation sites, their training status and forecasting training requirements.

(d) Coordinating with operations / training branch to vet the operational requirement for R&D projects and issue feasibility studies to No 108 AED / No 118 SED.

(e) Periodic definitization of Vision Systems' floats for spare support, planning and preparing budget estimates.

(f) Compilation, issue and revision of AFOs / AFIs for Vision Systems maintenance.

(g) Planning and arranging indigenous development of critical Vision Systems spares.

(h) Arranging functional check of newly developed / procured Vision Systems spares.

(j) Follow up / replacement of newly procured / developed and under warranty items through supplier / manufacturer.

(k) Preparation of Air Staff Presentation data and carrying out defect trend analysis and plan remedial measures.

(l) Issuing tasks to No 118 SED / No 108 AED for hardware and software maintenance of Vision Systems.

(m) Ensuring efficient functioning of the orderly room of the Dte.

(n) Raising of draft indents for procurement of spares.

(p) Processing of Hi-value item cases for PSO's approval.

(q) Maintaining the accounts of funds allocated to directorate under various heads.

(r) Follow-up all RRCs and keep up-to-date records of all purchase orders.

30. **Assistant Director Vision Systems (AD VS)**. He is responsible to DD VS for:

(a) Formulation / implementation of policies governing following Vision Systems and their ancillary equipment:

(i) ADOC / SMCCs

(ii) GMCCs

(iii) ECS

(iv) UPS / MUPS

(v) Generators

(b) Monitoring operational serviceability and performance of above-mentioned equipment installed / held at Air Defence field Units.

(c) Liaison with No 302 ALD and Project Vision for speedy actions on ROCP / AROCP demands for quick recovery of un-serviceabilities.

(d) Ensuring timely preventive maintenance to enhance reliability support equipment.

(e) Forecasting components / assemblies by analyzing periodic reviews, for the future requirements.

(f) Issuing of SOPs for the above equipment and ensure follow up actions.

(g) Keeping an up-to-date record of all personnel (officers and airmen) trained on Vision Systems equipment.

(h) Analysis of serviceability and other RCNs received from field Units.

(j) Training of officers / personnel for Vision Systems and ancillary equipment.

(k) Monitoring and implementation of directives issued for Vision Systems maintenance.

(l) Keeping an updated record of authorized pre-issue / bench stock and monitoring the deficiencies of support equipment.

(m) Monitoring the progress of special tasks issued to No 108 AED and No 118 SED.

(n) Provisioning of adequate amount of stationery and other office requirements.

(p) Arranging security clearance of individuals visiting the directorate.

(q) Up-keeping of all warranty claims and their consistent follow up.

(r) Maintenance of complete record of technical documents related to Vision Systems and to ensure incorporation of latest changes.

(s) Ensuring up-keeping of furniture record and their replacement / repair.

(t) Carrying out periodic check of directorate inventory.

31. **Deputy Director System Integration – DD SI**. He is responsible to Dir ADAS for:

(a) Integration issues related to all Sensors, Vision Systems with NCW systems / elements.

(b) Act as a contact point for resolving system integration problems.

(c) Maintain a close liaison with all Directorates and field Units related to sensors and Vision Systems.

(d) Foresee, proactively monitor and address system integration issues.

(e) Issue system integration related tasks to Centres of Excellence and Engineering Depots.

(f) Preparation of presentations, part cases, briefs and summaries on system integration.

(g) Coordinating with No 118 SED / 108 AED and OEM for software development to interface newly inducted equipment with Vision Systems, Sensors and NCW systems.

(h) Monitoring of integration related software configuration, control and documentation for Sensors and Vision systems.

32. **Director Long Range Radars (Dir LRR)**. Director is responsible to ACAS (RE) for:

(a) Organizing, controlling, directing and monitoring all echelons of maintenance for TPS series of radars (TPS-43G, TPS-43J, TPS-77) and their ancillary equipment including AC Plants and Power Generators.

(b) Planning and formulating policies, AFOs, AFLs for maintenance, new inductions and upgrading of existing systems related to HL Radars.

(c) Planning and forecasting, necessary upgrades to existing Systems and induction of new systems to support Air Defence operations.

(d) Forecast spares requirement for budget estimates and follow-up actions in this regard. Subsequently initiate / raise case with concerned agencies for procurement of same.

(e) Technical evaluation of radars systems, whenever requirement arises.

(f) Review of dormant spares, their disposal and purification of ALMS data.

(g) Carrying out Defect Trend Analysis and scrutiny / analysis of RCNs.

(h) Obsolescence management.

(j) Reviewing and definitizing the requirement of spares for providing defined Air Defence Operations.

(k) Planning and forecasting budget estimates.

(l) Ensure ALMS utilization by all concerned Units for supply and maintenance management.

(m) Management of technical orders, SOPs and maintenance instructions.

(n) Analysis of critical defects which affect system performance and developing their solutions.

(p) Ensure actions on CAS directives.

(q) Ensuring implementation of Air Staff directives, IGAF and staff visit reports.

(r) Conduct of cyclic System Maintenance Reviews and finalization of ERAB / UAL and TO&E cases of the mandated assets.

(s) Any other task assigned by ACAS (RE).

33. **Deputy Director YLC-2 & YLC-2A – (DD YLC-2 & YLC-2A)**. Deputy Director YLC-2 & YLC-2A is responsible to Director LRR for:

(a) Coordinating with Depots and field Units for smooth operation / up-gradation of YLC-2 & YLC-2A Radars and ancillary systems including generators and air conditioners.

(b) Tracking the serviceability and availability of radars and ancillary equipment and initiating remedial actions for recovery of prolonged un-serviceabilities.

(c) Planning YLC-2 & YLC-2A systems training requirements and coordinating with different agencies for timely conduct of planned training courses.

(d) Periodic definitization of rotables for YLC-2 & YLC-2A radars and ancillary equipment for planning the spares / budget requirements.

(e) Compilation and revision of AFOs for the maintenance of YLC-2 & YLC-2A radars and ancillary equipment.

(f) Planning and arranging indigenous development of critical YLC-2 & YLC-2A and ancillary equipment spares.

(g) Follow up with supplier / OEM on technical issues related to YLC-2 & YLC-2A radars, IFF systems and other ancillary equipment.

(h) Preparation of Air Staff Presentation data and carrying out defect trend analysis and plan remedial measures.

(j) Issuing tasks to No 108 AED / No 104 AED for maintenance / up-gradation of YLC-2 & YLC-2A and ancillary equipment.

(k) Providing necessary input to DCM & DRRCs for raising of indents and processing of procurement orders.

(l) Processing of Hi-value item cases for PSO’s approval.

(m) Follow-up all RRCs and keep up-to-date records of all purchase orders.

(n) Arranging functional check of random samples selected from the spares delivered through RRCs.

(p) Monitoring the allocation / utilization of funds allocated under different heads.

(q) Ensuring efficient functioning of the orderly room of the Dte.

(r) Management of technical orders, SOPs and maintenance instructions. Coordinating with other depots / Units for support and resources required for smooth operations / upgrade of Vision Systems.

(s) Follow-up all RRCs and keep up-to-date records of all purchase orders.

(t) Liaison with No 104 AED along with exploration of local market to finalize modalities pertaining to the induction of new and maintenance of existing Generators.

(u) Exploring indigenous manufacture and / or repair capability and maintaining liaison with DED, APF, No 108 AED, Local Academia and other stakeholders.

(v) Liaison with APF for Overhauls / Refurbishments.

(w) Liaison with OEM, APF, DED and other stockholders to finalize modalities pertaining to ToT and System's upgrades.

(x) Liaison with OEM and other stack holders to finalize modalities pertaining to induction of new systems.

(y) Formulation & review of documentation, PMIs Training Schedules (TRC, SEP etc.) for new inductions.

(z) Formulation of Indents and finalization of contracts in close liaison with DP (Air).

(aa) Any other task assigned by Dir LRR.

34. **Deputy Director TPS-77 & TPS-77 MRR – (DD TPS-77 & TPS-77 MRR)**. Deputy Director TPS-77 & TPS-77 MRR is responsible to Director LRR for:

(a) Serviceability and maintenance management of TPS-77 / TPS-77 MRR equipment.

(b) Obsolescence management for TPS-77 / TPS-77 MRR.

(c) Compilation of related AFLs / AFOs / Maint Advisory Notices and other policies for Dir LRR.

(d) Issuance of modification leaflets applicable to TPS-77 / TPS-77 MRR.

(e) Issuance of related tasks to No 108 AED.

(f) Forecasting budget requirements for TPS-77 / TPS-77 MRR.

(g) Ensuring timely preventive maintenance to enhance reliability support equipment.

(h) Analysis of serviceability and other RCNs received from field Units.

(j) Keeping an updated record of authorized pre-issue / bench stock and monitoring the deficiencies of support equipment.

(k) Initiating spares requirement and requisite liaison with Dtes of Contract Management and RRC for provision of same.

(l) Plan training courses for officers and men on TPS-77 / TPS-77 MRR equipment.

(m) Coordinate with HQ ADC, HQ SMCCs, No 108 AED and No 118 SED to carry out functional testing of repaired LRUs, plan scheduled maintenance and EFMs on due dates.

(n) Vetting TPS-77 / TPS-77 MRR related reviews / reports, processing of cases for Hi value items and monitoring Site and Theatre spares deficiencies through use of ALMS.

(p) Exploring indigenous manufacture and / or repair capability and maintaining liaison with DED, APF, No 108 AED, Local Academia and other stakeholders.

(q) Technical evaluation of TPS-77 / TPS-77 MRR, whenever requirement arises.

(r) Liaise with OEM for all problems beyond the scope of skill level of local manpower. This includes repair of BCR items, replacement of warranty items till expiry of warranty.

(s) Liaison with No 104 AED along with exploration of local market to finalize modalities pertaining to the induction of new and maintenance of existing Generators.

(t) Liaison with APF for Overhauls / Refurbishments.

(u) Liaison with OEM, APF, DED and other stakeholders to finalize modalities pertaining to ToT and System’s upgrades.

(v) Liaison with OEM and other stakeholders to finalize modalities pertaining to induction of new systems.

(w) Formulation & review of documentation, PMIs Training Schedules (TRC, SEP etc.) for new inductions.

(x) Formulation of Indents and finalization of contracts in close liaison with DP (Air).

(y) Any other task assigned by Dir LRR.

35. **Deputy Director TPS-43 – (DD TPS-43)**. Deputy Director TPS-43 is responsible to Director LRR for:

(a) Serviceability and maintenance management of TPS-43G/J radars’ equipment.

(b) Obsolescence management for TPS-43G/J radars.

(c) Compilation of related AFLs / AFOs / Maint Advisory Notices and other policies for Dir LRR.

(d) Issuance of modification leaflets applicable to TPS-43G/J radars.

(e) Issuance of related tasks to No 108 AED.

(f) Forecasting budget requirements for TPS-43G/J radars.

(g) Initiating spares requirement and requisite liaison with Dtes of Contract Management and RRC for provision of same.

(h) Plan training courses for officers and men on TPS-43G/J equipment.

(j) Coordinate with HQ ADC, HQ SMCCs, No 108 AED to carry out functional testing of repaired LRUs, plan scheduled maintenance and EFMs on due dates.

(k) Vetting TPS-43G/J related reviews / reports, processing of cases for upgrades, Hi-value items and monitoring Site and Theatre spares deficiencies through use of ALMS.

(l) Explore indigenous manufacture and / or repair capability and maintain liaison with Engg Depots, Local Market, DGMP and Dte of Engineering Development.

(m) Technical evaluation of TPS-43G/J radars, whenever requirement arises.

(n) Plan / Evaluate new enhancements vis-à-vis Air Defence needs of PAF.

(p) Liaison with APF, No 108 AED, Local Academia and other stakeholders for planning and arranging indigenous development of critical TPS-43G/J spares and ancillary equipments.

(q) Liaison with No 104 AED along with exploration of local market to finalize modalities pertaining to the induction of new and maintenance of existing Generators.

(r) Liaison with APF for Overhauls / Refurbishments.

(s) Liaison with OEM, APF, DED and other stakeholders to finalize modalities pertaining to ToT and System's upgrades.

(t) Liaison with OEM and other stakeholders to finalize modalities pertaining to induction of new systems.

(u) Formulation & review of documentation, PMIs Training Schedules (TRC, SEP etc.) for new inductions.

(v) Formulation of Indents and finalization of contracts in close liaison with DP (Air).

(w) Any other task assigned by Dir LRR.

36. **Deputy Director YLC-6 & YLC-18A – (DD YLC-6 & YLC-18A)**. Deputy Director YLC-6 & YLC-18A is responsible to Director LRR for:

(a) Coordinating with Depots and field Units for smooth operation / up-gradation of YLC-6 & YLC-18A radars and ancillary systems including IFF JZ / XW629MS, generators and air conditioners.

(b) Tracking the serviceability and availability of radars and ancillary equipment and initiating remedial actions for recovery of prolonged un-serviceabilities.

(c) Planning YLC-6 & YLC-18A systems and ancillary equipment training requirements and coordinating with different agencies for timely conduct of planned training courses.

(d) Periodic definitization of rotables for YLC-6 & YLC-18A radars and ancillary equipment for planning the spares / budget requirements.

(e) Compilation and revision of AFOs for the maintenance of YLC-6 & YLC-18A radars and ancillary equipment.

(f) Liaison with APF, No 108 AED, Local Academia and other stakeholders for planning and arranging indigenous development of critical YLC-6 & YLC-18A spares and ancillary equipment.

(g) Follow up with supplier / OEM on technical issues related to YLC-6 & YLC-18A radars.

(h) Preparation of Air Staff Presentation data and carrying out defect trend analysis and plan remedial measures.

(j) Issuing tasks to No 108 AED for maintenance / up-gradation of YLC-6 & YLC-18A radars and ancillary equipment.

(k) Providing necessary input to DCM & DRRCs for raising of indents and processing of procurement orders.

(l) Processing of Hi-value item cases for PSO's approval.

(m) Follow-up all RRCs and keep up-to-date records of all purchase orders.

(n) Arranging functional check of random samples selected from the spares delivered through RRCs.

(p) Monitoring the allocation / utilization of funds allocated under different heads.

(q) Liaison with No 104 AED along with exploration of local market to finalize modalities pertaining to the induction of new and maintenance of existing Generators.

(r) Formulation & review of documentations, PMIs Training Schedules (TRC, SEP etc.) for new inductions.

(s) Liaison with OEM, APF, DED and other stakeholders to finalize modalities pertaining to ToT and System's upgrades.

(t) Liaison with OEM and other stakeholders to finalize modalities pertaining to induction of new systems.

(u) Liaison with APF for Overhauls / Refurbishments.

(v) Formulation of Indents and finalization of contracts in close liaison with DP (Air).

(w) Any other task assigned by Dir LRR.

37. **Assistant Director Secondary Surveillance Radar – (AD SSR)**. Assistant Director Secondary Surveillance Radar is responsible to Director LRR for:

(a) Coordinating with Depots and field Units for smooth operation / up-gradation all SSRs (IFF Sets) & IFF reply Simulators.

(b) Tracking the serviceability and availability of SSR, IFF Reply Simulators and ancillary equipment (Generators, AC Plants & ECUs) and initiating remedial actions for recovery of prolonged un-serviceabilities.

(c) Planning IFF system and ancillary equipment training requirements and coordinating with different agencies for timely conduct of planned training courses.

(d) Periodic definitization of rotables for IFF systems and ancillary equipment for planning the spares / budget requirements.

(e) Compilation and revision of AFOs, in consultation with respective system's DDs, for the maintenance of IFF Systems and ancillary equipment.

(f) Planning and arranging indigenous development of critical IFF / ancillary equipment spares, systems' upgrades and new inductions in consultation with all stakeholders.

(g) Follow up with supplier / OEM on technical issues related to IFF systems and ancillary equipment.

(h) Preparation of Air Staff Presentation data and carrying out defect trend analysis and plan remedial measures.

(j) Issuing tasks to No 108 AED for maintenance / up-gradation of IFF systems and ancillary equipment.

(k) Providing necessary input to DCM & DRRCs for raising of indents and processing of procurement orders.

(l) Processing of Hi-value item cases for PSO's approval.

(m) Follow-up all RRCs and keep up-to-date records of all purchase orders.

(n) Arranging functional check of random samples selected from the spares delivered through RRCs.

(p) Monitoring the allocation / utilization of funds allocated under different heads.

(q) Liaison with OEM, APF, DED and other stakeholders to finalize modalities pertaining to ToT and System's upgrades.

(r) Liaison with OEM and other stakeholders to finalize modalities pertaining to induction of new systems.

(s) Formulation & review of documentations, PMIs Training Schedules (TRC, SEP etc.) for new inductions.

(t) Formulation of Indents and finalization of contracts in close liaison with DP(Air).

(u) Any other task assigned by Dir LRR.

38. **Director Surface to Air Missile System – Dir SAMS**. He is responsible to ACAS (RE) for:

(a) Organizing, controlling, directing and monitoring all stratum of maintenance for Surface to Air Missile System (SAMS), auxiliary equipment and power generators.

(b) Formulating policies for efficient maintenance management of Surface to Air Missile System (SAMS) equipment.

(c) Analysis of critical defects which effect system performance and develop solution.

(d) In-depth study of existing systems for initiation of indigenization programme to overcome critical deficiencies and improve performance of existing systems.

(e) Plan and forecast necessary upgrades to existing systems and induction of new systems to support Surface to Air Missile System Operations.

(f) Technical analysis of proposals submitted for new systems.

(g) Ensure action on CAS' directives, Inspectorate and staff visit reports.

(h) Co-ordinate with other specialist Dtes for support and resources required for smooth operations of SAMS Systems.

(j) Plan and forecast budget estimates necessary for field and depot level maintenance and spares procurement required for Surface to Air Missile System (SAMS).

(k) Forecast budget required for critical deficiencies, replacement and new measures.

(l) Review and definitize the requirement of spares for providing defined Surface to Air Missile System Operations.

39. **Deputy Director SPADA – DD SPADA**. He is responsible to Director SAMS for the following:

(a) Serviceability and maintenance of SPADA system.

(b) Compilation and issuance of related AFLs / AFOs and other policies.

(c) Taking follow-up actions on inspection and staff visit reports.

(d) Assignment and management of tasks to No 108 AED and Up gradations / new measures

(e) Assigning tasks to No 108 AED related to said systems.

(f) Preparing budget forecasts.

(g) Managing procurements of spares through indents, contracts and requisite liaison with concerned agencies.

(h) Planning training courses for officers and airmen.

(j) Vetting SPADA systems reviews / reports, processing of Hi Value items and monitoring float / bench stock deficiencies through use of PALMS.

(k) Explore indigenous manufacture and / or repair capability and maintain liaison with repair depots, civil market, DGMP and Directorate of Engineering Development.

(l) Promote Quality Assurance Programme (QAP) and Quality Control (QC) awareness at user and depot level for maintenance activities.

(m) Coordinate with HQs ADC and respective sector Headquarters for planning and implementing annual and bi-annual inspection of SPADA system.

(n) Maintaining progress on related tasks issued to No 108 AED.

(p) Analysis and scrutiny of R/D returns and tasks completion reports submitted by No 108 AED.

(q) Analysis and scrutiny of all RCNs received from Depots / Units, carrying out requisite trend analysis and initiation of necessary remedial actions.

(r) Evaluation of new enhancements proposed by the manufacturer vis-à-vis the Air Defence needs of the PAF.

(s) Monitoring the strength and skill level of technical manpower in SPADA system Units and to liaise with ACAS (P-A) Sectt accordingly.

(t) Preparation of accurate data for Air Staff Presentation of SPADA System.

(u) Vetting of NFR reports and recommending items / quantities to be procured by log directorate.

(v) Meeting and co-ordination with OEM, M/s MBDA and M/s Thales related to SPADA system.

40. **Deputy Director Crotale Weapon System - DD CWS**. He is responsible to Director SAMS for the following:

(a) Serviceability and maintenance of Crotale Weapon system (CWS).

(b) Preparation and implementation of maintenance plans for CWS.

(c) Performance evaluation, siting, installation and calibration of Crotale Weapon systems (CWS).

(d) Compilation and issue of related AFOs / AFLs and other policies.

(e) Follow-up actions on inspections and staff visit reports.

(f) Liaison with concerned directorates for auxiliary equipment requirements.

(g) Coordinate with HQs ADC for earmarking of Crotale Weapon System (CWS) to be overhauled / upgrade at No 108 AED and its subsequent allotment.

(h) Coordinate with No 108 AED of timely overhaul/ upgrade of CWS.

(j) Preparation of budget forecasts.

(k) Planning spares procurement through indents, contracts and RRCs and requisite liaison with concerned agencies.

(l) Monitoring the strength and skill level of technical manpower in Crotale Weapon System (CWS) Units and to co-ordinate with ACAS (P-A) Sectt accordingly.

(m) Planning training courses for officers and men in accordance with ATTS.

(n) Coordination with respective bases and regional commands for issues related to Crotale Weapon System (CWS).

(p) Vetting of Crotale Weapon System (CWS) related reviews / reports, processing of Hi Value items and monitoring of pre-issue / bench stock deficiencies.

(q) Explore indigenous manufacture and / or repair capability and maintain liaison with concerned agencies, civil market, DGMP and Dte of Engineering Development.

(r) Promote Quality Assurance Programme (QAP) and Quality Control (QC) awareness at user level for maintenance activities.

(s) Preparation of accurate data for Air Staff Presentation related to Crotale Weapon System (CWS).

(t) Vetting of NFR reports and recommending items / quantities to be procured by log directorate.

(u) Meeting and co-ordination with OEM and M/s Thales related to Crotale weapon system (CWS).

41. **Deputy Director EWTTR & WTR – DD EWTTR & WTR**. He is responsible to Director SAMS for the following:

(a) Serviceability and maintenance of EWTTR & WTR.

(b) Preparation and implementation of maintenance plans for EWTTR & WTR.

(c) Performance evaluation, siting, installation and calibration of EWTTR & WTR.

(d) Preparation of budget forecasts.

(e) Planning spares procurement through indents, contracts and RRCs and requisite liaison with concerned agencies.

(f) Explore indigenous manufacture and / or repair capability and maintain liaison with concerned agencies, civil market, DGMP and Dte of Engineering Development.

(g) Coordination with respective bases and regional commands for issues related to EWTTR & WTR systems.

(h) Vetting of EWTTR & WTR related reviews / reports, processing of Hi Value items and monitoring of pre-issue / bench stock deficiencies.

(j) Meeting and co-ordination with OEM and M/s Havelsan related to EWTTR.

(k) Preparation of accurate data for Air Staff Presentation.

(l) Coordinating with other depots / Units for support and resources required for smooth operations / upgrade of EWTTR & WTR systems.

(m) Periodic definitization of EWTTR & WTR systems, floats for spare support, planning and preparing budget estimates.

(n) Follow up / replacement of newly procured / developed and under warranty items through supplier / manufacture.

42. **Assistant Director EWTTR & WTR – AD EWTTR & WTR**. He is responsible to Deputy Director SAMS for the following:

(a) Serviceability and maintenance of EWTTR & WTR.

(b) Compilation and issue of related AFOs / AFLs and other policies.

(c) Follow-up actions on inspections and staff visit reports.

(d) Liaison with concerned directorates for auxiliary equipment requirements.

(e) Monitoring the strength and skill level of technical manpower in EWTTR & WTR Units and to co-ordinate with ACAS (P-A) Sectt accordingly.

(f) Planning the training courses for officers and men in accordance with ATTS.

(g) Promote Quality Assurance Programme (QAP) and Quality Control (QC) awareness at user level for maintenance activities.

(h) Liaison with No 302 ALD for speedy actions on ROCP / AROCP demands for quick recovery of un-serviceabilities.

(j) To ensure timely preventive maintenance to enhance reliability support equipment.

(k) Forecasting components / assemblies by analyzing periodic reviews, for the future requirements.

(l) Issuance of SOPs for the EWTTR & WTR systems and ensure follow up actions.

(m) Analysis of serviceability and other RCNs received from field Units.

(n) Monitoring and implementation of directives issued by Directorate for system maintenance.

(p) Keeping an updated record of authorized pre issue / bench stock and monitoring the deficiencies of support equipment.

(q) Up-keeping of all warranty claims and their consistent follow up.

(r) Maintenance of complete record of technical documents related to EWTTR & WTR systems and to ensure incorporation of latest changes.

43. **Director Short Range Radars – Dir SRR**. He is responsible to ACAS (RE) for:

(a) Organizing, controlling, directing and monitoring all echelons of maintenance for Air Defence Radars (MPDRs, DRAD) and Air Field Radars (ATCRs, PAR, MSS-PAR & DWR) along with ancillary equipment and Air Defence power generators.

(b) Formulating policies, AFOs, AFIs and AFLs for maintenance of system related to SRR equipment.

(c) Analysis of critical defects which effect system performance and develop solution.

(d) In-depth study of existing systems for initiation of indigenization programme to overcome critical deficiencies and improve performance of existing systems.

(e) Plan and forecast, necessary upgrades to existing systems and induction of new systems to support Air Defence Operations.

(f) Technical analysis of proposals submitted for new systems.

(g) Ensure action on CAS' directives, Inspectorate and staff visit reports.

(h) Co-ordinate with other specialist Dtes for support and resources required for smooth operations of SRR systems.

(j) Plan and forecast budget estimates necessary for field and depot level maintenance and spares procurement required for SRR systems.

(k) Forecast budget required for critical deficiencies, replacement and new measures.

(l) Review and definitize the requirement of spares for providing defined Air Defence Operations.

44. **Deputy Director MPDR – DD MPDR**. He is responsible to Director SRR for the following:

(a) Serviceability and maintenance of Air Defence Radars (MPDR & GRAD).

(b) Compilation and issuance of related AFLs / AFOs and other policies.

(c) Taking follow-up actions on inspection and staff visit reports.

(d) Assigning tasks to No 108 AED related to said systems.

(e) Coordinate with HQs ADC for earmarking of sensors to be overhauled / upgrade at APF PAC, Kamra and its subsequent allotment.

(f) Coordinate with APF PAC, Kamra of timely overhaul / upgrade of MPDR.

(g) Preparing budget forecasts.

(h) Managing procurements of spares through indents, contracts and requisite liaison with concerned agencies.

(j) Planning training courses for officers and airmen.

(k) Vetting MPDR systems reviews / reports, processing of Hi Value items and monitoring float / bench stock deficiencies through use of PALMS.

(l) Explore indigenous manufacture and / or repair capability and maintain liaison with repair depots, civil market, DGMP and Directorate of Engineering Development.

(m) Promote Quality Assurance and Quality Control awareness (in coordination with Directorate of QA) at user and depot level for maintenance activities.

(n) Coordinate with HQs ADC and respective sector Headquarters for planning and implementing annual and bi-annual inspection of MPDRs.

(p) Maintaining progress on related tasks issued to No 108 AED.

(q) Analysis and scrutiny of R/D returns and tasks completion reports submitted by No 108 AED.

(r) Analysis and scrutiny of all RCNs received from Depots / Units, carrying out requisite trend analysis and initiation of necessary remedial actions.

(s) Evaluation of new enhancements proposed by the manufacturer vis-à-vis the Air Defence needs of the PAF.

(t) Monitoring the strength and skill level of technical manpower in Air Defence Units and to liaise with ACAS (P-A) Sectt accordingly.

(u) Preparation of accurate data for Air Staff Presentation.

(v) Vetting of NFR reports and recommending items / quantities to be procured by log directorate.

45. **Deputy Director ATCR & PAR**. He is responsible to Director SRR for the following:

(a) Serviceability and maintenance of ATCR / PAR system.

(b) Preparation and implementation of maintenance plans for ATCR / PAR.

(c) Performance evaluation, siting, installation and calibration of ATCR and PAR systems.

(d) Compilation and issue of related AFOs / AFLs and other policies.

(e) Follow-up actions on inspections and staff visit reports.

(f) Liaison with concerned directorates for ancillary equipment requirements.

(g) Preparation of budget forecasts.

(h) Planning spares procurement through indents, contracts and RRCs and requisite liaison with concerned agencies.

(j) Monitoring the strength and skill level of technical manpower in Air Defence Units and to liaise with ACAS (P-A) Sectt accordingly.

(k) Planning the training courses for officers and men in accordance with ATTS.

(l) Coordination with respective bases and regional commands for issues related to ATCR and PAR.

(m) Vetting of ATCR / PAR related reviews / reports, processing of Hi Value items and monitoring of pre-issue / bench stock deficiencies.

(n) Explore indigenous manufacture and / or repair capability and maintain liaison with concerned agencies, civil market, DGMP and Dte of Engineering Development.

(p) Promote Quality Assurance and Quality Control awareness at user level for maintenance activities.

(q) Preparation of accurate data for Air Staff Presentation.

46. **Deputy Director PG / ESC & Spl Veh**. He is responsible to Director SRR for the following:

(a) Serviceability and maintenance of generators and Environmental Control systems pertaining to MPDR and ATCR / PAR system.

(b) Coordinate with HQs ADC and respective sector Headquarters for planning and implementing routine maintenance schedules of generators.

(c) Planning and implementing Phased-replacement program of AD generators.

(d) Liaison with OEM for issues related to newly procured generators.

(e) Planning and implementing generator overhaul program in coordination with No 104 AED.

(f) Coordination with APF PAC, Kamra for planning and implementation of ECS refurbishment program.

(g) Forecasting spare requirement in coordination with Air Defence Units and concerned depots.

(h) Spare procurement through central purchase / local purchase and requisite liaison with concerned agencies.

(j) Liaison with Air Defence Units to resolve maintenance issues.

(k) Monitoring progress on overhaul and special tasks issued to No 104 AED.

(l) Analysis and scrutiny of R/D Returns and tasks completion reports submitted by concerned agencies.

(m) Coordination with No 104 AED for indigenous manufacture of different generator parts.

(n) Liaison with manufacturers for clarification on various technical aspects of the system.

(p) Coordination with OEM for evaluation and implementation of system improvement program.

(q) Monitoring the strength and skill level of technical manpower (MT Fitter and Electric Fitter trade) in Air Defence Units and to laise with ACAS (P-A) Sectt accordingly.

(r) Preparation of requisite data for Air Staff Presentation.

**CHAPTER - 4**

# HEADQUARTERS AIR DEFENCE COMMAND

## Introduction

1. Headquarters Air Defence Command is the highest Air Defence formation, which exercises effective control over all Air Defence activities of the PAF during peacetime and in war. The Air Defence Command is commanded by an Air Officer who is directly responsible to the Air Defence Commander (Chief of the Air Staff) and through him to the Supreme Commander of all Armed Forces (President of Pakistan).

2. Headquarters Air Defence Command (HQ ADC) maintains its own operations room from where, it supervises and directs Air Defence operations of the country being conducted by Sector Headquarters. In addition to the conduct of Air Defence operations, HQ ADC is also responsible to implement Air Defence policies formulated by Air Headquarters. The organization of HQ ADC is given at Appendix 'C' to this AFM. The duties and responsibilities of personnel engaged in Air Defence operations at HQ ADC are given in succeeding paragraphs.

## Duties and Responsibilities

3. **Air Officer Commanding, Air Defence Command**. He is responsible to the CAS for operational and functional control over the Air Defence elements through respective Sector Headquarters, Bases and Units as laid down by Air Headquarters. The control and direction of these Units are to be to the extent and within the limits described below:

(a) On all matters related to the AD system which require policy guidance of AHQs, on behalf of the CAS, AOC is to deal and work in close coordination with DCAS (Ops) and DCAS (AD).

(b) Operational application of Air Defence component in coordination with DCAS (Ops) for surveillance of Pak airspace, border violations, scrambles etc. In all such operational matters, AOC ADC is to keep necessary liaison with DCAS (AD).

(c) Responsible to handle entire range of AD Ops / Airspace Management in coordination with DCAS (Ops).

(d) Ensures formulation and implementation of War Plans of entire AD setups through respective SMCCs.

(e) In peacetime, coordination of an effective training programme to prepare fighter Units for AD / Air Superiority role and to this end the most efficient utilization of air effort allocated for the purpose by AHQs.

(f) In war time or period of tension / emergency, to employ and direct the fighter effort in such a manner so as to support the overall PAF mission as assigned by AHQs.

(g) In war and during period of tension/emergency AOC ADC will, on behalf of the AD Commander (CAS), act as chief of the agency responsible for airspace management which is to extend over whole of Pakistan territory.

(h) In peacetime, AOC ADC is responsible to ensure that all deficiencies and shortcomings which exist in AD system are brought to notice of the CAS.

4. **Senior Air Staff Officer (SASO)**. The Senior Air Staff Officer, Headquarters Air Defence Command is responsible to AOC, ADC for the following:

(a) Ensuring that all directives and instructions issued by AOC Air Defence Command are disseminated, coordinated and executed by all the elements of Air Defence.

(b) Advising the AOC on all matters relating day to day functions of Air Defence Command as well as any desirable changes / improvements in the prevailing operational, tactical and organizational concepts in the Air Defence system.

(c) Planning of realistic Air Defence exercises and overall conduct of these exercises. To evaluate the deficiency of every participating Air Defence element as well as the overall effectiveness of the Air Defence system with respect to the assessed enemy threat and render necessary reports to the AOC.

(d) Advising the AOC on all matters relating day to day functions of Air Defence Command as well as any desirable changes / improvements in the prevailing tactical and organizational concepts in the Air Defence system.

(e) Ensuring that effective initial operational, continuation and on-the-job training syllabi are drawn up and followed by all Air Defence training and operational Units.

(f) Maintain liaison and coordination with the following:

(i) Regional Air Commands for the best utilization of allocated interceptor effort for the training of air crew and GCI controllers.

(ii) Regional Air Commands for the provision of administrative support by the bases to Air Defence Units.

(iii) Pakistan Army Air Defence and Army Aviation Commands on all matters of airspace management and deployment of Army Air Defence assets.

(iv) Naval Headquarters on all matters concerning airspace management over the sea and integration of Pakistan Navy Air Defence Sensors and Weapons.

(v) Director General Civil Aviation on matters concerning integration and utilization of Civil Aviation sensors and common facilities for Air Defence in time of emergency and war.

(vi) Director General Civil Defence for Civil Defence matters.

(g) Carrying out operational reviews of war books of individual Units of the Air Defence Command to ensure preparedness of these Units for fulfilling their wartime role.

5. **Senior Staff Officer Air Defence (SSO AD)**. He is responsible to SASO for the following:

(a) Ensuring that all orders and instructions issued by AOC, Air Defence Command and SASO are relayed, coordinated and executed by all the elements of Air Defence system.

(b) Ensuring selection of best possible war-sites of Air Defence Units and recommend the overall deployment plan.

(c) Planning and implementing the operational and training deployment of mobile Air Defence elements as per policy to ensure high state of operational readiness and field worthiness.

(d) Coordinating with respective Directorates at Air Headquarters for formulation / issuance of policy on various Air Defence related subjects.

(e) Continuously monitoring the Air Defence structure and assets and to highlight the deficiencies.

(f) Maintaining intelligence data received from various agencies in the intelligence room.

(g) Coordinating for all lectures to be delivered by the AOC to the visiting teams, Staff/War Colleges and NDC participants.

(h) Coordinating all visits to the Air Defence Command by foreign and local delegations.

(j) Maintaining Top Secret Registry.

(k) Coordinating all moves of Air Defence Units and personnel.

(l) Periodically reviewing war books of Air Defence Units for their accuracy and efficacy.

(m) Coordinating with Pakistan Army Air Defence on all matters concerning deployment of Army Air Defence assets.

(n) Coordinating with Naval Headquarters on all matters concerning integration of Pakistan Navy Air Defence sensors and weapons.

(p) Coordinating with Air Headquarters, Islamabad [Dte of Ops] for provisioning of airlift for AD assets/personnel.

(q) Coordination for participation of AD Contingent in 23rd March JS Parade.

(r) Coordination with sister services on various Tri-services matters.

(s) Processing of Military awards, CAS Commendation Certificate and Imtiazi Sanad in respect of AD Units.

(t) Coordination for provisioning of NOC for construction of Buildings / Towers.

(u) Planning Swift deployments of MMCC and associated sensors.

(v) Preparation of URD (User Requirement Document) in consultation of field AD Units.

(w) Preparation of ATP (Acceptance Test Procedure), Site Acceptance Procedures (SAT) for AD equipment.

(x) Planning Modifications / up-gradations and overhauling of AD systems.

(y) Preparation of draft ASRs / JSRs for AD systems.

6. **Staff Officer Air Defence (SO AD)**. He is to assist the SSO (AD) in the following:

(a) Activation, operational and training deployment of mobile Air Defence Units as per policy to ensure high state of readiness and field worthiness.

(b) Coordinating with Pakistan Army Air Defence on all matters concerning deployment of Army Air Defence assets.

(c) Coordinating with Naval Headquarters on all matters concerning integration of Pakistan Navy Air Defence sensors and weapons.

(d) Coordinating with Director General Civil Aviation on matters concerning integration and utilization of Civil Aviation sensors and common facilities for Air Defence in time of emergency and war.

(e) Preparing lecture scripts/presentations to be delivered by the AOC.

(f) Preparing lecture / presentation script for all visits to the Air Defence Command by foreign and local delegations.

(g) Planning and coordinating AOC visits to field Units.

(h) Authorizing moves of AD officers of AD Units.

(j) Provision of AD officers for exercises, courses and AD activities including DACT Camps etc.

(k) Air Defence development plans.

(l) Matters related to Air Defence Committee meetings.

(m) Organizational matters related to AD Units.

(n) Implementation of Directives of CAS / PSOs / AOC.

(p) Manning of Air Defence Units including establishment review, posting and attachment etc as per role and tasks.

(q) Matters related to AD officers Career Planning.

(r) Coordinating with Air Headquarters, Islamabad {Dte of Ops} for provisioning of airlift of sensors / GBADS / AD Equipment / changeover of personnel etc.

(s) Preparation of AD Contingent for participation in 23rd March JS Parade and related correspondence.

(t) Coordination with sister services on various Tri-services matters.

(u) Processing of Non-Ops Military awards and CAS Commendation Certificate, Imtiazi Sanad to Officers, Airmen, and Civilians of AD Units.

(v) Observations on issuance of NOC for construction of buildings/ towers in relation to employment/ deployment of radar and GBADS.

(w) Swift Deployments of MMCC and associated sensors.

(x) Revisions of Org Memo and Insignia of AD Units.

(y) Preparation of URD (User Requirement Document) in consultation of field AD Units.

(z) Preparation of ATP (Acceptance Test Procedure), Site Acceptance Procedures (SAT) for various radars / AD equipment in coordination with

(aa) Air Headquarters and AD Units.

(ab) Coordination for Modification / up-gradation and overhauling of AD sensors.

(ac) Preparation of ASRs for AD system.

7. **Staff Officer Plans (SO Plans)**. He is responsible to the SSO (AD) for the following:

(a) Maintenance of Top-Secret Registry and Secret Sensitive correspondence with AHQ and under command Units.

(b) Maintaining copy of the war books of SMCCs & Comforce HQs and ensuring that the plans are kept updated.

(c) Maintenance of record of inspections and approval of war books of all Air Defence Units.

(d) Keeping a record of post deployment plans of Air Defence Units and bringing any problems / recommendations by concerned Units to the notice of SSO (AD).

(e) All matters relating to war plans including deficiency of manpower, mob reserve and assets, which may affect Units' performance in field.

(f) All Operational, Technical and Administrative matters related to Balloon Barrage Units.

(g) Maintaining intelligence data / ORBAT received from concerned agencies.

(h) Propose revision of publications related to War Plans (War Plans Instructions, AFM 4-3, AFL 4-2 etc)

(j) All matters relating to budget forecasting / allocation and monitoring of funds utilization in respect of AD Units.

(k) To pursue acquisition of land cases pertaining to war sites.

(l) Allocation and deployment plans of Army Air Defence Weapons and Naval Terminal Defence Weapons and their surveillance radars.

(m) Tri-services level coordination with JSHQ on matters concerning AD Plans.

(n) Maintaining liaison with sector Headquarters for periodic review of war sites for deployment of AD assets.

(p) Maintaining track of all AD assets (Radars, GBADS, MOUs & BBUs) for deployment at designated places as per the war plan.

(q) Apprise SSO (AD) on war plan discrepancies / sensor shortfall / re-allotment and recommend alternate options for effectiv e execution of war plan.

8. **Staff Officer GBADS (SO GBADS)**. He is responsible to the SSO (AD) for the following:

(a) Maintenance of daily state of GBADS for morning brief and onward dispatch to AHQ.

(b) To monitor and ensures effective implementation of operational aspects related to GBADS.

(c) Actions of all concerned letters marked by SSO (AD), SASO & AOC.

(d) Timely submission of GBADS returns.

(e) To conduct / coordinate different seminar related to GBADS.

(f) Clearance of schedule / non-schedule maintenance of GBADS elements.

(g) To prepare and update record of GBADS for consultation.

(h) To attend AHQ / Tri Services meetings on GBADS.

(j) Directing and ensures implementation of CAS / AOC directives to SAM Squadrons.

(k) Update on various ongoing projects from SMCCs and forwarding to AHQ (VTWCC, Relocation of CWS, CCP project, Extended Ops etc).

(l) Timely submission of all GBADS related progresses on CAS directives to AHQ.

(m) Deployment / coordination of GBADS for AD Exercises.

(n) Compiling of SOPs for SWS, CWS MWS and TWCC for approval and ensures implementation of approved SOPs.

(p) Compiling of SOPs check list for ready reference.

(q) Audit of SAM Squadrons.

(r) Compiling and maintaining of audit reports and progress of audit reports of SAM Squadrons.

(s) Planning and conduct of Extended Ops.

(t) Preparation of training modalities and ensures their implementation (simulator, CBT and training against SAM evasive manoeuvring targets etc).

9. **Staff Officer Logistics (SO Log)**. He is responsible to SSO (AD) for the following:

(a) To coordinate with Regional Air Commands for the provision of administrative support to Air Defence Units by the respective bases.

(b) To monitor mob reserve deficiencies of Air Defence Units and take up cases with relevant agencies to fulfil the deficiencies.

(c) To coordinate with ACAS (Log) for pre-issue, bench stock and critical spares deficiencies of Air Defence Units.

(d) Scaling for Air Defence Units.

(e) To coordinate with ACAS (Log) and DAT for provision of rolling stock and airlift requirements of mobile Air Defence Units.

10. **Senior Staff Officer Operations (SSO Ops)**. He is responsible to SASO for the following:

(a) Supervising the conduct of day to day operations of Air Defence system and providing guidance to the SMCCs in handling abnormal Air Defence situations, as and when required.

(b) Keeping track of air violations, route deviations, and hijacking with dissemination of information of such incidents to the various agencies according to SOPs.

(c) Suggesting measures to the concerned agencies for the improvement of proficiency level of personnel engaged in Air Defence operations and to ensure combat effectiveness of all Air Defence elements.

(d) Coordinating and supervising special missions such as recce and escort missions.

(e) Formulation and implementation of various contingencies.

(f) Preparation of SOPs for the battle staff.

(g) Assigning watch timings to all Air Defence Units during peace and war in accordance with the policy.

(h) Implementing all operational tasks assigned to the Air Defence system.

(j) Ensuring continuous flow of operational data from Sector Mission Control Centres to Air Defence Operation Centre.

(k) Maintenance of the operational record in HQ Air Defence Command.

(l) To plan realistic Air Defence exercises, carrying out their analysis to highlight weak areas and recommend remedial actions.

(m) Issuing ADA readiness instructions to the concerned agencies.

(n) Coordinating conduct of probe missions and undertaking analysis of probe missions.

(p) Monitoring allocation / utilization of GCI effort and allocation of night GCI to various Units.

(q) Coordinating annual / bi-annual maintenance of sensors and scheduling their weekly maintenance.

(r) Ensuring accuracy of Air Staff Presentation Data.

(s) Coordinating with Dte of Air Defence for formulation / issuance of policy on subjects relating to Air Defence operations.

(t) To ensure the practice of ops shifting of all SMCCs & GMCCs to theirs / by locations in peacetime, carry out analysis to highlight weak areas and recommend remedial actions.

11. **Staff Officer Operations (SO Ops)**. He is responsible to SSO (Ops) for the following:

(a) Assisting the SSO (Ops) in efficient discharge of his duties.

(b) Ensuring smooth efficient operations and administration of the Air Defence Operations Centre (Ops Room).

(c) Coordinating annual / bi-annual maintenance of sensors and scheduling their weekly maintenance.

(d) Management of duty rosters and scheduling of lectures and training programs for Ops Crew.

(e) Conduct of SOPs / system Quiz on regularly basis to enhance knowledge of combat crew.

(f) Timely furnishing of Airspace violations, unusual occurrence and probe reports and tracings.

(g) Maintaining up-to-date records and status of the overall Air Defence system.

(h) Maintaining proper discipline and decorum of the Ops Room.

(j) Forwarding software system observation to Project Vision through Air Headquarters DAD (Ops) for software upgrades.

(k) Coordinating and analyzing Western flying activities and preparing slide for perusal of DCAS (Ops) / CAS.

(l) Maintenance of the operational record and returns in HQ ADC.

12. **Staff Officer Coordination (SO Coord)**. He is responsible to SSO (Ops) for the following:

(a) Assisting the SSO (Ops) in proper discharge of his duties.

(b) Planning and conduct of realistic Air Defence exercises.

(c) Coordinating with Air Headquarters and Regional Air Commands for the allocation and availability of flying efforts for Air Defence training and exercises.

(d) Coordinating with civil defence agencies for their participation in the Air Defence exercises.

(e) Preparing Morning / monthly / operational briefs and Air Staff Presentation Data.

(f) Coordinating with respective Sector Headquarters for smooth completion of GCI tasks.

(g) Maintains complete record of the GCI of Trg Institutes as well as Ops GCI of all sectors.

(h) To distribute GCI as per priority to bring Trg Units and Ops Crew at Parity.

(j) To keep the record of GBADS state, sensors watch timings, availability & operationalization.

13. **Duty Ops Officer**. He is responsible to SSO (Ops) for the following:

(a) The conduct of efficient and effective operations in ADOC.

(b) To continuously monitor the air activity and ensure that the SMCCs battle staff is responsive to any unusual/abnormal activity.

(c) To coordinate all inter-sector operational activities.

(d) To ensure that all Ops data inputs from the SMCCs are available in the system and are continuously updated.

(e) ADOC Ops data inputs are fed in the system and made available for instant display.

(f) To inform SASO and SSO (Ops) of all major un-serviceabilities that affects the ops performance and readiness of Air Defence.

(g) To immediately initiate all actions as per existing SOPs on various emergencies, contingencies and abnormal situations.

(h) To advise SMCCs in consultation with SSO (Ops), on scheduled / unscheduled maintenance of the equipment.

(j) To brief his reliever duty ops officer properly about all important occurrences during his watch and any matter that is pending.

14. **Tactical Ops Coordinator (TOC)**. During the tenure of his duty, he is responsible to SSO (Ops) for the following:

(a) Effective utilization and performance of personnel and equipment of the Ops Room.

(b) Ensuring that all operational situations which so demand, are brought to the notice of the Duty Ops Officer / SSO (Ops).

(c) Ensuring that all plotting and status boards are correctly and expeditiously updated.

(d) Monitoring the flow of information from all lower formations and painting out any delay or mistake to the appropriate source for rectification.

(e) Guarding against any unauthorized access or use of ADOC and its equipment.

(f) Air violations involving engagements between enemy and own aircraft are reported to the concerned outside agencies.

(g) Be vigilant and check SMCCs for any delay in identification and tactical actions in case of pending tracks.

(h) Keep vigilance on tracks across the border.

(j) Disposal of important mail received after office hours.

(k) Maintaining log of events

15. **Duties and Responsibilities of SSO Admin and Security (A&S)**. SSO (A&S) is responsible to SASO for the following:

(a) To address all admin related issues of HQ ADC, PAF.

(b) To maintain and update all admin related data of HQ ADC for smooth functioning of Command.

(c) To analyze and process all admin related issues arising out of admin reports.

(d) To disseminate and monitor the implementation of all administrative instructions, orders, directives, policies and procedures.

(e) To co-ordinate with specialist Directorates at Air Headquarters in relation to administration matters

(f) To prepare admin related data for Air Staff presentations and process staff comments on admin related reports and returns.

(g) To act as OC Admin and provide complete administrative support to all Sections within the command.

(h) To process cases regarding release / dismissal / removal of personnel from service of various AD Units.

(j) To offer staff comments on BOI, BOO, Formal investigation etc on matters relating to administration.

(k) To maintain high standards of discipline and morale of the Command personnel.

(l) To keep an updated record of all funds placed at disposal of Command Headquarters and utilize them as per directives of AOC / SASO.

(m) To process capital works submitted by AD Units and to ensure their timely submission to AHQs.

(n) Any other duty assigned by AOC / SASO.

16. **SSO (Admin & Security)**. He is also responsible to SASO for aspects related to security domain as follows:

(a) To supervise security mechanism of HQ Air Defence Command.

(b) To co-ordinate with specialist Directorates at AHQs for enhancement of security of HQ ADC and all AD Units.

(c) To oversee security plan and its implementation in Command HQs and suggest improvements / changes.

(d) To ensure implementation of all security Directives and Policies issued from AHQs and ensure upkeep of security plan in Command HQ area.

(e) To render advice to SASO on physical security matters concerning command HQ and under command AD Units.

(f) To provide guidance to AD Units regarding all security matters including plans, infrastructure and equipment.

(g) To assess / recommend security requirements (men & material) for all the under-command AD Units for maintaining desired level of security.

17. **Senior Engineering Staff Officer (SESO)**. Senior Engineering Staff Officer, HQ Air Defence Command is responsible to AOC Air Defence Command for the following:

(a) Advising the AOC on matters pertaining to Air Defence Units in the field of engineering including $C^2$ systems, radar, radios, SAM's, communication, generators and MT etc.

(b) Advising the AOC on engineering / maintenance matters of Air Defence equipment to ensure desirable ops availability of Air Defence assets.

(c) To update Air Officer Commanding on all Air Defence un-serviceabilities.

(d) Implementation of Air Headquarters policies on maintenance practices issued from time to time.

(e) Reviewing Secret Communication Orders from time to time.

(f) To maintain liaison and coordination with the Regional Air Commands / Air Defence Sector Commanders for the provisioning of technical support by the bases to Air Defence Units.

(g) Liaison with respective As CAS at Air Headquarters for timely recovery of unserviceable Air Defence equipment and supply of spares to the Air Defence Units.

(h) Highlight deficiencies in technical manpower of AD Units to Air Headquarters and OC AMO.

(j) Highlight weak areas and apprise Air Headquarters about deficiencies of Air Defence equipment.

(k) To ensure that timely and proper maintenance schedules of Air Defence equipment are followed by AD Units.

(l) Coordination with Engg depots for extending timely maintenance support in recovery of Air Defence assets.

(m) Monitoring all maintenance activities of AD Units to ensure quick recovery of unserviceable systems.

18. **Senior Staff Officer Engineering (SSO Engg)**. He is responsible to SESO for the following:

(a) Scheduling maintenance of Air Defence equipment and its timely accomplishment.

(b) Carrying out liaison with different Directorates at Air Headquarters for day-to-day maintenance issues.

(c) Analyzing data received from all Air Defence Units to highlight weak areas and apprise SESO about deficiencies of various weapon systems.

(d) Liaison with Engg and Logistics Depots for timely repair of unserviceable assemblies and supply of spares to the Air Defence Units.

(e) Detailing of personnel deployed in AD Units on courses.

(f) Forecasting and making arrangements of special funds through respective Directorates at Air Headquarters for Air Defence Units.

(g) Highlighting deficiencies in technical manpower.

(h) Monitoring pre-issue and bench stock of Air Defence Units and taking remedial actions.

(j) Coordinating and monitoring of the training activities of Engg officers and airmen of related technical trades to ensure availability of adequately trained manpower in Air Defence Units.

(k) Carrying out liaison with PTC / NTC authorities during peace, war and major exercises for communication requirements of Air Defence Units.

(l) Reviewing the Radar engineering & Networks plan of war book.

(m) To prepare Air Staff presentation data and ensure timely submission to Air Headquarters.

19. **Staff Officer Networks (SO Networks)**. He is responsible to SSO (Engg) for the following:

(a) Effective management of C&E resources for utilization.

(b) Collection / analysis of un-serviceability trends of communication equipment and to recommend suitable remedial measures.

(c) Maintenance of environmental conditions at all communication equipment sites.

(d) Availability of requisite tools / test equipment and relevant publications / procedures at all AD Units.

(e) Strict adherence to prescribed preventive maintenance at sites in accordance with maintenance policies.

(f) Monitoring skilled / unskilled manning level of Radio Fitter Trade in all AD Units.

(g) Ensuring that authorized pre-issue / bench stock level at all AD Units is maintained and replenishment action is initiated in time.

(h) Coordination with No 107 AED for the recovery of prolonged un-serviceabilities.

(j) Critical review of any modification / system discrepancy etc for submission to Air Headquarters.

(k) Liaison with AHQ [Dte of Networks (Command & Bases)] & PMB DEFCOMM for provisioning of ops / admin circuits for conduct of Air Defence exercises / deployments.

(l) To ensure that all AD Units implement instructions regarding communication/ computer security in accordance with AHQ policies.

(m) Updating of WP Comm Plan in co-ordination with under command AD Units.

20. **Staff Officer Radar & $C^2$ Systems (SO Rad & $C^2$ Systems)**. He is responsible to SSO (Engg) for the following:

(a) Effective management and utilization of resources to ensure optimum serviceability and operational availability of PAF vision systems / radars comprising Computers and Peripherals, Display and LSDs, Environmental Control, UPS and Power Supply Systems.

(b) Collection and analysis of the un-serviceability trends of all AD System and recommend suitable remedial measures.

(c) Strict adherence to prescribed preventive maintenance at sites in accordance with maintenance policies.

(d) Coordinating the knowledge and expertise of technical manpower of all sites for immediate solution of recurring or new / unusual problems of AD system.

(e) Liaison with AHQs (DADAS) to improve vision system software by studying operational systems.

(f) Visit of vision system / radar sites and pointing out any discrepancy regarding maintenance practices.

(g) Coordination with No 108 AED / Units for the recovery of prolonged un-serviceabilities.

(h) Critical review of any modification / system discrepancy etc. for submission to Air Headquarters.

(j) To collect and analyze data for pointing out un-serviceability trends of all radars and recommend suitable remedial actions.

(k) Maintenance of specified environmental conditions for all radar systems.

(l) Availability of requisite common / specialist tools, necessary test equipment and relevant technical publications / procedures in adequate number at all operational sites.

(m) To ensure that authorized stock level of all pre-issue / bench stock is maintained and replenishment action is initiated by Units well in time.

(n) To monitor adequate availability of pre-issue / bench stock spares at all radar sites.

(p) Critical review of any modification or system discrepancy etc. for submission to Air Headquarters.

(q) To coordinate with AHQ and APF for Preparation of annual plan for feeding of MPDRs to APF for overhaul / upgrade.

(r) Allotment of newly overhauled / upgraded radar to MCCs in coordination with SO (Plans) / SO (AD).

(s) Liaison with No 108 AED for induction of YLC-6MS radars for EFM.

(t) To coordinate with APF for overhauling of TPS-43G / J radars.

(u) To coordinate with AHQs regarding System Courses / Up-gradation / Relocation of AD systems.

(v) Preparation / Implementation of annual maintenance plan of high / low level radars and ADOCs / GMCCs / SMCCs in coordination with Sectors / AD Units.

(w) Coordination with SO Ops, SSO Ops and SASO for approval of ops clearances of high / low level radars and ADOCs / GMCCs / SMCCs.

(x) Preparation and compilation of monthly and quarterly Air Staff Presentations.

(y) To coordinate with O i/c Control Squadron (Sectors), O i/c FMSs and SEOs regarding un-serviceabilities of AD elements and apprise the same to SSO Engg and SESO.

(z) Review of QA reports of sector and AD Units and resolution / settlement of same.

(aa) Preparation / finalization and processing of AD Units requirements with AHQs / higher echelon.

(ab) Monitoring skilled / unskilled manning level of Engg Officers / Airmen on Air Defence elements to keep high skill level.

(ac) Review of technical proposals for newly inducted / already installed AD equipment.

(ad) To liaison with AHQ (DADAS), No 108 AED and AD Units for justified installation / usage of already installed / newly procured UPS/ MUPS.

(ae) Evaluate the quality of maintenances performed by AD Units and ensure corrective actions, if any.

(af) To review requirement of critical spares needed for smooth operation of radar and forward requirement to AHQ for replenishment.

(ag) To establish effective working relationship with all agencies.

(ah) Any other duty assigned by SESO / SSO (Engg) required to be carried out as per requirement.

(aj) Preparation and compilation of monthly and quarterly Air Staff Presentation.

21. **Staff Officer SAMS (SO SAMS)**. He is responsible to SSO (Engg) for the following:

(a) Liaise with SEOs regarding un-serviceabilities of SAM elements and update SSO Engg and SESO accordingly on Technical issues of SAMS.

(b) To ensure safe operations and maintenance of Spada / Crotale Batteries deployed at dispersed locations.

(c) Ensure all SAM Units to carry out scheduled / unscheduled maintenance in time.

(d) Ensure quick recovery in case of unscheduled maintenance of SAM equipment.

(e) Coordinate between different directorates for expeditious recovery of unserviceable equipment assemblies.

(f) Arrange / co-ordinate for provisioning of spares from Maint / Logistic depots or Units if necessary.

(g) Monitor manning / skill level of trained technical manpower in missile Units and take remedial actions, if required.

(h) Ensure proper training of untrained manpower in SAM squadrons.

(j) Arrange funds for SAM squadrons for early recovery of SAM equipment.

(k) Monitor progress on the courses for technical trades being conducted at No 451 SCU and No 459 SCU.

(l) Liaise with No 108 AED, No 115 AED, No 201 ALC and 303 ALC for provisioning of spares.

(m) Liaise with AHQs (Dte of SAMS) regarding Spada / Crotale related issues.

(n) Preparation and compilation of monthly and quarterly Air Staff Presentation.

(p) Critical review of all ongoing projects of Spada / Crotale (modification/system discrepancy etc.) to meet operational challenges.

(q) Any other duty assigned by SESO / SSO (Engg) required to be carried out as per requirement.

(r) Preparation / Implementation of annual maintenance plan in coordination with Sectors/ AD Units of Spada / Crotale through AD Ops.

(s) Collect and analyze data to arrest un-serviceability trends of all SAM Squadrons and recommend suitable remedial actions.

(t) Monitor adequacy / availability of pre-issue bench stock spares of SAM Squadron.

22. **Staff Officer Support (SO Support)**. He is responsible to SSO (Engg) for the following:

(a) Maintain record of establishment, holding and serviceability of power generators, common user and specialist vehicles.

(b) Work out the requirements of funds for special recovery and general maintenance of MT vehicles and forward it to Air Headquarters, (Dte of Grd Tpt) after necessary vetting.

(c) Coordinate with No 104 AED through Air Headquarters for overhauling of generators.

(d) To arrange allocation of funds from Air Headquarters (ACAS Works) for overhauling and recovery of MES generators being used by Air Defence Units.

(e) Monitor progress on recovery and overhauling of AD generators from No 104 AED and local market.

(f) Preparation of data for quarterly Air Staff presentation, about generators and MT vehicles.

(g) Preparation of data for weekly briefing regarding all air-conditioner, generators and MT vehicles.

(h) Assess future requirements of air-conditioner, MT and generators and process it with Air Headquarters for procurement and allocation.

(j) To ensure availability of requisite number of air-conditioner, MT vehicles and generators for efficient deployment and operations of Air Defence Units.

(k) Availability of trained manpower for maintenance of air-conditioner, MT vehicles and generators.

(l) Installation and maintenance of air-conditioning plants.

(m) Collect and analyze data to arrest un-serviceability trends of all Air Defence Unit and recommend suitable remedial actions.

(n) To coordinate with AHQs (Dte of LRR, SRR, SAMS and Grd Tpt) regarding relocation / re-allotment of AD generators and MT vehicles.

(p) To coordinate with Sector HQs for detailing of personnel (MTF, MTD & Elect Fitt) on system courses as per ATTS.

(q) To maintain MT accident record pertaining to AD setup.

(r) To coordinate with AHQs and AD Units regarding refurbishment of radio relay vehicles through APF.

(s) To establish effective working relationship with all agencies.

(t) Any other duty assigned by SESO/SSO (Engg) required to be carried out as per requirement.

23. **OC Engineering Wing (OC ENGG)**. He is responsible to SESO for the following:

(a) To ensure smooth functioning of ADOC Complex.

(b) To ensure serviceability of ADOC-I, ADOC-II and ADOC-III $C^2$ Systems.

(c) To manage SMT requirements of Air Defence Command and ensure serviceability and reliability of HQ ADC MT pool.

(d) To ensure serviceability of PAFCOMM and DEFCOMM exchanges installed at HQ ADC.

(e) To ensure serviceability of NGN, Intranet and IT services at HQ ADC.

(f) To ensure smooth functioning of ADNOC.

(g) To supervise SEP and AST Program at HQ ADC.

(h) Liaison with Air Headquarters and No 1 Comm Wing for smooth activation / de-activation of COC communication circuits.

(j) To maintain and update technical orders and publications pertaining to Communication equipment installed at HQ ADC.

(k) Liaison with PTC/NTC authorities for maintenance of landline communication.

(l) Maintenance and operation Cypher equipment.

(m) Formulation and issuance of relevant SOP's and ensuring their implementation.

(n) To ensure that maintenance is performed efficiently and in line with quality standards to achieve maximum operational capability.

(p) To control the resources for effective utilization.

(q) To ensure conformance with policies, rules and regulations pertaining to maintenance / personnel.

(r) To ensure implementation of applicable technical directives issued by specialist directorates of Air Headquarters.

(s) To review all pertinent QA inspection reports and post defect reports and also to take necessary corrective actions.

(t) To coordinate with local audit officer for settlement of Audit objections pertaining to technical domain of HQ ADC.

(u) To ensure that all Osi/c (ADNOC, COMM, MT, $C^2$ systems, IT) execute their assigned tasks efficiently and with responsibility.

(v) To ensure efficient functioning of Tech Registry at HQ ADC.

24. **O i/c AD NOC**. He is responsible to OC (Engg) for the following:

(a) To ensure real time monitoring of the services and serviceability status of all devices in area of responsibility.

(b) To ensure configuration of Router and Switches during newly deployed sensors from GMCCs.

(c) To ensure configuration of tunnel for E1 and Ethernet to establish secure traffic between GMCCs to SMCCs and further to ADOC.

(d) To ensure configuration of Router and Switches for additional requirements like SATCOM and DMZs.

(e) To ensure record and update of Access Control List (ACL) of each router and switches.

(f) To ensure installation / up-gradation of Network monitoring software like Solar wind, OP Manager and syslog etc.

(g) To maintain the record of IP traffic of all the routers and switches.

(h) To maintain the record of passwords for all the I/O devices as per AFM 12-1.

(j) To coordinate with AHQ Islamabad (Dte of NWC and Dte of NWB) for the integration of WAN Interfaces of NGN and DCN routers / switches with AD I/O devices.

(k) To coordinate with AHQ Islamabad (Dte of ADAS), Project Vision and No 108 AED for timely recovery of unserviceable I/O devices.

(l) To ensure and implement policies of AHQ Islamabad related to IT and Cyber warfare.

(m) To ensure all unnecessary / unused ports are blocked for security purpose in order to block entry of any intruder into the network.

(n) To implement well-segregated and sufficiently hidden IP addressing scheme and concept of ‘IP Masquerading / Readdressing.

(p) To ensure all types of routers and switches log must be enabled.

(q) To ensure entry of static / OSPF routes for new sensors / DCG servers.

(r) To ensure no extra entry in access ACL of routers and switches.

(s) To ensure the serviceability of resources allocated to the AD NOC for the data communication facilities.

(t) To ensure scheduled maintenance as per TOs / publications.

(u) To ensure real time monitoring of the services and serviceability status of all NGN routers and switches.

(v) To ensure configuration and management of Core, Aggregation and Access switches installed at HQ ADC, HQ Norsec and backup locations.

(w) To ensure maintenance, configuration of the system and diagnosis / analysis of networks elements.

(x) To ensure smooth functionality of Fault Control to provide services for the secure net users at HQ ADC, Norsec and their backups.

(y) To ensure availability of reserve resources in store for smooth functioning of NGN equipment.

(z) To ensure the availability of data and voice services (IP phone) to authorized users.

(aa) To ensure the availability of updated record of NGN nodes.

(ab) To ensure availability and updating of OFC / UTP cable utilization plan for NGN equipment.

(ac) To ensure maintenance and repairing of 20 ODFs installed at multiple locations of HQs ADC, HQs Norsec and backup locations of ADOC/ SMCC (North).

(ad) To ensure splicing of OFC in case of breakage within vicinity of HQs ADC and HQs Norsec.

(ae) To ensure maintenance of UTP cable network and future growth of access network to newly established offices and locations.

(af) To ensure availability of ISP and OSP plans availability of NGN equipment.

(ag) To ensure network availability for Video conferencing sessions of HQs ADC and HQs Norsec on NGN.

(ah) To ensure maintenance, monitoring, management and operation of the entire data networks on round the clock basis.

(aj) To ensure monitoring / management of power supplies / UPS installed with switches and routers.

(ak) To coordinate with AHQs Islamabad (Dte of NWB/ NWC) for functioning / recovery of unserviceable NGN equipment.

(al) To liaison with AHQ (Dte of NWC) for implementation of policies and securities in NGN setup.

(am) To ensure implementation of port security such as 802.1X and Mac Binding.

(an) To ensure availability of Network Management systems (NMS).

(ap) To ensure smooth operations of cameras along media converters, switches and other network accessories installed at different locations of ADC, PAF.

25. **O i/c Comm Flt**. He is responsible to OC (Engg) for the following:

(a) Efficient Maintenance and smooth operation of DEFCOMM system / PAFCOMM, EPABX installed in HQ ADC and Lalkurti.

(b) Discipline and training of all personnel employed in PAFCOMM / DEFCOMM Centres.

(c) Resolving all queries and complaints of subscribers tactfully and speedily.

(d) Ensuring that all faults pertaining to DEFCOMM / PAFCOMM systems are rectified as expeditiously as possible.

(e) Ensuring that all exchange charts, records and diagrams are properly maintained and kept up to date.

(f) Ensuring that a copy of SOPs is available in the DEFCOMM / PAFCOMM centres and all personnel are fully conversant with their duties.

(g) Ensuring efficient / smooth functioning and maintenance of all the switching equipment installed at DEFCOMM / PAFCOMM Centre.

(h) Ensure efficient smooth functioning of Fibre Optic / UG cables laid for HQ ADC.

(j) Ensure smooth functioning of SATCOM equipment installed at HQ ADC.

(k) Quick recovery of the unserviceable telephone extensions.

(l) Liaison with NTC / PTC and mobile phone companies for settlement of telephone bills.

(m) Liaison with sister services, NTC / PTCL and Mobile telephone companies for activation and deactivation of telephone connections and communication circuits with HQ ADC.

(n) Liaison with NTC for any communication works which are in progress through AHQ {Dte of Networks (Bases)}.

(p) Liaison with NTC for settlement / payment of telephone bills.

(q) To ensure that all the bills are paid in time and to liaise with AHQ {Dte of Networks (Bases)} in this regard.

(r) To ensure that all Cypher channels function efficiently and operate according to their schedules.

(s) To ensure that all messaging traffic is cleared in accordance with the precedence of time of origin of the signals vide AFO 102-4.

(t) To ensure that proper discipline of all the channels is maintained and control station instructions are complied with.

(u) To ensure that mail of appropriate security classification is transmitted through Cypher.

(v) Collection, accounting and safe custody of crypto material.

(w) Ensure instructions of Cabinet Division (NTISB) and Air Headquarters.

(x) Ensure serviceability of MEHFOOZ, SECTEL and Post Guard system.

26. **O i/c MT Flt**. He is responsible to OC (Engg) for the following:

(a) Maintaining optimum serviceability state of MT vehicles of HQ ADC.

(b) Ensuring timely accomplishment of all scheduled inspections of MT fleet of HQ ADC.

(c) Ensuring effective management of MT operations of HQ ADC.

(d) Enhancing the level of expertise of MTD's, MTF's and Elect Fitts.

(e) To maintain all files, documents & record, pertaining to maintenance activities.

(f) Conduct of OJT and maintenance of record concerning OJT & CTTB examinations.

(g) To incorporate necessary changes in AFO's, AFI's and technical publications when received from Air Headquarters.

27. **O i/c C$^2$ Systems**. He is responsible to OC (Engg) for the following:

(a) Utilization of resources for smooth operation and optimum serviceability of Vision System.

(b) To review daily optimum performance of the Vision system.

(c) To ensure high degree of serviceability of the Vision equipment.

(d) To ensure prompt attendance of persons on any un-serviceability.

(e) To ensure maintenance of environment condition for Vision systems.

(f) To ensure conduct of schedule/ un-scheduled maintenance.

(g) To ensure that all tasks are carried out according to TOs and given publications.

(h) To review all relevant technical publication / tools of the section on monthly basis.

(j) To review all relevant documents of the section on regular intervals.

(k) To review training program regularly.

(l) To review all quality reports and resolution/ settlement of same.

(m) To ensure the implementation of policies.

(n) To Coordinate with No 108 AED / Units for the recovery of prolonged un-serviceabilities.

(p) To ensure that Earthing values for Vision equipment is well within the prescribed limit.

28. **O i/c IT Section**. He is responsible to OC (Engg) for the following:

(a) To ensure serviceability of all PC's, printers, scanners, multimedia and peripheral equipment held on the inventory of HQ ADC.

(b) To ensure all IT equipment are properly maintained.

(c) To ensure provisioning of secure net services at HQ ADC.

(d) To provide efficient and timely IS services to users of HQ ADC.

(e) To ensure dispatch of unserviceable hardware to IT Centre for repair / rectification.

(f) To ensure installation of authorized software on all PCs.

(g) To ensure registration of new hardware from of IT center.

(h) To ensure round the clock software / hardware serviceability and rectification of IT equipment installed at HQ ADC Crises Control Centre.

(j) To ensure serviceability of equipment allotted for Command Operations Centre (COC) during peace time.

(k) To provide IT support to COC during Ops contingencies.

(l) To maintain and upkeep record of all the computers, printers, scanners, multimedia held on the inventory of HQ ADC.

(m) To co-ordination with IT Centre Nur Khan for implementation of necessary secure net and ICT policies.

(n) To ensure collection of issued IT equipment from IS Bases after coordination with Log Sqn and IT Centre Nur Khan.

(p) To co-ordinate with AHQ NOC and Dte of IS (Bases) for latest security patches of windows / antivirus.

(q) To ensure publishing of security advisories in CRO.

(r) To ensure resolving of complaints regarding Smart Verification System.

(s) To provide IT support in delegations, farewells and other official parties of HQ ADC.

(t) To provide necessary IT support for AD NOC of HQ ADC, PAF.

(u) To update antivirus on monthly basis for all stand-alone PCs installed at HQ ADC.

(v) To implement policies issued against Cyber Security.

(w) To ensure authorized IT support to all users of HQ ADC with available resources and to forward authorization revision case of PCs and Printers if required.

(x) To disseminate Cyber Security instructions to personnel of HQ ADC, PAF.

# CHAPTER - 5

# SECTOR MISSION CONTROL CENTRE

## Introduction

1. The duties and responsibilities of personnel working at Sector Headquarters and Sector Mission Control Centre (SMCC) are enumerated in succeeding paragraphs.

## Duties and Responsibilities

2. **Sector Commander (SC)**. He is responsible to AOC ADC for the efficient working and operational preparedness of the Sector Mission Control Centre and other Air Defence elements under his command. He is the Battle Commander of his Sector during periods of hostility and Air Defence exercises. In addition to these, he is responsible for the following duties:

(a) Ensuring highest standards of operational performance, maintenance, and serviceability of all operational assets of SMCC.

(b) Supervision and coordination of the overall operations of the Sector Headquarters and under command Units.

(c) Overall control of operations, both in peace as well as during war, within the area of his responsibility.

(d) Threat evaluation and generation of matching response by employing appropriate weapon.

(e) Coordination and control of activities of Air Defence elements drawn from other arms & services etc.

(f) Judicious utilization of Air Defence resources placed at his disposal.

(g) Ensuring that all orders, instructions issued by higher echelons are relayed, coordinated and executed by all Units under his command.

(h) Liaison with parallel Air Defence formations.

(j) Ensure that under command Units have carried out war site visits as per laid down procedures.

3. OC Ops Wing. He is Staff Officer to the Sector Commander. He is responsible for efficient operations of the elements of Sector Headquarters and also for control of operations in the whole Sector. In the absence of Sector Commander, he assumes the duties and responsibilities of Sector Commander. In addition to these, he is responsible for the following duties:

(a) Assisting Sector Commander for efficient operational working of SMCC and under command Units.

(b) Highest operational vigilance is maintained by SMCC battle staff.

(c) Coordinating with OC Engg for timely and accurate flow and display of the Air Defence data.

(d) Coordinating with the ADOC, adjacent SMCCs, and under command Units regarding all operational matters.

(e) Ensuring that orders and instructions issued by the Sector Commander are relayed, coordinated and carried out by all concerned.

(f) Reviewing LOPs / SOPs / Ops order.

(g) Maintaining liaison with fighter bases for availability / turnaround of Air Defence standby aircraft.

4. **Senior Controller**. He is responsible to OC Ops Wing for following duties:

(a) Assist OC Ops Wing in smooth and efficient running of operations.

(b) Ensure administration, discipline and deployment of personnel in the Ops Room.

(c) Ensure required availability of GCI effort for all the under-command Units as allocated by Dte of AD (Training & Evaluation).

(d) Conduct the training of officers arriving on cross training visits.

5. **Staff Operations Officer (SOO) / SO Plans**. Staff Officer Plans is responsible to OC Ops Wing / Sector commander for the following duties:

(a) Custodian of TSR room.

(b) Assist Sector Commander in formulating the war plans.

(c) Carry out periodic review of war plans.

(d) Vet war / contingency plans of all under command AD Units of Sector.

(e) Liaise with military and civil agencies with regard to his duties such as development of operational plans and conduct of exercises etc.

(f) Render quarterly war plans / war sites visit report to HQ ADC.

(g) Conduct ORI / Audit / inspections and submit follow up progress reports to Air Headquarters (IG's Branch) and HQ ADC.

(h) Disseminate Air staff directives, monitor their implementation and send confirmation report to HQ ADC.

(j) Conduct VIP visits.

(k) Carry out any other task assigned by Sector Commander

6. **Sector Training Officer (STO)**. At discretion of Sector Commander, a senior CCQ officer is to be designated as 'Sector Training Officer', who as a staff officer to Sector Commander shall monitor the implementation of Cyclic Training Programme (CTP) at SMCC and under command Units. His duties are mentioned below:

(a) Ensure training of AD officers / radar operators and categorization / currency of AD officers of Sector Headquarters.

(b) Review the practical / academic training procedures and syllabus for officers and recommend the required changes.

(c) Ensure conduct of SIM exercises as per CTP in conjunction with under command Units.

(d) Implement training policies for training of AD officers and radar operators as received from higher echelons.

(e) Update and maintain individuals training record.

(f) Ensure effective utilization of simulator by each controller to complete his allocated quota.

(g) Ensure procurement and maintenance of training aids as required.

(h) Plan and publish academic training programme for the officers and airmen as per schedule outlined in the master training folder.

(j) Ensure timely categorization of AD officers.

(k) Maintain constant check on the professional proficiency of AD officers and radar operators.

7. **Sector Duty Controller (SDC)**. On behalf of Sector Commander, the SDC leads battle staff and is responsible for the following:

(a) Assess / evaluate current air situation and take decision for initiating tactical action (This implies selection of appropriate weapon type (aircraft / SAM, etc) and its location if effort is available from multiple locations).

(b) Supervise and detail battle staff for different duties in Ops room and ensure that duties are carried out properly.

(c) Conduct and coordinate Air Defence operations with ADOC and adjacent SMCCs as per SOPs.

(d) Alert / assist search and rescue services, when necessary.

(e) Issuance of air raid warning.

(f) Monitor the status of Air Defence elements within the Sector.

(g) Provide FFS to the' aircraft desiring it through suitable GCI Unit.

(h) Keep track of ADA turnaround time and ensure availability of ADA elements.

(j) Allocate GCI missions to suitable GCI Unit as per given quota by Dte of AD (Trg & Eval) in coordination with Senior Controller.

(k) Liaison with respective Army / Navy formations in area of responsibility (AoR) for necessary coordination.

(l) Give decision for foreign military flights to enter / hold / operate in PAK airspace after verification of received flight plan in accordance with available Military Clearance (MC) and coordinate with ADOC in case of any ambiguity.

8. **Assistant Duty Controller (ADC)**. The ADC is an assistant to the SDC. He is responsible to the SDC for the following:

(a) Disseminate scramble orders / control order to SAM as dictated by SDC and update the status data.

(b) Coordinate different agencies to get ESM ID for threat evaluation and assist SDC in generating a matching response.

(c) Ensure accuracy and completeness of all status displays.

(d) Coordinate with external agencies to update operational records.

(e) Give time check to all concerned agencies as per laid down procedure and ensure synchronization of time at all tiers.

(f) Maintain duty controllers log book.

(g) Maintain close liaison with adjacent SMCC for smooth conduct of operations.

(h) Give update on unusual occurrences to higher echelons i.e. Radar N/A, U/S, SMC, ADA scrambles, border violations, and any important activity.

(j) Carry out any other task assigned by SDC.

9. **Air Surveillance Officer (ASO) SMCC**. The ASO SMCC is responsible to SDC for the following duties:

(a) Coordinate and carry out the air surveillance functions.

(b) Monitor current air picture and report to ADC any irregularity observed.

(c) Warn any friendly aircraft getting close to the international border and guide it through suitable GCI Unit to stay clear of the border.

(d) Supervise the actions of SMCC trackers.

(e) Coordinate with the subordinate Units for meaningful track management.

(f) Designate back told areas for each GMCC and apply Sector Overlap Area (SOA).

(g) Establish the location of Jammer based upon the jamming strobes reported by radar / GMCC and AEWACS**.**

(h) Assist MIO in correct identification of tracks.

(j) Authorize use of Non-Automatic Initiation (NAI) tracking areas in consultation with SDC.

(k) Ensure proper forward telling of tracks to ADOC.

(l) Check with appropriate sensors about an expected track not being picked up within their coverage.

(m) Hand over suitable track to concerned GMCC for performance check of different radars.

10. **Movement Identification Officer (MIO)**. The Movement Identification Officer is responsible to the SDC for the correct and prompt identification of all air movement within the area of Sector's responsibility in stipulated time. His specific duties include:

(a) Supervision of complete movement identification section.

(b) Maintenance of air movement information received from all the sources.

(c) Prompt analysis and timely identification of all tracks displayed in SMCC.

(d) Prompt SDC in case of any unusual track (UFG, Airspace / Flight plan violation, track of interest etc)

(e) Coordination with appropriate agencies on pertinent flight plan discrepancies.

(f) Defining the location and use of auto-identification zones.

(g) Ensuring compliance with all directives and procedures pertinent to aircraft identification.

(h) Maintaining close liaison with the MIOs of adjacent sectors.

11. **Assistant Movement Identification Officer (AMIO)**. He is responsible to SDC for correct and prompt identification of all air movements on the western border / area not covered by MIO. His specific duties include:

(a) Ensure availability of latest ATO's.

(b) Ensure identification of all aircraft operating near border according to ATO / NBL list.

(c) Prompt SDC and MIO in case of any unusual track (UFG, Airspace / Flight plan violation, track of interest etc)

(d) Ensure every collation aircraft operating inside Pakistan is flying as per permitted parameters i.e. height

(e) Raise query, if any, to ADOC supervisor and inform SDC.

12. **DLI Manager**. He is responsible to SDC for maintaining active link with AEWACs. His duties include:

(a) Carry out link brief with AEWACs.

(b) Ensure availability of GES and DLI.

(c) Ensure that recording is ON and aircraft specific settings are selected before the mission.

(d) Establish FTMs, Voice and Data Link with the A/C.

(e) Select following areas from presentation menu for display at DLI:

(i) AEW&C coverage / plot area.

(ii) Select back told area as desired.

(iii) Select plots, air and ESM tracks from presentation menu for display at DLI.

(f) Coordinate with Link Manager and PIF Manager for performance check on a suitable track.

(g) Carry out de-brief with the Link Manager.

(h) Maintain Log of DLI operations in a separate Log book and convey discrepancies of link operations through AEWACs After Mission Report (AMR) to Sector Headquarters.

13. **GBADS Coordinator**. GBADS coordinator assists the SDC by performing the following duties:

(a) Assisting SDC in dissemination of appropriate alert status / control orders for SAMs / Guns.

(b) Coordinating with SAMs / Guns Units within the limits prescribed by Sector Commander.

(c) Defining missile / gun safety lanes in coordination with concerned ATC's.

(d) Defining safe passage for friendly forces through missiles / guns defended areas in coordination with Terminal Defence Control Centre (TDCC) coordinator.

(e) Coordinating with SC, SSC and SDC for target allocation to SAMs / Guns.

(f) Monitoring SAM / Gun engagements.

(g) Maintaining correct SAM/Gun status.

(h) Disseminating target data by digital and voice means to the batteries to the DSCR / VTWCC.

14. **Fighter Marshal (FM)**. Fighter Marshal is responsible for the following duties:

(a) On identification of target by SMCC, coordinate with area GMCCs on hot lines to build requisite situational awareness.

(b) Monitoring and maintenance of interceptor tracks. Assist respective Unit commanders, intercept controllers and SMCC battle staff in CAP management as well as vital committal decisions.

(c) Controlling aircraft in safety corridors.

(d) On directives of SDC, assisting aircraft in emergency.

(e) Assisting in the identification of selected tracks i.e. decipher the altitude bracket through interpolation of available height information and detection ranges of various sensors.

(f) Handing over CAPs to suitable GCI Unit.

(g) Keeping a continuous track of the raid as well as interceptors and warn GMCC Fighter Marshals regarding any emerging threat or flight safety concern.

(h) Managing CAP stations as per requirement. It will be undertaken efficiently if correct fuel and ammo state is communicated to the SMCC FM by GMCCs.

15. **Filtration Officer**. The Filtration Officer is responsible to the SDC for the following duties:

(a) Coordinate with MIO in identifying the MOU reports.

(b) Carry out accurate decoding of MOUs symbology on MOU reports.

(c) Resolve ambiguities arising out of multiple reports and to discontinue reports, if not required.

(d) Establish MOU tracks based upon MOU reports.

16. **Tracker**. Trackers in manual ops room are responsible to ASO for the following:

(a) Maintaining the current and clear air surveillance picture.

(b) Monitoring the tracking functions and taking corrective actions to improve tracking.

(c) Entry of manual tracks reported on speech circuits.

(d) Resolving ambiguities arising on identical reports on the same track by two or more sensors.

17. **Manual Entry Operator (MEO)**. The MEO is responsible to MIO and the supervisor for the following:

(a) Recording and maintenance of data received on voice data links.

(b) Recording and entering flight plan information into data base ADS.

(c) Feeding hourly weather information.

(d) Entry of MOU reports.

18. **SMCC Supervisor**. He is the senior most radar operator on duty and is responsible to SDC for the following duties:

(a) Overall discipline and control of ops room staff and to ensure maximum operational efficiency.

(b) Furnish his crew with the necessary equipment needed for their duties.

(c) Rotation of personnel for various positions in the ops room and arranging for relief periods.

(d) Giving exposure to the under-training crew for operational duty.

(e) Maintaining status summaries of PAF and ARMY radars. Detail of Live / SIM interceptions, record of allied data system, summary of x-raids, RQs etc.

(f) Reporting to SDC for any operational difficulties.

(g) Ensuring that the noise level and other causes of distraction are kept to Minimum.

(h) Ensure proper cleanliness of the Ops Room.

(j) Maintaining the Supervisor's Log Book.

(k) Adjustment of SMCC digital wall clock.

(l) Preparation of tracks summaries.

(m) Filling in the interception reports proforma.

(n) Ensuring entry of MOU reports.

(p) Passing essential information / instructions to associated CRCP for flying activity in restricted and prohibited area, reception of unusual activity, any activity along the LoC is being communicated to CRCP.

(q) Keeping track of all AD / MT moves.

19. **Plotters (Radar Operators)**. In Manual Ops Room of SMCC there are number of plotters behind the plotting board connected to readers / tellers at GCI Units or the receiving and reporting flight, as appropriate. The plotters are directly responsible to the Floor Supervisor for the following:

(a) Plotting the position, direction and amplifying data on all tracks, immediately and correctly, as received by him.

(b) Allocating next serial number to the first plot of a new, or reappeared track, over riding the serial number passed by GCI Units etc if necessary, and informing the source about the same.

(c) Maintaining the serial number and amplifying data along with the is received by him that same has been transferred to the board.

(d) Maintaining only the track number and identification alongside each track after transfer of data has been affected, except when a change occurs.

(e) Telling downwards the identification of each track to the source from where he received the track.

(f) Acknowledging each plot or any other detail from the reporting sources, with single word 'Roger'.

(g) Following the laid down plotting procedures in letter and spirit.

(h) Bringing to the notice of Floor Supervisor any difficulties faced by him.

(j) Carrying out communication checks with the other end at reasonable intervals to ensure continuous contact and bring to the notice of Floor Supervisor any break in contact.

20. **Recorders (Radar Operator)**. The recorders are to:

(a) Monitor the plotting line and record all information passed according to the laid down pattern and on the appropriate form.

(b) Write the time in minutes and seconds, except in case of initial plot and when track fades, with six figures time group, indicating hour also.

21. **Tellers (Radar Operators**). The tellers transmit the required surveillance and tactical information to the appropriate agencies:

(a) **Forward Teller**. Forward and upward teller passes the required surveillance information to a higher echelon in the Air Defence system. He reports all tracks displayed on the surveillance plotting board in a clock-wise direction.

(b) **Downward Teller**. Downward teller is to pass all required information to the Units, at lower level of command, for the successful completion of mission assigned to these Units.

(c) **Status Board Teller**. He is connected to the status board operators and is responsible for passing information from plotting board on all identified tracks including all changes, and is also to ensure correct display of the same.

22. **Status Board Operators (Radar Operators)**. They are connected independently to the Assistant Duty Controller, Floor Supervisor and Status Board Teller. They are responsible for displaying the tactical action data, interceptor and ADA state, ancillary information of all tracks, the electronic and weather state on the appropriate status and display boards.

23. **Tracers (Radar Operators)**. Tracers are responsible for reproducing accurate tracing of any required track, with the help of recording sheets. They are to ensure that such recorded plot is reproduced on the tracing sheet even if the track is known to have covered straight line path:

24. **OC Engineering Wing**. He is responsible to the Sector Commander for the following duties:

(a) Control and direction of supply and maintenance organization of the sector.

(b) Maintenance of Technical Orders and publications.

(c) Liaison with concerned authorities for maintenance of land line communication.

(d) Maintenance and operation of electronic, communication, power, air-conditioning equipment and MT vehicles.

(e) Maintenance and operation of equipment inventories and Unit Authorization List (UAL).

(f) Implementation and supervision of Unit Skill Enhancement Programme (SEP).

(g) Formulation and issuance of relevant SOPs and ensuring their implementation.

(h) Ensure that maintenance performed on assigned equipment is timely and of high quality and is able to achieve maximum operational capability.

(j) Lay down standard maintenance policies and procedures or Unit servicing orders for effective administration, supervision and operation of the maintenance organization.

(k) Delegate the authorities necessary for staff and maintenance production activities to perform assigned responsibilities.

(l) Control the assignment and use of all available maintenance personnel.

(m) Ensure conformance with policies, rules, and regulations pertaining to personnel assigned to the maintenance complex.

(n) Ensure implementation of applicable technical directives issued by specialist directorate of Air Headquarters and Headquarters Air Defence Command.

(p) Ensure that requirements necessary to support the maintenance mission are included in appropriate plans, and programmes.

(q) Control assignment and utilization of allocated maintenance facilities. Submit necessary documentation to appropriate agencies for new construction and modifications or alterations to existing facilities, as required.

(r) Review and take necessary corrective actions on all pertinent quality control inspection reports and post defect reports. Ensure that effective safety programs are established throughout the maintenance complex.

(s) Manage the logistics, transport (MT), and communications functions of the maintenance complex along with the requisite financial program.

(t) Ensure that a comprehensive training program is established throughout the maintenance complex. Ensure that training of newly assigned maintenance personnel and key supervisors to enable them to carry out their responsibilities.

(u) Assuring that personnel do not remain in staff positions for excessive periods and thereby lose their trade proficiency.

(v) Ensuring that all personnel assigned to the maintenance complex are used to accomplished critical wartime tasks such as Air Defence Exercises, Deployments of the Units, and other operational duties.

(w) Co-ordinate with Officer Commanding Operations to maintain a balance between operational and maintenance capabilities.

25. **O i/c Automation**. He is responsible to OC Engg for the following:

(a) Control and direction of supply and maintenance organization of the sector.

(b) Expeditious recovery of unserviceable assemblies.

(c) Availability of pre-issue and bench stock items.

(d) Annual and routine maintenance of equipment.

(e) Enhancing the level of expertise of his section.

(f) Ensure maintenance of environment condition for system operation.

(g) Review all relevant technical publication / tools monthly.

(h) Ensure implementation regularly training programs.

26. **O i/c Communications**. He is responsible to OC Engg for the following duties:

(a) Maintain communication circuits and equipment in the highest state of serviceability.

(b) Expeditious recovery of unserviceable assemblies.

(c) Ensure conduct of schedule maintenance of equipment with TOs / Publications.

(d) Liaison with concerned agencies for provision of communication streams / circuits.

(e) Timely activation of different tactical and operational circuits as and when directed.

(f) Regular updating of UAL, pre-issue and bench stock lists and ensure availability of spares.

(g) Availability of special tools and test equipment for system maintenance.

(h) Review and ensure availability of SOPs.

(j) Ensure implementation of QAP / QC programs.

(k) Enhancing the level of expertise of section through supervising training activities.

(l) To assist under command Units in communication related issues.

27. **O i/c Training / Engg Control**. He is responsible to the OC Engg Wing for the following duties:

(a) Conduct of Familiarization, Induction and Continuation training of the Personnel and maintenance of necessary training record.

(b) Indoctrinating the trainers and trainees on the objective of training programs.

(c) Conduct of OJT and maintenance of record concerning OJT and CTTB examinations.

(d) Ensure implementation of training policies, maintenance orders/SOPs.

(e) Maintain historical record of the Unit equipment.

(f) Take necessary measures to improve skill level of personnel and ensure correct use of tools / test-equipment by them for the fault rectification.

(g) Monitor the effectiveness of training program through regular quiz tests.

(h) Availability and circulation of study material.

(j) Cross training of Engg Officers and airmen of technical trades.

(k) Procure and maintain necessary training aids.

(l) Consolidate and process requirement of technical publications of various sections.

(m) Maintain latest serviceability state of all equipment held on the Inventory of Engg Wing.

(n) Liaison with other Units to arrange availability of trainees / instructors as and when required.

(p) Incorporate necessary changes in AFOs, AFLs, AFMs and Technical Publications when received from Air Headquarters and bring it to the knowledge of all concerned.

(q) Maintain all files, documents and record pertaining to maintenance activities.

(r) Efficient working of Supply Section.

(s) To ensure upkeep of concerned sections and flights.

(t) To closely monitor the activity of complete section / flight.

(u) To keep record and ensure implementation of orders and instructions given by OC Engg Wing and Sector Commander.

(v) To ensure timely submission of all relevant correspondence, returns and RCNs.

(w) To ensure that all NCO i/c assigned with different tasks are very well familiar with their respective jobs.

(x) To keep OC Engg informed about various matters of his sections.

(y) To ensure timely replies of all correspondence of concerned sections.

(z) To maintain close liaison with Under Command Units, AHQs and ADC to meet the requirement of UC Units.

(aa) To guide Maint Control regarding drafting of letters and preparation of data.

(ab) To ensure timely maintenance of AD assets in coordination with UC Units and ADC.

(ac) To maintain close coordination with under-command Units regarding routine correspondence.

(ad) To stay in touch with Oi/c FMSs / SEOs / EOs regarding their un-serviceabilities

28. **O i/c R & R Flight**. He is responsible to OC Engg Wing for the following:

(a) Efficient functioning and administration of R&R Flight.

(b) Efficient functioning of all W/T, ops and admin channels as per schedule.

(c) Maintenance of all ground radio equipment to ensure minimum 'down time' and optimum performance.

(d) Frequency and call sign allocation to MOUs participating in operations.

(e) Ensuring timely distribution of Aircraft Recognition and Authentication (ACRA) and Station Identification Challenge Code (SICC) cards to the representatives of MOUs deployed.

(f) Periodic and scheduled maintenance of all electronic equipment of the flight.

(g) Take anti-corrosion preventive measures for installed antennas.

(h) Ensure proper discipline on channels and report any breach of security to OC Engg Wing.

(j) Clearance of all traffic in accordance with the precedence.

(k) Ensure personnel on duty are conversant with operating procedures.

# CHAPTER - 6

# GENERIC MISSION CONTROL CENTRE

## Introduction

1. Generic Mission Control Centres (GMCCs) are connected with a network of high, medium, and low-level radars operating under respective SMCC. GMCC operates in the automated Air Defence system in netted or at times in autonomous modes like remote radars. They pass the required data to SMCC on data links / voice circuits depending upon mode of operations. The duties and responsibilities of different personnel working at GMCC are given in succeeding paragraphs.

## Duties and Responsibilities

2. **Officer Commanding**. The Officer Commanding is responsible to Sector Commander for efficient operations, maintenance, security and administration of his Unit. He is to ensure the following:

(a) Highest operational vigilance by GMCC Ops crew.

(b) Optimum performance of on-watch sensors (Radars, PDS, Pantacle etc).

(c) Maintenance and serviceability of all operational assets (deployed / Storage).

(d) Visit of War sites, ensuring their readiness for deployment and recommend changes (if any) to higher command.

(e) Unit is ready for deployment at short notice.

(f) Organize and execute an effective professional training programme for all under command personnel as per existing policies.

(g) Develop individual’s initiative, sense of responsibility, leadership qualities and maintain high morale amongst Unit personnel.

(h) QA programme is implemented in the Unit through effective coordination with all concerned.

(j) Security of Unit and its assets through effective coordination with all concerned.

(k) All orders instructions and directives issued by the Sector Commander and higher echelons are known and complied with by all concerned.

(l) Any other duty assigned by the higher echelons.

3. **Second-in-Command (2 i/c)**. 2 i/c is responsible to Officer Commanding Wing for overall operations, administration, training, and war readiness of his Unit. His responsibilities are as following:

(a) Supervision of overall Air Defence operations.

(b) Correct surveillance / interception techniques are followed by all controllers under his command.

(c) Ensure all controllers possess valid Ops categories.

(d) Preparation / updating of SOPs and LOPs.

(e) Ensure that orders and instructions issued by the OC Unit and other higher commanders are relayed, understood and implemented by all personnel under his command.

(f) Ensure implementation of physical security policies and procedures at Wing Headquarters and radar sites.

(g) Carry out supervision of all activities of Unit in the absence of Officer Commanding.

4. **Flt Cdr (Operations)**. He is responsible to 2 i/c regarding all operational matters. His specific duties include the following:

(a) Supervision of the operations.

(b) Ensuring timely and accurate flow of the surveillance and other Air Defence data to the SMCC.

(c) Preparing duty roster for the officers.

(d) Checking / Signing of all log books, proforma, reports and returns etc regarding day to day operations.

(e) Ensure orders and instructions issued by the OC / 2 i/c Unit and other higher commanders are relayed, understood and implemented by all personnel under his command.

5. **Flt Cdr (Training)**. The Officer Commanding Wing will detail a qualified AD officer holding at least operational category 'B' as the Unit training officer. He will be responsible to 2 i/c for the following points:

(a) Implementing training policies for the training of AD officers and Radar Operators.

(b) Maintaining individuals training record.

(c) Ensuring adequate utilization of the simulator by each controller.

(d) Procuring and maintaining training aids as required.

(e) Planning / publishing academic training program for the officers and airmen as per schedule outlined in the master training folder of the Unit.

(f) Ensuring timely categorization of AD officers.

(g) Maintaining constant check on professional proficiency of AD officers and Radar Operators.

(h) Conducting the training of officers on cross training.

(j) Reviewing the practical / academic training procedures and syllabus for AD officers and Radar Operators and advising the 2 i/c Unit about the required changes.

6. **O i/c Top Secret Registry (TSR) / Staff Operations Officer (SOO)**. The Officer Commanding will detail a qualified AD officer as O i/c TSR. He will be responsible to the 2 i/c for the following:

(a) Safe custody of Unit war book and Top Secret documents.

(b) Update all the war plans and highlight the anomalies and observations to OC Unit and SMCC (SOO).

(c) Ensure TSR correspondence / handling as per policy.

(d) Delegate O i/c for different war plans and take regular feedback.

(e) Ensure requisite assets and manpower is available to execute Unit's wartime role in designated time.

(f) Arrange quarterly war book review meetings.

(g) Ensure education / training to all the Unit personnel about their war time duties.

7. **Duty Controller**. Duty Controller is the senior most AD officer of the shift who is responsible to Flt Cdr (Ops) for the following during the tenure of his duties:

(a) Conduct of Air Defence operations at peak performance level.

(b) Ensure uninterrupted flow of data to SMCC.

(c) Safe handing / taking over and effective utilization of interceptors for practice / hot interceptions.

(d) Inform SMCC / OC / 2 i/c Unit of all abnormal air activities.

(e) Assess operational state of the equipment through performance control check procedures.

(f) Preparation of records concerning tracks of special interest.

(g) Assign specific Ops duties to the personnel of his shift.

(h) Ensure discipline of the personnel in the shift.

8. **DCA Leader**. DCA leader is responsible for the following:

(a) Broad employment of DCA game plan. He is to integrate CAP effort for generating a composite response to any threat.

(b) Coordination with HVAA Marshal / PDS for ESM ID and pass it to CAP controllers.

(c) Deciphering raids in exercises / contingencies.

(d) Allocating intercept controllers CAPs to intercept suitable raids.

(e) De-conflicting CAPs and ensuing positive identification of adversary and friendly aircraft within an FAOR.

(f) Coordinating with SMCC for de-confliction of interceptors and GBADs in Joint Engagement Zone (JEZ).

9. **Air Surveillance Officer (ASO) / Track Management Officer (TMO)**. He is responsible to the Duty Controller for efficient airspace management. His duties are as follows:

(a) Update / delete the false tracks.

(b) Send recommended ID on all established tracks.

(c) Apply NAI in consultation with Sector ASO.

(d) Monitor air situation and report any violation of flight plan or area to the duty controller and SMCC.

(e) Take appropriate remedial actions to counter Electronic Attack (EA).

(f) Report all aircraft emergencies / unusual occurrence to the duty controller.

10. **Intercept Controller**. Intercept Controller is detailed by the duty controller. His duties are as follows:

(a) Conduct interception against the target assigned to him, using approved tactics and techniques.

(b) Provide Interceptor recovery services.

(c) Provide Flight Following Services (FFS).

(d) Aiding aircraft in distress.

(e) Provide weather or any other information requested by the aircraft.

(f) Obtaining pilot reports of weather aloft.

(g) Remain thoroughly acquainted with publications and procedures concerning his duties.

11. **$C^2$ Electronic Warfare Officer (EWO)**. Electronic Warfare Officer at GMCC is an Air Defence officer who is responsible to incorporate EW information accrued through various air and ground active/passive systems. EWO is responsible to GMCC Duty Controller for the following:

(a) To ensure implementation of GMCC/Sqn level EMCON Policy as per prevalent peacetime / contingency / wartime scenario.

(b) To assess electronic attack and subsequently apply GMCC/Sqn level EP plan in coordination with SMCC as per prevalent environment.

(c) To coordinate with DA-20/ GBJs and GBDTs for mitigation of un intentional radio / radar interference.

(d) To infer ESM information from airborne / ground-based systems for further integration into defensive / offensive operations.

(e) During execution of operations, he is responsible to provide tactical ESM information to Lead DCA / Offensive controller in facilitation of Combat ID.

(f) To coordinate and integrate COMINT information (Monitoring flights / DAI) for effective conduct of operations.

(g) To ensure communication security through specified codes.

(h) To ensure R/T authentication codes while guarding against spoofing

12. **Performance Control Officer (PCO)**. He is responsible to the duty controller for the following:

(a) Determining the equipment status by employing the standard performance control procedures on selected tracks.

(b) Record all the performance checks on appropriate proforma.

(c) Notify the duty controller, in case performance of the equipment falls below the operational standards.

13. **High Value Airborne Asset Marshal**. HVAA Marshal is responsible to the duty controller for the following duties:

(a) Smooth flow of AEWACS data to GMCC.

(b) Maintain R/T with AEWACS for coordination and threat information.

(c) Remain standby to hand over ACI missions of AEWACS to suitable Intercept Controller in case AEWACS is unable to take over the mission.

(d) Pass continuous threat updates to HVAAs during retrograde.

(e) Give join up to interceptors for air to air refuelling.

14. **Detachment Commander (Field Radars)**. He is responsible to the OC GMCC for following duties and responsibilities:

(a) Overall in charge of radar Dett on the matters of operations, security and administration.

(b) Ensure that the radar is timely switched on/off as per the received instructions of GMCC.

(c) Ensure the Ops equipment is providing operations at its best possible capacity. He is to regularly carry out Radar Quality Control (RQC) from site and pass the status to GMCC.

(d) Ensure that the Ops equipment at Dett is maintained and serviced as per laid down procedures.

(e) Ensure that all communication lines are serviceable.

(f) In case of adverse weather conditions, take appropriate decision regarding ops equipment (as per TOs / SOPs / LOPs) and inform higher command.

(g) Ensure all Ops manuals / SOPs / LOPs are available and updated from Headquarters.

(h) Ensure no unauthorised personnel enters the technical camp of Dett.

(j) Maintain good relations with surrounding area notables.

(k) Ensure availability and serviceability of firefighting equipment and first aid box for technical and domestic camp both.

(l) Maintain list of following nearest setups:

(i) Police station / Police check post

(ii) Sister services

(iii) Basic Health Unit (BHU) / hospital / medical facility

(iv) Fire brigade

(v) Rescue 1122 / Edhi etc

(m) Ensure general cleanliness of technical and domestic camp.

(n) Ensure general discipline, keenly observe all personnel of the Dett on matters of discipline and report to Headquarters in case of any lapse.

(p) Ensure harmony among all personnel by arranging regular sports and social activities and combined meals (Bara Khana).

(q) Detail personnel for security duties. Carry out surprise check and ensure security duties are carried out properly. He is to ensure security personnel have carried out Ground Defence Training.

(r) Keenly observe security related issues and possible lapses and apprise OC GMCC.

(s) Ensure Service Mechanical Transport (SMT) is not misused and driven by authorised personnel only possessing valid service driving licence.

(t) Ensure security weapons are placed under lock (mini armoury) and a proper mechanism exists for issuance and recovery of weapons to concerned personnel.

15. **Officer-in-charge Field Maintenance Squadron (FMS)**. He is responsible to the Officer Commanding Wing for the following:

(a) Control and direction of supply and maintenance organization of the Unit.

(b) Maintenance of technical library and implementation of technical orders issued by Air Headquarters.

(c) Liaison with concerned authorities for maintenance of landline communications.

(d) Maintenance and operation of electrical and mechanical equipment including radar, automation, communication, MT vehicles and power generators.

(e) Maintenance of cypher security and communication orders.

(f) Maintenance of tools and test equipment.

(g) Record of maintenance accomplished.

(h) Maintenance and up-dation of equipment inventories and UAL.

(j) Maintenance, review and updating of the bench stock and pre-issue holding of spares.

(k) Introduction, indoctrination and encouraging ground safety programmes self-reliance programme and cost consciousness in maintenance accomplished by the personnel.

(l) Implementation and supervision of Units Skill Enhancement Programme (SEP).

(m) Formulation and issuance of relevant SOPs.

(n) During operational moves / deployments / redeployments of the Unit or its detachments, he is to work out conveyance requirements including MT vehicles / railway or air transport and arrange procurement. Subsequently, he is to supervise dismantling / loading and unloading/installation of all the technical / operational equipment.

(p) Workout and procure the fuel for the move and operational deployment.

16. **O i/c Radar / GMCC**. He is responsible to O i/c FMS for the following duties:

(a) Handle all maintenance problems of Radar / GMCC.

(b) Maintain liaison with fault control sections and standby maintenance party and ensure their availability.

(c) Liaise with radars for effective reporting of faults.

(d) Ensure the availability of spare equipment for all ranges of equipment.

(e) Ensure prompt action for rectification of fault in shortest possible time.

(f) Ensure timely dispatch of equipment at sites if required.

(g) Inform O i/c FMS about any un-serviceability or damage to the equipment at Wing Headquarters / deployed radar sites.

(h) Keep track of all maintenance activities at Wing Headquarters and radar sites.

17. **O i/c Communication**. He is responsible to O i/c FMS for the following duties:

(a) Look after all the communications related problems of GMCC and radar sites.

(b) Liaise with PTCL / SATCOM Hubs / relay sites for effective reporting of faults.

(c) Ensure the availability of spare support of all communication equipment.

(d) Ensure prompt action for rectification of communication fault in shortest possible time.

(e) Inform O i/c FMS about any un-serviceability or damage to the equipment at Wing Headquarters / deployed at sites.

(f) Keep track of all maintenance activities at GMCC and relay sites.

18. **O i/c MT & MTR&I**. He is responsible to the O i/c FMS Unit for the following duties:

(a) Ensure serviceability and availability of all MT / specialist vehicles and generators.

(b) Ensure the availability of spare equipment of all MT / Specialist vehicles and generators.

(c) Ensure prompt action for rectification of fault in shortest possible time.

(d) Keep a track of all the movements of MT vehicles.

(e) Regularly brief MTDs and CDs about safe driving techniques.

(f) Ensure all MTDs / CDs have valid driving licence and TMAs.

(g) Keep track of all maintenance activities at Wing Headquarters and radar / relay sites.

(h) Visit the Wing Headquarters in case of any eventuality.

(j) Brief on maintenance activities in the next daily morning brief.

19. **Surveillance Supervisor**. Surveillance Supervisor at a Wing is responsible to Duty Controller for the following:

(a) Maintaining a constant check of the air surveillance equipment and ensuring a current and accurate display of the air situation and its flow to SMCC.

(b) Ensuring that information concerning, AP, clouds, unusual weather conditions and jamming, is brought to the notice of the Duty Controller. When precipitation echoes are observed at the radar scope, he is to designate a PPI operator to monitor the echoes.

(c) Guiding his crew in all cases to ensure the maximum use of the equipment even in case of AP and jamming etc.

(d) Ensuring that his crew adheres to the established procedures of air surveillance.

(e) Calling the duty technicians for rectifying the defects.

(f) Operational training of his crew.

(g) Ensuring the operational status information and changes thereafter are passed to appropriate authorities.

(h) Maintaining proper log of occurrences and changes in duties.

20. **Search Radar Operator**. The Search Radar Operator is responsible to ASO / supervisor for the following:

(a) Familiarizing himself with the scope picture and current air situation prior to relieving the operator on duty.

(b) Ensuring that the communication circuits are serviceable.

(c) Making necessary adjustments of the scope controls for optimum performance.

(d) Ensuring track generation on all positive plots.

(e) Remaining vigilant and to be constantly on the lookout for new tracks that may appear in his designated area of the scope.

(f) Deletion of false tracks.

(g) Informing the supervisor about all cases of AP, jamming or defects in the scope.

(h) Reporting to the ASO / supervisor immediately, in case any track of special interest.

21. **Intercept Control Technician (Radar Operator)**. He is responsible for the following points:

(a) Get mission details from ICs prior to the mission.

(b) Fill in the tactical engagement report forms during the mission.

(c) Get the form signed from ICs (along with remarks).

(d) Update the GCI in record folder.

(e) Pass the record to Ops Office and SMCC.

22. **Plotters (Radar Operators)**. In Manual Ops Room of GMCC there are number of plotters behind the plotting board connected to readers / tellers at GCI Units or the receiving and reporting flight, as appropriate. The plotters are directly responsible to the Surveillance Supervisor for the following:

(a) Plotting the position, direction and amplifying data on all tracks, immediately and correctly, as received by him.

(b) Allocating next serial number to the first plot of a new, or reappeared track, over riding the serial number passed by GCI Units etc if necessary, and informing the source about the same.

(c) Maintaining the serial number and amplifying data along with the received by him that same has been transferred to the board.

(d) Maintaining only the track number and identification alongside each track after transfer of data has been affected, except when a change occurs.

(e) Telling downwards the identification of each track to the source from where he received the track.

(f) Acknowledging each plot or any other detail from the reporting sources, with single word 'Roger'.

(g) Following the laid down plotting procedures in letter and spirit.

(h) Bringing to the notice of Surveillance Supervisor any difficulties faced by him.

(k) Carrying out communication checks with the other end at reasonable intervals to ensure continuous contact and bring to the notice of Surveillance Supervisor any break in contact.

23. **Recorders (Radar Operator)**. The recorders are to:

(a) Monitor the plotting line and record all information passed according to the laid down pattern and on the appropriate form.

(b) Write the time in minutes and seconds, except in case of initial plot and when track fades, with six figures time group, indicating hour also.

24. **Tellers (Radar Operators**). The tellers transmit the required surveillance and tactical information to the appropriate agencies:

(a) **Forward Teller**. Forward and upward teller passes the required surveillance information to a higher echelon in the Air Defence system. He reports all tracks displayed on the surveillance plotting board in a clock-wise direction.

(b) **Downward Teller**. Downward teller is to pass all required information to the Units, at lower level of command, for the successful completion of mission assigned to these Units.

(c) **Status Board Teller**. He is connected to the status board operators and is responsible for passing information from plotting board on all identified tracks including all changes, and is also to ensure correct display of the same.

**CHAPTER - 7**

# HIGH LEVEL RADAR SQUADRON

## Introduction

1. High level radars are netted with GMCCs in automated Air Defence system. They pass required data to GMCCs on data links / voice circuits depending upon the mode of operations. The duties and responsibilities of different personnel working at High Level Radars are given in the succeeding paragraphs.

## Duties and Responsibilities

2. **Officer Commanding**. The Officer Commanding is responsible to Sector Commander for efficient operations, function and administration of his Unit. He is to ensure the following:

(a) Highest standards of performance, maintenance and serviceability of all operational assets.

(b) Visit of War sites, ensuring their readiness for deployment and recommend changes (if any) to higher echelons.

(c) Unit is ready for deployment at short notice.

(d) Correct surveillance / interception techniques are followed by all Fighter Controllers under his command.

(e) Organize and execute an effective professional training programme for all under-command personnel in-line with existing policies.

(f) Develop individual's initiative, sense of responsibility, leadership qualities and maintain high morale amongst Unit personnel.

(g) QA programme / practices are implemented in the Unit through effective coordination with all concerned.

(h) Security of Unit through effective coordination with all concerned.

(j) All orders instructions and directives issued by the Sector Commander and higher echelons are known and complied with by all concerned.

(k) Any other duty assigned by the higher echelons.

3. **Second-in-Command (2 i/c)**. He is responsible to Officer Commanding for overall operations, administration, training and war readiness of his Unit. His responsibilities are as follows:

(a) Supervision of overall Air Defence Operations.

(b) Orders and instructions issued by OC Unit are relayed, understood and implemented by all personnel under his command.

(c) Correct interception techniques are followed by all Fighter Controllers under his command.

(d) All Fighter Controllers possess valid Ops categories.

(e) Ensure implementation of training policies for training of AD Officers and Radar Operators.

(f) Implement physical security policies and procedures at Unit Headquarters and radar sites.

(g) Ensure safe custody of Unit war book and Top-Secret documents.

(h) Supervise overall activities of Unit in absence of Officer Commanding.

4. **Officer In-charge Training**. The Officer Commanding Unit will detail a qualified AD officer (preferably CCQ / Minimum Cat 'B') as Unit training officer. His duties would include following:

(a) Implementing training policies for the training of fighter controllers (FCs) and radar operators of Unit as per policies in vogue.

(b) Maintaining an updated training record of the individuals.

(c) Ensuring adequate utilization of the simulator by all FCs.

(d) Ensuring availability of training aids to meet training requirements of Unit personnel.

(e) Planning / publishing academic training programme for FCs and Radar Operators as per schedule.

(f) Ensuring timely categorisation of FCs.

(g) Maintaining constant check on the professional proficiency of FCs and Radar Operators.

(h) Planning and conducting cross training of FCs and Radar Operators.

(j) Reviewing practical / academic training procedures and syllabus for Ops crew and advising OC Unit about the required changes.

(k) Procuring and maintaining training aids.

5. **Duty Controller**. Duty Controller is senior most AD officer of shift who is responsible to Second-in-Command for following during tenure of his duty:

(a) Conduct of Air Defence operations at peak performance level.

(b) Uninterrupted flow of data to GMCC / SMCC.

(c) Safe handing / taking over and effective utilization of interceptors for practice, ID and hot interceptions.

(d) Inform SMCC / GMCC and OC Unit of all abnormal air activities.

(e) Assess operational state of equipment through performance control check procedures.

(f) Preparation of records concerning tracks of special interest.

(g) Assign specific Ops duties to personnel of his shift.

(h) To ensure discipline of the personnel in the shift.

6. **Senior Engineering Officer (SEO)**. He is responsible to Officer Commanding Unit for the following:

(a) Control and direction of supply and maintenance organisation of the Unit.

(b) Maintenance of technical library and implementation of technical orders issued by Air Headquarters.

(c) Liaison with concerned authorities for maintenance of landline communications.

(d) Maintenance and operation of electrical and mechanical equipment including radar, automation, communication, MT vehicles and power generators.

(e) Maintenance of cipher security and communication orders.

(f) Maintenance of tools and test equipment.

(g) Record of maintenance accomplished.

(h) Maintenance and updating equipment inventories and UAL.

(j) Maintenance, review and updating of the bench stock and pre-issue holding of spares.

(k) Introduction, indoctrination and encouraging ground safety programmes self-reliance programme and cost consciousness in maintenance accomplished by the personnel.

(l) Implementation and supervision of in vogue training policies.

(m) Formulation and issuance of relevant SOPs.

(n) During operational moves / deployments / redeployments of the Unit or its detachments, he is to work out conveyance requirements including MT vehicles / railway or air transport and arrange procurement. Subsequently, he is to supervise dismantling / loading and unloading / installation of all the technical / operational equipment.

(p) To work out and procure the fuel for the move and operational deployment.

7. **Search Radar Operator (Remote Radar Site Only)**. Search Radar operator is positioned in front of radar scope and is responsible for following:

(a) Familiarizing himself with scope picture and current air situation prior to relieving radar operator on duty.

(b) Ensuring that communication circuits are serviceable.

(c) Making necessary adjustments of scope controls for optimum performance.

(d) Remaining vigilant and to be constantly on lookout for new tracks that may appear in his designated area of scope.

(e) Deletion of false tracks.

(f) Informing surveillance supervisor about all cases of AP, jamming or defects in scope.

(g) Prompting supervisor immediately, in case any track is observed indicating emergency or any unusual track.

(h) Manually reporting of all tracks to GMCC / SMCC in a clock-wise direction when required.

8. **Intercept Control Technician (Radar Operator)**. He is responsible for the following points:

(a) Get mission details from ICs prior to the mission.

(b) Fill in the tactical engagement report forms during the mission.

(c) Get the form signed from ICs (along with remarks).

(d) Update the GCI in record folder.

(e) Pass the record to Ops Office and SMCC.

9. **Recorders (Radar Operator)**. The recorders are to:

(a) Monitor the plotting line and record all information passed according to the laid down pattern and on the appropriate form.

(b) Write the time in minutes and seconds, except in case of initial plot and when track fades, with six figures time group, indicating hour also.

10. **Forward Tellers (Radar Operators**). The tellers transmit the required surveillance and tactical information to the appropriate agencies. Forward and upward teller passes the required surveillance information to a higher echelon in the Air Defence System. He reports all tracks displayed on the radar scope in a clock-wise direction.

**CHAPTER - 8**

# AIRBORNE EARLY WARNING AND CONTROL SYSTEMS

## Introduction

1. Historically, military planners have found situational awareness of hostile targets and friendly forces to be a key component in obtaining and sustaining military superiority over adversaries. With the introduction of AWACS, the situational awareness of potential targets and friendly aircraft over hundreds of square miles of airspace became a reality. In offensive and in defensive campaigns, AEW is often the first in and the last out, providing essential Surveillance, Command and Control capabilities throughout the duration of operation. The duties and responsibilities of different personnel working in AEWACS Squadron are given in the succeeding paragraphs.

## Duties and Responsibilities of Airborne Early Warning & Control (AEW&C)

2. **Officer Commanding**. The Officer Commanding is a qualified Captain and EMS operator who is responsible to Officer Commanding flying wing for efficient operations, function and administration of his squadron. He is to ensure that:

(a) Highest operational standards are maintained by his squadron.

(b) All pilots / controllers possess valid Ops categories.

(c) Operations are being conducted as per laid down procedures.

(d) OJT and training of personnel under his command are carried out properly.

(e) All orders instructions and directives issued by the Air Headquarters, Regional Air Command and other higher commanders are known and complied with by all concerned.

3. **Flt Cdr Ops GD (P)**. The Flt Cdr Ops GD (P) is responsible to Officer Commanding of Squadron for all operational matters pertinent to flying operations. His specific duties include the following:

(a) Supervision of the flying operations.

(b) Orders and instructions issued by the OC squadron and other higher commanders are relayed, understood and implemented by all flying crew under his command.

(c) Preparation / updating of SOPs / Ops Orders pertaining to flying operations.

(d) Preparing flying program while ensuring flying crew system currency.

(e) Signing of all log books, proforma, reports and returns etc regarding day to day operations.

4. **Flt Cdr Ops (AD)**. The Flt Cdr Ops (AD) is responsible to Officer Commanding of the squadron for all operational matters pertinent to AEW operations. His specific duties include the following:

(a) Supervision of the operations.

(b) Orders and instructions issued by the OC squadron and other higher commanders are relayed, understood and implemented by all AD crew under his command.

(c) Preparation / updating of SOPs / Ops Orders pertaining to $C^2$.

(d) Preparing flying and ACI planner of AD crew in order to meet the operational and training requirements.

(e) Signing of all log books, proforma, reports and returns etc regarding day to day operations.

5. **Flt Cdr Training**. The Officer Commanding squadron details a qualified GDP and AD officer as the squadron training officer. Both the Flight Commanders are responsible to their respective Flt Cdr Operations for the following:

(a) Implementing training policies for the training of GDP and AD officers.

(b) Maintaining individuals training record.

(c) Ensuring adequate utilization of the simulator by each flying and $C^2$ crew.

(d) Procuring and maintains training aids as required.

(e) Planning / publishing academic training programme for the officers as per schedule outlined in the master training folder of the squadron.

(f) Ensuring timely categorization of officers.

(g) Maintaining constant check on the professional proficiency of officers.

(h) Conducting the training of officers through cross training visits.

(I) Reviewing the practical/academic training procedures and syllabus for officers and advising the Flt Cdr Ops about the required changes.

6. **AEW Commander**. He is overall in-charge of the platform entrusted with the responsibility of safe mission accomplishment through judicious use of resources. He is the senior supervisor onboard, regulating and monitoring the activity of Ops and Aircrew and would act as the link between Higher Command and the Airborne Platform.

7. **Captain of the Aircraft**. He is responsible for the safe conduct of flight and employment of aircraft in line with the instructions of the AEW commander. He will be assisted by the co-pilot in mission accomplishment while remaining within the system limitations.

8. **Tactical Director (TAC-D)**. The tactical director during the tenure of his duty is responsible to the AEW commander for efficient conduct of the assigned mission by best exploitation of available man and machine resource of AEW&C. In this regard his duties include:

(a) To plan an optimized AEW&C orbit in coordination with captain of the aircraft.

(b) To carry out task assignment of the ops cabin crew commensurate with their exposure and seniority.

(c) To conduct thorough briefing prior to launching of the mission as per the mission crew detailing.

(d) To assist the mission commander in safe and efficient conduct of assigned task.

(e) To monitor the conduct of Tactical managers and assist them in correct execution of game plan.

(f) To keep close liaison with SDC in SMCC via the data link and perform the emerging operational tasks in consultation with the mission commander.

9. **Link Manager**. The AEW&C data link would remain an integral part of its operations during war as well as peacetime. It would be critical for an AEW&C to remain connected with designated SMCC for down telling own picture and receiving identified picture from SMCC. In the same context, the primary duties and responsibilities of link manager are as given below:

(a) To plan and brief data link related aspects with designated DLI operator in SMCC.

(b) To establish primary and secondary GESs during the conduct of the mission.

(c) To identify requirement of connecting as MLU or RLU in the link setup.

(d) To ensure that the mission DTM contains all data concerning the link setup during the mission.

(e) To conduct the data link in the most efficient manner and to ensure that the AEW&C remains connected with concerned SMCC during the entire duration of the mission while receiving identified picture.

10. **Sensor Manager**. The Sensor Manager during his tenure of duty is responsible to tactical manager for efficient conduct of both (PSR & SSR) the active sensors on board the AEW&C. The duties of sensor manager include the following:

(a) To recommend a suitable sensor management plan to the tactical director depending upon area of operations, primary and secondary tasks, nature of threat etc.

(b) To implement the sensor management plan in most efficient and effective manner.

(c) To periodically monitor the sensor performance data (fault location in system) and advise the tactical director regarding any deterioration in sensor hardware status.

(d) To keep critical sensor parameters (Radar Cycle and Radar Load) within allowable limits so that the sensor performance does deteriorate during operations.

(e) To inform the tactical director if the sensor performance is likely to degrade due to excessive load on PSR.

(f) To ensure modifications in sensor management plan based on emerging operational requirements.

11. **Tactical Manager**. Following are the duties and responsibilities of tactical manager:

(a) Preparation and Loading of mission in $C^2$ system.

(b) During surveillance operations perform ASO duties.

(c) Act as intercept controller whenever required.

(d) Log writing in a chronological order and note the timings of all important events.

(e) Assist link manager in acknowledgement of pointers received from DLI / SMCC.

(f) Retrieval of recorded data for debrief and analysis.

**Duties and Responsibilities of Airborne Warning and Control System (AWCAS)**

12. **Officer Commanding**. The Officer Commanding is a Qualified AEW Commander who is responsible to OC Flying Wing for maintaining a high standard of operational effectiveness and combat readiness of his squadron. His specific duties are:

(a) Ensure smooth conduct of routine operations as well as special operations as directed by Air Headquarters.

(b) Organize and execute an effective operational training program for Squadron Pilots, Controllers, Air and Ground Crew.

(c) Strive and achieve the highest standard of operational efficiency and discipline.

(d) Check and eradicate weaknesses, faulty techniques and unsafe practices in operations.

(e) Maintain high standard of maintenance and ensure that QA program / practices are implemented in the squadron.

(f) Develop flight safety consciousness amongst his personnel.

(g) Pay regular visits to FOBs / Detts concerned and ensure that squadron is ready for deployment at short notice.

(h) Keep war plan updated as per war book.

(j) Maintain high morale amongst his personnel and ensure sound standards of physical fitness.

(k) Any other duty assigned by the Base Commander / OC Flying Wing or required to be carried out as per Air HQ policy.

13. **Flight Commander Operations (FCO)**. Flight Commander (Ops) is a Qualified AEW Commander that is deputy to the Officer Commanding and responsible to him for the duties enumerated below:

(a) Conduct and ensure efficiency of operations and training of all Squadron pilots as laid down in flying policies directed by Air Headquarters.

(b) Effectively plan and execute cyclic training for the Squadron pilots as approved by Air Headquarters.

(c) Ensure all flying related documents i.e. Authorization book, Log Books etc and Post Exercise Reports are prepared and updated.

(d) Ensure all flying activity is conducted within the bounds of flight safety and air force flying policies.

(e) Ensure effective training of the pilots for specialist role.

(f) Monitor closely the performance of squadron pilots and brief Squadron Commander on anyone's shortcomings.

(g) Ensue regular updating of war plans.

(h) Plan and execute Air Defence / Operational exercises, ensuring participation of all available pilots.

(j) Supervise and accomplish of secondary duties by the officers who report through him to the Officer Commanding.

(k) Conduct the Squadron day-to-day affairs as "Acting Squadron Commander" in his absence.

14. **Flight Commander Air Defence (Operations) {FCAD (Ops)}**. FCAD is a Qualified Combat Controller directly responsible to Officer Commanding for the duties enumerated below:

(a) To plan and execute AWACS $C^2$ Ops / exercises.

(b) To chalk out a broad training plan based on cyclic syllabi.

(c) To monitor closely the performance of $C^2$ fighter controllers.

(d) To ensure that highest operational standards are maintained in his area of responsibility.

(e) To ensure correct AWACS procedures are followed by all controllers under his command.

(f) To ensure all controllers possess valid Ops categories.

(g) Safe custody of TSR and regular updating of war plans.

(h) Continuation training of personnel under his command.

(j) Orders and instructions issued by the OC Unit and other higher commanders are relayed, understood and implemented by all personnel under his command.

(k) Preparation / updating of SOPs / Ops Orders.

(l) Detailing of $C^2$ mission crew as per the requirement of AWACS mission.

(m) Signing of all log books, proforma, reports and returns etc regarding day to day operations.

(n) To supervise and accomplish of secondary duties by the officers who report through him to the Officer Commanding.

15. **Flight Commander Flying Training {Flt Cdr (Trg)}**. Flight Commander Flying Training is directly responsible to Flt Cdr (Ops) for the following activities to be accomplish on monthly basis:

(a) Prepare a forecast of monthly ground training and plan the lectures / phase briefs and other tools for managing the requisite ground training.

(b) Plan morning sessions for the month which is to include platform sub-systems, $C^2$ sub-systems, ESM and one-minute safety talk.

(c) Ensure conduct of monthly quizzes, essential exercises (blind fold, evacuation drill etc) and ground training lectures.

(d) Prepare ATT program for the newly inducted flight crew in co-ordination with Flight Commander Trg (AD).

(e) Review log books, training folders (pilots / FE), ground training register, and maintenance of told card booklet every month.

(f) Monitor of FE training activities in co-ordination with CFE.

(g) Keep track of training aids availability and aircrew bag essential items.

16. **Flight Commander Training (AD)**. Flight commander Air Defence training is directly responsible to Flt Cdr Ops (AD) for the duties enumerated below:

(a) Implement training policies and CTS for $C^2$ Officers.

(b) Plan, publish and conduct monthly ground training program including morning sessions, quizzes, exercises (fire and evacuation drill) and ground training lectures.

(c) Maintain a constant check on the professional proficiency of AD Officers.

(d) Ensure adequate utilization of the KETS by each controller.

(e) Ensure update individuals training record including $C^2$ log books, training folders, ground training register.

(f) Ensure timely categorization of AD officers.

(g) Conduct the training of officers on AEW exposure / currency.

(h) Procure and maintain training aids as required.

17. **Senior Engineering Officer (SEO)**. SEO is responsible to the Officer Commanding for the technical supervision and maintenance of the squadron. The SEO, assisted by EOs manages the assigned resources to accomplish maintenance tasks. The SEO is responsible for following duties:

(a) To assign EOs, a production superintendent, line chief, trade chiefs and NCO section based on their ability to manage as well as their technical knowledge.

(b) To work out the squadron flight line aircraft utilization and maintenance schedule.

(c) To apply the remedial measures advised by OC Engineering Wing, on weak areas in the organizational maintenance.

(d) To ensure scheduled inspections, life components change and un-scheduled maintenance as per the relevant technical orders / publications.

(e) Ensures aircraft serviceability, ground support equipment and base support equipment availability.

(f) Monitors rectification of all serious discrepancies reported on the aircraft.

(g) Monitors the number and condition of composite tool kits (CTKs) and equipment and takes action to correct deficiencies.

(h) Makes the decision to cannibalize from squadron resources and requests cannibalization from other squadrons through Engg Wing maintenance control.

(j) Monitor the squadron bench stocks program and ensure technical orders and regulations are available and properly maintained.

(k) Reviews and evaluates QA reports for proper corrective actions by subordinates.

(l) Ensures that the EQs are monitoring the quality control of scheduled and unscheduled maintenance in their respective areas of specialty.

(m) Make sure that aircraft documentation is being carried out as per the laid down AHQ policies and procedures.

(n) To develop mobility plans for deployment to Forward Operating Bases.

(p) To ensure training of subordinate EOs.

18. **Chief Flight Engineer**. Chief Flight Engineer is responsible to Officer Commanding through Flight Commander (Ops) while discharging his duties. He is responsible for the duties enumerated below:

(a) To update Officer Commanding and Flight Commander Operations on various technical aspects related to operations.

(b) To look after all affairs related to flight engineers and their welfare.

(c) To detail flight engineers as per the requirement.

(d) To coordinate with Flt Cdr Trg for the conduct of training of Pilots / AD courses related to Platform Technical Training.

(e) To prepare ground and flying training syllabus for flight engineer courses in coordination with Flt Cdr Trg (Flg) & Flt Cdr Trg (AD).

(f) To ensure timely completion of ground and flying training courses of flight engineers.

(g) To ensure operational clearance of flight engineers as per the set criteria.

(h) To ensure disciplined conduct by flight engineers.

(j) To ensure update of personal folders and flying log books of flight engineers.

(k) To ensure flight safety indoctrination and compliance of SOP's and flying orders.

(l) To train Crew Chiefs in accordance with the requirements for flight in co-ordination with SEO.

19. **AEW Commander**. He is a Qualified AEW Combat Pilot who is overall AEW Commander and is responsible for the following:

(a) Overall AWACS employment.

(b) Decisions matrix.

(c) AWACS employment manoeuvre vis-a-vis Fighter Force Ratio (Persist, Tactical Retrieval, Retrograde, and Defensive Manoeuvre).

(d) Flow of information to Tac-D and Captain of the aircraft.

20. **Tactical Director**. He is a Qualified Combat Controller who is responsible for the following:

(a) Anchor management and supervision of $C^2$ brief / debrief.
(b) Overall supervision of AWACS $C^2$ Operation.

(c) Coordination with AEW Cdr.

(d) Dissemination of AEW Cdr decisions to other crew.

(e) Assistance of AEW Cdr in resolving ID Matrix.

(f) Keeping AEW Cdr abreast with outcome of intercept flow.

(g) Informing AEW Cdr about fighters' endurance / weapon / recce state.

(h) Assigning mission crew duties.

21. **Senior Weapon Director / Weapon Director (SWD / WD)**. He is a System Qualified Fighter Controller who is responsible to Tactical Director for following duties:

(a) Anchor management and supervision of $C^2$ brief / debrief.

(b) Senior Weapon Director is to coordinate with external agencies for mission briefs and debriefs.

(c) Weapon Director is to ensure serviceability of allocated radio sets before the mission.

(d) Weapon Director is to ensure desired scope selection as per the exercise setup.

(e) Weapon Director to ensure monitoring of his allocated intercom net.

(f) Senior Weapon Director is to monitor Radio Channels of both ICs.

(g) Senior Weapon Director is to coordinate with Tac-D for safe conduct of ACI missions.

22. **Sensor Officer**. He is a System Qualified Fighter Controller who is responsible to Tactical Director for following duties:

(a) Ensure completion of Log book / Sensor proforma / F-781 formalities.

(b) Switching ON/OFF radar / sensor as per SOP.

(c) Trouble shooting of radar / sensor sub system.

(d) Trouble shooting of MCS sub system.

(e) Ensure the radar surveillance sector as per the Tac-D instructions.

(f) To ensure evaluation of radar performance in each mission.

23. **Link Officer**. He is a System Qualified Fighter Controller who is responsible to Tactical Director for following duties:

(a) Ensure prior coordination with DLI, GES & area GMCC.

(b) Establish data link on C-band / R&S.

(c) Apply / update back told area & plot area as per mission requirement.

(d) Ensure monitoring of data link status.

(e) Ensure Prioritizing of data in case of degraded data link.

(f) Trouble shooting of communication system.

(g) Ensure close coordination with DLI / SMCC / GES.

(h) ATM / Flight plan information from SMCC within AOR.

(j) Allocation of target for radar evaluation to DLI & GMCC in coordination with ASO / Sensor officer.

24. **Electronic Warfare Officer (EWO)**. He is a ESM System Qualified Fighter Controller or Engineer who is responsible to Tactical Director for following duties:

(a) Ensure preparation of mission on EWGSS.

(b) EWO is to ensure loading of current TL in ESM USB.

(c) EWO is to plan search strategy table according to mission route plan and ESM tasking.

(d) Help ASO in carrying out the “Associate & De-Associate” function.

(e) Inform the mission crew in case of threat development.

(f) EWO is to consider EOB for Narrow Band search strategy table.

(g) EWO is to assist AEW Commander in executing defensive manoeuvre.

25. **Air Surveillance Officer (ASO)**. He is a System Qualified Fighter Controller who is responsible to tactical director for following duties:

(a) Ensure updated information in ADA table.

(b) Ensure loading / recording of mission data.

(c) Air surveillance of AOR.

(d) Ensure close coordination with link officer for resolving ID matrix.

(e) Correlation of back told tracks with live tracks.

(f) Association and de-association of ESM tracks with live tracks.

(g) Prompting the Tac-D in case of any unusual air activity.

(h) Sending of post mission data & ZDK status report to command & control centre in coordination with Tac-D.

(j) Activating the received flight plan as ATM and active flight plan as per mission requirement.

(k) Observe the track data table to ensure that all concerned tracks are down told to DLI / SMCC and has Sector Track Number.

# CHAPTER - 9

# PASSIVE DETECTION SYSTEM

## Introduction

1. Project Eternal Vigil (PEV) System is a ground-based stationary passive Electronic Intelligence (ELINT) System (known as VERA-E System). System is intended for detection and identification of airborne, seaborne and ground based emitters. It is best known for identification of non-cooperative targets using its strong capability of Electronic Intelligence (ELINT). It operates on the principle of Time Difference of Arrival (TDOA).

## Duties and Responsibilities

2. **Officer Commanding Detachment (OC Dett)**. Officer Commanding Dett is responsible for all operational, maintenance and administrative activities of the Dett. He is responsible for following:

(a) To ensure smooth operations / maintenance of PEV system in line with established SOPs.

(b) To ensure security of Central Receiving Station (CRS) / Side Receiving Stations (SRSs).

(c) To ensure dissemination of data / reports to all concerned agencies.

(d) To ensure execution of operational tasks related to PEV system as and when directed by Air Headquarters.

3. **Electronic Warfare Officer (O i/c Ops & Maint)**. He is to ensure the security of the emitter data collected by the system along with following duties:

(a) Proper calibration of the System.

(b) Training of personnel working on System.

(c) Ensure smooth operation of entire system.

(d) Emitters’ data management and archiving.

(e) Fortnightly dispatch of ELINT Report.

(f) Management of EW Threat Library (TL).

(g) Proper manning of Ops Room.

(h) Installation of updated software on PCs and update the image back up on compact flash card for each PC.

(j) Availability of pre-issue / bench-stock items for smooth operation.

(k) Conduct of maintenance related activities of system on maintenance days.

(l) Updated backups of all relevant software with require security protocols.

(m) Availability of mission planning software / hardware key as and when required.

4. **NCO i/c Ops & Maint**. He is responsible to O i/c Ops Room for the following:

(a) Organization of Ops room, maintenance of discipline and efficient functioning of Ops room.

(b) Manning and maintaining of shift roster of Ops room as per operational requirement.

(c) Cleanliness and housekeeping of Ops room and CRS including surrounding area and to maintain good working environment.

(d) Ensure that required spares, test equipment, maintenance publications, cleaning material and other items, are available in Ops room to meet routine maintenance.

(e) Get the log books signed regularly for system.

(f) Maintain and update record of all maintenance activities on Form-43C series.

(g) Maintain SOP folder and ensure that all concerned personnel are fully conversant with these orders.

(h) Ensure that concerned personnel sign the SOP folder after reading relevant SOPs.

(j) Implementation of SOP in true letter and spirit.

(k) Maintain leave record and notice board with other necessary information.

(l) Periodic maintenance of all SRS / CRS inspection and maintaining system MTBF / Ops hours with documentation.

(m) Strict implementation of all SOPs related to security of operational data.

(n) Ensure stable input power supply of system and smooth operation of UPS at SRS / CRS.

(p) Inform and keep OC and O i/c Ops room updated on the matters related to his section.

5. **Ops Room Shift Worker**. He is responsible to NCO i/c Ops Room for the following:

(a) Fill under mentioned documents during his duty.

(i) Pick / Drop proforma.

(ii) Log Book (43C).

(iii) ERM Excel, folders and register.

(iv) Recordings on found RL.

(v) Daily ESM.

(b) Ensure serviceability of CRS and all SRS's and inform any un-serviceability to his chain of command.

(c) Brief comprehensively to next shift worker for any new / updated orders.

(d) Maintain proper house-keeping of Ops room.

# CHAPTER - 10

# SURFACE TO AIR MISSILES

## Introduction

1. PAF holds various radar and Infrared (IR) guided Surface to Air Missiles (SAMs) which are deployed for defence of VAs / VPs. Skillful employment and effectiveness of these weapons greatly depends on the expertise of the operators. For smooth functioning and efficient operations of these weapon systems, duties and responsibilities of personnel working in different PAF SAM Units are given in succeeding paragraphs.

## Duties and Responsibilities (Spada Weapon System)

2. **Officer Commanding**. Officer Commanding Unit is responsible to the Sector Commander for efficient operations, maintenance, security and administration of his Unit. He is to ensure the following:

(a) Highest standards of operational vigilance are maintained by on duty battle staff.

(b) Highest standards of performance, maintenance and serviceability of all operational assets.

(c) Regular visits of war sites, ensuring their readiness and recommending changes (if any) to higher command.

(d) Unit readiness for deployment at short notice.

(e) High morale amongst Unit personnel.

(f) Selection of Spada Shot-Doc and ensuring its implementation.

(g) Categorization of AD officers.

(h) Adherence to laid down procedures by all Unit personnel.

(j) SEP and continuation training of all personnel under his command.

3. **Second-In-Command (2 i/c)**. He is responsible to the Officer Commanding. His duties during field deployment and peace time location are as follows:

(a) **Field Deployment**. During field deployment he is responsible to OC Unit for the following:

(i) Ensure efficient administration and highest operational readiness of the flight / squadron.

(ii) Keep a track of proficiency check of SAM controllers.

(iii) Ensure that proper procedures are followed during the conduct of operations.

(iv) Ensure equipment performance check.

(v) Ensure that all instructions issued by the OC Unit and other higher commanders are known and complied with by all personnel of the flight.

(vi) Ensure manning of all positions during the operations / alert.

(vii) He is to sign the logbooks, proforma, reports and returns etc regarding day to day operations.

(b) **At Peace Time Location**. He is responsible to OC Unit for following:

(i) Supervision of operational activities.

(ii) Ensure all orders and instructions of OC Unit and other higher commanders are understood and followed by the Unit personnel.

(iii) Recommend and incorporate desired changes in SOPs.

(iv) Ensure that equipment and personnel earmarked for operational deployment are ready to move at a short notice to the designated operational site.

4. **Officer In-charge Training (O i/c Trg)**. Officer In charge Training is responsible to OC Unit for the following:

(a) Implementation of all training directives and policies.

(b) Training of Unit personnel.

(c) Carry out additional tasks as and when detailed by OC Unit.

5. **Tactical Control Officer (TCO)**. TCO is the overall battery commander. During the tenure of his duty, he is responsible for the following duties:

(a) Carry out Identification of automatic / manual tracks.

(b) Designate the target to the Firing Sections (FS).

(c) Select the firing criteria for FS (how many times a threat can be shot e.g. shoot leave / shoot look shoot).

(d) Execute the Shot doctrine through FS.

(e) Give firing authorization to FS.

(f) Control the engagement once the target is handed over to FS.

(g) Ensure that north alignment of DC, FCCs and MLs is correct.

(h) Activate and integrate all FCCs.

(j) Activate safety lanes.

(k) In case of communication failure TCO is to monitor GMCC channel on R & S radio.

(l) Designate jammer track to FCC for interception of jammer.

(m) Ensure that targets **should not be allocated** for practice to FCCs loaded **with live missiles**.

(n) Before commencing practice firing sequence, he is to ensure that missile simulator is installed and arming cables of live missiles are physically disconnected.

(p) Declare the status of the system to SMCC duty controller.

6. **Surveillance Officer (SO)**. He is primarily responsible for management of frequencies and ECCM functions related to RAC / 3D radar. Additionally, he is responsible for the following:

(a) Select operating mode of the Search Illumination Radar (SIR).

(b) Create a system jammer track for the TCO to further designate to FS.

(c) Delete false tracks.

(d) Generate manual tracks.

(e) Create special tracking zones [Non-Automatic Initiation (NAI)], Fast Automatic Initialization (FAI) and Track Transmission Inhibited (TTI) etc).

(f) Define surveillance sectors, ECCM sectors and radar silence sectors.

(g) Manage radar frequencies.

(h) Adjust RPM of the radar accordingly.

(j) Apply sector blanking in the direction of jammer.

(k) Generate a radar silence sector if needed.

(l) Perform north alignment of DC.

(m) Monitor closely that target designated to FCCs by TCO is positively hostile.

(n) Maintain temperature and pressure values in prescribed limit.

7. **Tactical Officer (FCC)**. Tactical Officer is responsible to Tactical Control Officer for the following:

(a) Ensure that the targets designated by TCO are locked by FCC. He is responsible for the correct tracking and launch of missile against the target designated by DC.

(b) Initiate firing sequence in time against designated target as per selected shot doctrine.

(c) Perform north alignment of FCC and MLs.

(d) Ensure that **firing sequence practice is not** carried out with live missiles.

(e) Before commencing practice firing sequence, ensure that missile simulator is installed.

(f) Apply / change FCC frequencies and MTI mode.

(g) Evaluate the intercept after launch.

8. **Assistant-I (DC)**

(a) He is to assist TCO and SO in surveillance, coordination and record keeping at DC.

(b) He is to ensure smooth flow of combat data to and from other elements involved in operations.

(c) He is to provide information received from various agencies to TCO / SO.

(d) Ensure that system recording / replay facility is on and serviceable.

(e) Recording of Air Activity as directed by TCO / SO.

(f) Assisting in Quality and Radar performance checks.

(g) Filling of target engagement proforma.

(h) Noting un-service-abilities, actions taken and rectifications carried out.

(j) Ensure serviceability of all communication lines.

(k) Ensure proper housekeeping of items inside DC.

9. **Assistant-II (FCC)**

(a) He is to assist TO in tracking and target acquiring functions.

(b) Assist in missile loading as per missile loading procedures.

(c) Recording sequence of operations, missile status and control orders.

(d) Ensure that system recording / replay facility is on and serviceable.

(e) Mentioning un-serviceabilities and rectifications measures.

(f) Assist in carrying out north alignment of FCCs and Missile launchers.

(g) Filling of target engagement proforma.

(h) Ensure serviceability of all communication lines.

(j) Ensure proper housekeeping of items inside DC.

10. **Senior Engineering Officer**. He is responsible to Officer Commanding Unit for the following:

(a) Implementing maintenance policies / directives issued by Air Headquarters and formulations / issuance of relevant SOPs.

(b) Maintaining technical orders, publications, tools, test equipment and record of maintenance accomplished.

(c) Maintenance and operation of electrical, electronic and mechanical equipment of Spada Weapon System, communications, MT vehicles and power generators.

(d) Maintenance, review and updating of the bench stock and pre-issue holding of spares.

(e) During operational moves / deployments / re-deployments of the Units or its detachments, he is to work out conveyance requirements including MT vehicles / railway or air transport and arrange procurement. Subsequently, he is to supervise dismantling / loading and unloading / installation of all the technical / operational equipment.

**Duties and Responsibilities (Crotale Weapon System)**

11. **Officer Commanding**. Officer Commanding Unit is responsible to the Sector Commander for efficient operations, functions and administration of his Unit. His duties are the same as mentioned in Para 2 of this chapter.

12. **Officer In-charge Operations**. Officer-in-charge Operations is second-in-command to the Unit commander and is also O i/c No 1 Flight of the Squadron. His duties during field deployment of his flight and while the squadron is at peace time location are as following:

(a) **Field Deployment**. During field deployment he is responsible to OC Unit for the following:

(i) Ensure efficient administration and highest operational standards of the flight.

(ii) Carry out proficiency check of AD officers.

(iii) Ensure proper procedures are followed during the conduct of operations.

(iv) Ensure equipment performance checks are carried out.

(v) Ensure that all instructions issued by the OC Unit, Sector Commander and other higher commanders are known and complied with by all personnel of the flight.

(vi) Ensure manning of all positions during alert.

(vii) Ensure that respective O i/c sites endorse signatures on the logbooks, proforma, reports and returns etc on daily basis. However, O i/c Ops to sign the logbooks on fortnightly basis.

(b) **At Peace Time Location**. He is responsible to OC Unit for following:

(i) Supervising operational activities.

(ii) Ensure that all orders and instructions of OC Unit and other higher commanders are understood and followed by the Unit personnel.

(iii) Recommend and incorporate desired changes in SOPs.

(iv) Ensure that the equipment of his flight and personnel earmarked for it are ready to move at a short notice to the designated site of the flight.

13. **Officer In-charge Training**. Officer In charge Training is also O i/c No 2 Flight and is responsible to OC Unit for the following:

(a) During field deployment his duties are the same as of O i/c No 1 Flight.

(b) Ensure implementation of all training directives and policies.

(c) Carry out Training of Unit personnel.

(d) Ensure readiness of the equipment and personnel of his flight for operational deployment at the designated site.

14. **Chief Controller (AU)**. During the tenure of his duty Chief Controller (AU) is responsible to O i/c flight for the following:

(a) Carry out familiarization with area of operations.

(b) Manning of AU.

(c) Constant surveillance within the area of responsibility.

(d) Supervision of all operational activities and coordination with other elements of the flight for peak performance.

(e) Assessment of prevailing air situation for appropriate tactical action as per the control order given by SMCC.

(f) Effective control over FU and optical sight.

(g) Reloading of missiles as appropriate and within specified time.

(h) Identify and counter the ECM.

15. **Chief Controller (AU)**. During the tenure of his duty Chief Controller (AU) is responsible to O i/c flight for the following:

(a) Effective control on FU operations and its crew.

(b) Carry out the necessary checks to ensure availability of FU for target engagement.

(c) When instructed:

(i) Feeding in of soft / hard prohibitions.

(ii) Selection of missile codes.

(iii) Selection of Single / Salvo fire.

(d) Discern between actual and false lock-on, and take remedial actions in case of the latter, and inform Chief Controller accordingly.

(e) Identify and counter the ECM.

(f) Take prompt actions for the expeditious recovery of un-serviceability.

(g) Supervise and ensure ejection of empty containers and missile reloading within stipulated time.

16. **Assistant-I (AU)**. He is responsible to Chief Controller for the following:

(a) Assist the Chief Controller to take prompt and correct tactical action by providing him information received from various agencies.

(b) Ensure serviceability of all the land lines.

(c) Ensure smooth flow of combat data to and from other Units involved in operations.

(d) Ensure recording of air activity.

17. **Assistant-II (AU)**. He is responsible to the Chief Controller for the following:

(a) Filling target engagement proforma.

(b) Recording sequence of operations, missile status and control orders.

(c) Noting un-serviceability’s, actions taken and rectifications carried out.

(d) Assist AU Chief in vehicle alignment.

18. **Assistant-I (TV Opt)**. He is responsible to the FU controller for the following:

(a) Identify the target and determine its strength.

(b) Inform FU controller in cases of false lock-on.

(c) After the launch of missile, monitor its track on TV camera.

(d) Inform FU controller on allied aircraft recovering in emergency.

(e) Assist in missile loading.

19. **Assistant-II (Optical Sight Operator)**. He is responsible to FU controller for the following:

(a) Visual surveillance and sector scanning in expected enemy approach and as directed by FU controller.

(b) Acquisition and tracking of the allocated targets.

(c) Assist to achieve lock-on, on the targets not detected by AU radar.

(d) Reporting aircraft recovering in emergency.

20. **Senior Engineering Officer**. He is responsible to the Officer Commanding Unit for the following:

(a) Ensure Implementation of maintenance related policies / directives issued by Air Headquarters and formulation of relevant SOPs.

(b) Ensure Maintenance technical orders, publications, tools, test equipment and record of maintenance accomplished.

(c) Ensure maintenance and smooth operations of electrical, electronic and mechanical equipment of Crotale weapon system, communications, MT vehicles and power generators.

(d) Ensure maintenance, review and updating of the bench stock and pre-issue holding of spares.

(e) During operational moves / deployments / re-deployments of the Units or its detachments, he is to work out conveyance requirements including MT vehicles / railway or air transport and arrange procurement. Subsequently, he is to supervise dismantling / loading and unloading / installation of all the technical / operational equipment.

**Duties and Responsibilities (Mistral Weapon System)**

21. **Officer Commanding**. OC Unit is responsible to the Sector Commander for efficient operations, functions and administration of his Unit. His duties are same as mentioned in Para 2 of this chapter. In addition, he is to liaison with Staff Ops Officer of the Base / satellite (VP) for availability and upkeep of ops sites.

22. **Officer In charge Operations**. He is second-in command and is responsible to the Officer Commanding for day to day operations. He is to ensure that all the operational policies issued are implemented. In addition he is to ensure that:

(a) Proper procedures are followed for the conduct of operations.
(b) All orders and instructions issued by SMCC and OC Unit are complied with.

(c) SOPs are formulated and updated as and when required.

(d) Men and equipment are kept ready for deployment at short notice.

(e) The requisite Mob Reserve is available for immediate use.

(f) Planning, coordination and conduct of local exercises to practice the deployment concept of Mistral and enhance operators' efficiency.

23. **Officer In-charge Training**. He is responsible to OC Unit for the following:

(a) Implementation of directives/policies pertaining to training of personnel.

(b) Conduct training courses for officers as well as for airmen.

(c) Conduct of training of the Unit personnel.

(d) Supervise CTTB preparations.

(e) Scrutinize operational hand books.

(f) Attending basic / advance level of conceptual queries about the system.

(g) Readiness of his flight for field deployment at short notice.

24. **Officer In charge Engineering**. He is responsible to OC Unit for following:

(a) Ensure maintenance and highest serviceability state of all operational, communication / electronic equipment and MT.

(b) Ensure continuation training of technical staff.

(c) Liaise with Base Armament Squadron for periodical and functional checks of firing stations and missiles.

(d) Liaise with base authorities for additional requirements of MT vehicles in case of urgent requirement/deployment.

(e) Liaise with Base Electronic Squadron for the availability and serviceability of communication circuits at MOB / FOB.

(f) Ensure first line maintenance of ops equipment.

(g) Ensure availability of all tools / test equipment as per UAL and their serviceability.

(h) Ensure upkeep of stores and records.

25. **Squad Leader**. During field deployment, Squad Leader is responsible for following:

(a) Carry out site preparation.

(b) Establish communication with Terminal Weapon Control Centre (TWCC) and receive manning state and control orders.

(c) Carry out deployment of the Mistral weapon system.

(d) Receive target information from TWCC.

(e) Give target position and approach direction to the gunner.

(f) Visually scan the area to help acquire the targets.

(g) Select inhibition, non-inhibition of proximity fuse on gunner's call.

(h) Maintain log of time checks, important information, mUnitions left and results etc.

26. **Gunner**. Gunner is responsible for the following:

(a) Carry out site preparation.

(b) Setting up firing station and loading of the missile.

(c) Execute engagement orders given by Squad Leader.

(d) Initiate Battery and Coolant Unit (BCU), achieve lock-on and fire the missile on order by TWCC when target is within range.

(e) After firing reload the available missile immediately.

**Duties and Responsibilities (FN-16 Weapon System)**

27. **Officer Commanding**. OC Unit is responsible to the Sector Commander for efficient operations, functions and administration of his Unit. His duties are the same as mentioned in Para 2 of this chapter in addition to the following:

(a) During war / exercises timely deployment of squadron detachment at pre-designated sites as per the war plan and conduct operations as directed by HQ Air Defence Command.

(b) Coordinate and liaison with respective MCC / SMCC on the performance of FN-16 weapon system during AD exercises and war.

28. **Flight Commander Operations**. He is responsible to OC Unit for the following:

(a) Ensure standard procedures are followed for the conduct of operations.

(b) Ensure implementation of orders and instructions issued by SMCC and OC Unit.

(c) Formulation and updating of SOPs and Ops orders when required.

(d) Ensure manpower and equipment are kept ready for any AD deployment at pre-designated sites at short notice.

29. **Flight Commander Training**. He is responsible to OC Unit for the following:

(a) Ensure implementation of training policies for the training of Unit personnel.

(b) Ensure maintenance of training record of individuals.

(c) Carry out review of practical / academic training procedures and syllabus for personnel.

(d) Conduct of regular training of the personnel posted in the squadron.

(e) Conduct courses for attached officers / airmen as directed by Air Headquarters.

30. **Senior Engineering Officer**. He is responsible to the OC Unit for following:

(a) Ensure maintenance and highest serviceability state of all operational, communication/ electronic equipment and MT.

(b) Ensure continuation training of technical staff.

(c) Liaison with concerned Depot for periodical and functional check of missiles and allied equipment.

(d) Ensure first line maintenance of Ops equipment.

(e) Ensure serviceability of missiles through periodic inspection in Comprehensive Test Cabin (CTC) at No 109 AED.

(f) Ensure testing of FN-16 weapon system by OEM as and when advised by Air Headquarters.

(g) Ensure timely inspection of missiles and pursue quick recovery of missiles and allied equipment.

(h) Ensure implementation and supervision of Unit OJT / CTTB program.

31. **Detachment Commander / Shooter**. During field deployment, senior most person to act as Dett Commander, he is responsible for following:

(a) Preparation of site and deployment of FN-16 Weapon System at pre-designated site.

(b) Ensure establishment of communication through technical team with TWCC / MCC / SMCC for receiving control orders.

(c) Brief all personnel at site about expected threat cone and safety lane.

(d) Ensure that missile posts have all necessary technical and administrative facilities as required.

(e) Ensuring that unfired missiles, as well as empty launch tube of fired missiles are returned to concerned depot.

(f) Ensure proper storage and 'O' level maintenance of FN-16 weapon system.

(g) Promptly communicate all the un-serviceability's occurring in FN-16 missile system to TWCC Commander and Flight Commander Ops of that particular Unit.

(h) Ensure, upon termination of contingency that FN-16 weapons are re-stored as laid down in relevant SOPs.

32. **Auxiliary Shooter**. He is responsible for the following:

(a) Assist Dett Commander in preparation of site, deployment and operations of FN-16 missile system.

(b) Keep the weapon system in a good readiness state.

(c) Keep himself familiar with expected threat cone, different types of enemy aircraft and their likely tactics.

(d) Search out and correctly identify the enemy aircraft.

(e) Evaluate accurately all targets information (velocity, height, shortcut course and distance).

(f) Keep himself familiar with the different firing zones and lethal ranges for approaching / receding targets.

(g) Undertake a proper firing sequence.

(h) Adhere to the control orders passed by TWCC / SMCC.

# CHAPTER - 11

# TERMINAL WEAPON CONTROL CENTRE

## Introduction

1. TWCC exercises full operational control, over terminal weapons deployed at a VP for safe, effective and economical utilization of weapons including firepower distribution function. It basically exploits the resolution property of an MPDR on MIS along with the surveillance data of SMCC on WCS and provides a centrally located place for all the weapon controller executives to employ their weapon system in coordination with each other. This center houses PCs configured for various roles and tasks and act as communication node for all the weapon system operators as well as controlling and coordinating agencies.

## Duties and Responsibilities

2. To accrue maximum operational benefit from Air Defence weapons deployed at a VP, the duties and responsibilities of TWCC personnel are defined in the ensuing paragraphs.

3. **TWCC Commander**. TWCC commander is responsible to HQ ADC through respective SMCC for operational and functional control and to MOB / FOB commander for administrative functions. He is to ensure that the highest operational standards are maintained at TWCC and is responsible for:

(a) Ensure battle staff at TWCC is formally trained and carrying out training as per laid down policies.

(b) Ensure manning level is being maintained by all weapon systems, as directed by SMCC.

(c) Liaison with all terminal defences / agencies for smooth running of TWCC. He is also to ensure that all the positions at TWCC are manned as per SOPs.

(d) Liaison with Base authorities for all administrative requirements for establishing and maintaining TWCC Ops room and manual TWCC setup.

(e) Ensure availability of necessary communication links at underground ATC or any other appropriate place for operations under degraded TWCC environment.

(f) Ensure that standard time check is received from SMCC and passed to all weapon systems.

(g) Maintaining close liaison with MGC at SMCC.

(h) Ensure that correct weapons and ammo status is passed to SMCC by all weapon coordinators and the same has been maintained at TWCC.

(j) Receiving control orders from SMCC and passing to all weapon systems as per SOPs.

(k) Allocating the target to the appropriate weapon system as per laid down SOPs.

(l) Ensuring ground defence of SAM sites and TWCC through base authorities.

(m) Maintaining close liaison with TDCC at ATC to ensure safety of allied traffic in MDA.

(n) Shifting the operations from auto to manual TWCC within stipulated time, whenever required.

(p) Ensuring that a/c recovery and safety lane procedures are worked out and known to all weapon operators.

(q) Ensuring that WCS is setup correctly, all necessary communication circuits / lines are available and positions of all deployed weapon systems and important points are marked on WCS.

4. **Mistral / FN-16 Coordinator**. He is responsible to TWCC Commander for the following:

(a) Ensuring that MIS is setup correctly, all necessary communication circuits / lines are available, all weapons are marked and common reference point is displayed on MIS.

(b) Ensuring readiness and operational status at each firing stations is maintained as directed by SMCC / TWCC commander.

(c) Ensuring that SOPs are followed while allocating the target to individual mistral site. He is also responsible to ensure that the rest of the sites are only tracking the targets.

(d) After allocation of target, he is to manually generate / update a target.

(e) Passing accurate cueing information to appropriate firing station and for ensuring that all squad leaders acknowledge the same by repeating the range and bearing with site suffix. For example, 120 / 7 ‘A’.

(f) Ensuring that primary and secondary arcs of each Mistral / FN-16 site have been worked out and known to all the personnel at the site. Also, he is to ensure that primary and secondary arcs are displayed at auto and manual TWCC.

(g) In case of MI by forward Mistral / FN-16 site, he is to allocate the target to next appropriate site.

(h) Ensuring that safety lane is accurately marked at all Mistral / FN-16 sites and all MWS / FN-16 operators are aware of safety lane / emergency recovery corridors.

(j) Ensuring that a synchronized time is maintained at all Mistral / FN-16 sties.

(k) Ensuring that the live Mistral / FN-16 missiles are not loaded on firing station during exercises.

5. **Spada Coordinator**. The Spada coordinator is responsible to TWCC Commander for the following:

(a) Ensuring that MIS is setup correctly, all necessary communication circuits / lines are available and common reference point is displayed on MIS.

(b) Ensuring alert status as directed by SMCC is maintained by all Spada elements.

(c) SOPs are followed while allocating the target to each DC.

(d) After allocation of target, he is to manually generate / update the target.

(e) Passing accurate cueing information to appropriate DC and for ensuring TCO acknowledges the same by repeating the range and bearing with battery suffix. For example, 120 / 7, Battery-1.

(f) Ensuring that primary and secondary arcs of each FCC have been worked out and known to all the personnel at the site. He is also to ensure that the primary and secondary arcs are displayed at TWCC.

(g) In case of MI by a DC he is to allocate the target to subsequent DC.

(h) Ensuring that safety lanes are fed at all the DCs and all weapon operators are aware of safety lane / emergency recovery procedures.

(j) Ensuring that communication circuits of all DCs are available at manual TWCC.

(k) Ensuring that synchronized time is maintained at all Spada elements.

(l) Continuous update to TWCC Commander regarding current weapon / ammo state Alert order and any other special / unusual occurrence.

(m) Collecting the results from Spada DC for preparing consolidated results.

(n) Removal of arming cables before the exercises.

6. **Crotale Coordinator**. The Crotale coordinator is responsible to TWCC Commander for the following:

(a) Ensuring that MIS is setup correctly, all necessary communication circuits / lines are available and common reference point is displayed on MIS.

(b) Ensuring alert status as directed by SMCC is maintained by Crotale flight / FU.

(c) SOPs are followed while allocating the target to each flight / FU.

(d) After allocation of target, he is to manually generate / update the target.

(e) Passing accurate cueing information to appropriate flight and for ensuring AU / FU Chiefs acknowledges the same by repeating the range and bearing with flight suffix. For example 120 / 7, Flight-1.

(f) Ensuring that primary and secondary arcs of each flight / FU have been worked out and known to all the personnel at the site. He is also to ensure that the primary and secondary arcs are displayed at TWCC.

(g) In case of MI by a flight / FU he is to allocate the target to subsequent Flight / FU.

(h) Ensuring that safety lanes are fed as soft / hard prohibition at all the FUs and all weapon operators are aware of safety lane / emergency recovery procedures.

(j) Ensuring that communication circuits of both the flights are available at manual TWCC.

(k) Ensuring that synchronized time is maintained at Crotale flights / FUs.

(l) Continuous update to TWCC Commander regarding current weapon / ammo state alert order and any other special / unusual occurrence.

(m) Collecting the results from Crotale Weapon for preparing consolidated results.

(n) Removal of arming cables before the exercises.

7. **Ack Ack (Army / Navy) Coordinator.** Ack Ack Coordinator is responsible to TWCC Commander for the following:

(a) Ensuring that all deployed Ack Ack and SAMs are maintaining Alert / manning status as directed by SMCC / TWCC Commander.

(b) Ensuring that MIS is setup correctly, all necessary communication circuits / lines are available and common reference point is displayed on MIS.

(c) Manual generation / up-dation a target after it has been allocated.

(d) Handing over target to appropriate weapon i.e. SAMs / MAA or LAA.

(e) Passing accurate cueing information to appropriate Command Post and ensuring receipt of acknowledgement of the same correctly i.e. Command Post repeats the range and bearing. For example, 120 / 7.

(f) Ensuring that primary and secondary arcs of all weapons systems have been worked out and known to all the concerned. He is to also ensure that the primary and secondary arcs are also displayed at MIS.

(g) Ensuring that safety lanes are marked at all guns and missiles posts and all weapon operators are aware of safety lane / emergency procedures.

(h) Availability and serviceability of communication circuits between concerned Command Posts and TWCC.

(j) Ensuring that live ammUnition is not loaded on guns during exercises.

# CHAPTER - 12

# MOBILE OBSERVER UNITS

## Introduction

1. Mobile Observer Units (MOUs) are an important element of the Pakistan Air Defence System. They are deployed in the direction of expected threat. MOUs see the aircraft visually and report it to the appropriate controlling agency for analysis and tactical decision. MOUs are also used as gap fillers where the radar coverage is not available. They are very useful particularly in hilly terrain. The duties / responsibilities of personnel working in MOUs are given in succeeding paragraphs.

## Duties and Responsibilities

2. **Officer Commanding**. He is responsible to Sector Commander for the following:

(a) Optimum operational efficiency of the Squadron.

(b) Recce of sites and taking approval of the Sector Commander about the recommended sites before deployment.

(c) Visit war sites every six monthly and recommend change (if any).

(d) Ensure all Flight Commanders, Section Commanders and MTDs are familiar with the recommended sites.

(e) On receipt of warning order for field deployment, brief Flight Commander about:

(i) General area of deployment.

(ii) Number of Units to be deployed.

(iii) Date of commencement of deployment.

(iv) Expected period of exercise/operation.

(f) Prepare the Ops Order with emphasis on:

(i) Type of mission and its execution.

(ii) Arms/ammUnition to be carried and its security / handling.

(iii) Route plan and time for initial contact after reaching the sites.

(iv) Details of frequencies, channels and call sign.

(v) Any change in aircraft reporting procedures.

(vi) Special instructions about exercise / operation.

(g) Visits of MOS / MOUs during field deployment.

(h) Liaison with Army Area Commander for MOUs advance or withdrawal during war.

(j) Post deployment debrief after the deployment/exercise is terminated, highlighting weak / problem areas and remedial actions needed to avoid recurrence in future.

(k) Checking and monitoring daily personnel and equipment operational state of squadron.

(l) General briefing regarding camouflaging, security, ground defence, discipline and public relations.

(m) Ensuring general welfare and high morale of airmen.

(n) After the termination of deployment, debriefing his squadron and discussing the weak areas, or any special problem faced in the field and also taking remedial action to avoid recurrence in future.

(p) Submission of post exercise report to Headquarters Air Defence Command through Sector Headquarters.

3. **Flight Commander**. He is responsible to Officer Commanding for the following duties:

(a) Detail personnel, equipment and MT Vehicles for deployment of Flt HQs / MOS / MOUs.

(b) Issue of the following according to the scale and entitlement:

(i) General Stores.

(ii) Signal Equipment.

(iii) F-748 Stores.

(iv) First Aid Kit.

(c) Brief MOS Cdrs and NCO i/c MOUs about location of their posts in the field.

(d) Detail Convoy Commanders and brief them about convoy procedures.

(e) Ensure operational readiness, discipline and organization of MOS / MOUs.

(f) Ensure training of personnel.

(g) During advance, retrieval and shifting of MOS/MOUs, no signs are left which could be of any assistance to the enemy.

(h) Carry out frequent visits of MOS / MOUs to ensure better discipline and performance during deployment.

(j) Arrange issuance of communication and aircraft codes and ensure that men are quite familiar with their use.

(k) Brief MOS Cdrs and NCO i/c MOUs regarding the route to be followed. MOS Cdrs are to be made responsible to lead convoy and deployment of their respective posts.

(l) Brief all personnel to operate communication channels, frequencies, authentication systems and ensure strict adherence to the laid down SOPs / LOPs.

(m) Ensure that 100% serviceable equipment (HF set and SMT etc) has been issued to his men.

4. **Engineering Officer**. He is responsible to Officer Commanding for the following:

(a) Scheduling personnel and equipment for the functioning of Squadron Headquarters in the field area.

(b) Prepare Ops Order.

(c) Coordinate with Oi/c Comm in SMCC for allocation of frequency circuits required for communication during the exercise / operation.

(d) Conduct detailed communication briefing to MOUs about call signs, arrangement of operational and standby channels and aircraft reporting procedures etc.

(e) Brief all concerned about maintenance of MT vehicles and other equipment in the field.

(f) Arrange issue of necessary spares of MT vehicles and charging sets for field servicing and maintenance of equipment.

(g) Submit demand and arrange collection of POL for MOS / MOUs.

(h) Raise authority Proforma and submit it to Base Armament Squadron for the issue of arms / ammUnitions to the personnel detailed for deployment.

(j) Arrange payment of SRRA / TADA advance to the men taking part in Operations.

5. **Section Commander**. He is responsible to Flight Commander for the following:

(a) Operational readiness and discipline of his section.

(b) Training of personnel in the field according to SEP policy.

(c) On the receipt of instructions for deployment, brief the personnel and arrange collection of Mob Reserve items / Equipment and POL from Supply Points.

(d) Line up his section in convoy order with convoy flags on first and last vehicles.

(e) Prior to move ensure proper DI of all MT vehicles and all drivers hold valid driving category endorsed on F- 1629.

(f) Collect route plan and brief his personnel accordingly.

(g) Lead the convoy and observe strict convoy procedure. Make re-planned halts and check the vehicles for refuelling etc.

(h) Before leaving, ensure serviceability of wireless sets, charging sets, power source (Generator / Solar Panel) and after deployment of MOUs on its specified location, initial contact with RRF.

(j) Report equipment and man power state to Flight Commander and Squadron Headquarters as per instructions.

(k) On receipt of MOU “OFF Air” signal from RRF, section commander is to immediately visit the respective MOU site.

(l) Maintain good living standard and liaison with local population without jeopardizing security of information and material.

(m) During contingency, liaise with local army commanders in case of any withdrawal, retrieval or advance and inform SMCC about latest situation.

(n) Ensure availability of 50% fuel reserves at all times after re-fuelling the vehicles.

(p) During re-deployment / retrieval ensure no sign is left behind which could be of any use to the enemy.

6. **NCO i/c MOU**. He is responsible to Section Commander for the following:

(a) Serviceability of MOU equipment, training of personnel and ensuring operational efficiency of his MOU.

(b) Maintaining high standard of discipline and security, which includes protection of documents, equipment and personnel in the field.

(c) Making necessary arrangements to safeguard against ground and aerial attacks.

(d) Ensuring his MOU remains in contact with Auto RRF all the time.

(e) Ensuring that all reports are passed in time.

(f) Keeping the Section Headquarters informed about any emergency and confusion.

(g) Ensuring physical and mental health of all crews of his MOU.

## CHAPTER - 13

# BALLOON BARRAGE UNITS

### Introduction

1. The concept of using balloons for Air Defence purpose is two centuries old. This concept always remained debatable due to the balloons being vulnerable to shooting and making the target area prominent. But still they proved to be an effective passive Air Defence measure. When balloons are hoisted, these distract the pilot's attention and compel the attacker to gain height thus, become vulnerable to the surface-to-air weapons. Attack accuracy is also adversely affected. For effective operations, the personnel deployed for balloon barrage duties need to be well conversant with their duties and operational requirements. The duties / responsibilities of personnel working in Balloon Barrage Units (BBUs) are given in the succeeding paragraphs.

### Duties and Responsibilities

2. **Officer Commanding**. He is responsible to respective Base Commander for the following:

(a) Implementation of all policies and operational tasks assigned to Balloon Barrage Units.

(b) Operational deployment of balloon barrage elements.

(c) Training of personnel on balloon barrage.

(d) Inspection of newly manufactured balloons and accessories.

(e) Ensuring proper storage of balloons.

(f) Review of integrated Balloon / Air Defence Gun / SAMs deployment plan for all the airfields in consultation with respective Bases.

(g) Visiting of war sites.

(h) Preparation / review of balloon barrage SOPs.

3. **Officer In-charge Operations**. He is responsible to Officer Commanding BBU for following duties:

(a) Ensuring that all orders regarding field deployment given by the OC Wing are relayed, coordinated and carried out by all concerned.

(b) Ensuring deployment of required number of balloon barrage posts at a given airfields / VPs.

(c) Formation of Balloon Barrage Control Centre (BBCC), maintenance party and balloon barrage posts before move.

(d) Coordination with the respective Bases for admin and technical support and services required.

(e) Dispatch of equipment required for airfields / VPs.

(f) Ensuring availability of the reserve stock at each balloon barrage post / Dett.

(g) Training and record keeping of personnel detailed for the courses.

4. **O i/c Training**. He is responsible to OC BBU for the implementation of all training directives and policies issued by higher authorities. He is to ensure the following:

(a) Conduct training as per training schedule approved by Air Headquarters {Dte of Air Defence (Trg & Eval)}.

(b) Ensure that the training programme is prepared for incoming balloon barrage courses.

(c) Coordinate well in advance with all concerned agencies about hoisting of the training balloon.

(d) Ensure that sufficient Hydrogen gas cylinders are available for inflation of the training balloon.

(e) Ensure the maintenance of an up-to-date record of all balloon barrage trained personnel.

(f) Ensure that F-292s in respect of the balloon barrage trainees are dispatched to concerned agencies.

(g) Implement Continuation Training Programme as approved by Air Headquarters {Dte of Air Defence (Trg & Eval)}.

(h) Ensure training of the newly inducted personnel of BBUs.

(j) Conduct refresher training as per approved syllabus.

(k) Maintain all the training aids.

5. **Engineering Officer**. He is responsible to OC BBU for the following:

(a) Control, discipline and efficient functioning of all administrative and technical services.

(b) All technical equipment used with balloon barrage is serviceable for operational use at all times.

(c) Proper maintenance and storage of balloon is being carried out.

(d) MT is kept at a high state of serviceability.

6. **Unit Log Officer**. He is responsible to OC BBU for following:

(a) All matters pertaining to supply of equipment.

(b) Availability of all equipment pertaining to balloon barrage at the Wing Headquarters.

(c) Initiating case for procurement of balloons, if required.

(d) Initiating case for hydrogen gas cylinders as and when required.

(e) Maintaining record of technical and general stores.

(f) Ensure Storage and security of hydrogen gas cylinders and other equipment at Base through respective Base Log Officer.

7. **WO i/c Training / Ops**. He is responsible to O i/c Training / Adjutant for the following duties:

(a) Maintaining of master training folder.

(b) Updating of individual training folders of controllers and radar operators.

(c) Preparation and record keeping of all training record including training programme for controllers and radar operators.

(d) Familiarization and induction tests for radar operators.

(e) Ensuring all training returns are prepared and forwarded in time.

(f) Keeping up to date record of all operational activities.

(g) Preparing duty roster for radar operators.

(h) Signing of supervisor and activity log book.

(j) Compilation and preparation of returns and reports.

8. **NCO i/c MT / Equipment**. He is responsible to O i/c Engineering for following:

(a) Submitting the serviceability state of balloon equipment held on the charge of MT Section.

(b) Ensure signing of Section Order Book and other orders by drivers.

(c) Maintenance, discipline and cleanliness of MT section.

(d) Routine correspondence and submission of returns.

(e) New demand / replacement of unserviceable equipment.

(f) Tools for maintenance of equipment are available and technical staff is fully aware of its use and maintenance procedure.

9. **WO / NCO i/c Balloons Maintenance / Repair Section**. He is responsible to Unit Log Officer for following:

(a) Periodical inspection and serviceability of all balloons held on inventory.

(b) Maintaining log book and relevant records of all balloons.

(c) Repairing unserviceable balloons.

(d) Replacing or demanding accessories of balloons through supply officer.

(e) Training of personnel on balloon maintenance.

(f) Proper storage of balloons.

**Duties and Responsibilities during Deployment at MOBs / FOBs**

10. **Dett Commander**. He is responsible to the Base Commander of respective Base through Staff Ops Officer (SOO) for the following duties:

(a) Operational / Administrative activities at the Dett.

(b) Briefing the Dett crew about exercises / operations etc.

(c) Arranging and issuing of all field equipment to the balloon post commanders through Balloon Barrage Control Centre (BBCC).

(d) Ensuring that general layout of posts is away from the working area.

(e) Ensuring that proper procedures for balloon operations are followed.

(f) Ensuring timely provision of all admin facilities (food, water etc) to posts.

(g) Liaison with Base SOO for Ops Instructions.

(h) Ensuring balloon readiness state passed to SOO on daily basis.

(j) Liaison with Base authorities for admin / log / MT support.

(k) Liaison with O i/c Log Sqn for collection of mob reserve items, hydrogen gas cylinders and other essential equipment.

(l) Preparation of a comprehensive deployment plan and briefing the personnel regarding the exercise / operation on their arrival.

(m) Nominating the personnel as Post Commanders.

(n) Passing all the relevant orders to post commanders as and when received from DATCO / Sector HQ / OC BB Wing and SOO.

(p) Visiting all the posts at least once a day with making observations in the post log book.

(q) Ensuring hoisting / lowering of Balloons on instructions of ATC.

11. **Shift Incharge (Rad Opt) at BBCC**. He is responsible to Dett Commander for the following duties:

(a) Maintaining log books at BBCC and balloon posts by noting down all the Ops / Maintenance / Admin activities and other occurrences of important nature.

(b) Ensuring Air Defence Standard Time (ADST) check is received from ATC and passed to all the Balloon Posts. (As per procedure mentioned in ADST Chapter).

(c) Ensuring serviceability of Balloon Barrage equipment.

(d) Timely passing of hourly weather warnings received from ATC to all the balloon posts.

(e) Ensuring filling / hoisting / lowering / re-filling orders are passed to the balloon posts well in time.

(f) Ensuring timely provision of all admin facilities (food, water etc) to the posts.

(g) Ensuring manning adjustment at the posts.

(h) Ensuring preparation of parade state (BBCC / balloon posts) and forwarding it to the Dett Cdr.

(j) Detailing a maintenance party to convey filled hydrogen gas cylinders and collect empty cylinders to / from balloon posts.

(k) Ensuring a proper mechanism to replenish the gas promptly from the respective supply squadron.

(l) Ensuring availability of serviceable equipment and retrieval of unserviceable equipment / Walkie Talkie sets and other balloon barrage related accessories through maintenance party.

12. **Area Commander**. The senior most SNCO of the area is to act as area commander and is responsible to Dett Commander through BBCC for the following duties:

(a) Ensuring that all personnel in his area are well conversant with the balloon SOPs / Local Orders.

(b) Visiting all balloon posts in his area once a week or whenever detailed by Dett Commander for carrying out thorough inspection of the personnel and equipment.

(c) Keeping Dett Commander / BBCC in picture about all the important events / happenings in the area.

(d) Ensuring proper discipline of his area by all crew.

(e) Ensuring proper manning of each post in his area.

(f) Carrying out surprise check of the posts.

13. **Post Commander**. He is responsible to the Dett Commander through Area Commander for the following duties:

(a) All crew members of his post are fully conversant with SOPs / LOPs.

(b) Ensuring hourly weather information is taken from the BBCC and logged in the log book.

(c) Ensuring all the crew under his command is briefed about the balloon operation and its ancillary equipment.

(d) Ensuring synthetic gloves and rubber soled shoes are used by the crew during handling of the balloon / equipment.

(e) Ensuring Walkie Talkie set and SOPs are safeguarded and the Walkie Talkie sets are kept in fully serviceable condition with standby battery.

(f) To ensure refilling is done an hour before the first light and last light with the permission of BBCC (if required) during winter and at low temperatures. If the balloon is contracted, ensuring refilling of the balloon with the required quantity of hydrogen gas with the permission of BBCC.

(g) Timely conveying un-serviceabilities to the BBCC for their remedial action.

(h) Ensuring hoisting / lowering of the balloon by the crew of the post as per SOPs.

(j) Ensuring duty roster is maintained, night duty is equally shared by all the personnel and the on duty crew is alert and vigilant outside the tent during the tenure of his duty.

(k) Ensuring R/T discipline is maintained on Walkie Talkie by the post personnel.

(l) Approaching BBCC through area commander for:

(i) Manning adjustment at the post.

(ii) Replacement of empty hydrogen gas cylinders with filled ones.

(iii) Provision of admin facilities if required.

(m) Ensuring sufficient quantity of filled hydrogen gas cylinders available at his post.

(n) Ensuring at least one ‘L’ shaped trench is available at balloon post.

(p) All occurrences of important nature are logged and reported to BBCC.

**Note**: 1. Air Defence balloons are centrally procured by AHQs and distributed to concerned BBUs through respective echelon. Similarly, hydrogen gas cylinders are also authorized / procured by AHQs and provided to BBUs through related Base authority. O i/c POL / S&T of respective Bases are responsible for their safe custody and replenishment to BBUs as per requirements.

2. Storage of hydrogen gas cylinders at an appropriate place / environment is the responsibility of O i/c POL / S&T of the concerned base.

# CHAPTER - 14

# CRITERIA AND PROCEDURE FOR SITING AND DEPLOYMENT OF AIR DEFENCE ELEMENTS

## Introduction

1. Air Defence Elements are often deployed in the field areas during peace as well as war. Operational suitability of the site, technical considerations of the equipment and properly laid out camps ensure safety and efficiency of the Unit. These Air Defence Elements can be categorized into following types.

(a) **MMCCs**

(b) **Radars**

(i) High level Radars (TPS-77, YLC-II A, YLC-II, TPS-43G)

(ii) Medium Level Radars (TPS-77 MRR, YLC-18)

(iii) Low Level Radars (YLC-6, MPDR-45-60 / 90)

(c) **GBADs / Composite SAM Units**

(i) HIMADS

(ii) LOMADS

(iii) E-SHORADS / SHORADS

(d) **Balloon Barrage Units**

(e) **Mobile Observer Units**

(f) **PDS (Passive Detection System)**

## Recce of Sites

2. The recce task assigned to the Units, should be met systematically for selection of the sites. Following steps are to be kept in mind during conduct of recce:

(a) General area of interest should be selected and studied on the map / Google map / Falcon View.

(b) Different probable sites should be marked on falcon view or milli / quarter inch map, while ensuring secrecy of information.

(c) Comparison of Theoretical RCIs of the shortlisted sites calculated via DTED based software (Falcon view, Site Adaptation software of TPS-77 etc)

(d) Approach and route plan of these sites should be studied.

(e) If possible, aerial recce be carried out before ground recce or the topography be studied through satellite imagery.

(f) During ground recce following points should be worked out:

(i) Co-ordinates of all sites through GPS or from map readings for site registration in GMCC or console / PPI map.

(ii) Screening angles from 0 degree to 359 degrees around the site (by using theodolite, compass and binocular). Theoretical RCI on Falcon view may also be prepared for final selection of the site.

(iii) Identification of nearby manmade screening objects and probable interference sources.

(iv) To avoid screening, provision of ramp or high lift should be considered.

**Recce Report**

3. Recce report be prepared after considering different sites and finally selecting the best out of them, depending upon the operational requirement, security considerations viz-a-viz equipment limitations and logistic support. The recce report should include the following:

(a) General route plan of the area.

(b) Exact location of the site.

(c) Operational / technical / administrative / security considerations for selecting the primary site.

(d) Selection of alternate sites, if possible.

(e) Disadvantages / requirements, if any.

(f) Requirement of radome to cater environment effects / limitations.

(g) Requirement of fencing / security special equipment (NVGs, metal detectors, scanners, solar Batteries).

(h) Theoretical RCI for shortlisted sites.

**Siting and Deployment of Radar Units**

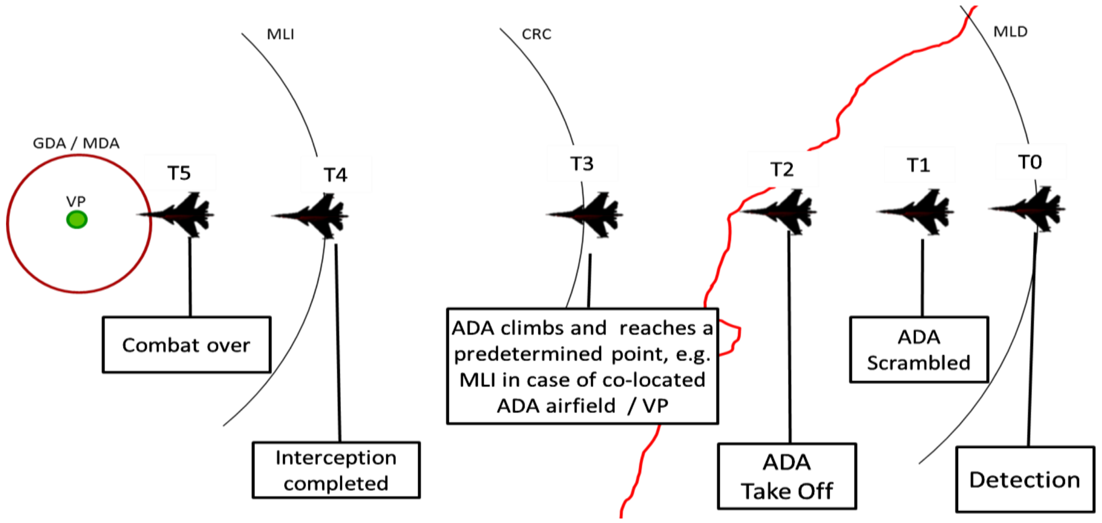
4. The phrase "Siting of Ground Radar Equipment" means determination of the precise location of a radar station so that it may best perform its function and fulfil the assigned task effectively with due regard to its operational limitations and economy of effort. The difference between siting and deployment of radar is to be clearly understood before initiating the process of siting a radar. In the first place, general deployment is to be considered, once the area of deployment has been generalized, specific location is to be selected to obtain maximum performance from the radar. The factors involved in radar siting are many and some of them are always conflicting. These problems are solved by balancing the various factors concerned and making a compromised solution which will meet the operational requirements of the immediate situation and also suit the overall plans of coordinated radar system. The various operational, technical, administrative and security factors have been discussed in the succeeding paragraphs in order to provide guidance to the siting team.

5. **Operational Considerations**. Operational deployment of a radar is basically governed by following considerations:

(a) Enough early warning for alerting the defensive weapons i.e. Minimum Line of Detection (MLD).

(b) Adequate coverage and control capability i.e. Control Radar Coverage (CRC).

(c) Sufficient combat time after successful ground-controlled intercept beyond the operational radius of point defence weapons i.e. Minimum Line of Intercept (MLI).




MLD, CRC and MLI are the minimum calculated distances (from the VP to be defended) at which enemy's detection will ensure certain events to take place timely for shooting down of intruder before entering GDA / MDA by the interceptor (scrambled from ground). It is important here to understand that MLD, CRC and MLI are the calculated minimum required distances from the VP for detection, control capability and interception. However, actual pickup of early warning sensors may vary owing to their siting e.g. calculated MLD may require siting of an early warning radar at a point which is either beyond the border or no appropriate site is available in close vicinity of such calculated point. These variations may warrant either positioning of standing CAPs instead of ground scrambles to cater for actual early warning less than MLD or illumination of such critical areas by AEWACs. Vice versa, detection beyond MLD, control capability beyond CRC and interception of raider before MLI are supplementary advantages. Moreover, for protection of a VP against standoff weapon threat, these calculated lines have to be further extended away from VP corresponding to range of standoff weapon threat.

6. During war, decision datum for alerting the weapons for defence of a VP must conform to the above mentioned minimum calculated distances from the VP. However, this may not be possible during peacetime.

7. If the interceptor base is located some distance north (for example) from the target the MLD will be closer to the airfield for a raid from the north, and extended for a raid from the south. When such variables enter in the calculation, these lines will not be perfect circle but irregular shaped pattern. However, when the interceptor base is co-located with the VP/VA, the MLI, CRC and MLD would be perfect circles around the VP / VA. For ease of understanding we shall take co-located interceptor base and VP for subsequent paragraphs.

8. Various factors interact to define the limits of minimum early warning requirements. These factors and their cumulative effect on determining the minimum early warning requirements for interceptor operations from a particular site are discussed below:

(a) **Minimum Line of Interceptions (MLI)**. The minimum line of interception (MLI) is that point of critical penetration at which the enemy forces must be intercepted. It is determined by two following factors:

(i) **GDA / MDA**. The gun or missile defended area is the area around the vital point. This is generally calculated by taking into view the maximum firing ranges of anti-aircraft guns and missiles to be deployed and adding to it some safety factors.

(ii) **Combat Time**. Combat time is the time required by the interceptors to destroy the enemy. It is a calculated time, required for interceptors to successfully attack and destroy the enemy. It is converted into distance, based on the speed of hostile aircraft so the MLI would be this distance plus the guns/missile defended area.

**Example**: If the GDA / MDA is = I0 NM in radius; the combat time is 2 minutes and the speed of Hostile a/c is = 480 Knots, ie, 8 NM / Minute, converted to distance according to the combat time it would be = 8 x 2 = 16 NM.

As MLI=GDA/MDA + Combat time (in distance).MLI = 10 + 16 = 26 NM.

(b) **Control Radar Coverage (CRC).** It is an area or areas within the Air Defence system where solid radar search, detection and tracking capabilities exist and in which positive control of interceptors can be maintained. The control radar coverage (CRC) is a requirement line which establishes the radius of positive control. CRC range is dependent on range, azimuth and altitude data provided by the radars. It consists of the radius of minimum line of interception (MLI) plus the manoeuvre time. The latter is the calculated time needed to position the interceptors for attack. Manoeuvre time, like the combat time, is an arbitrary figure determined by higher Headquarters.

**Example**: If manoeuvre time is 3 min, the CRC would be = MLI + Manoeuvre Time. MLI had been calculated to be 26 NM in the previous example. Considering the same speed of the hostile a/c, i.e, 8 NM in a minute the manoeuvre time converted to distance comes to be 24 NM.

∴ Therefore CRC = 26 + 24 = 50 NM

It is clear in this case that the interceptor positioning on the hostile target must begin when it is 50 NM away from its intended target (VP or VA) to allow the interceptors to begin firing by the time the hostile a/c reaches the MLI.

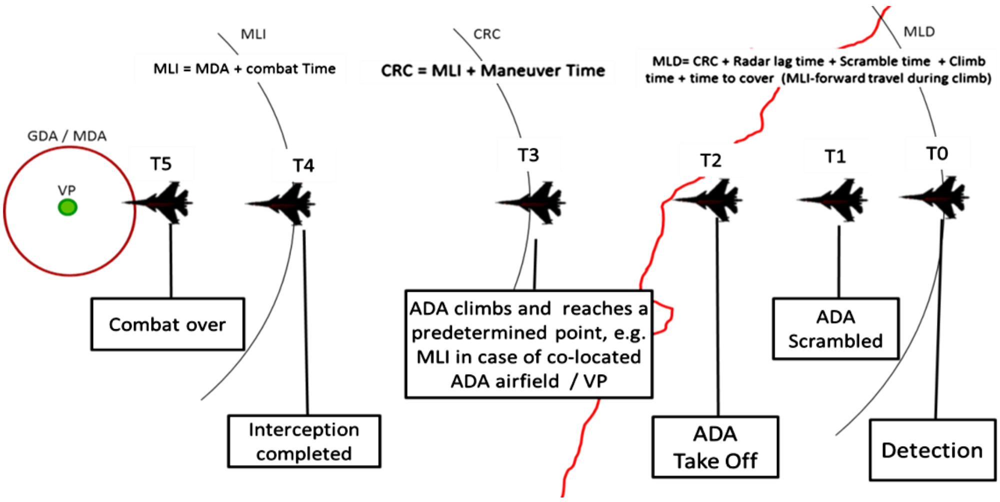
(c) **Minimum Line of Detection (MLD).** The minimum distance from a VP / VA at which a particular type of enemy aircraft must be detected so as to provide enough time for tactical action and for disposing off at safe distance from VP, is known as MLD or simply it is early warning of radar, or other sensors, coverage requirement line, which Is determined by the distance from the hostile aircraft during the radar lag time, the interceptor scramble time, the interceptor's scramble time, climb to the required altitude and the interceptor's level flight time to the MLI.

(i) **The Radar Lag Time**. It is time required for initial detection and identifying the plot of the hostile aircraft. In Air Defence operation on manual mode, a scope operator observes a hostile aircraft on his scope and after observing it for some time to be sure that it is a moving object passes it to the Sector Operations Centre where identification takes place or GMCCs tracker has some tracking logic that after these many returns tracker would generate the track (usually 03 hits). This manual / automatic track generation activity and subsequent identification by SMCC requires some time while the enemy aircraft, of course is still flying towards the target. The radar lag time is also an arbitrary figure spelled out by the higher authorities or the pre-defined tracking logic of the GMCC.

(ii) **The Interceptor Scramble Time**. It is the time taken by the Interceptors to get airborne after the scramble order has been given. This time depends upon the alert status of the interceptor aircraft and is specified for each alert status for each type of aircraft.

(iii) **The Interceptor Climb Time to Specified Altitude**. This time would depend on the type of interceptor aircraft. For planning purposes, the time of the slowest interceptor aircraft is taken.

(iv) **The Interceptors Level Flight Time to the MLI**. This again would depend on the type of aircraft and should be figured for the slowest interceptor being employed.




9. The three perimeter lines must be well understood to comprehend radar deployed and are a pre-requisite to the choice of the area. Following example will make it clear in which a simple interception sequence is given from the moment of detection until the completion of interception against a high-level target. Assume that the following conditions exist for conducting interception:

| | |
|---|---|
| GDA-MDA | 10 NM |
| Combat Time | 2 minutes / 16 NM (based on raider speed of 8 NM / min) |
| Manoeuvre Time | 3 minutes / 24 NM (based on raider speed of 8 NM / min) |
| Raider Altitude | 30,000 feet |
| Raider Speed | 480 Knots (8 NM per min) |
| Radar Lag | 2 minutes |
| Scramble Time | 3 minutes (from cockpit standby alert status) |
| Interceptor Rate of Climb | 10,000 feet/minute |
| Interceptor Average Speed During Climb | 360 Knots (6 NM per Min) |



Revised on 29 Jan 21

(a) For the purpose of this example, it is assumed that the controlling GMCC of the interceptor and the interceptor airfield are at or near the VP / VA, the intended target of the raider.

(b) As surveillance scope operator observes a blip at the minimum line of detection (MLD). Two minutes later it is displayed and identified in the Sector Mission Control Centre. During those two minutes of radar lag the raider has flown 16 NM.

(c) Interceptors are scrambled and take three minutes to get airborne. During this time the raider is 24 NM closer.

(d) The interceptors climb to 30,000 feet at the rate of 10,000 feet / minute which requires three minutes and the raider is 122 NM (Radar lag time 2' + scramble time 3' + interceptor climb time 3'+ interceptor's short of MLI 7’ 15”; total 15’ 15” x 8 NM = 122 NM)) nearer to his target: During climb to 30,000 feet the interceptors travel at the rate of 6 NM / minute and cover a distance of 18 NM towards the raider. They level off and attain a speed of 480 Knots i.e. 8 NM / minute (480 knots for slowest interceptor aircraft). They are to fly 58 miles more to reach MLI ((MLI - Distance by Interceptors during Climb i.e. 76 - 18 = 58). It would take them seven minutes fifteen second (7’ 15”) and the raider would have further travelled 58 NM.

**MLD** = MLI + Radar lag / ID Time + Scramble Time + Interceptors’ Climb Time + Interceptors’ short of MLI

**MLD (Time)** = MLI + 2min + 3min + 3min + 7min 15sec = MLI + 15’ 15”

**MLD (NM)** = MLI + 16 NM + 24 NM + 24 NM + 58 NM = MLI + 122 NM = 76 NM + 122 NM = 198 NM

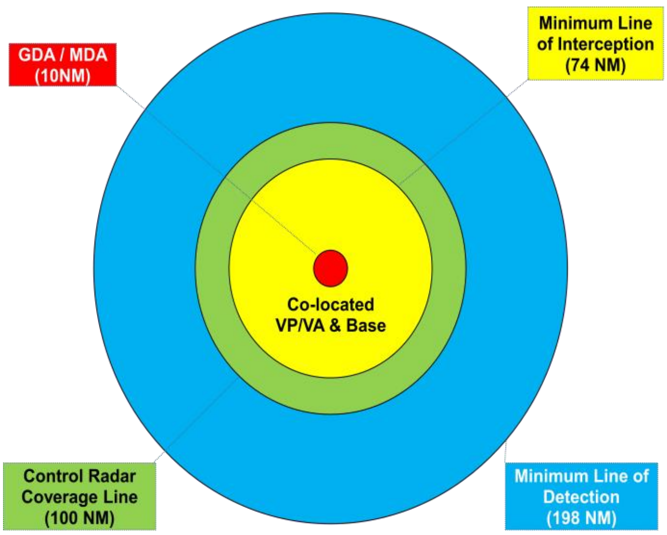
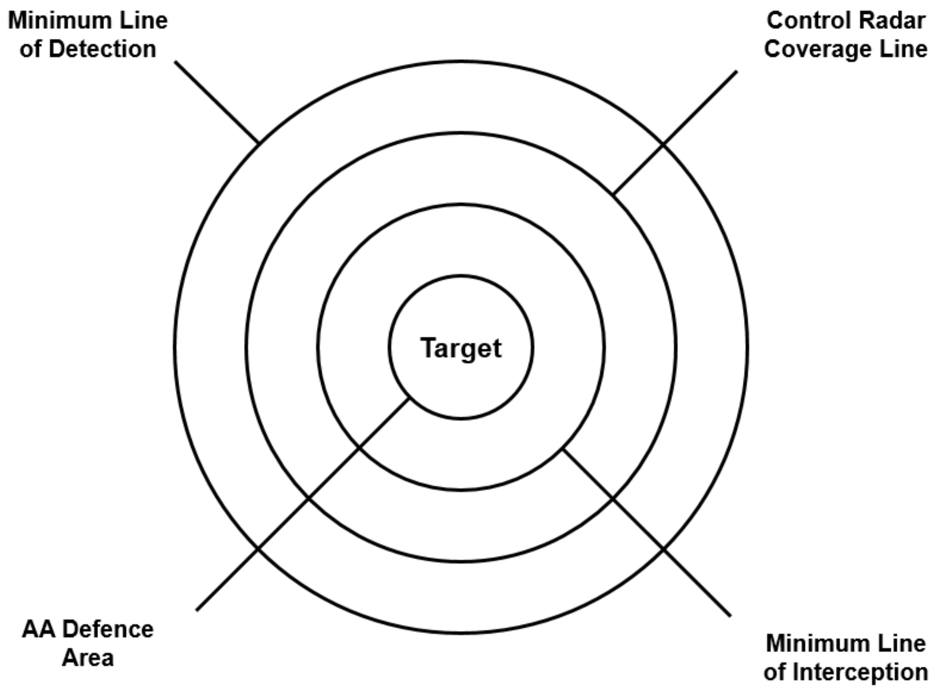



**Figure - 1**

(e) As stated earlier when the interceptors are at MLI, the raider should be at CRC, i.e, 3 minutes away (manoeuvre time which converted into distance would be 8 x 3 = 24 NM). At this stage, if not earlier, arrangements must exist to let the interceptors know the precise information about the raider's height so that they can position for attack. There are three minutes for the interceptors to make necessary adjustment for altitude and position so as to be able to fire at the raider when it reaches MLI at the end of this time.

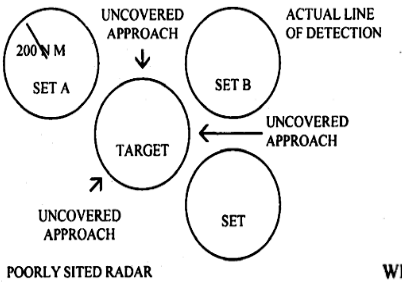


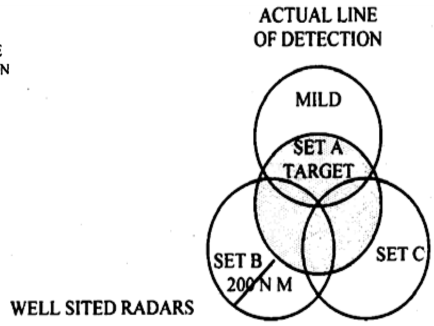

(f) The interceptors may need two minutes of combat time, if at the end of the two minutes they are notable to destroy the raider HIMADS / LOMADS can take over as the raider.

(g) In the above case the MLD would be 198 NM (76 + 122) meaning that to be able to intercept a raider at MLI, with the parameters as given in the example, the detection must take place at least 198 NM and 148 NM away respectively.

10. Effect of MLD, CRC and MLI on the positioning of a radar in any area of responsibility are discussed below:

(a) **MLD**. When the depth of a Country permits, the radar defences can be broken down Into Surveillance stations and control radar stations. This is a cheaper way than to have all sensors as the control stations. Surveillance stations are placed to cover the entire perimeter of the MLD. Since Surveillance radar function is to sound alarm, they are the first to detect approaching aircraft. You can note in the illustration, "Well Sited Radars" how three Surveillance radars have been placed to provide total coverage of MLD of a VP / VA. Also look at the illustration "Poorly Sited Radars". Here the same sets have been deployed far away from the target with no consideration for the MLD and their effectiveness has been destroyed. You can observe the corridor through which aircraft could slip undetected. Another method of placing surveillance radars consists of deploying them in depth as indicated in the illustration, it is clear that to place the Surveillance station, first the MLD is to be determined.




(b) **CRC**. A similar technique is to be applied for placement of control radar stations (GCI Units, GMCCs) with the exception that they will cover the entire perimeter of control radar coverage requirement line instead of MLD. Control radars are carefully sited to ensure good coverage in the maneuver control areas. An interceptor controller will become more familiar with this type of Unit than any other. When the radar Units are deployed to cover VP/VAs and approaches to them, it is highly desirable that there be an overlapping of coverage between Units so that in the event when one Unit is off the air, another can assume operations in its area of responsibility.

11. **Technical Considerations**. There are number of technical considerations which are to be kept in mind before siting radar. The satisfactory performance of radar equipment depends, to a great extent on the height above the ground at which the antenna is placed. So that line of sight of radar is clear to provide detection of low flying aircraft at maximum possible ranges. The immediate surrounding area is also an important aspect of siting. An uneven and mountainous area around the radar will affect the radar performance adversely because the radar indicators will be cluttered by the echoes from surroundings and weak echoes from aircraft may be completely obscured. The main points as discussed below are to be considered by the technical staff:

(a) **Electrical Characteristics**. The electrical characteristics of a radar set must be studied thoroughly as these are directly related with the performance of radar. Some of these characteristics are variable within the prescribed limits and the effects on the performance of radar by varying these characteristics are to be fully comprehended. This will enable the technical staff in pre- calculating the maximum expected performance at a particular site.

(b) **Ground Reflection**. Ground reflection depends on the nature of ground and determines the vertical coverage of radar. Ground reflection occurs when the beam radiated from the antenna strikes the surface of the earth and bounces upward. The vertical coverage of the reflected waves may arrive at the target in a manner that will either aid or oppose the direct waves striking the target. The reflected waves cause a decrease in the overall amount of power striking a target at certain altitudes and an increase in power striking the target at other altitudes. This subtraction and addition between waves create a vertical pattern consisting of areas of maximum power called "NULLS" and minimum power called "LOBES". At the lower radar frequencies, the nulls and lobes are widely spaced and an aircraft flying through this pattern will be seen to appear and disappear on the scope. At the higher radar frequencies, the nulls and lobes are closely spaced so that the tracking of the aircraft by the radar is improved. Ground reflection is a variable factor depending mainly on the type of terrain. Reflection is more when the reflecting surface is smooth and it decreases when the reflecting surface is uneven. Uneven land areas, trees or grass may absorb a large portion of the radiated energy to cause scattering of the energy, thus reducing the amount of reflected energy being added to or subtracted from the direct waves.

(c) **Screening**. Within the range of radar set, there is an area in which a target may be present but remains undetected. This area is created by the screening effect of terrain features of the earth. When the earth is smooth, the curvature of the earth causes the area beyond the horizon to be invisible to the radar beam. If there are hills and mountains in the radar path above the horizon, the screened area is increased and radar coverage is reduced. The amount of screening associated with the radar site depends upon the screening angle. This is the angle formed by the intersection of radar line of sight and the horizontal plane at the antenna. The angle may be positive or negative depending upon the site. The effect of screening is expressed as a relationship between screening angle and radar range at a given altitude. The effect of elevation on the radar line of sight is shown graphically in Fig-6. In the diagram, the antenna site is at sea level and radar line of sight range for a screening angle of 0 at 10,000 feet is 123 NM. If the radar line of sight is changed to a screening angle of +1, the line of sight limit for 10,000 feet is reduced to 67 NM, a reduction in range of 56 NM.

(d) **Site Elevation**. High sited radar has the greater low angle coverage and is useful for detection of low flying aircraft. Low sited radar provides less low angle coverage, thus limiting the detection of low flying aircraft, but at the same time ground clutter is decreased and high-altitude coverage is increased. A Low sited radar will also provide a more solid track as compared to the high sited radars.

(e) **Effective Antenna Height**. The effective height of an antenna is a significant factor in calculating the effect of the earth on the radiation pattern. The effective height of an antenna is its height above the local terrain or reflecting surface. The factor of effective antenna height has significance only where reflection from earth surface materially effects the structure of the radiation pattern. If the radar is sited at a place where the surrounding area is smooth the variation in the effective height of antenna will produce an appreciable difference in performance. When the surrounding surface is rough, the variation in effective height of antenna will have no effect on the performance. This is due to scattering of radar energy striking the rough surface.

(f) **Provision of Communication Circuits**. The provision of reliable and secure communication circuits is also to be considered as these will ensure uninterrupted and smooth conduct of operations. Where several alternative sites are available. Consideration is to be given to time and cost involved in the provision of essential land line tele-communication circuits for the various sites.

(g) **Availability of Power Supply**. The proximity of local Supply sources are to be considered when alternate sites are available. The performance of the radar is most satisfactory when operated from a well regulated power supply. The power generators are to be used as standby power supplies, if local supply is available.

(h) **Distance from Adjacent Radar**. Distance from adjacent radars to be ensured to avoid any interference and damage to receivers. Detail instructions as per AFM 103-1 may be followed in this regard.

(j) **Electromagnetic Spectrum Analysis**. Spectrum analysis may be carried out before deployment of radar at a new location, especially for the RF interferences incurred from civil area. In this context, PR-100 may be utilized.

12. **Administrative Considerations**. Siting of radar and its operation from a particular site involves the consideration of admin factors. A most suitable site may entail admin problems of such proportions that available resources may be inadequate to overcome them. The following administrative considerations are to be kept in mind:

(a) Site is to be accessible.

(b) Available for immediate occupation.

(c) Source of water.

(d) Camouflage.

(e) Ground Defence.

(f) Security, Engineering and Medical requirements for a camp site are met.

13. **Procedure for Radar Siting**. AOC (Air Defence Command) will allot the task of siting radar to the siting team. The siting team is to be composed of at least one Engg officer, one AD officer and one MES representative. The siting team shall assemble at a convenient place and proceed in the following manner:

(a) **Collection of Information**. The siting team is to collect comprehensive information about the operational requirements and technical characteristics of the radar. All available topographical and engineering data for the area under investigation is to be obtained.

(b) **Initial Selection of Sites**. Complete study of all available information is to be carried out. All possible sites in the proposed area are to be studied in the context of operational, technical and administrative considerations discussed earlier.

(c) **Aerial Reconnaissance**. An aerial reconnaissance of the area is to be carried out. This provides a rapid method of reviewing the proposed sites and may assist in eliminating the least desirable ones. Objectionable features of a site, not shown on a map, will be seen in this manner. The remaining sites are to be further investigated.

(d) **Final Selection**. The siting team is to proceed to the remaining sites to conduct investigation and collect specific data about each site for evaluation and comparison of sites under consideration. A final choice from the sites actually visited is made by ruling out unsatisfactory locations. Sometimes, it is desirable to measure the relative severity of permanent echoes at each of two or more perspective sites by operating the radar at each site in sequence. In practice it is unusual to obtain the ideal conditions and one has to accept a site which offers the best conditions keeping in view the area of threat.

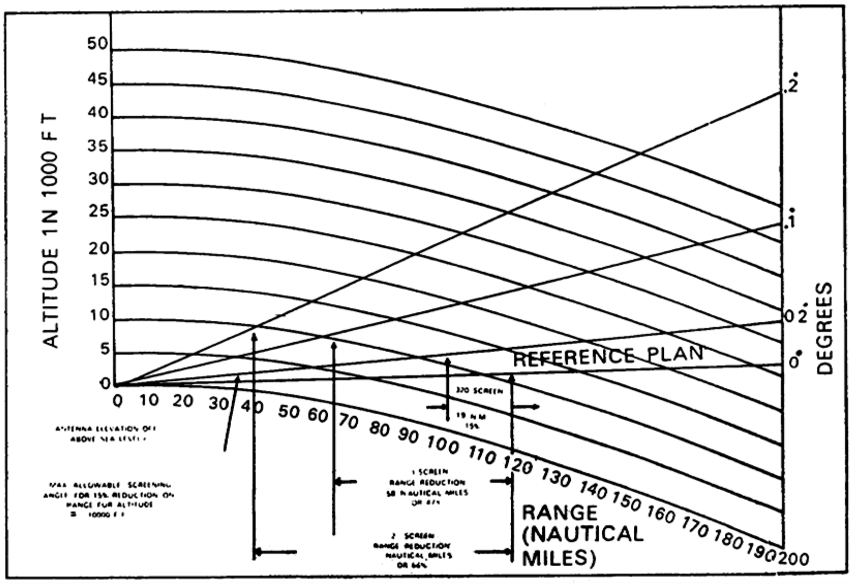
14. **Radar Calibration**. Proper siting as discussed above is essential to good radar operation but their actual effectiveness will be determined after testing and calibration. Radar factors such as radiated power and absolute receiver sensitivity can only be accurately measured in the field by radar calibration. The first few flights of a performance test provide the only certain method for determining whether or not the equipment is operating at optimum performance level. The calibration is an extremely specialized task requiring close co-operation between ground and airborne sections. It is normally performed by radar calibration flights. A good radar site might deteriorate with the passage of time due to construction in the area, which would present more clutter on the radar scope.

15. **Azimuth Range and Height Accuracy Tests**. Accurate knowledge of range, azimuth and height correctness is a prerequisite to good performance testing. Azimuth range and height accuracy tests can easily be performed in GMCC by comparing the radar data with reference radar. For standalone radar, these tests be carried out in following manner:

(a) **Azimuth Accuracy**. Azimuth accuracy is determined by a series of tests. Orientation of the antenna by means of compass provides a reference point for the azimuth dial setting. Permanent echoes or fixed targets that may be located accurately on a reference map may be used for an orientation check. However, the best check is by means of a prearranged flight where the azimuth of the aircraft is known or controlled. This is to be accomplished throughout the area of responsibility and at least at every 30° in azimuth.

(b) **Range Accuracy**. Range accuracy may be checked by using the PEs or known fixed targets and by accurately locating points on a reference map. Generally speaking, radar measures range more accurately than any other factor. Aircraft flights during performance tests may serve as further check.

(c) **Height Accuracy**. Height accuracy can be checked by using a single aircraft flying at different heights. Flights can be carried out at every 5000 feet. If the site is considered to be bad, flights at every 1000 feet may be tried. Should the flights at different bearings show discrepancies




at a given height, the results obtained on the runs differ from the actual heights the error is to be corrected on the scope. In this regard GPS / ADS-B may also be utilized as a reference.

**Siting and Deployment of a SAM Unit**

16. The siting of a SAM Unit means determination of an exact location of a SAM Unit so that it may best perform its function and fulfils the assigned task effectively. The deployment of a SAM Unit varies according to the types of system, i.e. short range or medium range and nature of the target to be defended, i.e. whether a Vulnerable Point (VP) or a Vulnerable Area (VA). The various

operational, technical and administrative considerations for SAM deployments are as under.

17. **Operational Considerations**. A SAM Unit includes the missiles launching equipment or firing Unit and acquisition Unit (not in case of shoulder mounted missiles). Additional radar could also be used for tracking and guidance of missile after the missile has been launched. Different missile systems have different kinds of launchers and missiles to cover a variety of VAs / VPs. Kill zone of a missile is an important factor which needs due consideration while selecting a site. The SAM Units can be deployed in the following manner:

(a) Short range missile be deployed for point defence.

(b) Combination of short, medium and long-range missiles is deployed to defend any VA.

(c) A medium range missile Unit be deployed in the direction of probable approach of threat. Similarly, long range weapons when used for area defence can be deployed well away from the area towards the ingress routes of the enemy.

(d) A ramp of 08 feet diameter and 05 feet height may be added for MWS / FN-16, whereas, for CWS 10 feet height of ramp may be added**.**

(e) Site is to provide clear field of view / LOS**.**

18. **Technical Considerations**. The satisfactory performance of SAM equipment depends, to a great extent on the ground on which the Unit is deployed. The missile launcher should be deployed on a hard level ground capable of sustaining the weight and jerk of the launcher at the firing of missile. The acquisition radar should preferably have clear line of sight. Other technical considerations are as follow:

(a) Provision of communication circuits.

(b) Site elevation offering minimum screening all around.

(c) No radar in the close vicinity operating on the frequency of tracking radar (It would hamper the guidance of missiles).

(d) Environmental considerations like temperature / Dust / average-rainfall and wind patterns.

19. **Administrative Considerations**. The administrative considerations for missile deployment are given below:

(a) Missile storage should be away from main road and civil population but close to the site for quick replenishment

(b) Camouflage.

(c) Ground defence of the Unit.

(d) Administrative facilities should be easily accessible.

20. **Security and Ground Defence Considerations.** Due to internal and external security threat environment in the country, security considerations for

21. Human and material at any short listed Air Defence Site is of prime importance. The following security considerations are to be included in siting and deployment mechanism. These considerations must be separately identified for sites falling in military compounds / cantonments / PAF & PN bases and sites in open areas:

(a) Requirement of security troops.

(b) Liaison with Law Enforcement Agencies [en-route and in vicinity of site].

(c) Availability of Site parameter / fencing / boundary wall.

**Deployment of Balloons**

22. The balloons when floating high in the air pose a great danger to the attacking aircraft. Balloons attract the attention of the raider pilots and cause them to pull up exposing them to the ground fire. The balloon deployment is of two types:

(a) To defend VP (Runway etc).

(b) To defend VA.

23. **Operational Considerations / Requirements (VP Defence)**. For deployment of balloons on a VP, following operational considerations and requirements are to be kept in mind:

(a) There should be no high obstacles, buildings, trees, overhead electric/ telephone lines etc around the balloon post.

(b) Valleys and expected threat approaches around runway should be covered with mobile balloon posts.

(c) Fixed/trolley mounted mobile balloon sites for the airfields/VPs. The number of posts will depend upon the length of runway, taxi track and size/dimensions of VPs. The general guide-lines to work out the number of balloon post is:

(i) Four mobile posts are established on the main runway.

(ii) Three mobile posts on the taxi track.

(iii) Fourteen posts are spread around the airfield.

(d) The balloon posts should not fall in the firing arc of SAMs and Air Defence guns deployed at / around airfield.

(e) Mobile posts on runway and taxi track should be cemented / metallic to avoid FOD.

(f) Surrounding area of at least 50 meters in radius around hoisting point should be levelled and cleared of bushes.

(g) Balloon posts should be in between VP (Runway) and SAMs or anti-aircraft guns deployed around VP.

24. **Operational Considerations / Requirements (VA Defence).** The following points are to be considered while deploying the balloons for VA defence:

(a) All valleys and approaches to the VA should be covered with balloon posts.

(b) Balloons should not be deployed in the direction of high hills closer to VP.

(c) Balloons sites are not to be selected in the lower areas.

(d) Distance between sites should be enough that the balloons do not inter-mingle during high winds.

25. **Post Deployment Precautions**. For all kinds of deployment, following precautions are to be observed for safe operations:

(a) No fire or inflammable material is brought within the radius of 200 meters of inflation point.

(b) Balloon is properly placed on tarpaulin and hydrogen gas cylinders are placed at proper and convenient place for inflation.

(c) To prevent any accident, fire precautions are strictly observed before inflation of hydrogen gas and 'No Smoking' board is erected.

(d) Inflation and deflation to be carried out as per SOPs.

(e) Hydrogen gas cylinders are kept in dug-in-pits and under shade.

**Deployment of MOUs**

26. Wireless observers are the pioneers of Air Defence and aircraft reporting system and still they are an essential element of modern system. Their efficiency depends upon their ability to correctly observe and promptly report the aircraft movement. For the observation purposes they are normally deployed close to border areas and where no radar coverage is available. Such deployment is in the form of two belts, which are called outer and inner belts. Outer belt is called reporting belt and the inner one confirmatory belt. Various considerations / requirements which need attention in the deployment of mobile observer Units are discussed in succeeding paras.

27. **Operational Considerations / Requirements**. Deployment of MOUs depends upon the following:

(a) Enough early warning could be provided to affect lane control interception.

(b) No obstacles in the surroundings to have best all-around visibility.

(c) Posts be well away from gun and missile defended areas.

(d) Posts should be close to enemy border but beyond enemy ground fire.

(e) Posts should be equally spaced so that no aircraft passes unnoticed in between the two MOUs. Establish MOS (Mobile Observer Section) after every five MOUs to provide any assistance to the MOUs.

(f) The posts should fill up the gaps of radar coverage.

28. **Technical Considerations / Requirements**. The technical considerations for MOUs deployment are listed below:

(a) Site elevation to achieve line of sight for the communication.

(b) Antenna installation away from high voltage. No interference on Ops communication circuits.

(c) Availability of sufficient number of standby operational equipment such as radio sets, Charging sets and accumulators in MOS and Flt HQ.

(d) Availability of suitable transport at MOU and MOS according to the area requirement.

(e) Establishing contact with control station, i.e. RRF and Flight / Squadron.

(f) Headquarters soon after reaching their pin points.

(g) Round the clock uninterrupted operations.

29. **Administrative Considerations**. An operationally suitable site may pose some problems of administrative nature. Following points are to be kept in mind to reduce such problems while selecting MOU deployment sites:

(a) The post should be located at a reasonable distance from the populated area.

(b) Free access to water source.

(c) Camouflage requirements.

(d) Safety of equipment and men against sabotage activities.

(e) Availability of trenches, snake channels for the safety of men.

(f) Ground Defence.

(g) Easy access to the site for visits, attendance, evacuation or medical assistance.

30. **Security Consideration and Liaison with LEAs**. Keeping in view the generic volatile security situation, special emphasis should be laid on safe move & deployment of men and material. On receipt of deployment order, close coordination with PAF and civil LEAs should be carried out thorough local P&S Units.

**CHAPTER - 15**

# DATA HANDLING AND REPORTING PROCEDURES

## Introduction

1. To ensure rapid response to any hostile air attack / activity within the Air Defence System, there is an obvious and paramount need for maximum coordination for employment of all Air Defence means without loss of time. The successful accomplishment of any Air Defence mission primarily depends on how expeditiously and accurately Air Defence data about hostile air activity can be processed and passed to the appropriate tactical operation centre.

## Threat Recognition (Quantum & Classification)

2. Threat recognition is the determination of hostile air activity. It is termed as warning. It is to be accomplished, if proper and timely Air Defence measures are to be taken. Threat recognition requires Current air surveillance data. With the availability of composite air picture at GMCC, filtered air picture is up told to SMCC for timely identification and generating Recognized Air Picture (RAP) which helps in timely initiation of appropriate tactical action. Threat recognition comprises of quantum of threat and threat classification.

(a) **Quantum of Threat**. Quantum of threat can be determined through detections by active (ground based, aerial and ship borne) sensors and inputs by passive means through intelligence sources (COMINT / HUMINT etc), ESM systems, on ground troops and MOUs belt deployed at selected sites.

(b) **Classification of Threat**. Threat classification can be done by the use of ESM systems, Non-Cooperative Target Recognition (NCTR) methods and different intelligence sources (COMINT / HUMNT etc).

## Combat Activities

3. Combat activities embrace the total information on which tactical decisions are based. It consists of informing subordinate, lateral and higher commanders about the enemy activities. These activities include enemy capabilities and intentions gathered through intelligence, surveillance, tactical information, battle damage, nuclear detonation and deployment / re-deployment of their weapon system. Combat activities are to be passed to all echelons as expeditiously as possible.

## Combat Data

4. **Definition**. The total information which an operation center requires for the proper execution of its mission, in order to complete its assigned task, is to be referred as combat data. Its broad categories are given below:

(a) Air Surveillance data.

(b) Tactical action data.

(c) Status data.

(d) Battle damage information.

(e) Nuclear detonation information.

(f) Electronic attack information.

(g) Intelligence information.

5. **Sources of Combat Data**. Combat data can be gathered through variety of sources. These sources may include but not limited to all military, civilian and government activities affecting air defence. The data is gathered active and passive sensors, interceptors/recce squadrons, UAVs, monitoring flights, MOUs, VHF monitoring sets and satellites. From the data collecting agencies the combat data is forwarded to higher echelons where evaluations are made. Appropriate to the level of command.

6. **Air Surveillance Data**. Air Surveillance Data consists of the systematic observation of the air space within the area of responsibility by electronic, visual, photographic or other means for the purpose of detecting and identifying all air borne objects. The air situation is to be transmitted expeditiously to each level of command. It is to be displayed in the operation centers, for the use of the commanders in making timely decision and taking appropriate Air Defence action. It is, therefore, essential that each Unit complies with the air surveillance procedures set-forth in this manual and employs all measures necessary to ensure timely and accurate submission of all air surveillance data. Sources of surveillance report will include, but are not limited to the following:

(a) Air Surveillance data.

(b) Tactical action data.

(c) Status data.

(d) Battle damage information.

(e) Nuclear detonation information.

7. **Tactical Action Data**. Tactical action data consists of the action taken on each Surveillance report and the result of that action. It represents a complete picture of the Air Defence battle which will help the higher commanders in evaluating the battle situation. Sources of tactical action data are the operation centers authorized to take tactical action. Information concerning local engagements is to be

forwarded to the higher Headquarters on a specimen form given as **Appendix 'K'** to this AFM.

8. **Status Data**. The successful conduct of air battle is partially dependent on the timely and accurate submission of status reports to SMCC. It is mandatory that status data including the availability of all types of weapons employed under the operational control of the Sector Commander i.e. land based radars, interceptor aircraft, SAGWs and Air Defence artillery be made readily available to SMCC. Specimen form for the interceptor state report is given at **Appendix 'L'** to this Chapter.

9. **Battle Damage Information**. Battle damage information embraces reports of damage from enemy action, to the Air Defence weapon system, augmentation forces that are available, airfields and any other activity or equipment which is likely to adversely affect the Air Defence commander's ability to defend against subsequent air attacks. It is practicing of assessing damage caused on a target. Assessment is executed using many techniques like footprints by UAV's, cameras, and forces on the ground near the target, satellite, imagery and follow up visit to the targets. For nuclear weapon special technique for battle damage assessment may be required to ascertain extent of damage caused.

10. **Nuclear Detonation Information**. Nuclear detonation includes all nuclear and thermonuclear explosions occurring in or adjacent to Pakistan as a result of any enemy action. Nuclear detonation reports are to be immediately forwarded by all Air Defence agencies to their respective SMCC through normal surveillance reporting channels.

11. **Nuclear Detonation Report (NUDET)**. NUDET Report is to be prepared by SMCC and submitted to ADOC when detonations are detected (other than test detonations). Duplication in the reporting of detonation is to be eliminated at the SMCC by evaluating the reports received from other agencies, prior to forwarding to higher Headquarters. A specimen form of the report is given at **Appendix 'M'** to this Chapter.

12. **Electronic Attack (EA) Information**. Electronic Attack information includes all reports on electronic countermeasures used by the enemy to reduce the effectiveness of the radar, communications of other Air Defence techniques dependent for their operation on electromagnetic radiations. Reports are to form part of both surveillance data related to hostile tracks and intelligence summaries listing the effectiveness of enemy tactics. Electronic attack reported tracks fall into the classification of unknown' or 'hostile'.

13. **Electronic Attack (EA) Reporting**. The presence of EA is to be known to SMCC and ADOC through available communication channels. Radar sites are to report un-correlated EA in the form given at Appendix 'D' to this Chapter. One-time jamming reports are to be filed by the affected radar Units and forwarded to area SMCC.

14. **Intelligence Information**. It is the information collected by various means regarding enemy's plans, procedures, deployment of forces, degree of preparedness, types and capabilities of weapons. These are to be passed to the agencies concerned by the fastest means.

**CHAPTER - 16**

# AIR DEFENCE STANDARD TIME

## Introduction

1. For co-relation of different Air Defence activities and synergetic employment of tri-services AD elements, uniform and accurate time keeping is essential. Following procedures are to be adopted by various elements of Air Defence organization to disseminate time throughout the Air Defence Network.

## Standard Time

2. The Air Defence Standard Time (ADST) is to be maintained on the same basis as Pakistan Standard Time (GMT + 5 hours, day light saving alterations, if any). For all reference purposes, the time maintained at ADOC (Vision console / digital wall clock) is to be treated as Air Defence Standard Time. The digital wall clock of ADOC will be termed as “Master Clock”.

## Agencies / Formations to Maintain ADST

3. The following agencies / formations are to maintain the ADST:

(a) ADOC.

(b) All SMCCs, GMCCs and VTWCCs.

(c) Ground based sensors of PAF and sister services.

(d) GBADs of PAF and sister services.

(e) All AEWACS.

(f) Active fighter ADA huts.

(g) All PAF ATCs and personnel on duty.

(h) Air Space Coordination Cells.

(j) All flying Bases / Units / Wings Operations Rooms.

(k) All Pilots / Fighter controllers on duty, whether in the air or on ground.

(l) All wireless observer Headquarters down to each MOU post.

(m) Any other setup / personnel actively engaged in or connected with Air Defence activities

**Broad Categories for Maintenance of ADST**

4. For maintenance of ADST, the above mentioned agencies can be divided into two following categories:

(a) Vision Based Systems.

(b) Non-Vision Systems.

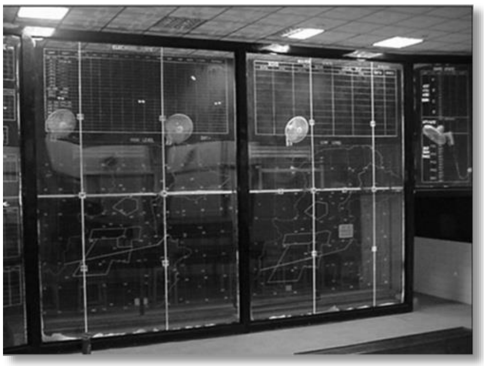
**ADST in Vision Based Systems**

5. All vision automation systems i.e. ADOC, SMCCs, GMCCs and V-TWCCs are provided time through a Network Time Protocol (NTP) server. The server takes time from GPS in UTC and converts into PST as (GMT+5). There are four NTP servers, one in each SMCC. Various vision-based agencies can be connected as a client to any of the NTP server. Hence the time displayed on all vision-based systems is synchronized at all tiers. Nevertheless, there can be instances when single / multiple Ops room gets out of sync with rest of the vision system due technical malfunction. In this regard, it is important for the battle staff to maintain a manual routine check, which timely identifies time discrepancies.

6. In this context at ten minutes past every odd hour, ADOC is to initiate time check with under command SMCCs. The SMCCs within ten minutes of receiving time check will pass the ADST to under command GMCCs and V-TWCCs. The passing of time check among vision systems will be a quick procedure using following words “ADOC to Sector- passing you time check now – time is e.g. 14 hour 50 minutes, seconds 30-31-32-33-34-35-36-37-38-39-40”.

7. Any discrepancy more than fifteen seconds is to be brought in the notice of technical staff. Upon identification of de-sync more than fifteen seconds, the single / multiple Ops rooms are to inform its higher echelon and shift to digital wall clock till availability of synced vision system time.

**Note**: In addition to the reasons mentioned above, shifting of automated operations to manual mode would also necessitate a shift to Digital Wall clock.



(Operations in Manual Ops room)

**Procedure for Maintaining and Relaying Time via Digital Wall Clock**

8. In case of single / multiple Ops room shifting to digital wall clock, it is to maintain time in sync with its higher Ops room. In case all Vision based systems shift to digital wall clock same procedure will be followed as mentioned in Para 6.

9. Master clock is to be set according to the time signals of Pakistan radio or Pakistan television at any convenient hour and is to be subsequently checked for accuracy with similar time signals, as frequently as possible, but in no case later than six hours. In case master clock shows a discrepancy of fifteen seconds or more in a period of twenty-four hours, it is to be considered unserviceable and be replaced with standby digital wall clock.

**ADST in Non-Vision systems**

10. Within ten minutes of receiving time check, SMCCs, GMCCs and V-TWCCs either maintaining time on vision system or digital wall clock are to initiate time check to under command non-vision systems. Within ten minutes of receiving the time check, all non-vision systems will further down-tell time till the last AD element / tier.

11. Non-Vision systems can maintain ADST on system scope (where available) or digital wall clock. Preference is to be given on maintaining time on system scope as it useful during playback of recording for subsequent analysis. For all non-vision systems, a time discrepancy of more than fifteen seconds in system clock / digital wall clock is to be treated as unserviceable. Under mentioned vision systems will pass ADST to agencies as mentioned against each:

(a) **SMCC**

(i) Fighters on ADA in AOR.

(ii) Associated ATCs which will further down tell the ADST to BBUs and TDCC.

(iii) Pak Army / Navy CRCPs, which will downwards pass the ADST till individual element of shooter tier and to any auxiliary / parallel setups.

(iv) Wireless Observer HQs which will downwards pass the ADST till each MOU.

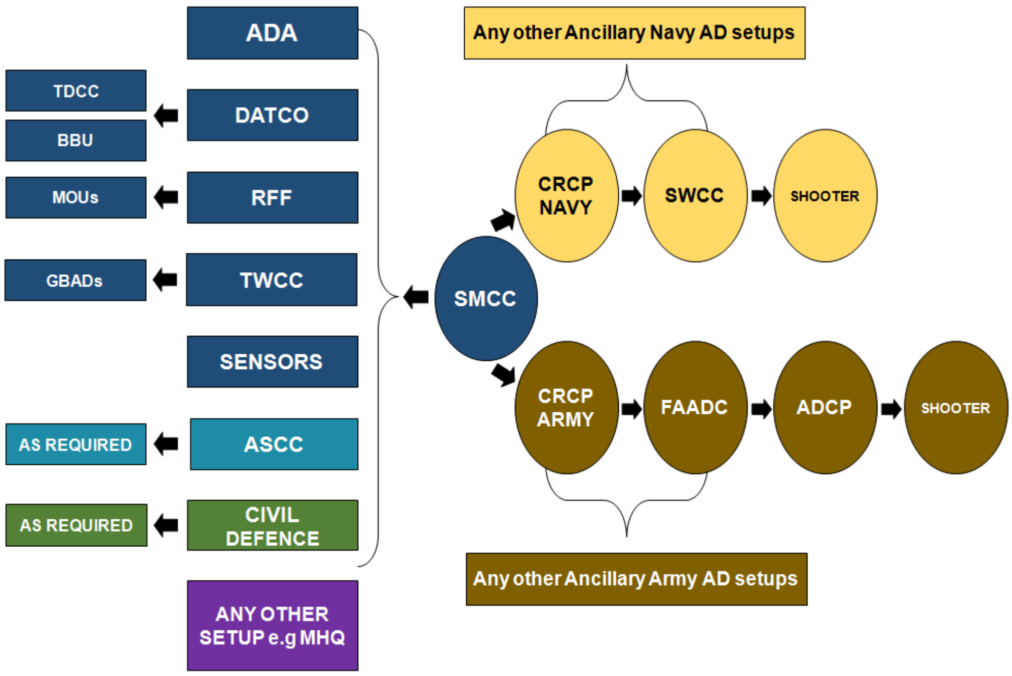
(v) Any sensors hooked with SMCC directly.

(vi) Air Space Coordination Cell (ASCC) when established.

(vii) Associated TWCCs which will further down tell the ADST to associated GBADS.

(viii) Civil Defence Setups (When required).

(ix) Any other setup engaged in Air Defence activities or associated with SMCC.




(b) **GMCC**. All sensors of PAF and sister services in netted mode.

(c) **V-TWCC**. All GBADS of PAF and sister services in netted mode.

**Exceptions**

12. The above-mentioned procedure of passing ADST flows from the higher formations towards the lower formations. However, few exceptions are elaborated below for some elements / individuals:

(a) **AEWACS**. Link officer is to take time check from respective SMCC before takeoff and correlate with respective mission systems.

(b) **Flying Bases / Wings / Units Operations Rooms**. They will take time check from the SMCC under which AOR they are associated as and when deemed necessary.

(c) **Pilots / Fighter Controllers / ATC personnel**. They will take time check from area SMCC as and when required.

## CHAPTER - 17

# TRANSFER AND DISPLAY OF SURVEILLANCE DATA (AUTOMATION AND MANUAL MODE)

### Introduction

1. Air surveillance is the systematic observation of a given sector of airspace by electronic, visual or other means for the purpose of detecting, tracking, identifying and analysing airborne objects for subsequent tactical actions and transferring whatever data is obtained, to other concerned Air Defence agencies. This transfer of data is done either through automated, or through manual system depending on the connectivity available between the reporting sensor, GMCC and the respective SMCC.

### Transfer and Display of Auto Data

2. In automated Air Defence System, the data from all the deployed-on watch sensors including high, medium and low level is transferred to GMCC. Visions based GMCCs received data from all radars in the form of plots and generate tracks on these plots. These plot and track data are displayed on Air Control Station (ACS) in GMCC. The auto and manual biasing feature is available in GMCC software to avoid positional error of these plots in overlap areas. The tracks fusion is also done in overlap sensor coverage areas to avoid excessive clutter on the GMCC scope. These tracks are further transferred to SMCC from computer to computer on different available data links to provide real time Recognized Air Picture. Likewise, MOUs are linked up to SMCCs (RRF) through Portable Radio Communication PRC-786 on DT -200 terminals. The MOU reports are also fed in SMCC by the operators. This data is in the form of tracks (received from Sensors) or reports (received from MOU sites) which are digital depictions of the information regarding the flying object being picked up are displayed on the 'Surveillance and Tactical Station' or the STS which is a digital console. This complete information (tracks & reports) is automatically transferred and displayed at the ADOC in automated mode.

### Track Display at GMCC

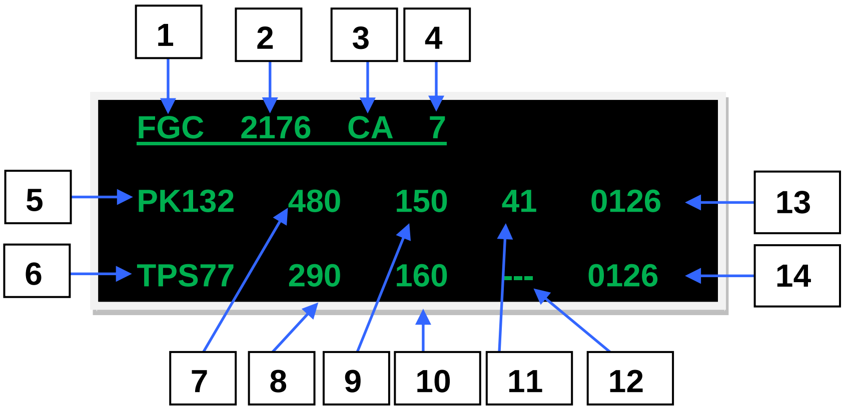
3. An example, track displayed at GMCC from an automated sensor is explained below:

| | |
|---|---|
| <br>**FGC 1234 P 7** | In the symbol,  the indicates the position of the aircarft, while the direction of the line shows the heading and its length indicates the approximate speed of the aircraft |
| FGC | Track identification given by SMCC (FGC indicates Friend General civil) |
| 1234 | Track No allotted to the aircraft by the SMCC Tracker |
| P | Suffix of the reporting radar |
| 7 | Track quality (1 – 7). Here it shows the maximum track quality, meaning by that the track is being picked up consistently by the reporting sensor |

4. Provision of additional information is also available in ACS with the track including speed, radar and C height, Mode 1, 2, 3 subject to the aircraft selection of these parameters and Sensor capability.

**Track Display at SMCC**

5. The Track representation on the Surveillance and Tactical Station (STS) at SMCC describes track label and track related data. The track label detail is tabulated below:




| Index No | Description | Specifications |
|---|---|---|
| 1 | TRK ID | UP, UE, UH, HG, FG, FH, SM, SS, SI |
| 2 | TRK No | 1001-9999 |
| 3 | GMCC Suffix | CA [C=Censec, A=Suffix (481)] |
| 4 | TRK Quality | 0-7 |
| 5 | Call sign | CS as per FPN eg. Hataf-1 |
| 6 | Sensor Painting with best Quality | TPS-77, YLC-6,ATCRetc |
| 7 | Speed | 0001-4000 Kts |
| 8 | Heading | 001-360 |
| 9 | Radar Height | 0-999 (Thousands of feet) |
| 10 | C Height | 0-999 (Thousands of feet |
| 11 | IFF Code 1 | NN (N=0-7,N=0-3) |
| 12 | IFF Mode 6 | A or B etc |
| 13 | IFF Code 2 | NNNN (N=0-7,N=0-7, N=0-7,N=0-7) |
| 14 | IFF Code 3 | NNNN (N=0-7,N=0-7, N=0-7,N=0-7) |

**Transfer and Display of Manual Data**


6. **From Manual Radars**. The data of all sensors may not be connected to the GMCC in automated mode due to certain limitations of media links or any other technical reasons. Therefore, the data of these sensors would also be not available at SMCC and subsequently at ADOC. In this regard, there has to be a way of transferring the data manually from the sensors to respective GMCC and then to SMCC. Such sensors pass the data to the GMCC on voice link for generation of manual tracks. These manual tracks would be passed to SMCC in automated mode and any ID given by SMCC to these tracks would also be passed to GMCC and ADOC in automated mode. GMCC is responsible for updating the manual tracks. The difference between an automated track, as discussed earlier and the manual track is that the manual track has “**m**” for manual indication and it does not indicate any track quality. Hence it can be easily distinguished at the GMCC, SMCC and ADOC. The display of manual track at STS is as shown here.


7. There could be instances when Sensor may not be connected to GMCC but connected to SMCC on voice links. In this case, Sensor would manually pass the data to SMCC on voice link and an operator at SMCC would feed this data in the system through a designated console generating manual tracks for its display on all the STS. This manual track would be available at all STS at SMCC and would be passed to ADOC in automated mode. Any subsequent changes in the heading, speed and other parameters of the manual track have to be updated manually by the respective sensor for its subsequent feeding at SMCC, or else it will continue with the initial parameters. The display of data at SMCC from manual radar is as shown.

8. The difference between an automated and the manual track is that it has “**M**” for manual indication and it does not indicate any track quality. Hence it can be easily distinguished at the SMCC and ADOC.

9. **From MOUs**. MOU data is based on visual sighting of the airborne object or contrail and hearing of the engine noise. This data is automatically passed by the MOU operator through a portable terminal called DT – 200 on PRC-786 media to Receiving and Report Flight (RRF). The RRF operator would pass this information to SMCC operator which subsequently fed this information for generation of MOU report at STS. The report so fed is displayed automatically at all STS in the format shown below:

| | |
|---|---|
| | In the symbol, the indicates the position of the reported target at the time of sighting / hearing, while the direction of the line shows the heading and its length indicates the approximate speed of the target |
| **KB 007 F** | |
| KB | It is the suffix of the reporting MOU site |
| 007 | It is the Report Number as generated by SMCC computer |
| F | It is the ID of the aircraft |

10. Apart from the immediate information available through Auto / Manual tracks, the detailed information regarding any track or MOU report can be seen on the tabular display area. In case of Auto / manual radars, the tabular display provides information on speed, heading, altitude (if available) and IFF code and strength. Whereas, in case of MOUs, information regarding time of report, aircraft type, position, heading, height, ID (U, F, E, H), Mode (Hearing or Sighting) and strength is available at the tabular display area. The tracks and reports would be displayed in different colours / Shapes at STS based on their IDs. All this is transferred to the ADOC automatically from the four SMCCs, thereby, providing complete air picture of entire air space to ADOC.

11. **Track Classifications**. Once the data is transferred to SMCC, pending tracks are identified on the basis of the information available at the SMCC from various agencies including ACCs and ATCRs. Subsequently, the tracks are labelled as per the following classifications:

| Main | Description | Sub Category | Description |
|---|---|---|---|
| **UP** | Unknown Pending | UPG | Not Specified (General) |
| **UE** | Unknown Evaluated | UEG | Not Specified (General) |
| | | UEB | Bomber |
| | | UEF | Fighter |
| | | UEH | Helo / Transport |
| | | UEA | AWACS / Recce / EW / Decoy |
| | | UEP | Missile Platform |
| | | UEZ | Zombie |
| **UF** | Unknown Friend (Assumed Friend) | UFG | Not Specified (General) |
| | | UFB | Bomber |
| | | UFF | Fighter |
| | | UFH | Helo / Transport |
| | | UFA | AWACS / Recce / EW / Decoy |
| | | UFP | Missile Platform |
| **UH** | Unknown Hostile (Assumed Enemy) | UHG | Not Specified (General) |
| | | UHB | Bomber |
| | | UHF | Fighter |
| | | UHH | Helo / Transport |
| | | UHA | AWACS / Recce / EW / Decoy |
| | | UHX | X-Ray |
| | | UHP | Missile Platform |
| | | UHD | Drone/RPV |
| **HG** | Hostile General | HGG | Not Specified (General) |
| | | HGM | Missile |
| | | HGB | Bomber |

| Main | Description | Sub Category | Description |
|---|---|---|---|
| | | HGF | Fighter |
| | Hostile General | HGA | AWACS / Recce / EW / Decoy |
| | | HGH | Helo / Transport |
| | | HGP | Missile Platform |
| | | HGJ | Jammer |
| **FG** | Friend General | FGB | Military Training |
| | | FGC | Non Military (Civil) |
| | | FGG | Not Specified |
| | | FGJ | Joker |
| | | FGK | Neutralized Faker |
| | | FGN | Neutral |
| | | FGR | RTB |
| **FH** | Friend Hello | FHG | Not Specified (General) |
| | | FHA | Anti Submarine Warfare |
| | | FHS | Search and Rescue |
| | | FHN | Gunship |
| | | FHR | Recce / Rec |
| | | FHL | Logistic |
| | | FHT | Troop Lift |
| | | FHM | Med-Evac |
| **SM** | Special Mission | SMG | Not Specified (General) |
| | | SMA | AEW/ARP/ABCC |
| | | SMS | Search & Rescue |
| | | SME | EW |
| | | SMR | Recce |
| | | SMJ | Faker Jammer |
| | | SMK | Faker |
| | | SMY | Kilo |
| **SS** | Strike Support | SSA | Anti-Submarine Warfare |
| | | SSB | Tanker Boom |
| | | SSC | Close / Direct Air Support |
| | | SSD | Tanker Drogue |
| | | SSG | Not Specified (General) |
| | | SSI | Interdiction |
| | | SSL | Logistic |
| | | SST | Tanker General |
| **SI** | Special Interceptor | SIC | CAP |
| | | SIG | General |

| Main | Description | Sub Category | Description |
|---|---|---|---|
| | | SIR | RTB |
| | | SIS | Rescue CAP |
| | | SIU | Unavailable |

12. The unidentified and identified tracks can be distinguished at GMCC and SMCC from the following table:

| Location | Unidentified Track | Identified Track |
|---|---|---|
| **GMCC** | **UPG 1234 P 7** | **FGC 1234 P 7** |
| **SMCC** | **UPG 1234 CA 7** | **FGC 1234 CA 7** |

**Telling Procedures (Manual Mode)**

13. **Search Radar Operator**. Search radar operator using the search radar scope is connected to the plotting board operator and recorder. He is to adopt the following procedure while passing plots to the vertical plotting board:

(a) **Initial Plot**. When a new track is observed it is to be reported in the following sequence:

(i) Initial Plot

(ii) Track Number - As individual digits

(iii) GEOREF pin point

(iv) Strength - As a whole number

**Example**: Initial plot, track one two (12) KH 5326 two (2)

(b) **Second Plot**. Within one minute or after ten nautical miles, whichever is earlier, confirm movement of track using the following sequence:

Track / Serial Number - Direction - GEOREF pin point - Height (as individual digits in thousands of feet) - Estimated Speed (as a whole number).

**Example**: Serial one five (15), South, Hotel kilo five three two zero (HK 5320), At three zero (indicating height 30,000 Speed three hundred (300 N. Miles per hour)

(c) **Cancellation of Track / Initial Plot**

(i) Scrub

(ii) Track designation / GEOREF Position

**Example**: Scrub; Track 12 / Scrub initial Plot HK 2020

(d) **Subsequent Plot**. On all established tracks, plots are to be told once a minute or after every 10 NM, whichever is earlier, in the following sequence:

(i) Track designation (furnished by the plotter of the vertical plotting board) - as individual digits.

(ii) Direction

(iii) GEOREF pin point

**Example**: Friendly one five (15), South HK 5315

(e) **Unusual Track Report**. When a track appears to be Unusual in size, flight behaviour, or other characteristics, the supervisor is to be called to observe the scope for further analysis.

(f) **Orbit Report**. When a track appears to be orbiting, it is to be reported in the following sequence:

(i) Track designation

(ii) GEOREF position

(iii) Orbiting left (or right)

**Example**: Friendly two zero (20), HK 1656, orbiting left

(g) **Amendment Report**. Amendment report is to be passed in the following sequence and the specific words shown against each ancillary information are to be used before individual change is passed.

(i) Track designation

(ii) Direction

(iii) GEOREF position

(iv) Strength - (Word "Now" and give change)

(v) Height - (Word "Now at" and give change)

(vi) Speed - (Word "Now speed" and give change)

**Example**: Strength Change:

"FGC0020, SE, HL 1926, Now 3" - **Height Change**

"FGC0020, SE, HL 1926 Now at 26" - **Speed Change**

"FGC0020, SE, HL 1926, Now speed 500"

(h) **Change in More Than one Ancillary Information**. When there is change in more than one ancillary information of a particular track, the word "Now" is to be used only once before the first change.

**Example**: "FGB0020 SE, HL 1926 Now 3 at 36" (Note that word "Now has been used once only before passing the first change and has not been used before the change in Height. The word "AT" is sufficient to indicate the change in height as well).

(j) **Split Track Report**. PPI Observer is to be alert to the possibility of an observed track splitting. When a track splits into two or more segments the track nearest to the Unit reporting is to retain the assigned track designation and a new designation is to be assigned to the other segment by the observer, using the initial report sequence.

**Example**: "UHX one-zero (10) splitting, initial plot (for the new track) track one-two (12) HK 5346 two (2) ".

(k) **Re-appearance of Track**. Re-appeared track is to be told in the following sequence:

(i) "Re-appeared".

(ii) Track designation.

(iii) Direction.

(iv) GEOREF pin-point.

**Example**: "Re-appeared, FGC 0020 South, HK 3926".

(l) **Mix up / Combined Track Report**. The word 'Mix up' is to be used only when an interceptor's track mixes up with hostile, X-raid or a friendly track acting as target. In all other cases where hostile mixes up with another hostile track or an unknown track, it is to be reported as "Combined" track. When a friendly track mixes up with another friendly track, it is also to be reported as a 'Combined' track. The serial number with the greater strength (or in case of equal strength the one with the higher serial number) is to be maintained while reporting. Mix-Lip Combined track can be a combination of the following categories of tracks.

(i) Hostile or X-raid with interceptor - Mix-up.

(ii) Hostile or X-raid mutually or with each other - Combined.

(iii) Friendly with another friendly - Combined.

(iv) Friendly a/c acting as target and the interceptor - Mix-up.

(v) The following sequence of telling is to be used:

(A) Mix-up / Combined.

(B) Track designation.

(C) Direction.

(D) GEOREF pin-point.

(E) Strength.

(F) Height.

(G) Speed.

$\left.\begin{array}{l}\text{(E) Strength.}\\ \text{(F) Height.}\\ \text{(G) Speed.}\end{array}\right\}$ If any change occurs

**Examples**:

1. "Mix-up 28, South East, HK 1536 Now 6, Hostile 26 Fighter 28 Mixed-up" (Assuming that Hostile 26, 2 at 30 with speed 500 and fighter 28, 4 at 30, with speed 500 mix-up).

2. "Combined H26. South West, HK 3040, H25 and H26 Combined" (Two Hostile combined. In case of a combined track of a "Hostile" and "X-raid" the "Hostile" prefix is to be retained).

3. "Mix-up 26, South West, HK 3040 T25 and F26 Mixed-up" (Assuming both tracks of equal strength).

4. "Combined 28, South West, HK 3040, B 27 and F 28 Combined. (Case of equal strength or Fighter in greater strength).

OR

5. "Combined 32, South East, HK 2910, Fighters 32 and Bomber 34 Combined" (case of Fighter in greater strength).

(m) **Contact Lost Report**. If an established track does not appear on the scope for one minute, it is to be reported as "Contact lost". Plotter, however, is to retain the track on the plotting board for a minimum period of three minutes after the contact lost report before it can be declared as a "Fade Track".

During the period between "Contact Lost" and "Fade Report" the plotter is to "Dead Reckon" the track to maintain track continuity, at intervals of at least one minute. At the end of these three minutes period a "Fade" display is to be plotted on the track and the reporter is to be told.

**Example-1**

**Contact Lost Report**. "HGG two-six (2626), Contact Lost". (Dead reckon may be discontinued if instructed by MIO. This may be in the case of track fading due to landing at an airfield etc. Otherwise the plotter is to declare the track as "Faded", provided it does not appear within three minutes of DR tracking).

**Example-2**

**Fade Report**. "HGG two-six (26), Faded".

14. **Upward Telling from Radar Units to SMCC and Higher Ops Room**

(a) **Initial Plot**. The upward telling is to be carried out on all tracks unless otherwise instructed. The reporting procedures are to be the same as for the raid- reporting Unit to the track production board and the examples given in sub-para 3 (a) to (m) are to be followed.

(b) **Change in Upward Telling**. Upward teller is to use the following procedure in passing the change in ident or ancillary information:

(i) Old track designation.

(ii) "Now".

(iii) New track designation.

(iv) Direction.

(v) GEOREF pin-point.

**Example**: "UHX2626 Now HGG 2626, SE, HK 1629".

OR

"FGB2020, SE, HK 1828 Now 2 AT 35". (Here the changes are in strength and height. Note the word "NOW" has been used once).

15. **Frequency of Search Report**

(a) The initial report from the reporting station is to be forwarded as soon as a track has been picked up. This report is not to be held up for lack of information on strength, height or speed.

(b) Follow up report is to be originated within a maximum of one minute or after 10 NMs whichever is earlier.

(c) Amendment Report is to be passed when applicable.

(d) "Contact Lost" report is to be passed when an established track does not appear for one minute, but the "Contact Lost" will be shown on the last reported position.3

(e) If the plotter at ADOC receives instructions from his Duty Controller to discontinue plotting on a track, he is to pass the word "Discontinue" followed by track designation to the upward teller e.g. "Discontinue FGC 34".

16. **Frequency of Search Report**. The initial report from the reporting station is to be forwarded as soon as a track has been picked up. This report is not to be held up for lack of information on strength, height or speed.

**Plotting Procedure**

17. **Vertical Plotting Board**. Plotters are located at the back of a vertical plotting board and are provided with communications to either search radar operators of their Unit or of subordinate station. A plotter is responsible for the following:

(a) Acknowledging all information and plotting it on the plotting board.

(b) Connecting the plots to form tracks.

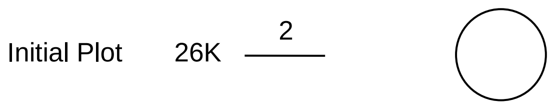
(c) Giving the serial number and station suffix to the track reported from the list provided on the corner of the plotting board and crossing out the used serial number to avoid duplication.

(d) Displaying ancillary in formation next to the track until it is transferred to mission data board.

(e) Indicating track designation alongside a track at all times.

18. **Plotting Display**. The plotter is to use the following display procedures:

(a) **Initial Plot**. Locate the GEOREF position received and mark it with a dot inside a circle of one-inch diameter or less if instructed locally. Write the serial number and strength adjacent to the plot along with station suffix if any.



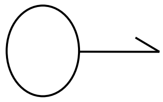

(b) **Second Plot**

(i) Mark the second position with a dot. Connect the position with the initial plot thereby marking a track. Direction is to be indicated by a half arrow-head at the new position.

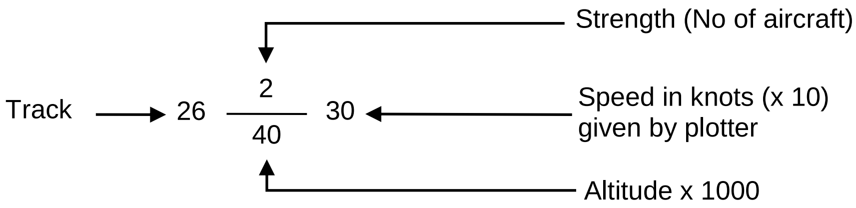


Revised on 29 Jan 21

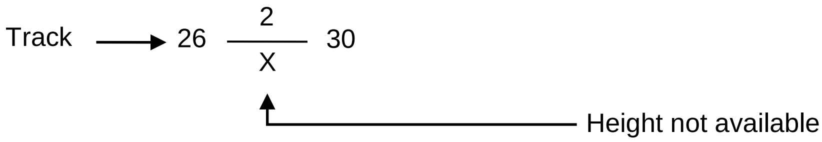
Plot 26K



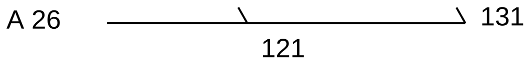
(ii) The strength, height and speed are to be displayed in the following manner:



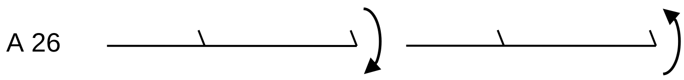

(iii) When data for an item is not available immediately, the space for that particular data is to be left blank and the report immediately transmitted to the appropriate agency. Required items of the initial report will be passed to the appropriate agency as soon as they become available.



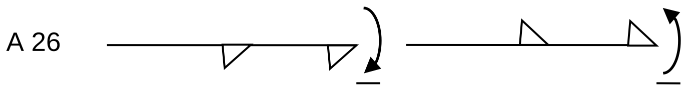

(c) **Subsequent Plot**. Mark the GEOREF position with a dot and connect this position with the last received position and indicate direction with a half arrow-head at the new position. Two latest plots of a track are to be retained on the vertical plotting board unless otherwise directed by the MIO.




(d) **Orbiting Plot**. Indicate on the plotting board by a curved arrow, the head of which is to show the direction of turn.



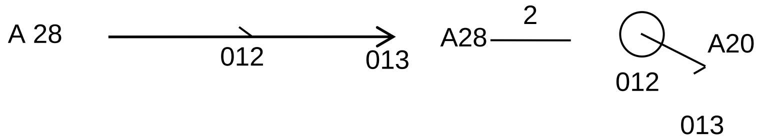

(e) **Distress Pattern Plot**. Indicate on plotting board by a triangle denoting direction left or right by one curved arrow.



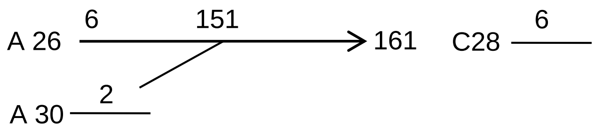

(f) **Split Track Plotting**. The segment of the track retaining the original track designation is to reflect change in ancillary track data along with one full arrow head plot. The segment of the track deviating from the original heading is to follow the procedure of initial plot.



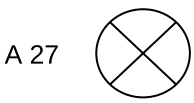
Revised on 29 Jan 21




(g) **Merging Track Plotting**. When the plotter receives word that two or more tracks are merging, he is to plot it as follows:

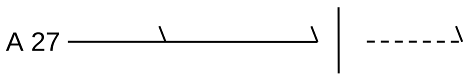



(h) **Re-appeared Track**. Locate the GEOREF position and mark it with a cross inside a circle.



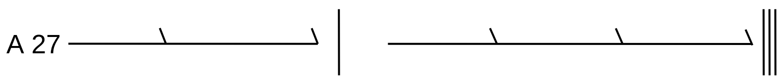

(j) **Contact Lost D/R and Fade Plotting**. The plotting procedure for contact lost and D/R is to be as follows:

(i) When the search radar operator reports "Contact Lost" on a track, the plotter is to display the information by a straight line perpendicular to the direction of flight at the point of contact lost.




(ii) Where the search radar operator reports a "Contact Lost" the plotter is to dead-reckon the track and display it by dotted line. The track is to be dead reckoned for a period of three minutes unless otherwise directed by the MIO.

(iii) If the track cannot be re-established within three minutes of D/R, it is to be faded and indicated by three perpendicular straight lines at the tip of the leading D/R arrow head, unless otherwise directed by the MIO.




(k) **Ancillary Information**. To indicate the change in the ancillary information two consecutive positions are to be plotted with full arrow-heads. The subsequent plot would be with half arrow head. The changes in ancillary information are to be indicated in the following manner:

(i) Indication on the track by two consecutive full arrow-heads plots.

(A) 

(B) Change in Strength 21 $\frac{6}{\quad}$

(C) Change in Height 21 $\frac{\quad}{26}$

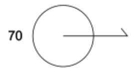
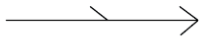
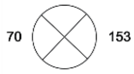
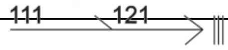
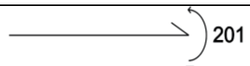
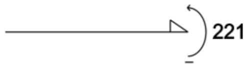
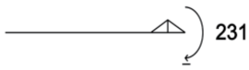
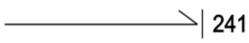
(D) Change in Speed 21 —— 35

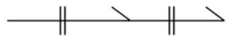
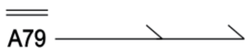
(E) Change in Strength and Height 21 $\frac{6}{26}$

(F) Change in Strength and Speed 21 $\frac{6}{\quad}$ 35

(G) Change in Height and Speed 21 $\frac{\quad}{26}$ 35

19. **Standard Plotting Symbols**. The plotting symbols used in plotting on vertical plotting board are given below:

| Description | Symbols |
|---|---|
| Initial Plot (New Track) |   |
| Track and direction |   |
| D / R Plot |  |
| To indicate change in ancillary information of the track |  |
| Re-appeared |   |
| Faded in PE |   |
| Orbiting left |   |
| Orbiting right |   |
| Distress left Hand |   |
| Distress right Hand |   |
| Contact lost |   |
| Faded |  |

| Description | Symbols |
|---|---|
| Specific track radiating ECM |  |
| Strength | $\underline{3}$ |
| Height | $\overline{301/40}$ |
| Speed | —— 35 |
| IFF Mode emergency (M E) | F56 —— $\frac{2}{40}$ —— 54ME |
| Both Radar and I FF / SIF Plots (ie the a/c picked up on radar is also being picked up on IFF / SIF |   |
| IFF / SIF plotting (ie the a/c is not being picked up on radar, it is only IFF / SIF plot) |   |
| Allied Ships |   (White Colour) |
| Hostile Ships |   (Red Colour) |

20. **Colours of Plotting**. Different coloured china graph pencils are to be used as follows:

| S No | Track Classification | Colour |
|---|---|---|
| (a) | Hostile | Red |
| (b) | X-Raid | -“- |
| (c) | Combined (non-bonafide) | -“- |
| (d) | Mix-up | Yellow |
| (e) | Fighter interceptor | Green |
| (f) | Allied | White |
| (g) | Bombers | -“- |
| (h) | Targets | -“- |
| (j) | Unidentified | -“- |
| (k) | Strike Force | -“- |
| (l) | Recce aircraft | -“- |

| S No | Track Classification | Colour |
|---|---|---|
| (m) | Transport | -“- |
| (n) | Special (V) | -“- |
| (p) | Combined (bonafide) | White or green (depending on the type of combination) |
| (q) | Mobile Observer Unit Plot | Normal colour code, as for radar plot, is to be applied |
| (r) | Allied Ships | White |
| (s) | Hostile Ships | Red |

21. **Display of Ancillary Information on Mission Data Board**. The mission data teller is to pass the ancillary information on all tracks to the mission board plotter in the following sequence:

(a) **Initial Report**. Initial report is to consist of the following:

(i) Track designation.

(ii) Strength.

(iii) Height.

(iv) Speed.

**Example**: FGB2020, 3 at 40, speed 500

(b) **Amendment Report**. Amendment report is to consist of the following:

(i) Track designation.

(ii) Change.

**Example**: Strength Change - FGB2020, Now 3

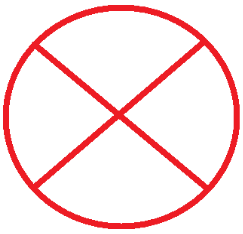
Height Change - FGB2020, Now at 26

Speed Change - FGB2020, Now speed 500

**Note**: If any two of the three or all three components of ancillary information need to be changed the word "Now" is to be used once with the first change only.

22. **Display of Nuclear Detonation Report**. Nuclear detonation report is to be displayed on the vertical plotting board by a two-inch diameter red circle

with crosshatched red diagonal stripes surrounding point of reported detonation. The point is to be retained for 15 minutes following the report.



23. **Telling Procedure Mobile Observer Plots**. When an MOU observed report is received in the Receiving and Reporting Flight (RRF) or any other agency deputed for this purpose, it is passed to the plotting board in the following sequence:

(a) Callsign of MOU.

(b) Heading of the aircraft.

(c) Strength of the aircraft.

(d) Type of the aircraft in code word.

(e) Height of the aircraft.

(f) Time of observation (four figures - Minutes and seconds of current hour).

**Example-1**: AL, WEST, 2PD, (PD Codeword for Mirages) Low / High / V High 1230. When secondary information, ie aircraft is Bombing, Rocketing Strafing or Para Dropping, is received with the Primary Report, it is passed after the time of observation.

**Example-2**: AL West 2 PD LOW 1230 Rocketing.

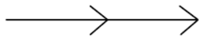
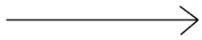
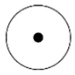
**Upward and Downward Telling of MOU Plots**

24. Upward and downward telling of MOU plots is to be carried out in the same manner as mentioned in paragraph 3 above.

**Display of Mobile Observer Unit Plots**

25. Mobile Observer Unit plots are to be plotted in the same colour as radar plots. The reported position is to be plotted with double full arrow-head for observed identified report, single full arrow-head for observed un-identified report, single half arrow-head for heard directional report and a circle for head non-directional report.

**Examples**:

| **Description** | **Plots** |
|---|---|
| Observed identified |  |
| Observed unidentified |  |
| Heard directional |  |
| Heard non-directional |  |
| Bombing | B |
| Rocketing | R |
| Strafing | S |
| Para dropping | P |

## CHAPTER - 18

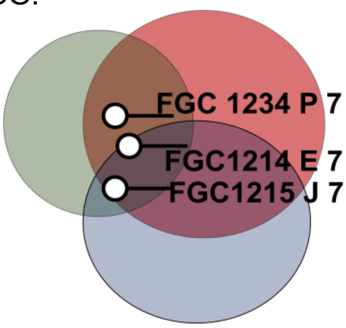
# TRACK REPORTING UP-DATION IN RADAR OVERLAP AREAS

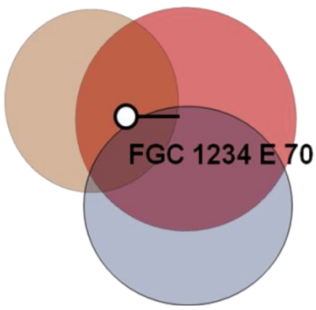
### Introduction

1. Aircraft flying through overlap areas are likely to be detected and passed to GMCC by multiple radars. The multiple track presentation at GMCC unnecessarily clutter the data display at GMCC and subsequently at SMCC console (STS). Presently, plot data of single radar is available with more than one GMCC; therefore, track would be generated from both GMCCs based on this plot data. These tracks would be passed to SMCC from both GMCCs in automated mode thus creating scope picture at SMCC more cluttered if not meeting the fusion criteria. For presenting a clearer picture of air activity at GMCC and SMCC, the procedures mentioned in succeeding paragraphs are to apply in case of multiple radar sites reporting the same track to single or multiple GMCCs.

### Auto Mode

2. To avoid generation of multiple tracks on single airborne object GMCC software is provided with '**Manual / Auto biasing'** to address the plot depiction error of single aircraft being picked up by multiple sensors at different positions. Plot biasing has the concept of Master radar to resolve the positional error of plots and remaining radars are slave to the Master radar. If more than one track is generated on this single aircraft due positional error of plots, GMCC software is also provided with fusion facility to correlate and to generate single track. The fusion facility is also available at SMCC Software to correlate different tracks received from Same GMCC or from different GMCCs. These tracks will be correlated to display single track by SMCC computer if they meet the fusion criteria. If system does not correlate, two / three different tracks will be displayed at GMCC consoles and same will be passed to SMCC if they are not meeting the fusion criteria. The figure below explains this fusion at GMCC:




3. The figure on the left side shows tracks of one aircraft as picked up by three different radars. If they are not fused, they will appear as three separate tracks at GMCC and subsequently at SMCC in automated mode, although they indicate the same aircraft. The figure on the right shows that all three tracks have been fused into one. Now, all the radars will have the same track number for this aircraft on their individual displays.

**Manual Mode**

4. **Area-wise Reporting**. The whole area of tactical responsibility is to be divided amongst different radar stations according to their operational performance. There will be a marked overlap area of pick up between any two stations. Responsibility and procedure for reporting is to be as follows:

(a) Within their own area of responsibility, each radar station is to report all tracks unless ordered by the SMCC to stop reporting on any particular track. The reporting station will, however, continue plotting locally and recording the disregarded track.

(b) Outside their own area of responsibility, they are to pass only initial plots on all tracks and also pass subsequent plots on the tracks asked for by the SMCC.

(c) To help the radar scope reader of the reporting station, the area of responsibility is to be marked on his scope.

5. **Code Words**. The following code words are to be used for passing control orders:

| Words | Meaning |
|---|---|
| Discontinue (Track Designation) | Discontinue passing information on the track mentioned till further order. Radar to maintain the track on their board and continue recording its positions for further reference. |
| Continue (Track Designation) | Continue passing information till further order. |
| Check "GEOREF Position" | Report track if any at or around the position stated. |
| Priority Telling (Track Designation) | Give priority in telling the particular track. |

6. **Use of Control Orders**. Normally the plotters themselves are to decide when to pass these orders but overriding authority is to be the Duty Controller of the watch. The following example explains how the control orders are to be used:

(a) In the diagram, Red, Blue and Green are three reporting radars covering a particular area. Red is responsible for the area of his coverage contained within the angle ABC. Similarly, Blue is responsible for the angle DEF and Green for the angle POR.

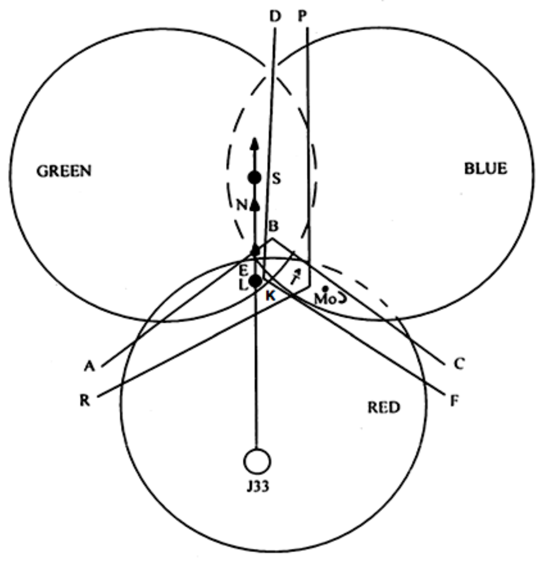
(b) Suppose an aircraft is picked up at position 'J' (within the RED area of responsibility). Red is to pass an initial plot (Say Serial 33) and will continue passing the track till position 'K' (At position 'K' RED area of responsibility ends). At position 'K' it is entering the Green area of responsibility. The plotter will ask Green 'GREEN CHECK AB 1234' (AB 1234 representing position 'K'). If Green has contact he will pass the initial plot 'GREEN Initial Plot Serial 15 AB 1236 strength One' (AB 1236 representing position 'L') Plotter will tell Red. 'Red discontinues Serial 33' and at the same time will allot Green Serial 33 for the track through its area of responsibility.

(c) Suppose Green does not pick up Serial 33 at position 'L' then Red may be asked to continue passing the track till it is reported by Green or fades from the scope of Red.

(d) Supposing Blue passes an initial plot Serial 35 at position 'M' outside its area of responsibility but within Red's area of responsibility. There are two possibilities:

(i) In Case Red is already passing the Plot. If Red is already having contact and passing the plot at position 'M', Blue is to be asked to discontinue.

(ii) In Case Red is not passing the Plot. If Red is not passing this particular plot, plotter is to ask Red 'Red Check AB 3515' (AB 3515 representing position 'M'). If Red gets contact, he is to pass an initial plot at position 'M'. Plotter is to ask Blue to discontinue and Red is to continue tracking through its area of responsibility. If Red does not pick up serial 35 at position 'M', Blue is to be asked to continue tracking till it is reported by Red or fades from scope of Blue.



## CHAPTER - 19

# RADAR QUALITY CONTROL PROCEDURES

### Introduction

1. Continuous RQC is broadly defined as the process of determining radar's performance. The objective of the quality control procedures is to establish and maintain, on a continuous basis, a high and consistent level of performance. To accomplish this, the capability of radar in its operational environment to produce a detectable signal is sampled, at least once every hour, and compared with an expected level of performance as derived from the Radar Coverage Indicator. Performance limits are established so that abnormal performance may be identified, investigated and corrected. Because of the several factors which contribute to variations in radar performance (weather lobing, target aspect, variations in receiver sensitivity and transmitter power output), performance will normally vary about an average level. Whenever, radar performance falls below average level as indicated by quality control results, an investigation is launched. An honest and vigorous application of the techniques out lined in this chapter will serve following important purposes:

(a) It will inform maintenance personnel that radar performance has degenerated below established limits so that an investigation and appropriate corrective action may be initiated.

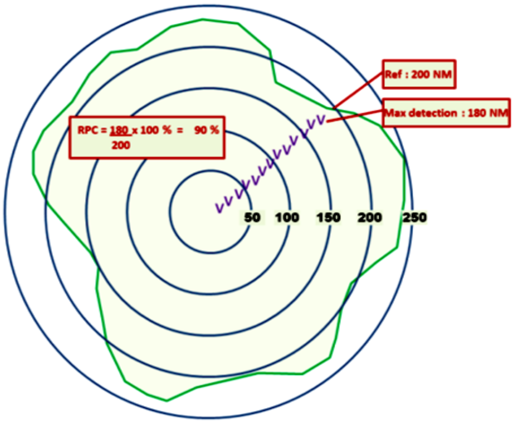
(b) It will assure all echelons of command that radar equipment and personnel are maintaining a maximum average level of effectiveness.

### Operational Status

2. It is the indication of the equipment's capability to perform its operational role at a particular time. For different types of equipment / capabilities it is determined by the following factors:

| Equipment / Capability | Factors to be determined |
|---|---|
| Surveillance Radars | (a) Radar Performance Control (RPC)<br>(b) Blip to Scan Ratio (BSR) |
| Height finding capability of 3D radars | (c) Difference of indicated altitude with actual altitude (C -Height)<br>(d) Discrepancy between relative height indication of two a/c |
| Secondary Surveillance radars (IFF) | (e) Practical range is to be compared with predicted range |
| Data links | (f) Data loss between sending and receiving equipment |

**Operational Status**




3. **Radar Performance Control (RPC)**. RPC determines the performance of a radar viz-a-viz the maximum actual pick-up range as compared with the expected pick-up range on an aircraft at a particular azimuth, height and antenna tilt. The performance control figures are to be expressed as percentage of an expected (reference) range of detection. Reference range is the maximum distance from radar center taken as a reference at a particular azimuth, height, RCS and antenna tilt at which the aircraft is detected during a calibration mission to validate theoretical RCI. If due to any reason practical RCI of radar is not formulated, the theoretical RCI (calculated by using theodolite or any DTED based software) may be taken as a reference for the first time only. However, max detection range observed on an aircraft during such missions is to be taken as reference range for all future calculations. RPC is the quotient of the range at which the radar produces the first visible signal and the reference range multiplied by one hundred to convert to percentage:

$$RPC = \frac{\text{Detect Range}}{\text{Ref Range}} \times 100\ \%$$

4. **Blip to Scan Ratio (BSR)**. The second factor which determines the operational status of radar is the blip to scan ratio. It is the depiction of radar's performance in terms of pickup probability. Radars have an inherent tendency of decrease in pickup probability with increase in range. A 50% BSR (for example) at maximum range of a radar as compared to 50% BSR at minimum range of the same radar do not signify its true pickup probability. This effect is pronounced in long range radars and negligible in medium / short range radars. In order to depict factual pickup probability of radars, it is essential to adopt such a method of calculating the pickup probability that takes in to account the factor of distance from radar. The concept of "Resultant BSR" provides a realistic depiction of pickup probability while taking in to account the range factor. Therefore, for short / medium range radars calculation of blip to scan ratio is sufficient, however for long range radars Resultant BSR is to be calculated. For this purpose, the radars have been divided into two categories as follows:

(a) Long range radars - Range 101 NM or more

(b) Med / Short range radar - Range 100 NM or less

5. The following procedure is to be applied for the calculation of BSR for the two different categories of radars mentioned above:

a) **BSR Calculation for Long Range Radars**. For the purpose of ascertaining the pickup probability of long-range radars, resultant BSR has to be calculated. The process involves following steps.

(i) **Step-1**. Divide range of the radar in 10 equal segments. Standard BSR for each segment as given in table below is to be taken as reference for max pickup probability for that particular segment.

| Range % of instrumented (max theoretical) range | Standard Blip to Scan Ratio |
|---|---|
| 90% – Max - segment No. 10 | 50% |
| 80 – 90 % - segment No. 09 | 64% |
| 70 – 80 % - segment No. 08 | 76% |
| 60 – 70 % - segment No. 07 | 85% |
| 50 – 60 % - segment No. 06 | 91% |
| 40 – 50 % - segment No. 05 | 96% |
| 30 – 40 % - segment No. 04 | 98% |
| 20 – 30 % - segment No. 03 | 99% |
| 10 – 20 % - segment No. 02 | 99% |
| Min – 10 % - segment No. 01 | 99% |

(ii) **Step-2**. For each segment, count total number of blips / plots and total number of scans. Calculate BSR by using following formula:

$$\text{Segment \# BSR} = \frac{\text{Total blips / plots Picked up x 100 \%}}{\text{Total number of scans}}$$

(iii) **Step-3**. Divide the BSR of each segment by the standard BSR of respective segment given in table above:

$$\text{Segment \# RBSR} = \frac{\text{Segment \# BSR x 100 \%}}{\text{Standard BSR}}$$

(iv) **Step-4**. Average out RBSRs by adding the results of all the segments and dividing total number of segments to get overall resultant BSR.

b) **BSR Calculation for Med/Short Range Radars**. For ascertaining the pickup probability of Medium / short range radar BSR is calculated by using following formula:

$$BSR = \frac{\text{Total blips / Plots picked up x 100 \%}}{\text{Total number of scans}}$$

6. **Selection of Tracks**. Performance Control is to be carried out only on tracks for which aircraft type is known. SMCC is to select tracks and allot them to GMCC / Radar along with the aircraft flight plans. While selecting tracks for performance control, SMCC is to bear in mind the following criteria:

(a) **Use of Single Aircraft**. To effectively utilize quality control, single aircraft is to be used. Formation of aircraft which appears as a single blip on PPI is not to be used for this purpose.

(b) **Aircraft Penetration**. The aircraft selected for calculation of RPC is to penetrate or exit from ranges corresponding with the radars RCI. In case of aircraft operating within the RCI (e.g. GCI aircraft) only BSR / RBSR is to be calculated. RPC cannot be calculated in this case.

(c) **Aircraft Type**. Normally performance control is to be done only on fighter aircraft or aircraft of comparable db ratings. Civil or military transport aircraft may be used for performance control purposes only when a fighter aircraft is not available for this purpose.

(d) **Aircraft Height**. Low looking radars are not to utilize aircraft flying above 5000 feet for quality control.

(e) **Arranging special flights**. When no tracks are available for performance control, SMCCs are to arrange special flights for this purpose.

(f) **Height Determination**. Performance Control checks are to be carried out using only those aircraft on which positive height information is available. In the case of civil aircraft, a close coordination between the SMCC and the FIC/ATC is required for determining the exact height of the aircraft. If the height finding facilities do not exist in certain radar squadrons, SMCC is to pass this information to them after obtaining it from other available sources.

**Quality Control Procedure of Karakoram Eagle-03 (KE-03)**

7. RQC checks can also be performed on KE-03 to gauge the performance of the sensor. The procedure for calculating the RPC/BSR remains the same as of procedure for Low Level radars. However, the following points are to be kept in mind:

(a) The performance of airborne sensors tends to degrade while the platform is in a turn. It must be ensured that KE-03 is flying a steady leg throughout the duration of calculation of BSR.

(b) The selection of track must be coordinated with SMCC in advance for provision of flight plan of the said track to the KE-03.

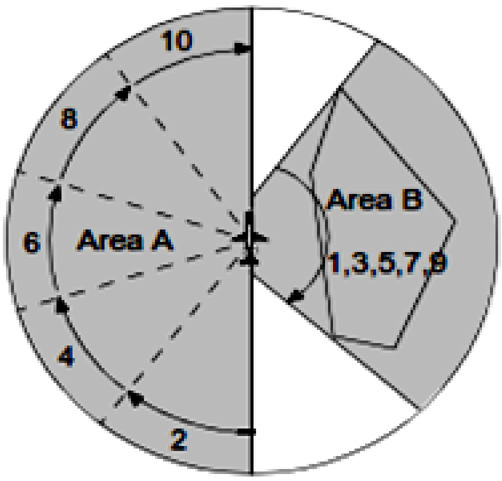
(c) The selected track should not be entering Blind Speed Area of sensor during the calculation. This will require an extrapolation of the track by C2.

(d) For KE-03 platform, the sensor must be operating in Air to Air mode.

(e) RPCs calculated at KE-03 and GMCC must be reconciled post mission for analysis at both ends.

**Quality Control of Saab Surveillance System (SSS)**

8. The above mentioned BSR calculation procedure is applicable to radars having Track While Scan (TWS) capability i.e. they scan mechanically with a known RPM, making it possible to count the total number of detections in a given number of scans. However, this procedure cannot be applied to determine the BSR of Erieye radar, as its working principle is different from the radars with TWS capability. By virtue of having rapid beam steering capability, Erieye radar provides “Adaptive Tracking” feature, which means that the radar will continuously change the pattern of its tracking illuminations, to meet the specified track update rates of user defined tasks.




9. The above figure shows the functional principle of search order with a Primary Air Surveillance (Area B) and a Support Air Surveillance area (Area A) defined. The search order depicted is basically for the search of new targets. Once the target is detected, the scans are also interrupted for track illuminations. Therefore, there is no regular sweep line visible on the scope and it is not possible to count the total number of scans. Subsequently, calculation of BSR is not possible. The Erieye Mission System (EMS) displays real time technical parameters of the radar, which are utilized by sensor manager to ascertain the radars performance.

**Standards of Operational Statii**

10. The operational status of radar and other equipment is to be determined according to the following standards:

(a) **Surveillance Radar**

(i) **Status-I**

(A) Both RPC and BSR / RBSR should be 90% or more.

(B) All anti-jamming facilities should be serviceable.

(ii) **Status-II**

(A) RPC or BSR / RBSR is between 75% - 89% of the Reference Range.

(B) Any "anti-clutter" and anti-jamming" facility is U/S.

(C) MTI cancellation is not upto operational standards or is U/S.

(iii) **Status-III**

(A) When conditions are not met for "Status-II", the operational status of equipment is to be downgraded to "Status-III”.

(B) When equipment is operational status-III due to any reason other than AP or weather, the equipment is to be handed over to the maintenance staff for rectification.

(b) **Height Finding Capability of 3D Radar**

(i) **Status-I**

(A) Difference of indicated altitude with actual altitude within the permissible error limit given in the TOs or ± 1500 feet (when no such indication is A/A in TOs).

(B) No discrepancy between the relative height indication of two aircraft.

(ii) **Status-II**

(A) Difference of indicated altitude with actual altitude is beyond the permissible error limit given in the TOs or it is more than 1500 feet but less than 2000 feet (when no such indication is A/A in TOs.

(B) Discrepancy up to ±500 feet between the relative height indication of two aircraft.

(iii) **Status-III**

(A) When conditions are not met for "Status-II", the operational status of equipment is to be downgraded to "Status-III".

(B) When equipment is operational status-III due to any reason other than AP or weather, the equipment is to be handed over to the maintenance staff for rectification.

(c) **Secondary Surveillance Radar (IFF)**

(i) **Status-I**. IFF/SIF should be fully serviceable and it should give a range of 90% or more of the predicted operational range.

(ii) **Status-II**. When the returns are weak and intermittent and the range is less than 90% of the predicted operational range.

(d) **Data Link**

(i) **Status-I**. ≤ 2 %

(ii) **Status-II**. >2 % & ≤5%

(iii) **Status-III**. > 5%

**Note**: Decisions to accept the quality of data link may be based on data redundancy (multiple sensors) to the end user vis a vis nature of assigned task. A less important mission may not require data of a sensor having even a lower level of data loss and handing over of the data link to maintenance staff for rectification may be afforded. However, on the other hand a more important mission may not allow handing over of a data link, even with higher level of data loss, to maintenance staff.

**Note**: Notwithstanding, the results accrued from Quality Control, the radar duty controller is fully authorized to use his discretion in revising the radar's status whenever in his options such a revision is warranted due to any deterioration or improvement in the general scope picture.

11. No operational status is to be declared on the following equipment. They will either be "Serviceable" or "Unserviceable":

(a) Anti-jamming facilities

(b) Radio Sets

(c) Ground to ground telecommunications

(d) Simulators

(e) Power generators

(f) Computers

(g) Manual Entry Data devices (MEDDs)

(h) Video maps

12. **Upgrading / Downgrading of Statii**

(a) Status of a radar may be downgraded due to AP, clouds, jamming, or when in the opinion of the controller, a sudden deterioration in the performance has become evident.

(b) Normally the operational status of radar is to be downgraded only after unsatisfactory result of performance control on two consecutive tracks have been obtained. However, the status may be lowered even after one performance control with unsatisfactory results if in the judgment of the Duty Controller the equipment performance so warrants or performance control on a subsequent track is not feasible due to unavoidable reasons.

(c) After rectification or technical check, the operational status of the radar is to be provisionally upgraded pending satisfactory results of performance control on one track or till the Duty Controller is satisfied with the equipment performance by observing other tracks.

(d) If after rectification the performance control still indicates unsatisfactory performance the equipment will continue to have the earlier lower status with previous timings.

13. During operations when there is a cause for lowering the operational status of radar equipment, the Duty Controller will give one copy of Quality Control Performa to the Unit Technical Officer, who will return same to the Duty Controller after necessary action.

14. **Guide for Investigations**. Following is a guide for Unit Technical Officer for investigation and making entries on investigation sheet:

(a) Re-compute the reference range, RPC and blip to scan percentage to ensure that no arithmetical errors have occurred.

(b) Check the PPI and interrogate the Performance Control Officer about the scope picture.

(c) Perform as many equipment checks as possible without shutting the equipment down. If deemed necessary by the senior controller and the maintenance officer, brief shut down may be requested in accordance with current regulations.

(d) Summarize the conclusions of the investigation. The conclusion is to be supported by evidence. Any corrective action taken as a result of the investigation is also to be recorded.

**Quality Control Records-Keeping etc**

15. Continuous Evaluation of radar equipment and associated facilities is to be recorded by operational and maintenance flights of all Air Defence Units on day to day and monthly basis. Local arrangements are to be made for maintaining the monthly records in a graphic form for presenting overall picture of Unit performance under different conditions.

16. All GMCC / High Level Unit commanders and Flt Cdr Ops (AD) in case of AEWACs are responsible for the implementation of procedures outlined in this chapter as well as for supervising and monitoring the quality control programme. Although, the criteria for different operational statii have been defined, the need for senior most controllers to use utmost care and discretion cannot be over emphasized. Efforts are to be made at all times to develop the programme into an efficient and true gauge of the operational capability of the Unit.

17. Higher commands are to make suitable arrangements for a continuous monitoring of the performance control programme within the areas of their operational control by conducting a periodic review of performance control records of each Unit. They are to ensure that all operational and maintenance personnel at the Units understand the theory purpose, procedures and benefits of performance control programme.

18. Each Unit is to prepare practical VPD / RCI, based on screening angle and site evaluation/ calibration data for each site, for the following equipment:

(a) Surveillance radar

(b) Height Finder radar

(c) IFF

(d) Radio (UHF and VHF)

**Radar Calibration and Preparation of RCIs for Air Defence Radars**

19. As far as possible the site for radar deployment should be selected as per the criteria laid down in Chapter 14, Section-III, AFM 355-2, however the criteria for site selection may have to be by passed if it is decided to position a radar in a particular area of interest because of operational reasons. Site selected should be such that best results are obtained through radar performance. Thus, ideally speaking, after evaluation of different sites, through theoretical RCIs, a particular site having the best theoretical RCI is to be selected for deployment and practical calibration. The following guidelines are to be followed for radar calibration and preparation of RCIs:

(a) Theoretical RCIs should be prepared by:

(i) Digital Terrain Elevation Data (DTED) based software's (e.g. Site Adaptation Software (SAS) of TPS-77 or Falcon View etc.).

(ii) By obtaining screening angles with the help of a theodolite or through trigonometric calculations, in case theodolite is not available, AFM 31-1-16 may be consulted.

(b) The calibration plan after vetting by the respective SMCC is to be provided to the aircrew at least two days in advance along with call sign and R/T channel of the controlling agency.

(c) A single fighter a/c should be selected for flying the calibration legs, and each 10˚ radial of the area of threat should be calibrated by flying in bound and out bound legs. The area other than area of threat may be calibrated at each 30˚ radial if adequate flying effort is not available to calibrate each 10˚ radial.

(d) The aircraft should be taken to the maximum pick up range of the radar as ascertained from the TOs on particular heights, selected for calibration.

(e) The pilot of the calibration A/C should be pre-briefed to turn after a specific time, in case R/T fades out. At no stage the A/C with fading R/T should be told to discontinue and turnabout, rather it should maintain heading for a specific time as pre-briefed.

(f) While guiding the A/C on different legs all safety precautions must be taken and border limitations etc, be adhered to.

(g) Particular emphasis should be laid on radials where screening exists, so that factual pick up is ascertained in these areas at different altitudes, and these radials should be preferred over radials clear of screening.

(h) Calibration must be done on the centred scope and maximum range selected on radar.

(j) Each leg is to be recorded separately and no pick-up points are to be marked in order to establish the blank areas in the total coverage.

(k) If required, different antenna tilts be tried to select the best tilt angle for the assigned task.

(l) Poor quality of R/T is also to be determined during each leg for establishing radio blank areas.

(m) The altitudes selected for calibrations should be best suited to the role and task of the Unit, however, following altitudes may be selected for different categories of radars:

| | | | |
|---|---|---|---|
| (i) | High Level | - | 10,000 feet<br>20,000 -“-<br>30,000 -“- |
| (ii) | Medium Level | - | 5,000 feet<br>10,000 -“-<br>15,000 -“- |
| (iii) | Low Level | - | 250’ AGL |

20. The RCI of a radar deployed at a particular site, is to be prepared after the calibration, and is to be updated in the following circumstances:

(a) The screening angle at the site has undergone a change due to any reason such as growth of trees, construction of new buildings etc.

(b) The radar is required to be operated on new antenna tilt angle, which has not been evaluated earlier.

(c) Results of continuous daily performance checks, over a considerable period of time, indicate a marked change in the radar performance.

21. The procedures to be followed on preparation, maintenance and custody of RCIs are as follows:

(a) All RCIs be prepared in both soft and hard copies as per a defined scale.

(b) All RCIs must indicate the Unit's location, type of radar, antenna tilt, date, time, height and type of aircraft.

(c) The RCIs must show the dots which would indicate the legs that have been calibrated and must also reflect no pick-up areas through shaded circles.

(d) The MCCs must prepare consolidated RCIs in addition to RCIs of individual radars.

(e) Headquarters Air Defence Command would keep a record of all the RCIs Unit wise for comparison of performance of each Unit. Headquarters Air Defence Command would also prepare a consolidated record area-wise for easy reference for any future deployments.

(f) All RCIs are to be serially numbered for each Unit and the Unit is to keep a record of all such RCIs along with their disposal.

(g) All RCIs are to be security graded as 'Top Secret' for war sites and 'Secret' for peace time sites.

(h) The RCIs are to be made in four copies and numbered as follows:

| | | | |
|---|---|---|---|
| (i) | AHQ (DAD) | : | 1 |
| (ii) | HQ ADC | : | 2 |
| (iii) | Sector HQs | : | 3 |
| (iv) | Unit | : | 4 |

(j) RCIs of war sites are to be dispatched to the Sector Headquarters in whose area of responsibility the Unit would be deployed during war / emergency.

(k) RCIs of war and peace time sites are to be placed in separate folders.

(l) Sectors Headquarters are to retain RCIs of only those sites, which actually lie in their area of responsibility.

(m) In case equipment is phased out, Copy No 1, 2 and 3 is to be returned to the Unit for destruction through a Board of Officers.

(n) RCI being a sensitive document is not to be shown to any unauthorized person.

# CHAPTER - 20

# MOVEMENT IDENTIFICATION

## Introduction

1. Movement identification of all aerial platforms is one of the major functions of SMCC. It involves collection, analysis and dissemination of aircraft movement data and associating such data with tracks that are detected for the purpose of positive identification.

2. **Functions of Movement Identification Section**. The functions of Movement Identification Section at SMCC are as follows:

(a) Collection, analysis and identification of all Air Movements within the area of responsibility.

(b) Prompt and accurate display of the identification information in auto / manual mode.

(c) Maintaining close liaison with concerned ACCs, ATCs and PBOs (Pilot Balloon Observatories) of Met department within the area of responsibility for the purpose of obtaining flight information.

(d) Passing flight plan information of appropriate / relevant air movement to under command AD Units / GMCCs for the purpose of information, building and enhancing situational awareness, and for the purpose of radar quality control.

(e) Liaison with adjacent SMCCs regarding air movements which are entering or likely to enter each other's AOR and transfer / receive of flight plans accordingly.

(f) Promptly reporting any irregularity in flight plan to battle staff in lateral and vertical tiers and concerned ACCs / ATCs or any other controlling agency.

3. **Movement Identification Procedures**. In the automated or manual Air Defence systems, the procedure mentioned in succeeding paragraphs are to be adopted for positive identification of all tracks detected in the area of responsibility.

4. **Movement Identification in Automated Air Defence System**. In auto-mated SMCC, Movement Identification Officer (MIO) is assisted by Assistant Movement Identification Officer (AMIO), Movement Liaison Clerks (MLCs) and Manual Entry Operators (MEOs). They are responsible for the collection and feeding of all air movement information into the system and using it for identification of all tracks being displayed at SMCC display screens. Following procedure is followed for movement identification:



Revised on 29 Jan 21

(a) MLCs are to receive all air movement information from ACCs / ATCs. They will record all such information in the proforma provided for this purpose and issue Air Defence clearance number to all movements to take place.

(b) MEOs are to feed this information into the system and update flight plans of all scheduled/non-scheduled flights and local flying aircraft.

(c) MIOs are to keep all active flight plans displayed on the console. Inactive flight plans can be displayed as and when required.

(d) MIOs are to closely monitor and check all tracks appearing or reappearing on the console, being automatically correlated for the purpose of guaranteeing correct identification.

(e) Deviation of any track from its flight plan is to be detected by displaying flight plan routes, correlation Tie-Lines and correlating track movement with it.

(f) AMIOs along with assisting MIO in all his actions and responsibilities will bear the additional responsibility of dedicated MLC and MEO for local fighters flying only.

5. **Movement Identification in Manual Air Defence System**. In manual SMCC, MIO is provided with movement record, track display board and time scales marked at various speeds most commonly observed. He is assisted by an AMIO / Dead Reckoning (DR) Officer and MLCs. For positive identification of all tracks being displayed on SMCC plotting boards, MIO and the personnel of Movement Identification Section are to adopt the following procedures:

(a) To obtain air movement information for subsequent two hours from ACC / ATCs.

(b) To record the information and the serial number (obtained from the MLC) in the appropriate columns of the boards.

(c) To consult the RCI of the concerned surveillance stations, calculate the estimated time and position of detection of the tracks and fill in the remaining columns of the movement record board.

(d) To draw the projected tracks of all air movements on the movements display board other than those confined to the local flying areas.

(e) To write the movement numbers and mark the direction of flight, with a single full arrow, at the estimated position of pick up on the track.

(f) To correlate the surveillance data with the aircraft movement data, identify the track and allocate the appropriate classification.

(g) To co-ordinate with the Air Surveillance Officer of the reporting stations / GMCC for electronic identification of the tracks.

(h) To keep close watch on the plotting board and re-classify all tracks that violate their flight plans, unless otherwise directed by the Sector Duty Controller.

(j) To maintain the air movement information on the board till the time aircraft lands or goes out of the coverage for landing outside the area of responsibility.

(k) To co-ordinate with the DR Officer for obtaining the DR position of new and reappearing tracks for the purpose of filtering and positive identification.

.

6. **Movements Record and Tracks Display Board**

(a) **Construction**. A Chart of 1:1000,000 scale covering the entire area of the Sector sandwiched between two Perspex sheets of appropriate size and thickness is mounted on a wooden or metallic frame. The map is illuminated by fixing low wattage electric bulbs inside the frame. The bulbs are fixed in such a way that a uniform illumination is achieved. The intensity of illumination is controlled by a combination of ON / OFF potentiometer. The frame is to be large enough to accommodate both the area map and the movement record sheet.

(b) **Area Chart**. Latest edition of 1:1000,000 scale charts are joined to cover the entire area of the Sector. The following information is marked on the chart before it is installed:

(i) Radar Coverage of all Units within the area of responsibility.

(ii) Local / Special flying areas of all active airfields.

(iii) Low flying areas.

(iv) International air corridors and the national air routes.

(v) Important pin-points.

(vi) Prohibited and danger areas.

(c) **Movements Record Sheet**. This sheet consists of two portions. One portion accommodates the schedule and non-schedule flying information and the second one contains the local and special flying information.

**Dead Reckoning Procedures**

7. DR Officer is provided with a stop watch, a DR board, and the speed scales.

The DR board is constructed exactly like that for the MIO with the exception of movement record column. The DR Officer consults the movement record column of the MIO board for obtaining the relevant flight plan information.

8. Following procedure is to be adopted for DR of assigned tracks:

(a) Obtain relevant information from MIO on tracks that are assigned for DR.

(b) Draw the tracks on the DR board and plot the expected minute to minute progress of the tracks based on the flight plan and other available data.

(c) Correlate the pre-plotted tracks with the actual detected tracks, and inform the MIO, if they tally with each other in all respects. Readjust the DR on receipt of actual position and reports.

(d) Assist MIO in identification if the detected tracks fall outside the available movement information.

(e) Reconcile divergent surveillance reports from different reporting stations and intimate the MIO about the true position of various tracks.

(f) Continue the DR even if the track is being picked up by radar and stop only when aircraft lands or goes out of the area of responsibility.

## CHAPTER - 21

# FLIGHT FOLLOWING SERVICE

**Introduction**

1. GMCCs / Radar Units are required to provide Flight Following Service (FFS) to aircraft flying within their respective AoRs during peace as well as during war time, at the discretion of ADOC, depending air situation at particular time.

**Definition and Scope**

2. FFS means 'provision of assistance in the form of information and guidance to enable an aircraft carry out its projected safely and expeditiously. This includes the following:

(a) Keeping continuous radar track

(b) Establishing position of the aircraft at intervals

(c) Assisting for diversion

(d) Circum-navigating weather

(e) Avoiding collision

(f) Alerting rescue services, if necessary

(g) Assisting aircraft in distress or emergency

3. Flight Following Service is purely advisory in nature and does not in any way imply positive control.

4. This service is to be provided as follows:

(a) **VIP Flights**. Continuous radar track is to be maintained of all known VIP flights, and the services mentioned in sub-para 2(b) and (c) are to be provided on request. The services mentioned in sub-paras 2(d) to (g) above are to be provided whenever in the opinion of the Fighter Controller these services will be of use to pilot irrespective whether a request has been received from the aircraft or not.

(b) **Non-VIP Flights**. On request made through flight plan or on R / T.

(c) **Aircraft in Distress**. Whenever a distress message either through R / T or IFF, is picked up by a GMCC / Radar Unit, it is mandatory for the controller on duty to record position of the aircraft, give 'Pigeons' for the nearest suitable airfield, follow the track of the aircraft and inform the controlling SMCC.

(d) **Practice FFS**. Aircraft requiring FFS for practice purposes are to indicate this by including 'Request practice FFS in their message'. The duty controller at SMCC is to accept this request if he can do so without interfering with the normal Air Defence commitments.

5. With the exception of the conditions specified in sub-paragraph 4(c) above, a GMCC / Radar Unit may not provide FFS if, at the specific time, the station is controlling a live interception on a hostile track, and provision of this service is likely to interfere with the interception.

6. Provision of FFS is to be restricted to PAF aircraft only except that civil aircraft may be provided assistance in DISTRESS or EMERGENCY.

**Procedure for Aircraft**

7. When planning operation under conditions in which need for FFS, is visualized, the captain of the aircraft is to incorporate the request for FFS in the flight plan. For flight where flight plan is not submitted in advance, this request is to be communicated to the appropriate SMCC through the ATC/FIC or departing airfield.

8. In addition to the pre-flight request for FFS all aircraft are to request on R/T whenever they want service to commence. The message requesting FFS is to include the following information:

(a) Call sign of the SMCC / GMCC / Radar Unit

(b) Position of aircraft if known, general area if position is not known

(c) Own call sign

(d) Service required

9. If the aircraft calls on R / T, SMCC is to hand over the aircraft to an appropriate GMCC / Radar Unit for the provision of FFS subject to the provision of sub-paragraph 11 (a) below. In case the a/c directly calls GMCC / Radar Unit, Fighter controller is to provide FFS and inform SMCC.

**Procedures for SMCCs**

10. All SMCCs are to maintain a continuous listening watch on the UHF and VHF frequencies specified for FFS. These frequencies will be known as 'GCI Common' and will be so annotated in the relevant confidential communication orders.

11. On receipt of request for FFS, the Duty Controller is to:

(a) Decide whether FFS could be provided

(b) If unable to provide the FFS, inform the pilot by a message containing the following information:

(i) Call sign of the aircraft requesting FFS.

(ii) Call sign of the SMCC.

(iii) Text containing the single word 'unable'.

(c) If in a position to provide service, he is to allocate this responsibility to an appropriate GMCC / Radar Unit.

(d) SMCC are responsible to alert rescue service through appropriate ATC / ACC, whenever required.

**Procedure for GMCC / Radar Unit**

12. On being allotted the responsibility for provision of FFS, the-Duty controller at the GMCC / Radar Unit is to detail a controller for this purpose who is to:

(a) Acknowledge the message.

(b) Identify the aircraft, if it has not already been done.

(c) Continue to provide the service requested.

(d) When unable to continue the service, indicate the same by transmitting a message containing the following information:

(i) Call sign of the aircraft.

(ii) Text containing the word 'unable' followed by the position of the aircraft at the time, and any other information considered useful for safe navigation.

**Distress and Emergency Procedures**

13. An aircraft in DISTRESS or EMERGENCY will indicate the same by following the standard procedures as laid down in AFM 96-5.

14. **Action by SMCC**. On receipt of any distress or emergency message, either on R/T from the aircraft or through GMCC / Radar Unit, SMCC is to take the following action:

(a) To find out the cause of distress and render assistance required.

(b) To alert rescue services through appropriate ATC / ACC.

(c) To hand over the aircraft to suitable GMCC / Radar Unit.

(d) To divert any suitable aircraft available in the vicinity towards the aircraft in distress (civil aircraft is to be diverted through ACC).

(e) To scramble interceptor to join the aircraft in distress for the purpose of rendering navigational or any other assistance required by the aircraft.

**Action by GCI Unit (GMCC / Radar)**

15. GCI Unit is to take following action:

(a) To record the position of the aircraft, give 'Pigeons' to the nearest suitable airfield, follow track of the aircraft and inform SMCC.

(b) Try to establish R/T Communication.

(c) Inform SMCC about the cause of distress and assistance required.

(d) Inform aircraft of its position at intervals not exceeding 5 minutes and give 'Pigeons' for destination or suitable diversionary airfield whichever is nearest.

(e) Join up the diverted aircraft or the interceptor with the aircraft in distress.

(f) Continue rendering assistance till emergency is over or no longer in a position to render assistance and inform SMCC.

## CHAPTER - 22

# AIR TRAFFIC CONTROL

### Introduction

1. Air Traffic Control (ATC) is a facility operated by appropriate authority to promote safe, orderly and expeditious flow of air traffic. The policy for Air Traffic Control in the PAF is that it shall provide, in time of peace and war, aeronautical facilities and a ground organization which would:

(a) Enable pilots to operate safely with tactical freedom in all weather conditions. The objectives shall be:

(i) To prevent collision between aircraft.

(ii) To maintain orderly and expeditious flow of air traffic.

(iii) To prevent collision between aircraft on the maneuvering area and obstructions on that area.

(b) Meet the requirements of Air Defence organization for the notification of aircraft movements.

(c) Provide advice and information useful for safe and efficient conduct of flights.

(d) Alert emergency services and initiate search and rescue activity when required.

2. To achieve above aim, certain rules have been laid down which are followed by the civil aircraft and the military aviation. Because the Fighter Controllers also use the air space for operational and training missions, it is imperative that they are conversant with these rules.

### Semi-Circular Rules

3. To reduce the hazard of collision between aircraft engaged in en-route flight, some form of separation is required. This is achieved by simple system of relating flight level chosen, to the magnetic track being flown, and by ensuring that when in level flight, at or above the transition level, aircraft fly at flight levels on standard altimeter setting (i.e. 29.92 inches and 1013.2 Mbrs).

4. The relation between the magnetic track of an aircraft and its appropriate flight level is obtained by dividing the compass rose into semi circles and by allocating each semicircle associated flight level as given in the table below:

| Tracks** | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| From 000 degrees to 179 degrees*** | | | | | | From 180 degrees to 359 degrees*** | | | | | |
| IFR Flights | | | VFR Flights | | | IFR Flights | | | VFR Flights | | |
| Standard Metric | Level | | Standard Metric | Level | | Standard Metric | Level | | Standard Metric | Level | |
| | Meters | Feet | | Meters | Feet | | Meters | Feet | | Meters | Feet |
| 0030 | 300 | 1000 | - | - | - | 60 | 600 | 2000 | - | - | - |
| 0090 | 900 | 3000 | 0105 | 1050 | 3500 | 0120 | 1200 | 3900 | 0135 | 1350 | 4400 |
| 0150 | 1500 | 4900 | 0165 | 1650 | 5400 | 0180 | 1800 | 5900 | 0195 | 1950 | 6400 |
| 0210 | 2100 | 6900 | 0225 | 2250 | 7400 | 0240 | 2400 | 7900 | 0255 | 2550 | 8400 |
| 0270 | 2700 | 8900 | 0285 | 2850 | 9400 | 0300 | 3000 | 9800 | 0315 | 3150 | 10300 |
| 0330 | 3300 | 1080 | 0345 | 3450 | 11300 | 0360 | 3600 | 11800 | 0375 | 3750 | 12300 |
| 0390 | 3900 | 12800 | 0405 | 4050 | 13300 | 0420 | 4200 | 13800 | 0435 | 4350 | 14300 |
| 0450 | 4500 | 14800 | 0465 | 4650 | 15300 | 0480 | 4800 | 15700 | 0495 | 4950 | 16200 |
| 0510 | 5100 | 16700 | 0525 | 5250 | 17200 | 0540 | 5400 | 17700 | 0555 | 5550 | 18200 |
| 0570 | 5700 | 18700 | 0585 | 5850 | 19200 | 0600 | 6000 | 19700 | 0615 | 6150 | 20200 |
| 0630 | 6300 | 20700 | 0645 | 6450 | 21200 | 0660 | 6600 | 21700 | 0675 | 6750 | 22100 |
| 0690 | 6900 | 22600 | 0705 | 7050 | 23100 | 0720 | 7200 | 23600 | 0735 | 7350 | 24100 |
| 0750 | 7500 | 24600 | 0765 | 7650 | 25100 | 0780 | 7800 | 25600 | 0795 | 7950 | 26100 |
| 0810 | 8100 | 26400 | 0825 | 8250 | 27100 | 0840 | 8400 | 27600 | 0855 | 8550 | 28100 |
| 0890 | 8900 | 29100 | | | | 0920 | 9200 | 30100 | | | |
| 0950 | 9500 | 31100 | | | | 0980 | 9800 | 32100 | | | |
| 1010 | 10100 | 33100 | | | | 1040 | 10400 | 34100 | | | |
| 1070 | 10700 | 35100 | | | | 1100 | 11000 | 36100 | | | |
| 1130 | 11300 | 37100 | | | | 1160 | 11600 | 38100 | | | |
| 1190 | 11900 | 39100 | | | | 1220 | 12200 | 40100 | | | |
| 1250 | 12500 | 41100 | | | | 1310 | 13100 | 43000 | | | |
| 1370 | 13700 | 44900 | | | | 1430 | 14300 | 46900 | | | |
| 1490 | 14900 | 48900 | | | | 1550 | 15500 | 50900 | | | |
| etc | etc | etc | | | | etc | etc | etc | | | |

## Reduced Vertical Separation Minima (RVSM)

5. RVSM has been introduced to some airspace in order to increase capacity of traffic by utilizing intermediate flight levels previously avoided. As the name implies, a reduced vertical separation is employed between aircraft. This change in procedures has been made possible due to improved accuracy in modern altimeter systems. In order to fly in RVSM airspace, the aircraft must have been awarded the status of 'RVSM compliant' (ie it meets specific requirements for altimeter equipment, engineering practices and crew training), or have been granted exemption from RVSM rules.

## Separation Standards (ICAO)

6. Detailed procedures regarding ICAO separation standards (vertical , lateral and longitudinal separations etc) are covered in Chapter 16, VOL-I of AFM 60-13 (ATC Manual).Readers are can refer to subject AFM for further information.

**Instrument and Visual Meteorological Conditions (IMC, VMC)**

7. IMC and VMC refer to weather conditions encountered during a flight. These terms are used to denote actual weather conditions as distinct from flight rules under which the flight is being conducted.

(a) VMC minimum requirement shall be:

(i) Flight visibility 8 km.

(ii) Horizontal distance from clouds 1.5 km.

(iii) Vertical distance from clouds 1,000 ft (300 M). Except that outside controlled air space at or below 3,000 ft (1000 M) AMS or 1,000 ft/300 M above terrain whichever is higher; flight visibility 1.5 km clear of clouds and in sight of ground or water.

(b) IMC exists when weather conditions are below the minimum standard prescribed for VMC.

**Instrument and Visual Flight Rules**

8. These are the rules under which flight is conducted and are applied as follows:

(a) In IMC, flight under IFR is mandatory.

(b) In VMC, Flight may be under IFR as follows:

(i) The captain may elect to fly under IFR.

(ii) An ATC authority may impose IFR.

(c) Unless authorized by ATC, VFR flights are not to be operated:

(i) During night.

(ii) Above flight level 150.

(d) Except when clearance is obtained from an ATC Unit, VFR flights are not to takeoff or land at an aerodrome within control zone of either the aerodrome traffic zone or .traffic pattern:

(i) When cloud ceiling is less than 1,500 ft.

(ii) When ground visibility is less than 8 km.

(e) Except in sub-paras (b) (i) & (ii) flights in VMC are normally conducted under VFR.

**Definitions**

9. When the following terms are used in the International Standards for Rules of the Air, they have the following meanings:

(a) **Advisory Airspace**. An airspace of defined dimensions, or designated route, within which air traffic advisory service is available.

(b) **Advisory Area**. A designated area within a flight information region where air traffic advisory service is available.

(c) **Advisory Route**. A designated route along which air traffic advisory service is available.

(d) **Area Control Centre**. A Unit established to provide air traffic control service to controlled flights in control areas under its jurisdiction.

(e) **Aerodrome**. A defined area on land or water (including any buildings, installation and equipment) intended to be used either wholly or in part for the arrival, departure and surface movement of aircraft.

(f) **Air Traffic Control Service**. A service provided for the purpose of:

(i) Preventing collisions between aircraft, and on the manoeuvring area between aircraft and obstructions.

(ii) Expediting and maintaining an orderly flow of air traffic.

(g) **Airfield**. A defined area on land used for take-off and landing of aircraft.

(h) **Airway.** A control area or portion thereof established in the form of a corridor equipped with radio navigational aids.

(j) **Controlled Airspace**. An airspace of defined dimensions within which air traffic control service is provided to IFR flights and to VFR flights in accordance with the airspace classification.

(k) **Control Area**. A controlled airspace extending upwards from a specified limit above the earth.

(l) **Control Zone**. A controlled airspace extending upwards from the surface of earth to a specified upper limit.

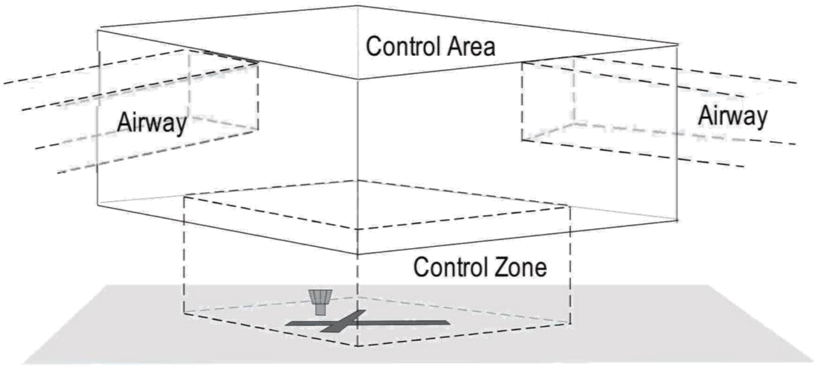



Illustration of The Relationship between Control Zones, Control Areas and Airways

(m) **Danger Area**. An airspace of defined dimensions over land areas or territorial waters of a state within which activities dangerous to the flight of aircraft may exist at specified times.

(n) **Flight Information Region**. Airspace of defined dimensions within which flight information service and alerting services are provided.

(p) **Prohibited Area.** Airspace of defined dimensions, above the land areas or territorial waters of a state, within which the flight of aircraft is prohibited.

(q) **Restricted Area**. An airspace of defined dimensions, above the land areas or territorial waters of a state, within which the flight of aircraft is restricted in accordance with certain specified conditions.

(r) **NOTAM (Notice to Airmen).** A notice containing information concerning the establishment, condition or change in any aeronautical facility, service, procedure or hazard, the timely knowledge of which is essential to personnel concerned with flight operations.

**Altimeter Settings for Use in Pakistan**

10. Following altimeter pressure settings are used by aircraft flying in Pakistan:

(a) All airfields - QNH

(b) En-route - (i) 1013.25 Hectopascal (HPa) or millibars (mb)

or

(ii) 29.92 inches of mercury

(c) Control Areas and Zone - A value of QNH as specified by the ATC Unit concerned.

11. **Standard Altimeter Setting (QNE)**. QNE is pressure value 1013.25 mbrs, 29.92 inches on which the standard altimeter setting system is based. The system is designed to ensure a safe minimum standard of vertical separation during en-route flying QNE is the setting used for all flying above the transition altitude.

12. **Airfield / Regional QNH**. Airfield QNH is the observed pressure at an airfield elevation corrected for temperature and reduced to MSL using the ICAN formula. When set on an altimeter on the ground, the altimeter is to give a reading equivalent to the elevation of the airfield. Air Traffic Services Units are to have the regional QNH for the altimeter setting region in which their area of jurisdiction lies and appropriate adjacent regions. These values are passed to pilots when requested or at the discretion of Air traffic controller.

13. **Terms Relating to Vertical Displacement**

(a) **Transition Level**. It is the level at which the pilot of a descending aircraft changes to QNH from the standard setting (1013.2 mbs). This may be regarded as the lowest flight level at which an aircraft can safely proceed on the 'en-route' altimeter setting.

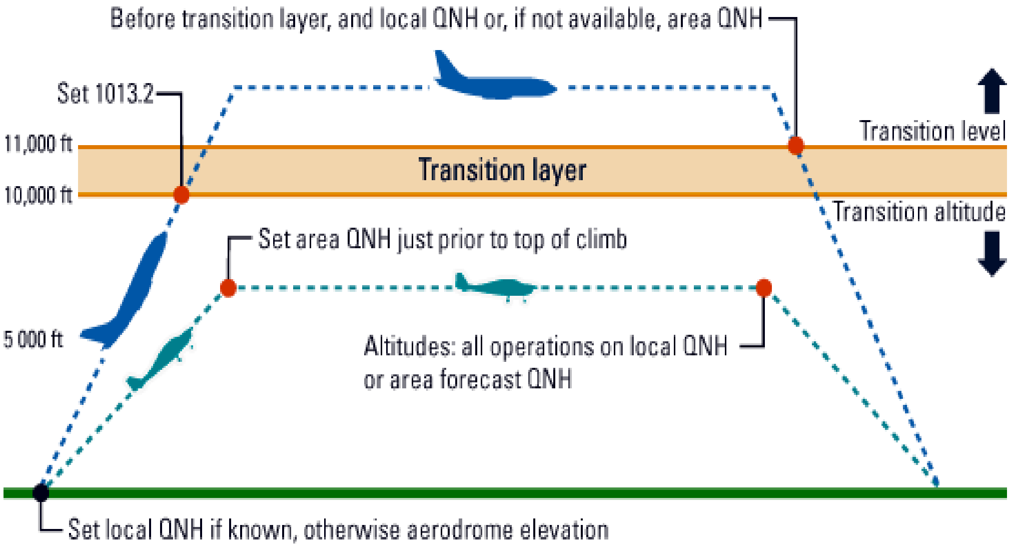
(b) **Transition Altitude**. It is the altitude at and below which local QNH setting is mandatory. Transition altitudes are published. They are local, regional or national and vary considerably.

(c) **Transition Layer**. It is the airspace between (a) and (b).

(d) Aircraft climbing change from QNH to 'en-route' setting after leaving transition altitude and before reaching transition level.

(e) The pilot of a descending aircraft when above transition level is to report his altitude as flight level (e.g. 6,500 ft. as FL 65,12,000 ft. as FL 120).

(f) In ascent, altitude is to be given in feet, up to the transition altitude and as flight level above it.




**Emergency States and Calls**

14. There are two states of emergencies, urgency and distress, which are internationally recognized.

| Emergency | R/T | When Used |
|---|---|---|
| **Urgency** | “Pan, Pan, Pan" followed by aircraft c/s once | Aircraft is in danger and in urgent need of assistance with the aid of which the danger may be overcome, ie aircraft lost, short of fuel, partial engine failure |
| **Distress** | "Mayday, Mayday, Mayday" followed by aircraft c/s three times | Aircraft is threatened by serious and imminent danger and the crew is in need of immediate assistance, ie ditching, crash landing or abandoning aircraft |

15. **Radar Assistance in Emergency and with U/S R/T**. When lost or in emergency and unable to make radio contact, the pilot is to try to alert radar control as follows:

(a) **No Transmitter but Receiver Operative**. The pilot is to fly triangular pattern to the right, holding each heading for one minute (two minutes for piston engine a/c). He is to complete at least two such patterns before resuming course and if possible will keep repeating the pattern at intervals.

(b) **Transmitter and Receiver Both U/S**. The pilot is to fly a left-hand triangular pattern holding each heading for one minute (two minutes for piston engine a/c). He is to complete at least two such patterns before resuming course and if possible will keep repeating the pattern at intervals.

(c) **If Fitted With IFF/SIF**. The pilot is to select mode emergency (7600).



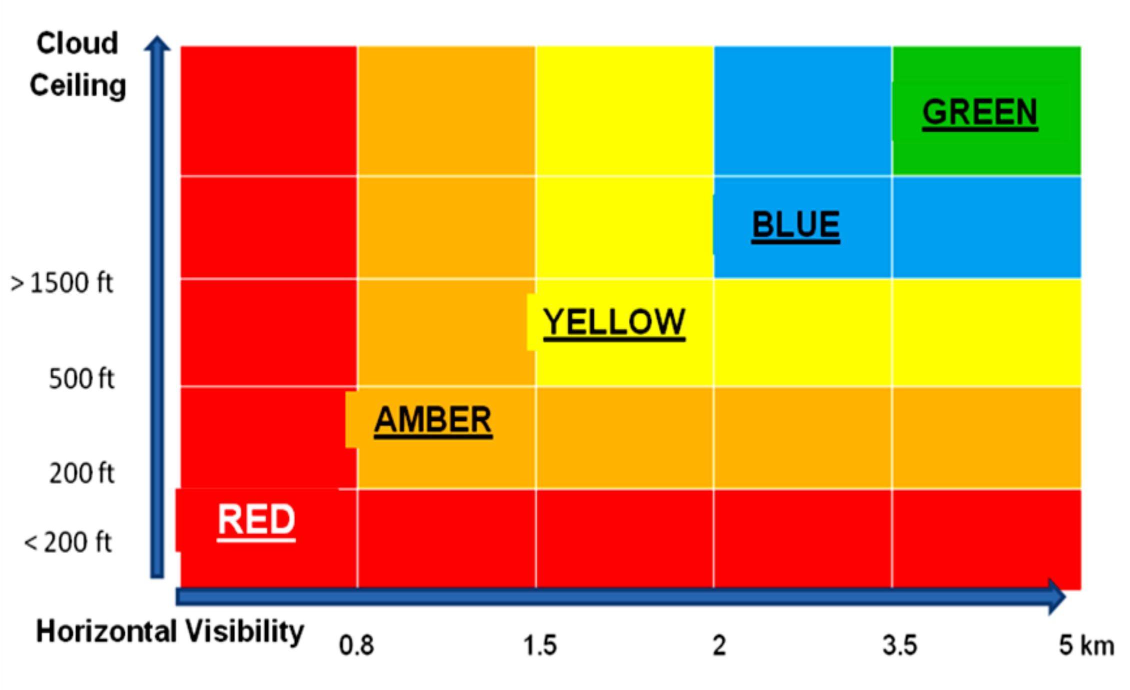
Revised on 29 Jan 21

16. **Weather Minima Applicable to Airfields**. The aircraft in PAF have been divided into the following four categories according to their performance for the purpose of operations at airfields with critical weather conditions:

| | | |
|---|---|---|
| **Category - I** | : | Transport, trainer, maritime and other light aircraft in use in PAF, Army or Navy. |
| **Category - II** | : | Sub sonic aircraft i.e K-8 aircraft. |
| **Category - III** | : | Supersonic fighter aircraft. |
| **Category - IV** | : | Helicopters. |

17. **Airfield (states)**. Ground visibility and clouds (base and amount) are two basic meteorological elements, which determine the airfield state at a PAF aerodrome. The criteria to declare the various airfields states and corresponding limitations imposed are explained below in the table:

| Airfield State | Prevailing Weather | Limitation imposed |
|---|---|---|
| **Red** | Clouds : More than 5/8 at or below 200' AGL,<br><br>Visibility : Less than 01 km | No take offs or landings |
| **Yellow-II** | Clouds: More than 5/8 at 200' to 500'AGL<br><br>Visibility: 1 to 2 kms. | Cat 1 Visual take offs and landings permitted. Controlled descent mandatory if in IMC Cat II. Controlled descent and GCA landings only. |
| **Yellow -I** | Clouds More than 5/8 at 600' to 1500'AGL<br><br>Visibility 2.1 to 3.5 kms | Cat I, II & III Visual landings and Take-offs permitted.<br>Controlled descent and GCA mandatory in IMC. |
| **Green IFR** | Clouds - Any cloud amount above 1500’ provided not more than 5/8 below 1500’ '<br><br>Visibility from 3.6 to 5 kms | All categories of aircraft may take off and land |
| **Green** | Clouds - Any cloud amount above 1500' provided not more than 5/8 below 1500',<br><br>Visibility More than 5 kms | All categories of aircraft may take off and land |




**Note**: PAF Base, Samungli is a peculiar airfield due to high terrain around it. Therefore above mentioned airfield states are replaced as 'Above Jet Minima' (AJM), 'Marginal' and 'Below Jet Minima' (BJM) which are explained below:

| Airfield State | Prevailing Weather | Limitation imposed |
|---|---|---|
| **Above Jet Minima (AJM)** | **Clouds -** Any cloud amount above 12000 ft AGL but not more than 4/8 at or below 12000 ft AGL<br>**Visibility -** 4 km or more<br>**Wind -** Less than 25 kts<br>**Cross wind comp -** Less than 15 kts | No restriction on aircraft operations |
| **Marginal** | Clouds - More than 4/8 at or below 12000 ft.<br>Visibility - 3.8 or 3.9 km<br>Wind - 25 kts or more<br>Cross wind comp: 15kts or more | Only Hot scramble and CAP aircraft may take off, if:<br>• PAF Shahbaz is having A/F state GREEN and<br>• ATC/Crash facilities are available at PAF Shahbaz.<br>• Radio and Navigation aids at Samungli and Shahbaz are serviceable. |
| **Below Jet Minima (BJM)** | Clouds<br>• More than 4/8 at or below 9000 ft AGL<br>Visibility:<br>• 3.7 km or less. | No take-off is allowed |

# CHAPTER - 23

# WEATHER AND ITS REPORTING IN AIR DEFENCE

## Weather

1. Weather is the state of the atmosphere, as determined by the simultaneous occurrence of several meteorological phenomena at a geographical locality or over broad areas of the earth. Weather is closely associated with the distribution of pressure in the atmosphere especially at the surface. Moreover, pressure differences provide the forces, which are responsible for the generation of winds and the consequent changes in weather. Weather comprises of various physical elements in the atmosphere. At a given locality, at least seven such elements may be observed at a time. These physical elements are clouds, precipitation, temperature, humidity, wind, pressure and visibility.

2. Radar Meteorology is the study of the scattering of radar waves by all types of atmospheric phenomena for making weather observations and forecasts of shorter duration (generally 3-4 hours). In general, radar is useful in Meteorology because it is capable of detecting water and ice particles in the atmosphere. It can also be used to observe storm developing regions in the atmosphere which have large temperature gradient and abundance of water-vapors. Applications of radar include radar reflectivity from water and ice particles; measuring the type, amount, and intensity of precipitation, study of clouds; observations of tornadoes and hurricanes; and radar echoes from targets other than water or ice particles.

3. Air Defence requires weather services, which is characterized by speed and accuracy. The radar weather observations are used not only by the radar Unit observing them but timely and accurate radar weather reports assist and greatly enhance the capability of meteorologist in making short term weather forecast accurately. Radar weather provides actual weather in the areas of which weather data is not available. Although Air Defence radars are designed and optimized for elimination of unwanted echoes, including those caused by weather, however, echoes from clouds can still be detected by most of the radars. Therefore, weather observations are to be reported to the concerned agencies by all radar Units whenever possible without compromising the primary mission of air defence. The radar which do not pick up weather on the normal video selections, should accomplish this task when specifically asked by respective SMCC. This section will deal with clouds and procedure for reporting echoes from clouds.

## Clouds

4. A cloud is formed by the condensation of water vapors into droplets, or occasionally into ice crystals. The immediate cause of condensation to water droplets is the reduction of air temperature below the dew-point.

**Constitution of Clouds**

5. Clouds consist of water droplets or ice crystals, or a mixture of both. At levels where the temperature is above 0°C, clouds consist almost entirely of water droplets, whereas at comparatively low temperatures prevailing at cirrus level clouds consisting of ice crystals predominate. A feature which when visible, helps to identify an ice cloud in the solar or lunar halo of 22°radius, tinged red of the inside and blue on the outside. Water clouds, on the other hand, are indicated when a solar or lunar Corona is visible; this is smaller than the halo and the sequence of colors is reversed, a bluish tint appearing on the inside.

6. Water droplets do not necessarily freeze immediately when the air temperature falls below 0°C. Even down to -10°C clouds frequently consist of water droplets, and small droplets can persist at temperatures as low as atleast-40°C. Such droplets are said to be super-cooled and are the commonest cause of air frame icing.

**Formation of Clouds**

7. Clouds are formed owing to moisture being cooled below its dew point. The cooling results mainly from expansion in air which, is rising to the regions of lower atmospheric pressure. When the rising air has expanded sufficiently for its temperature to fall below the dew point, water droplets or ice crystals are formed and supported by the rising currents. If the water droplets or ice crystals in a cloud grow so large that upward currents can no longer support them, they fall as precipitation.

**International Cloud Classification**

8. Clouds are classified into four main categories, each of which is then sub-divided into two or three classes as shown in the accompanying table. The four main families consist of high, medium, low and heap clouds (or clouds of vertical development). These names imply different heights, but it should be emphasized that the ranges of cloud heights vary considerably with latitude, as the following table shows:

| **Cloud Family** | **Classes (General)** | **Abbreviation** | **Cloud Heights at Various Regions** | | |
|---|---|---|---|---|---|
| | | | **Polar** | **Temperate** | **Tropical** |
| High | Cirrus<br>Cirrostratus<br>Cirrocumulus | Ci<br>Cs<br>Cc | 10000<br>to<br>25000 | 16500<br>to<br>45000 | 20000<br>to<br>60000 |
| Medium | Altocumulus<br>Altostratus<br>Nimbostratus | Ac<br>As<br>Ns | 6500<br>to<br>13000 | 6500<br>to<br>23000 | 6500<br>to<br>25000 |
| Low | Stratus<br>Stratocumulus<br>Cumulus | St<br>Sc<br>Cu | Surface<br>to<br>6500' | Surface<br>to<br>6500' | Surface<br>to<br>6500' |
| Heap | Cumulonimbus | Cb | 25000' | 45000' | 60000' |

**Types of Clouds**

9. Clouds can be divided into ten main types:

(a) **Cirrus**. Detached clouds of delicate and fibrous appearance, without shading, generally white in color, of ten of silky appearance.

(b) **Cirrostratus**. A thin whitish veil which does not blur the outline of the sun or moon. This type of cloud of ten gives rise to haloes.

(c) **Cirrocumulus**. A cirri form layer or patch composed of small white flakes, or very small globular masses without shadows and arranged in groups or lines, but more often in ripples like those sometime seen in sand on the sea shore.

(d) **Altocumulus**. A layer or patches composed of laminate or rather flattened.

(e) **Altostratus**. Striated or fibrous veil, more or less grey or bluish in colour. The sun or moon shines through the clouds vaguely, as through ground glass, and may eventually be completely obscured by the clouds. Alto stratus is like thick cirrostratus but does not produce a halo.

(f) **Nimbostratus**. A low, amorphous and rainy layer of a dark grey cloud and nearly uniform appearance.

(g) **Stratus.** A uniform but usually shallow layer of clouds resembling fog but with its base above the general ground level.

(h) **Stratocumulus**. A layer, or patches, composed of globular masses or rolls. The smallest of the regularly arranged elements are fairly large; they are soft and grey with darker parts. The clouds are often arranged in groups, lines, or waves, aligned in one or two directions and the rolls may be so close that their edges join.

(j) **Cumulus**. Thick clouds with vertical development; the upper surface is dome-shaped while the base is nearly horizontal.

(k) **Cumulonimbus**. Heavy masses of cloud with great vertical development, whose cumuliform summits rise in the form of mountains or towers. The upper part of a cumulonimbus cloud has a fibrous texture and often spread out in the shape of an anvil.

**Types of Cumulonimbus**

10. **Convection.** These clouds are created by heating of the layer of air close to the surface. This type of Cb commonly forms in the late afternoon after the peak diurnal heating. Thunderstorms of this type are a daily occurrence in many areas of

the tropics. The storms are usually single Cb cells rather than clusters of cells and so can generally be avoided by flying around them.

11. **Orographic Uplift**. Caused by rising ground (Mounts / Mountains) forcing the air upwards (Orographic Lift). These storms form when a general flow of moist unstable air passes over higher terrain e.g., a ridge line or mountain range. Such storms often form in a line along the ground feature and are therefore more challenging to avoid than single cells.

12. **Mass Ascent**. These clouds are caused when a weather front forces the air upwards. As with orographic lift, the Cb cells form in a line along the front, frequently embedded within wider frontal cloud. Thus, presenting is a challenge to the aircraft trying to navigate through the front.

**Cloud Amount**

13. Cloud amounts are reported in 'Octas', or the number of eighths of the sky covered by cloud. An Octa is easily estimated if one imagines the sky cut into four equal quadrants each quadrant, of course, being two Octas.

14. An observer on the ground usually notes the total amount of clouds and also, if there are several cloud layers present, the amount of each layer i.e. the amount of sky which would be covered by that layer if all other clouds were removed. The clouds are reported as follows:

| | | |
|---|---|---|
| (a) | 1 - 2 Octas | Few |
| (b) | 3 - 4 Octas | Scattered |
| (c) | 5 - 7 Octas | Broken |
| (d) | More than 7 Octas | Overcast |

**Classification of Echoes for Reporting**

15. For the purpose of reporting, the character or pattern of the precipitation echoes observed on the scope is classified as under:

(a) **Isolated Echo**. Individual solid echo either isolated or separated from other echoes by a distance greater than its diameter (See figure1A).

(b) **Scattered Area**. A group of echoes, not forming a line, covering less than 5/10 of the area within a circumscribing circle.

(c) **Broken Area**. A group of echoes, not forming a line covering 5/10 to 9/10 of the area within a circumscribing circle.

(d) **Sol id Area**. A group of echoes not forming a line, and covering greater than 9/10 of the area within a circumscribing circle. This is distinguished from an isolated echo in that area and is observed to be composed of a solid mass of individual echoes.

(e) **Scattered Line of Echoes**. A group of echoes covering less than 5/10 of the area within a circumscribing rectangle and forming a straight or slightly curved line.

(f) **Broken Line of Echoes**. A group of echoes covering 5/10 to 9/10 of the area within circumscribing rectangle and forming a straight or slightly curved line.

(g) **Solid Line of Echoes**. A group of echoes forming a solid, straight or slightly curved line (Figure1C).

(h) **Spiral Bands**. Echoes appearing in line with spiral towards a centre are associated with hurricanes or tropical storms. A spiral band may be composed of scattered, broken or solid curved echoes (See figure 3).

**Note**: Unless a tropical storm is expected or the formation remains evident for at least thirty minutes, some caution is necessary in describing echoes as spiral bands. In some cases the echoes may be elliptical or L-shaped or widely scattered. In these cases the operator will have to rely on his judgment in classifying the pattern.

**Components of a RAREPS**

16. Every radar weather observation report constitute following components:

(a) **Character or Pattern**. The echoes are to be passed according to their appearance on the scope. The classification of echoes for this purpose has been discussedinPara12above.

(b) **Echo Pattern**. This may be classified as a 'line' only if the echoes are arranged in a recognizable or organized line, such as might be reflected from a squalling or a front. This pattern is normally to have a length of at least five times that of its width and be normally at least 20 miles long.

(c) **Intensity**. Estimation and classification of the echo's intensity as weak, moderate, or strong is to be made from the appearance of the echo on the scope with normal setting of the receiver (video) gain. The most intense areas can be indicated by judiciously adjusting the RX gain when many echoes are present. 'Weak' echoes appear thin and grey oils the scope. Owing to the effects of range attenuation; echoes beyond100 miles are to be classified as moderate. Experience is necessary to determine the relative strength of echoes.

(d) **Tendency**. Several consecutive observations of the intensity of an echo are necessary to determine the change in the strength of the echo. If any-change is observed, it is to be noted whether the echo is slowly or rapidly increasing or decreasing or remaining steady. Again, the range attenuation factor is to be taken into consideration i.e. as the echo moves nearer to radar, it becomes brighter. Changes are reported as "Remaining Steady "or" Slowly / Rapidly" increasing / decreasing.

(e) **Position**. The position of the observed echoes is always to be passed in GEOREF. While reporting the position of isolated echo, solid echoes or area of echoes, the GEOREF position of the centre of the circumscribing circle along with its radius in nautical miles is to be given. To report position of lines of echoes, the GEOREF position of all four corners of the circumscribing rectangle is to be given. In case of spiral bands the position of the point about which it seems to be rotating is to be given along with four or her points on the edges.

17. While reporting position of the observed echoes, the reference to a prominent geographic pin point may be given.

18. **Movement**. The information about the movement of the precipitation echoes is constituted of following two parts i.e. Direction and Speed:

(a) **Direction**. The direction of movement is determined from the direction from which the echoes are moving and is to be reported to the nearest10 degrees. The direction of movement is to be determined by plotting past and present position of the echoes on the scope. This is best done by joining the centre of the past and present positions of the circumscribing circles or rectangles, as applicable.

(b) **Speed**. The speed at which the echo moves is calculated to the nearest knot. In case of longer lines of echoes, the speeds of both ends may be reported.

**Note**: The determination of movement and speed of the echoes may pose some problems as they often move very slowly. As caution is to be exercised not to mix up the increase in the size of the echo with its greater movement. A maximum of half an hour is to be sufficient to determine both speed and direction. This would only be possible if initial positions and outlines of the circumscribing circle so rectangles are retained on the scope and the shift compared with the new position of the center of the circles/rectangles and not the circumference or the outlines.

19. **Height**. Whenever possible the height of the echoes is to be measured and reported in thousands of feet above ground level. Both upper and lower heights may be given.

20. **Remarks**. Pertinent remarks not covered by the entries prescribed above are to be given at the end of the RAREPS. These remarks may include information concerning the freezing level, winds hear, or turbulence if observed on the HRI. Elaboration about Echoes apparently associated with a tornado or hurricane is to be noted in the remarks.

**Reporting Procedures**

21. When precipitation echoes are observed at a radar Unit, the PPI reader is to immediately contact the surveillance supervisor and inform him of the weather developments on the scope. The surveillance supervisor is then to designate a PPI reader to monitor the echoes. The Surveillance/GCI stations are to pass RAREPS directly to the controlling SMCC on existing plotting lines. The SMCC shall then, relay the RAREPS to appropriate ATCs and other agencies as mentioned in the local orders.

22. **Frequency of Reports**. The initial report is to be transmitted to the controlling SMCC as soon as any development is observed on the scope. The initial report may not contain the information about direction and speed but it is to be passed as soon as it is determined. The subsequent reports are to be originated very hour or whenever one of the following occurs:

(a) The intensity of an echo changes from weak to strong or vice versa.

(b) The direction of movement of previously reported echoes change by 45° degrees or more.

(c) The speed previously reported changes by 30 Kts or more.

(d) A new echo is observed at a distance of 100 NM or more from the position of previously reported echoes.

(e) The echo or echoes fade.

(f) Scattered echoes become a solid line or vice versa.

(g) Scattered area of echoes become solid area or vice versa.

(h) When directed by ADOC or SMCC.

23. The surveillance supervisor is to keep a record of the RAREPS transmitted to the SMCC. If the transmission of RAREPS is to be held up due to operational commitment, which is always to take priority, the fact is to be so annotated in his log book. He is to inform the duty controller of all such entries.

24. **The Sequence of Reporting**. The RAREPS are to be transmitted and logged in the following sequence:

(a) Character or pattern

(b) Intensity

(c) Tendency

(d) Position

(e) Movement (Direction and Speed)

(f) Height (Upper / Lower if possible)

(g) Remarks

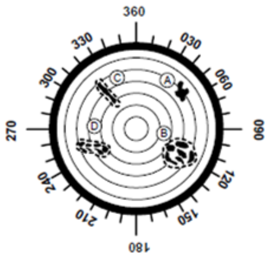


Fig 1

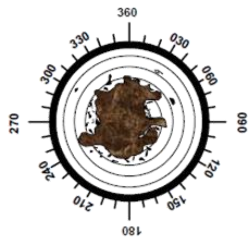


Fig 2

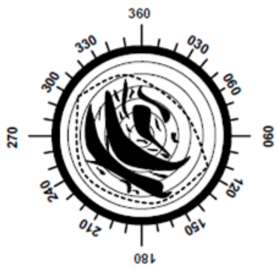


Fig 3

**Weather Hazards in Aviation**

25. There are many weather phenomena which, unfortunately, can prove to be hazardous in Aviation industry. Some of the significant weather hazards are discussed below.

**Turbulence and Wind Shear**

26. Wind shear is nothing more than a change in wind direction and/or wind speed over the distance between two points. If the points are in a vertical direction then it is called vertical shear, if they are in a horizontal direction than it is called horizontal shear. In the aviation world, the major concern is how abruptly the change occurs. If the change is gradual, a change in direction or speed will result in nothing more than a minor change in the ground speed. If the change is abrupt, however, there will be a rapid change of airspeed or track. Depending on the aircraft type, it may take a significant time to correct the situation, placing the aircraft in peril, particularly during takeoff and landing.

27. Mechanical turbulence is a form of shear induced when a rough surface disrupts the smooth wind flow. The amount of shearing and the depth of the shearing layer depends on the wind speed, the roughness of the obstruction and the stability of the air. Turbulence is the direct result of wind shear. The stronger the shear the greater the tendency for the laminar flow of the air to break down into eddies resulting in turbulence. However, not all shear zones are turbulent, so the absence of turbulence does not infer that there is no shear.

## Thunderstorm Hazards

28. No other weather encountered by a pilot can be as violent or threatening as a thunderstorm. The environment in and around a thunderstorm can be the most hazardous encountered by an aircraft. In addition to the usual risks such as severe turbulence, severe clear icing, large hail, heavy precipitation, low visibility and electrical discharges within and near the cell, there are other hazards that occur in the surrounding environment:

(a) **Gust front**. The gust front is the leading edge of any downburst and can run many miles ahead of the storm. This may occur under relatively clear skies and, hence, can be particularly nasty for the unwary pilot.

(b) **Downburst, Macroburst and Microburst**. A downburst is a concentrated, severe downdraft which accompanies a descending column of precipitation underneath the cell. When it hits the ground, it induces an outward, horizontal burst of damaging winds. There are two types of downburst, the “macroburst” and the “microburst”. On occasion, embedded within the downburst, is a violent column of descending air known as a “microburst”. Microbursts have an outflow diameter of less than 2.2 nautical miles and peak winds lasting from 2 to 5 minutes. Such winds can literally force an aircraft into the ground.

(c) **Lightening Hazard**. The effects of lightening on an aircraft are many. If lightning strikes a previously sound, metal bonded structure, the aircraft will remain structurally sound, and the passengers and crew will not be directly affected by the strike’s voltage and current, due to the Faraday Cage effect. However, entrance and exit burn marks will be evident on the skin of the aircraft. This result from the temperature of 3000 - 32000 K within the lightning channel. If the discharge is adjacent to or through structures such as aerials, then these structures may be destroyed. The effect of a lightning strike on both passengers and crew will induce shock, and possibly fear. At night a lightning strike may cause the crew to suffer temporary blindness, or degraded vision.

## Icing

29. One of simplest assumptions made about clouds is that cloud droplets are in a liquid form at temperatures warmer than 0°C and that they freeze into ice crystals within a few degrees below zero. In reality, however, 0°C marks the temperature below which water droplets become super cooled and are capable of freezing. While some of the droplets actually do freeze spontaneously just below 0°C, others persist in the liquid state at much lower temperatures. Aircraft icing occurs when super cooled water droplets strike an aircraft whose temperature is colder than 0°C. The effects icing can have on an aircraft can be quite serious.

## General Aviation Hazards of Cb Clouds

30. **Turbulence**. Vertical movement within a Cb can be as much as 50 kt. The interaction between strong updrafts and strong downdrafts causes wind shear and severe turbulence within the clouds. Strong surface winds, variable in direction and strength, are common at surface level in the vicinity of the Cb. These can be particularly hazardous to aircraft on take-off or landing. In-Flight Icing (of moderate to severe intensity) can be expected, especially in the higher levels of the cloud.

31. **Electrical Disturbance**. Aircraft flying in the vicinity of Cb clouds may experience electrical disturbances effecting communications and navigation systems. The electrical phenomenon is an indication of nearby Cb activity and an aircraft in the vicinity of a Cb is always at risk of being hit by Lightning.

32. **Hail and Precipitation**. Hail can cause significant structural damage to an aircraft. Other precipitation, such as snow, sleet, or rain, can contaminate airfield and runway surfaces creating a hazard to aircraft attempting to take-off or land.

33. **Extreme Weather**. Severe downdrafts, micro bursts, and funnel clouds such as Tornados are also features of Cumulonimbus clouds.

## How to Mitigate the Potential Impacts of Cb Cells?

34. Flight into a Cb is highly dangerous. The only sensible Defence against the hazards associated with a Cb is therefore to avoid flying into one in the first place.

35. Predicting an individual Cb cell is difficult but it is possible to predict the conditions which will trigger formation of a Cb. Forecasters are therefore able to advise flight crews and controllers of the likely timing, location, direction of movement, and height of cells and whether or not they may be embedded. Base authorities can plan aircraft movements to take into account the disruption to operations caused by storms, and approach controllers can consider how they will manage en-route, departing, and arriving traffic when storms are in the vicinity. Aircrew can alter their routings to avoid forecast Cb activity or decide to carry extra contingency fuel in case they have to re-route in flight to avoid the storms or burn additional fuel because of the potential use of aircraft de/anti icing systems.

## Awareness

36. Awareness of the conditions which lead to the formation of a Cb, recognition of a developing and mature Cb, and awareness of the signs which indicate the proximity of a Cb will help Air Defence controllers and aircrews to plan operations to avoid the associated hazards.

37. In addition to visual recognition, weather radar is a particularly valuable aid to avoiding Cb clouds. Airborne weather radar enables the pilots to identify the areas of the storm cloud which hold the largest water droplets, which indicate the areas with strongest updrafts. The area of the cloud with the most severe turbulence is where

the updrafts adjoin the downdrafts; therefore, pilot must avoid flying through the edge of the areas of cloud with the largest water droplets. It should be remembered that a large cloud will absorb a great deal of the radar pulse which may therefore not penetrate all the way through the storm. This can give a false impression to the Air Defence controllers that there are no Cb cells beyond the cell immediately ahead of the aircraft.

38. In certain circumstances, navigating through a line of Cb cells may be the only option opens to a pilot, either because his destination is beyond the line of cells or because he is unable to climb over them. In such circumstances, the aircraft may have to diverge from track by many, perhaps hundreds of miles, in order to find a gap in the wall of Cb clouds. The pilot will need to judge the least hazardous track to follow through the line of cells. The Weather Radar is invaluable in this situation.

39. If the Cb cell is situated over the destination aerodrome, then the pilot would be well advised to hold off or divert rather than attempt a landing. In case of any weather advisory or approaching inclement weather, Air Defence controllers must keep a close liaison with respective Met Sqn for clarity as well as dissemination of latest information to all concerned.

**Weather Effects and Air Defence Procedures**

40. **Lightning**. Needless to emphasis that safety of Air Defence systems vis-à-vis personnel working on these systems against lightning and thunderstorm has vital impact on Air Defence operations. In the absence of precautionary measures, lightening can seriously endanger life and damage or degrade vital Air Defence equipment. However, proper understanding of protection devices available in the system against electrical surges is also essential to avoid undue alarm and render the system unavailable for operational use. Hence there is need to implement safety measures against thunderstorms / lightning while being fully cognizant of the operational requirements.

41. **Self-Protection Devices**. The main purpose of the Self Protection Devices (SPDs) is to protect electronic assemblies against electric surge caused by lightning / thunder storms. In addition to this, lightening arrestors, installed on top of the buildings / equipment, provide safety against direct lightning strike. The systems also have additional features installed by OEMs, against induction of electromagnetic field.

42. **Actions by Duty Controller**. With the provision of safety features, the endeavor should be to ensure that operations are carried out in a safe and efficient manner. The equipment may be switched off when lightening activity is observed directly overhead / near and chances of direct lightning strike exits.

43. All the controllers are to be mindful of the developing weather condition, confirm the same from Met Squadron and keep SMCC updated on the weather and its developments. On observing and confirming the information regarding

thunderstorms / lightning activity at overhead / close vicinity of Squadron, duty controller is to take following actions:

(a) Order switching-off of radar, GMCC and communication equipment with prior clearance from OC Unit and intimation / clearance from SMCC.

(b) Inform 2 i/c and O i/c Electronics.

(c) Instruct supervisors of radar, radio and Ops Room to take required actions.

(d) Ensure that Control line, Auto, Defcom are shifted in shelter.

(e) Ensure switching-off of main WAPDA supply CB of GMCC installed at the entrance of Ops room.

(f) A comprehensive entry is to be made in the log book, covering the details of the weather, actions taken and the names of the persons informed.

(g) Maintain close liaison with Radar and Radio sections.

(h) Remain available at the site to assess the situation and ensure that on dissipation of inclement weather, radar and allied equipment is made available without any delay. He is to take a round of the technical area and report any unusual occurrence to OC Unit.

44. **Actions by O i/c Electronics**. On receiving the information regarding thunderstorms / lightning activity at overhead / close vicinity of Squadron, O i/c Electronics is to take following actions:

(a) Ensure that RF connectors of Radar and Radio equipment are removed from the systems for protection against electrical surges.

(b) Ensure that all under command sections are taking required actions as per laid down procedures.

**Conclusion**

45. The radar weather observations and analysis are used for awareness of fighter controller to read the scope display accurately and to preserve the Air Defence assets from hasty tactical actions. Additionally, timely and accurate radar weather warnings ensure safety of Air Defence equipment and enhance their life time.

# CHAPTER - 24

# AIR NAVIGATION - GENERAL

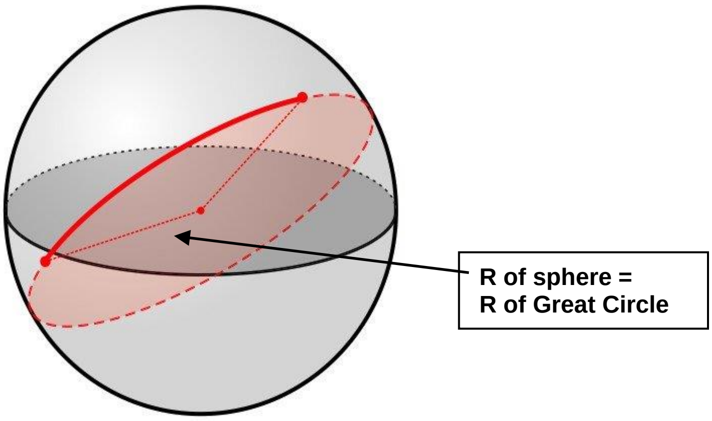
## Introduction

1. 'Air Navigation' is a science of conducting a flight safely from one place to another and determining its position with respect to the surface of earth. Air navigation forms an important part of Air Defence operations. Air Defence officer is required to control fighter aircraft with maximum efficiency, safety and confidence, in all conditions. Therefore, an understanding of the basic terminologies and principles of air navigation will not only greatly assist Air Defence officers in guiding the interceptor aircraft to a point, from where the attack can be effectively delivered but would also improve his overall operational understanding while providing assistance to any aircraft in distress. This chapter lays down the basics of air navigation.

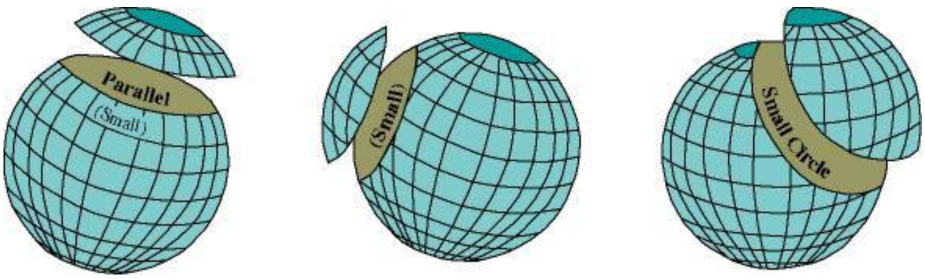
## Terminologies

2. Basic terminologies pertaining air navigation include the following:

(a) **Great Circle**. A great circle is a circle on the surface of a sphere whose center and radius are those of the sphere itself.

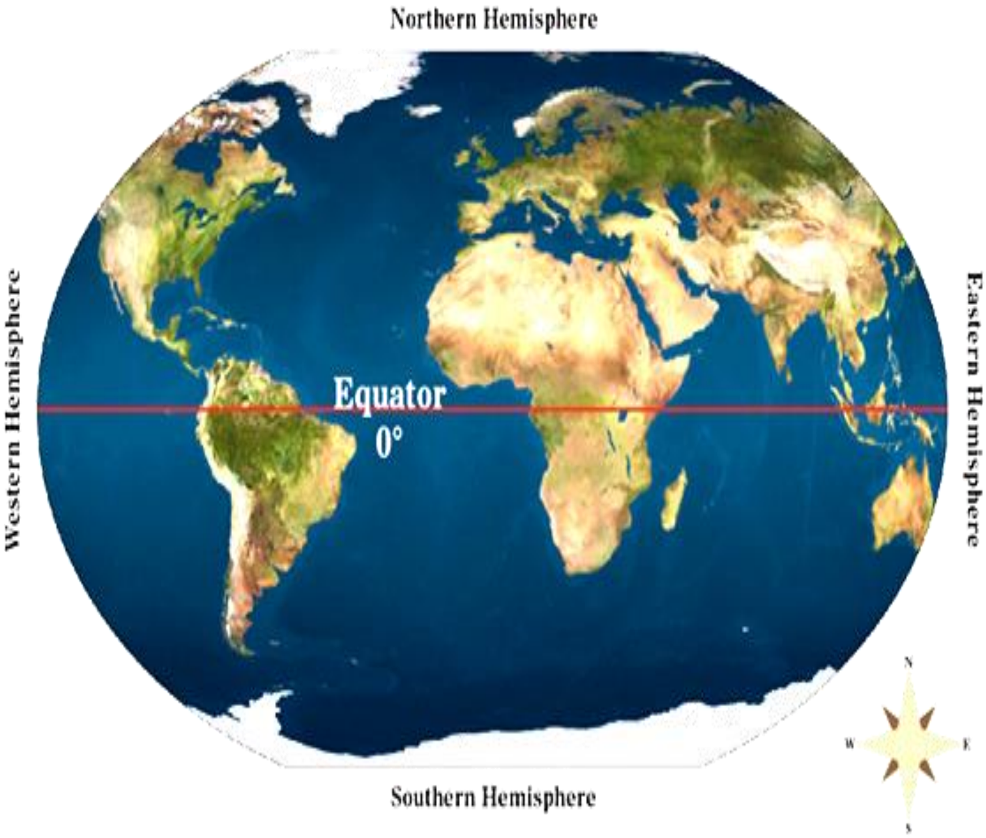



(b) **Small Circle**. A small circle is a circle on the surface of a sphere whose centre and radius are not those of the sphere.

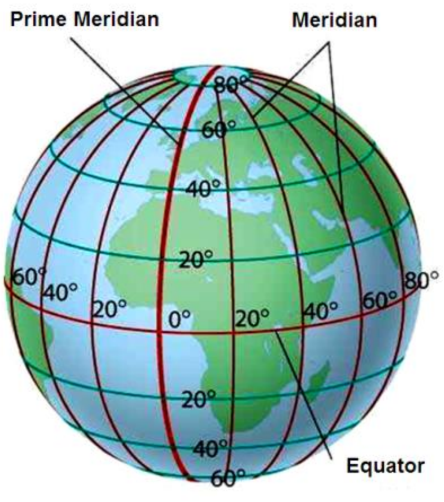


(c) **Parallels of Latitude**. Parallels of latitude are small circles on the earth whose planes are parallel to the plane of the equator. They, therefore, lie in East-West direction.

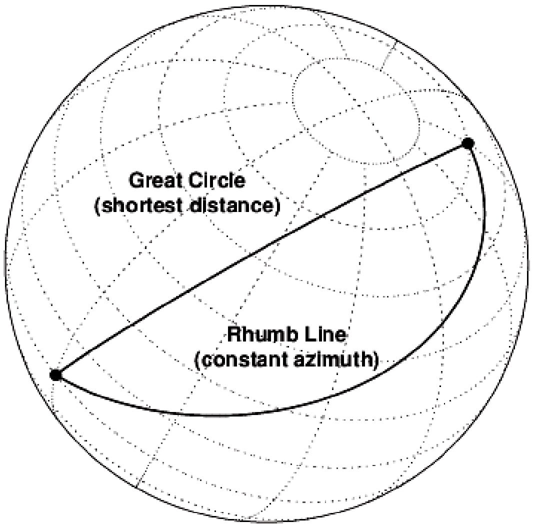
(d) **Equator**. Equator is a great circle whose plane is perpendicular to the axis of rotation of the earth.



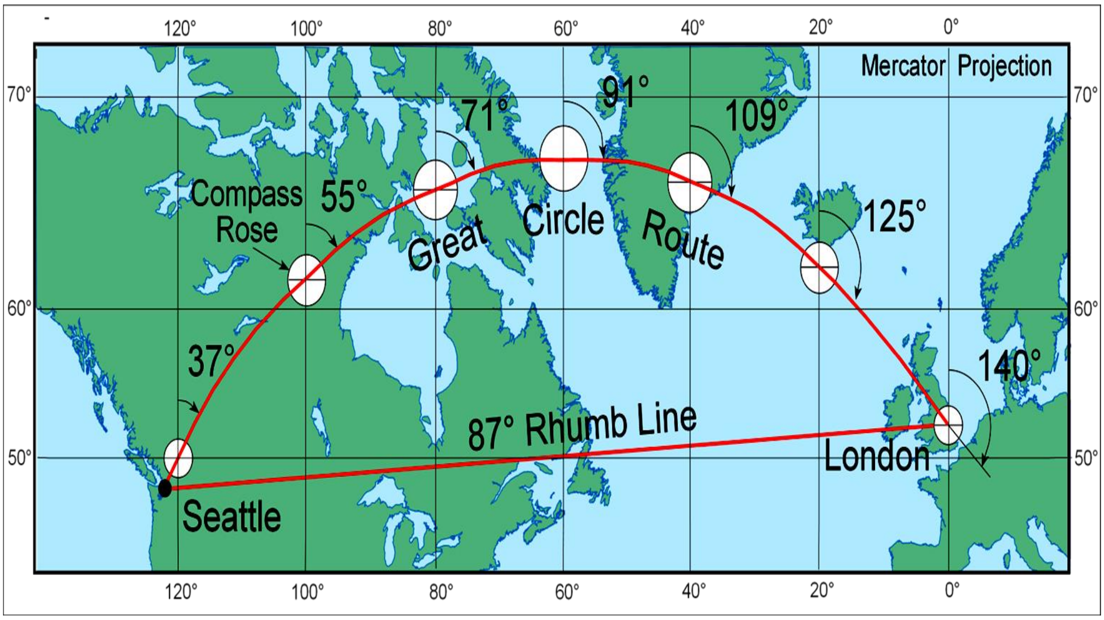

(e) **Meridians**. Meridians are semi-great circles joining the poles. All meridians indicate North-South direction. The meridian passing through the observatory at Greenwich near London is used as a reference from which the angles of other meridians are measured, and is called the Prime Meridian.



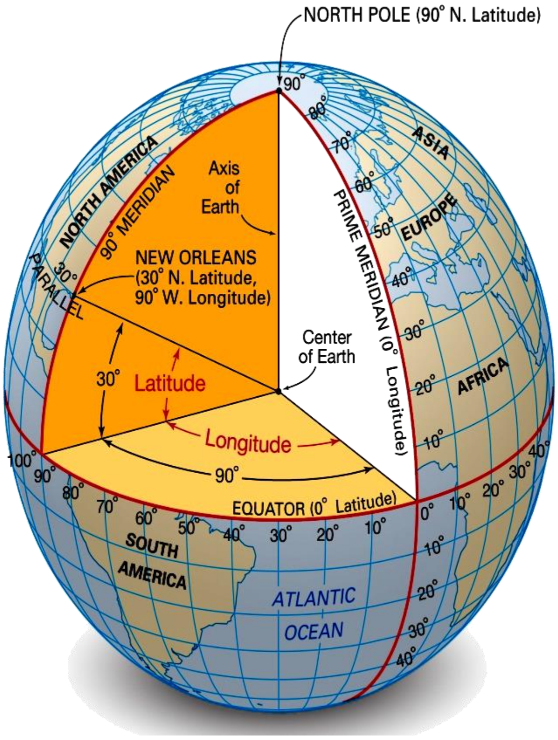
(f) **Rhumb Line**. Rhumb line is a regular curved line on the surface of the earth cutting all meridians at the same angle. By design, the direction of the rhumb line is constant; therefore the rhumb line between two points may be followed more conveniently than the great circle (other than meridians and equator where the path of rhumb line and great circle would be same) whose direction changes continuously. Moreover, on a Mercator projection where the sphere is stretched as a flat rectangle, this Rhumb line becomes a straight line cutting the meridians at same angle.




(g) **Latitude**. Angular distances on meridians are called latitudes. The latitude of a point is the angular along the meridian from equator to that point. It is expressed in degrees, minutes and seconds, and is annotated 'N' or 'S' depending upon the position of a point from equator. For example New Orleans is 30˚ North of the equator, hence at 30˚ N latitude.



(h) **Longitude**. Angular distances on Equator are called longitude. The longitude of a point is the angular distance along the equator from Prime meridian to that point. It is expressed in degrees, minutes and seconds, and is annotated 'W' or 'E' depending upon the position a point from Prime meridian. For example New Orleans is 90˚ West of the Prime meridian, hence at 90˚ W longitude.




(j) **Variation**. The pre-defined angular difference between the direction of true north and magnetic north at any given point is called variation. Variation is measured in degrees and is called east (+) or west (-) according to whether the north-seeking end of the needle of the magnetic compass in an a/c lies to the east or west of true north at any given point e.g.

| | | |
|---|---|---|
| Magnetic Heading | = | 100° |
| Variation | = | +10° |
| True Heading | = | +110° |

| | | |
|---|---|---|
| Magnetic Heading | = | 100° |
| Variation | = | -10° |
| True Heading | = | +90° |

(k) **Isogonal**. Lines joining the points of equal variation are called isogonals.

(l) **Deviations**. The angular difference between the direction of magnetic north and that of compass north is called deviation. Deviation is measured in degrees and is named east (+) or west (-) according to whether the north seeking end of the a/c compass lies to the east or west of the magnetic north e.g.

| | | |
|---|---|---|
| Compass Heading | = | 100° |
| Deviation | = | +4° |
| Magnetic Heading | = | +104° |

(m) **True Heading**. It is the heading measured from true north. True heading is found after correcting a/c compass heading for deviation and variation, as explained below:

| | | |
|---|---|---|
| Compass Heading | = | 225° |
| Deviation | = | -2° |
| Magnetic Heading | = | +223° |
| | | |
| Magnetic Heading | = | 223° |
| Variation | = | -12° |
| True Heading | = | +211° |

(n) **Wind**. Air in motion relative to the surface of earth is known as wind. Wind direction is that direction from which the wind blows. It is always expressed in terms of direction and speed, and is called as wind velocity. Thus, an easterly wind of 15 Kts would be expressed as Wind Velocity (W/V) 090°/15 Kts.

(p) **Heading**. The direction in which the longitudinal axis of aircraft is pointing is called its heading. It is the angle measured clockwise from reference north to the datum fore and aft axis of the aircraft. Since there are three north reference datum, heading may be expressed in three ways viz true, magnetic and compass heading.

(q) **Track**. The directional path of an aircraft over the ground is called track.

(r) **Drift**. The angular difference between the heading and the track of an aircraft is called drift and is expressed in degrees.

(s) **Bearing**. The relative direction of one point from another is called bearing. Like all other directions, bearing may be expressed as true magnetic or compass.

(t) **Ground Position**. It is the position on the ground directly beneath an aircraft.

(u) **Dead Reckoning Position**. It is the ground position of an aircraft deduced from the knowledge of the air position, its heading, speed, time and estimated wind effect.

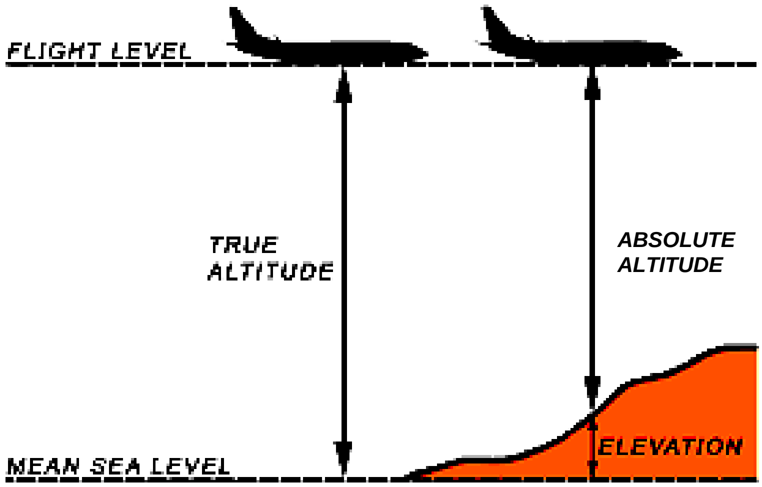
(v) **Altitude**. The vertical distance of a level, a point or an object, considered as a point measured from mean sea level:

(i) **True Altitude**. True altitude is the height above mean sea level.

(ii) **Absolute Altitude**. Absolute altitude is the height above the terrain directly below the aircraft.

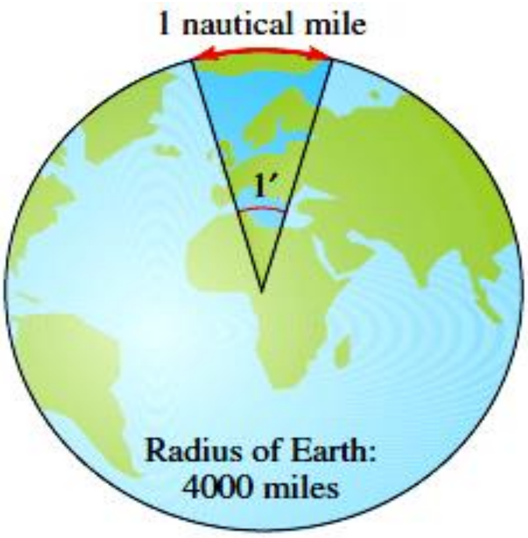
(w) **Height**. The vertical distance of a fixed point above ground level or some specified datum other than mean sea level.

(x) **Elevation**. The vertical distance of a fixed point above or below mean sea level.




3. **Measurement of Distance**

(a) **Nautical Mile**. It is the length of the arc of a great circle which subtends an angle of one minute at the centre of the earth. Since the curvature of the earth is less pronounced at the poles than at the equator, the length of one minute of latitude increases from 6046 feet at the equator to 6107.8 feet at the poles. For the purpose of navigation the length of a nautical mile is taken to be 6080 feet.



(b) **Statute Mile**. As decreed by a statute of Queen Elizabeth I, it is a length of 5280 feet.

(c) **Kilometer**. It is the length of 1 /10,000th of the average distance between the equator and either of the poles. It is equivalent to 3280 feet.

4. **Speed and its Measurement**. Speed is the rate of change of position. The speed at which an aircraft passes through the air is called its air speed and is measured by an air speed indicator which suffers from the following three defects:

(a) **Instrument Error**. It is an error caused by the constructional defects of an instrument. Such an error can however be removed by calibrating the instrument.

(b) **Position Error**. It is the error caused by the effects created by air eddies on the pitot tube.

(c) **Density Error**. The instrument is calibrated on the standard ICAN temperature and pressure (15°C, 1013.2 Mb or 29.92"). Since air density decreases with height the air speed indicator reads too low at great heights. Roughly, at about 10,000 feet the TAS is about 40% in excess and at 40,000 feet it is almost double that of IAS. The true air speed is derived as follows:

(i) IAS - Indicated air speed on the ASI Uncorrected for error.

(ii) CAS - Indicated air speed corrected for Installation effects.

(iii) EAS - CAS corrected for compressibility or Pressure effects.

(iv) TAS - EAS corrected for atmospheric Density.

(v) Ground Speed - TAS corrected for wind velocity.

(d) **Mach Number**. It is the ratio between the TAS and the local speed of sound, i.e.

$$M = \frac{\text{True Air Speed}}{\text{Local Speed of Sound}}$$

**Scale of Maps**

5. It is the ratio between a given length on the map and the actual corresponding distance on the earth's surface, i.e.

$$\text{Scale} = \frac{\text{Map Length}}{\text{Earth Distance}}$$

Where both numerator and denominator are expressed in the same Units.

6. Scale may be indicated on map in one of three ways viz:

(a) **By a Statement in Words**. For example, 1/4 inch to a mile. This statement indicates that one mile on the earth surface is represented by 1/4 inch on the map.

(b) **By Representative Fraction**. For example 1:500,000 indicates that 1 inch on the map is representing 500000 inches on the earth's surface. As there are 63360 inches in a mile, this scale may be expressed as:

$$\text{One Inch} = \frac{5{,}00{,}000 \;/\; 7.89 \text{ miles}}{63360}$$

(c) **By Graduated Scale Line, for the Particular Sheet**. On most of the topographical maps a scale of miles is constructed which is always mentioned in the margin for reference. The distances are usually expressed in statute miles, nautical miles or Kilometers.

7. The amount of detail which can be shown on a map depends on the scale. Maps which show large amount of detail are known as large scale maps e.g. 1 inch to 1 mile series. Small scale maps show a smaller amount of detail e.g. the 1:10,000 series.

**Map Projections**

8. The surface of the earth may truly be represented only on a globe. To depict detailed earth features, such a globe is likely to be of extraordinary size which would create handling problems. Consequently, navigational charts are always printed on flat sheets of paper.

9. The greatest problem encountered in chart making is that the surface of the earth cannot be perfectly represented on a plane surface. Since a spherical surface cannot be spread out on a plane, it is said to be non-developable. The surface of the earth cannot be perfectly presented on a flat surface without distortion (the misrepresentation of direction, shape and relative size of the features of the earth's surface).

10. In chart construction, distortion cannot be entirely avoided but it can be controlled by the use of several types of projections. If a map is required for a particular purpose, it can be projected in such a way as to minimize the type of distortion which is most detrimental to the purpose. If the AREAS are strictly comparable over the entire projection, such a projection is said to be an EQUAL-AREA projection. However, if the SCALE at any point over the entire projection is the same in all directions and the shape of small areas is preserved then such a projection is said to be orthomorphic.

11. **Wind and Drift Correction**. Any free object in air moves downwind with the direction and speed of the prevailing wind. An aircraft flying through a moving air

mass will also be displaced from its track in the direction and with the speed of movement of the air mass. This is true regardless of the speed of the aircraft. This displacement of the aircraft from its track is called drift.

12. **Vector Analysis for Solving Wind Problems**. A vector is a quantity having both direction and magnitude. A vector which results directly from two or more vectors is said to be resultant. A vector quantity may be represented graphically on paper by drawing a straight line in the form of an arrow. The direction of the vector is shown by the bearing of the line with reference to north. The magnitude of the vector may be shown by the length of the line in comparison with some arbitrary scale. If a controller knows components of the vector diagram, he may compute the resultant. By substituting track and ground speed as one vector component, and wind velocity as the other component, the resultant heading and TAS may be found.

13. **Dead-Reckoning with Radar Scope**. Dead reckoning position of an aircraft is found by applying elapsed time, ground speed, and distance along the track from a known point of departure. Heading is found by solving the wind vector problem. The time of arrival at any point is determined by dividing the distance to be covered by the ground speed of the aircraft. Drift can easily be estimated on a radar scope. The track of the aircraft is visually displayed and deviation from known heading may be readily determined. The ground speed can be determined more rapidly and easily than the pilot or the navigator. The distance travelled in one minute is accurately noted on the scope and multiplied by 60, the resulting figure is the ground speed. All critical tracks must be marked at regular intervals, preferably every sweep of the trace, and their tracks projected well ahead of their position. If at any time the radar contact is lost with them, their positions may still be marked progressively by using the previously obtained data on their ground speeds and tracks.

# CHAPTER - 25

# WORLD REFERENCE SYSTEMS

## Introduction

1. In the entire world, three Reference Systems are being used to define any position on the map. These systems are, Latitude Longitude Reference System (Lat, Long System), Geographical Reference System (GEOREF) and Grid System. During map study, aircraft reporting, recce of sites and deployment of Air Defence assets, the Air Defence officers will be required to refer to any one of these systems. Moreover, Conversion of position from one to another system would also be needed at times. This chapter provides a brief understanding of these three systems and also conversion methodology of position from one to another.

## Lat Long Reference System

2. It is the simplest reference system commonly used in air navigation. In this system, the whole globe is divided into 180° latitudes (90° N and 90° S) and 360° longitudes / meridians (180° W and 180° E). Zero latitude and longitude are called equator and prime meridian respectively. The origin of this system is the intersect point of equator and prime meridian which is south of Ghana (Africa), in the South Atlantic Ocean.


**Fig-1 Latitude and Longitude Reference system**

3. Any point, right of the prime meridian is in the east and left of it is in the west. Similarly, any point above equator is north and below is south. Pakistan and adjoining areas fall on the right of prime meridian and above equator. i.e. in the north east region.

4. Lat and Long are drawn on the map after every one degree. Degrees are further sub-divided into minutes, where one minute on face of the earth is equal to one nautical mile.

5. In this system, any point on the map will be read as elaborated in the following example:

**Example**: 3337N, 07306 E

It is read 3337 north and 07306 east, 33 and 073 are the degrees, 37 and 06 are minutes respectively.

Latitudes (N and E) are always written in two digits and longitudes (E and W) in three digits like 005,075,180 because of the greatest latitude and longitude being 90 and 180 respectively.

6. **Defining of Area**. Any circular area on the map can be defined by reading the Lat Long of its centre along with the radius. Similarly, area of irregular shape can be defined by reading the Lat Long of its corners as elaborated in Fig 5 and 6 in case of GEOREF.

7. **Disadvantages**. The latitude and longitude method of reporting position suffers from the following disadvantages:

(a) The possibility of confusion in areas close to the equator and the prime meridian.

(b) The necessity of giving a 10 or 11 figure group to obtain positional accuracy of 1 min (1 NM) e.g. 5136N 0125W or 5136N 10125W.

(c) One minute of latitude and one minute of longitude represent different distances on the earth, except at the equator, and distance represented by one minute of longitude decreases with increasing latitude.

**Geographic Reference System (GEOREF)**

8. GEOREF reference system is based upon the lines of latitude and longitude that encircle the earth. It may be applied to any map or chart regardless of projection. Its point of origin is the 180$^{th}$ meridian and the South Pole. Unlike Lat / Long reference system, GEOREF runs in one direction horizontally (East from the

180th meridian[1]) and one direction vertically (North from the South Pole). GEOREF is always read to the RIGHT and UP.

9. GEOREF divides the earth's surface into quadrangles of longitude and latitudes with a simple, brief, systematic code that gives positive identification to each quadrangle. The system and identification include following levels:

(a) Twenty-four longitudinal zones of 15° each which are lettered from A through Z (omitting I and 0) eastward from the 180th meridian and twelve bands of latitude, each 15° wide, lettered from A through M (omitting I) northward from the South Pole. This combination divides the earth’s surface into 288 primary 15° quadrangles, each identified by two letters. These two letters are the first two characters of a full GEOREF coordinate as shown in figure 2.

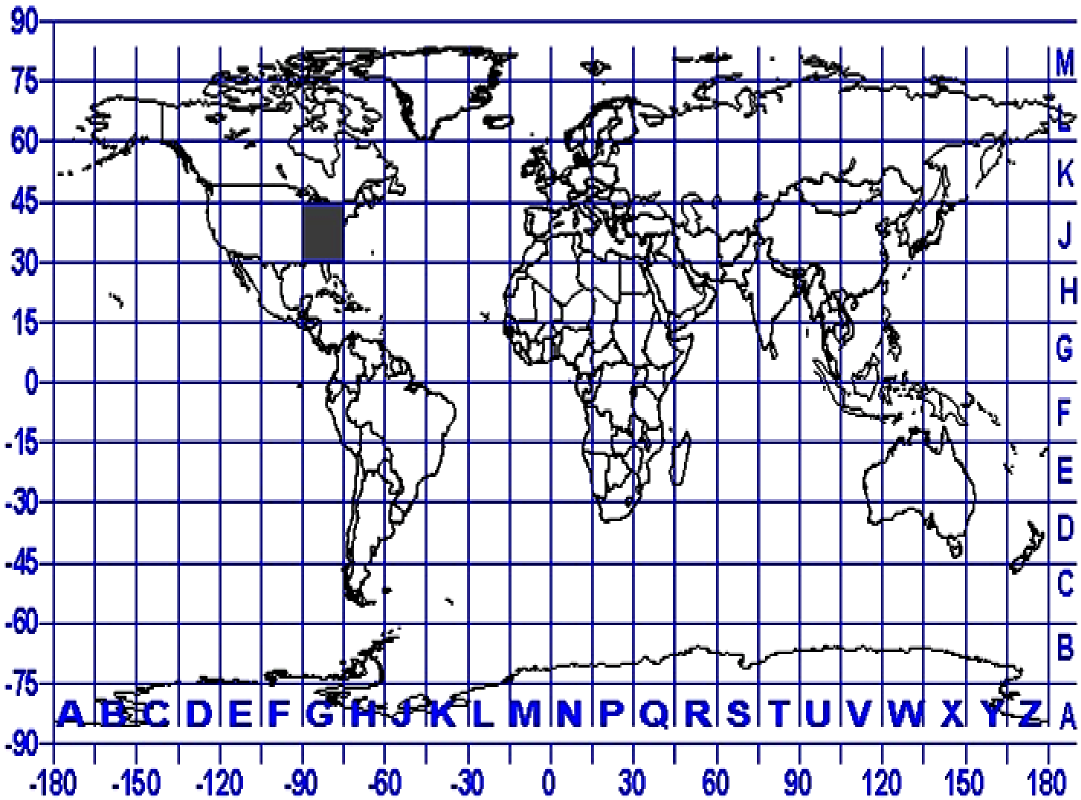



**Fig-2 Geographic Reference System**

[1] **180th Meridian** – Also called as Anti Meridian, is a meridian 180˚ both west and east of the Prime Meridian with which it forms a great circle dividing the earth into the Western and Eastern hemisphere.

(b) Each 15° quadrangle is further divided into fifteen 1° quadrangles eastward and fifteen 1° quadrangles northward. These secondary quadrangles are designated by letters A trough Q, omitting I and O. As in the illustration, (Fig 2) reading right and up the shaded 1° quadrangle is designated by letter GJPJ. Thus four letters will definitely identify any 1° quadrangle in the world. In local operations the 15° quadrangle letter GJ are normally dropped for abbreviations.

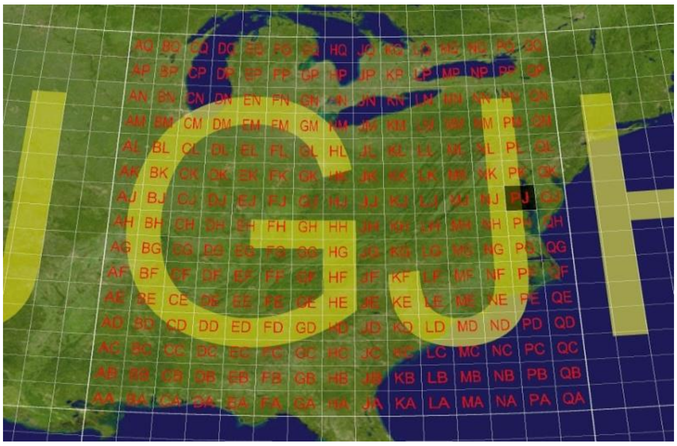


**Fig-3 Secondary Quadrangle**

(c) Each 1° quadrangle is divided into 60 numbered 'minutes' (Units) eastward and northward. Thus four letters and four figures will identify a one minute tertiary quadrangle. In Air Defence operations a 10 minutes breakdown of the degree area is used. The Unit minute values are obtained buy estimation as illustrated in Fig.3.

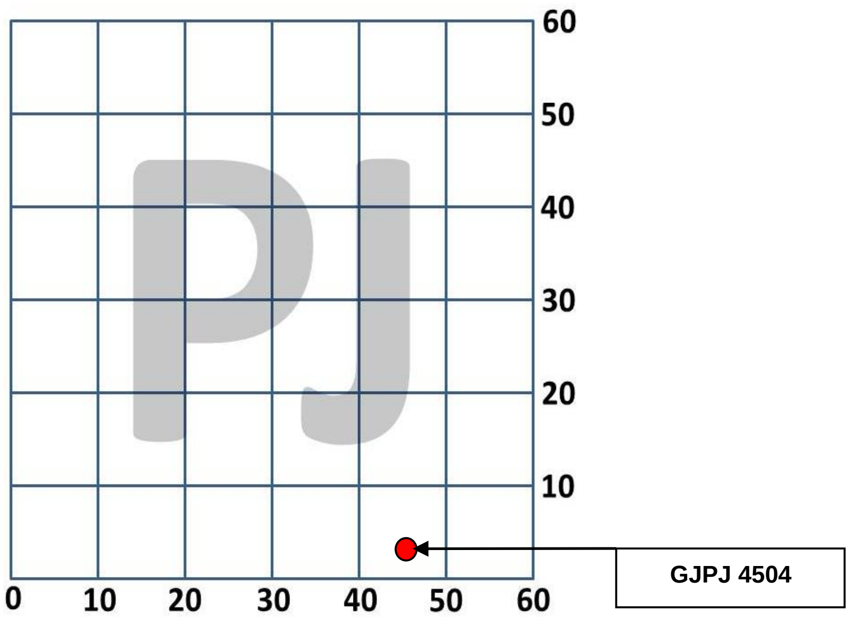



**Fig-4 Tertiary Quadrangle**

(d) If reference of greater accuracy than one minute is required, each one minute quadrangle may be further sub-divided into decimal parts eastward and northward. Thus, four letters and six figures will define a location to 1/10th part of a minute. Thus, a minute quadrangle would be divided into 100 Units rather than 60, if seconds were used.

10. **Indications of Areas**. In addition to basic position reference, it is often necessary to indicate areas of various shapes. An adaptation of the Georef System with respect to their requirements is as follows:

(a) **Rectangular Area**. To designate a rectangular area other than the Unit area referred to in the Georef System, the following procedure is used:

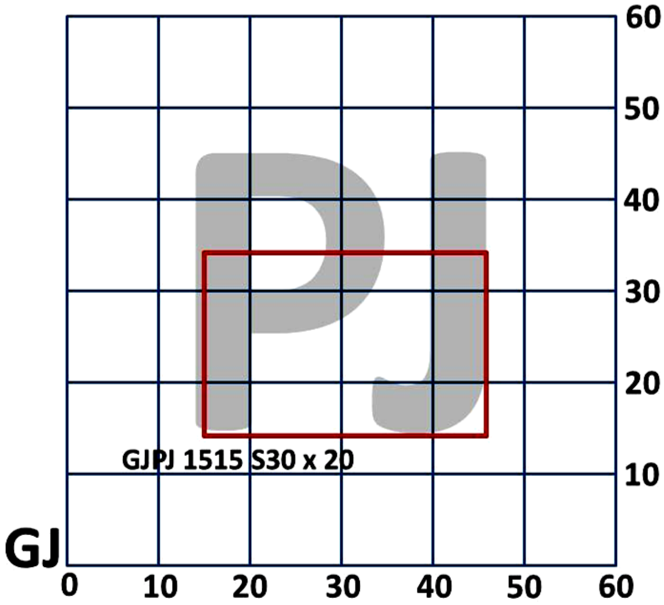
(i) Read the Georef code coordinates of the southwest corner of the area (nearest to the 180$^{th}$ meridian and the South Pole).

(ii) Immediately following the Georef code coordinates, add the letter ‘S’ (denoting side).

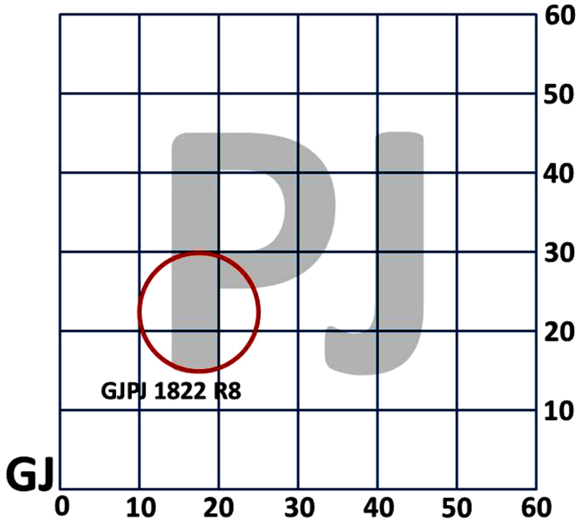
(iii) Add digits defining the east west extent of the area in nautical miles.

(iv) Add the letter ‘X’ (denoting ‘multiplied by’).

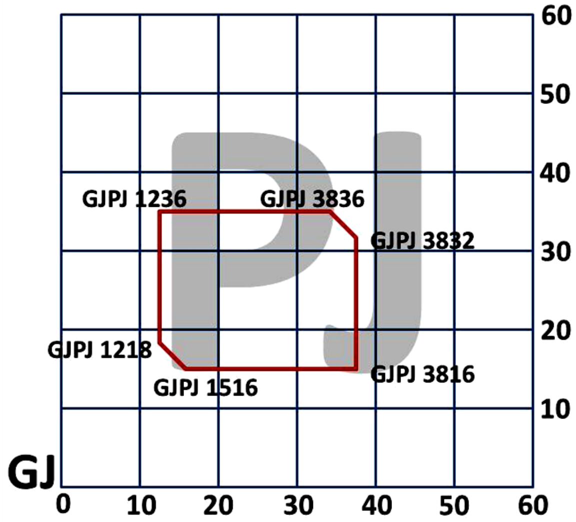
(v) Add digits defining the north-south extent of the area in nautical miles.



(b) **Circular Area**. To designate a circular area, locate the centre of its code coordinates, and add the letter ‘R’ (denoting radius) and digits defining the radius in nautical miles, as shown in the illustration below:




(c) **Irregular Area**. Since some irregular areas are not adaptable to coding procedures, each corner of an irregular area must be defined by its Georef code coordinates as in the sketch below:




11. **Advantages of GEOREF**. GEOREF is simple and accurate reference system for communication purposes. Each division is in harmony with every other division and with the spherical surface which it defines. All sub-divisions are contained completely in all previous divisions and there is no ambiguous overlapping. The code is simple and uniform because it follows a logical sequence with letters and numerals for identification. The Georef code may be applied to all maps and charts without disfiguring them or interfering with their normal use. For the purpose of security, it is comparatively simple to change the code letters from time to

time. It is to be noted that Georef System of reporting is contrary to navigation reporting where, latitude is reported before longitude.

**Conversion of Positions from one System to another System**

12. Lat, Long and GEOREF reference systems are mostly used in the PAF. Air Defence officers may be required to convert the position given in one system into the other. For such conversion, the procedures given in the following examples could be followed:

**Example-1**: Convert 5505N 1025W into GEOREF. We know the basis of the GEOREF system is the division and subdivision of the Earth's surface into 15° and 1° quadrangles. Remembering that origin for GEOREF is 90° S, 180° E/W we can derive a simple method for converting a latitude and longitude position into GEOREF co-ordinates. Using the conventional signs for N(+), S(-), E(+) and W(-), add 90 to the latitude and 180 to the longitude. For a position 5505N 1025W proceed as follows:

$$\begin{array}{rr} & 90^\circ \\ + & 55^\circ 05' \\ \hline & 145^\circ 05' \end{array} \qquad\qquad \begin{array}{rr} & 180^\circ \\ - & 10^\circ 25' \\ \hline & 169^\circ 35' \end{array}$$

Now divide the degree portions by 15

$$\begin{array}{r|r} & 9 \\ \cline{2-2} 15 & 145 \\ & 135 \\ \hline & 10 \end{array} \qquad\qquad \begin{array}{r|r} & 11 \\ \cline{2-2} 15 & 165 \\ & 169 \\ \hline & 4 \end{array}$$

Now add I to the quotients; making them 9+1 = 10 and 11+1 = 12 in the above example. Write down the letters in GEOREF corresponding to these numbers, omitting I and 0 in the count, ie, 10 = K (Lat), 12 = M (Long). As GEOREF is read from RIGHT to Up, along the longitude first. Similarly, add 1 into remainders i.e. 10+1 = 11, 4+1 = 5 and count corresponding letter in GEOREF, which are L and E. Writing longitude first we get EL. Now write them after primary quadrangles as MK EL. The minutes can be written directly from last 2 digits of Lat and Long but longitude first.

**Note**: 1 is always added in quotients and remainders to make them at least 1 or more, as the first corresponding count in GEOREF is A.

**Example-2**: Convert GEOREF MK EL 3505 into Lat and Longs. GEOREF is first read along longitude therefore M & E are the primary and secondary quadrangles whereas 35 are the minutes along longitude. Similarly K & L are the primary and secondary quadrangle, and 05 are the minutes along latitude. Now putting them under Lat and Long:

| Lat | Long |
|---|---|
| KL 05’ | ME 35’ |

Write corresponding digits of K, L, M and E omitting I and 0.

| Lat | Long |
|---|---|
| 10 11 05’ | 12 5 35’ |

Subtract 1 from 10, 11, 12 and 5 (Refer to the Note of Example 1)

9 10 05’ 11 4 35’

9 and 11 are 15° quadrangles. Multiply by 15 to convert them into degrees.

135 10 05’ 165 4 35’

10 and 4 are the degrees of 1 degree quadrangle add them into the degrees obtained from 9 and 11 (15° quadrangles).

145 05’ 169 35’

As origin of GEOREF is 90° South therefore, subtract 90 degrees from 145 to get Lat (Lat cannot be more than 90).

145 - 90 = 55

Lat is 55° and 05' North of origin which is written as 5505 N.

Similarly, 169° and 35' can be converted in longitude. As GEOREF origin is 180 west of prime meridian, 169° 35' which is less than 180, is also west of prime meridian by 10° 25' (180-169° 35') or longitude is 1025W.

GEOREF MK EL 3505 is 5505 N 1025W

**Grid System**

13. Grid Reference System was devised to overcome the disadvantages of Latitude and Longitude System. It is mostly used by the army. In this system the world map is divided into 100 KM squares which are further sub-divided into 10 KM, 1 KM and 100 Meters squares depending upon the scale of the map and plan for use. Presently, there are three grid systems in use:

(a) **The National Grid System**. It is a terrestrial coordinate system used by Britain.

(b) **The Universal Transverse Mercator Grid (UTM)**. It is a system for assigning coordinates to any location on the surface of earth and is used by NATO armed forces.

(c) **The Universal Polar Stereographic Grid (UPS)**. This system resembles the UTM Grid and has two separate grids for north and south poles.

14. National, UTM and UPS (North Pole) Grids are shown in the figures 8, 9 and 10 respectively.

15. **Cartesian coordinate system**. This system is based on a reference point which could be based on any position calculation system in GEOREF, Polar Coordinate Lat or Long etc.

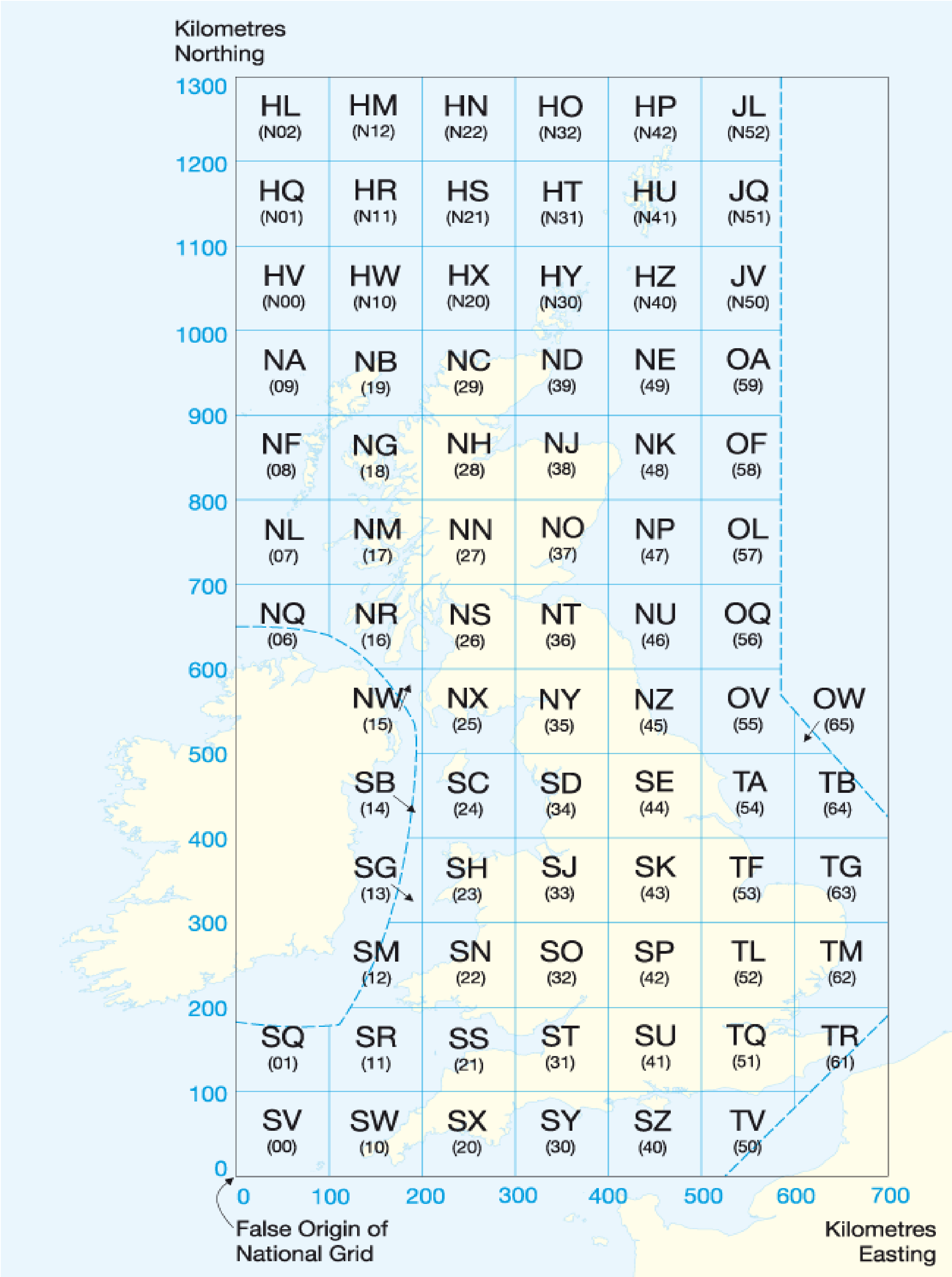



**Fig-5 National Grid System**

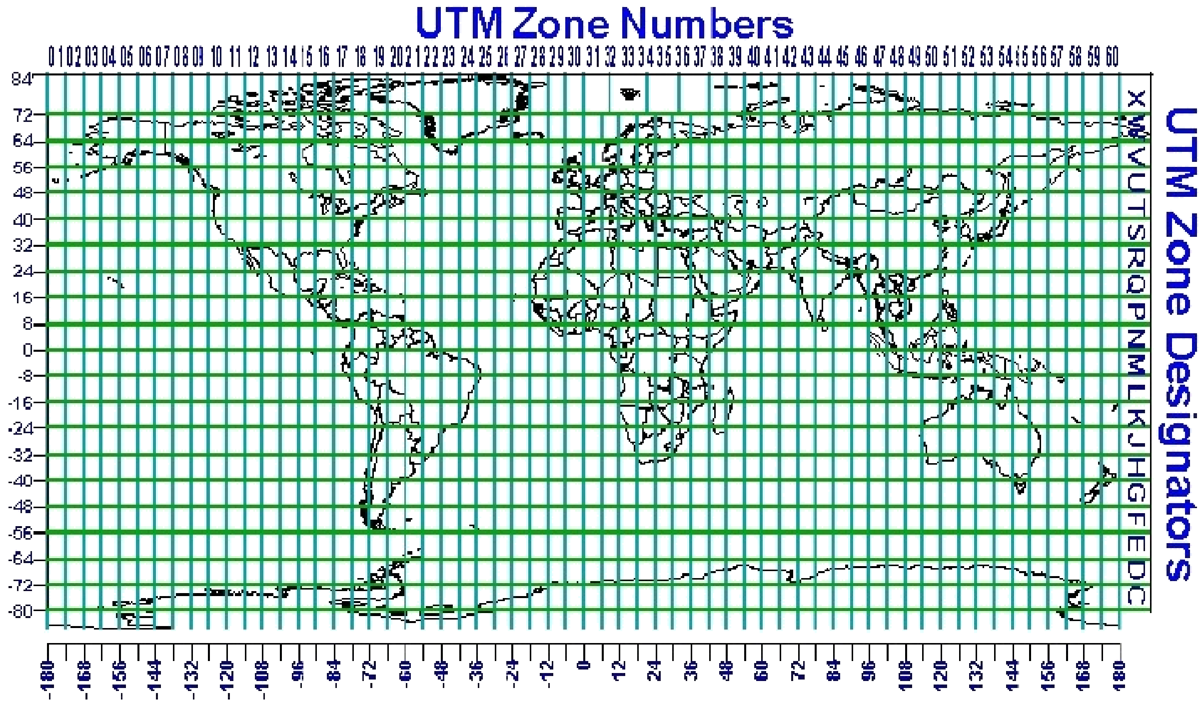



**Fig-6 UTM Grid System**

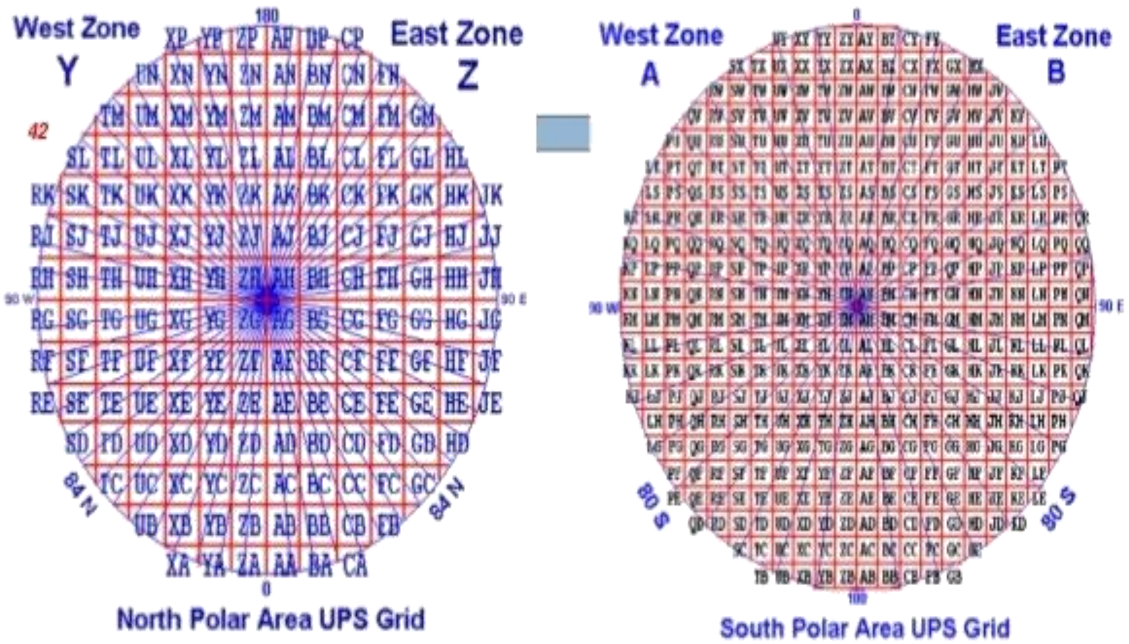



**Fig-7 Universal Polar Stereographic Grid System**

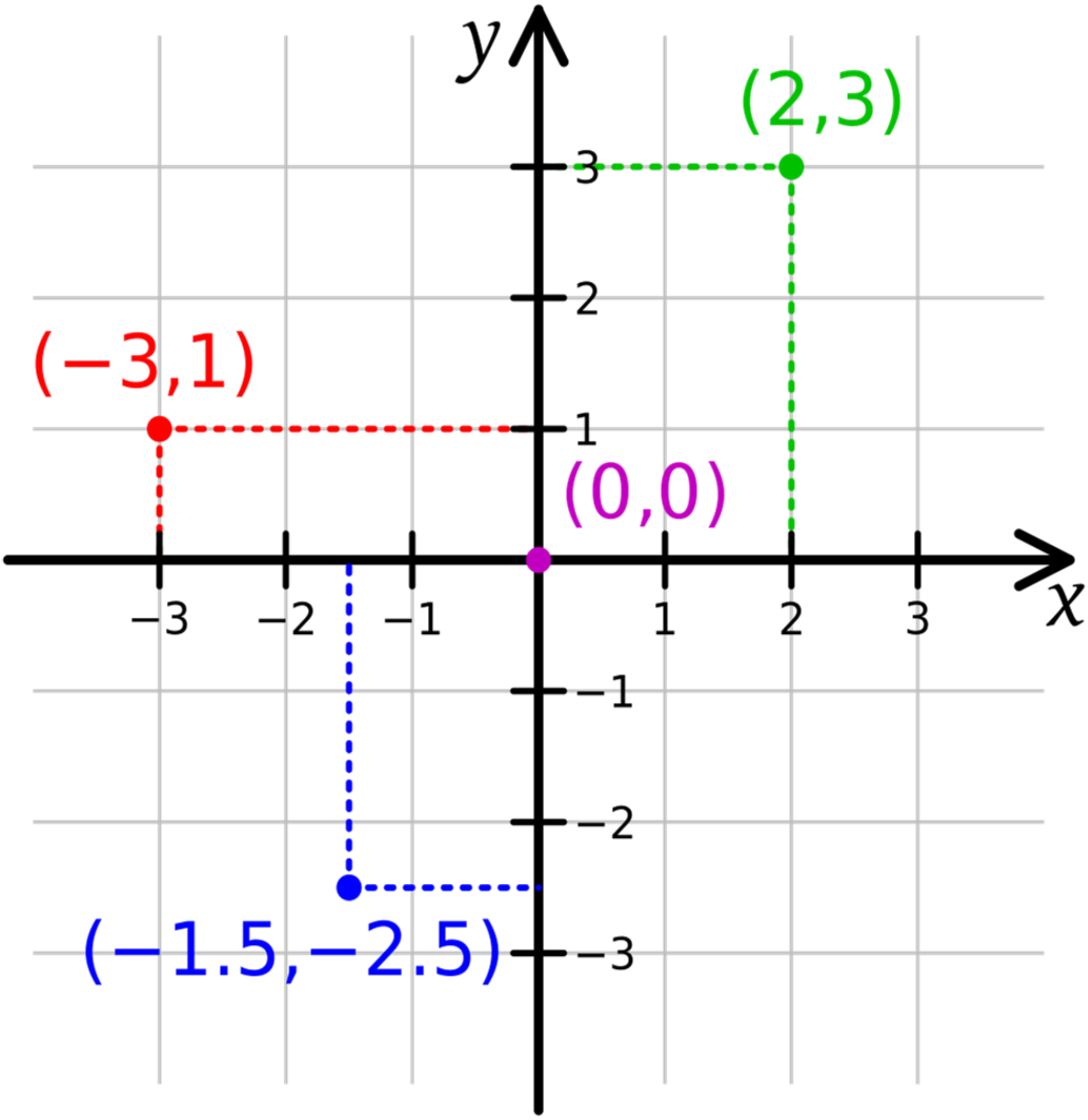



**Fig-8 Cartesian Coordinate System**

## CHAPTER - 26

# COMMONLY USED ABBREVIATIONS

### Introduction

In Air Defence operations official correspondence, Inter-services and International Exercises, certain terms are used which are to be abbreviated for sake of convenience and standardization. The authorized standard abbreviations of such common terms are given below. The terms and their abbreviations are directive in nature and be used either vocally or in writing by all elements of Air Defence, as required.

| Abbreviations | Description |
|---|---|
| **A** | Allied- A Track Classification to Denote a Friendly Aircraft |
| **A/C** | Aircraft |
| **A/F** | Airfield |
| **A2/AD** | Anti Access / Aerial Denial |
| **AAA** | Anti Aircraft Artillery |
| **AAC** | Army Aviation Corps |
| **AAD** | Army Air Defence |
| **AAD** | Army Air Defence |
| **AADC** | Army Air Defence Centre |
| **AADG** | Army Air Defence Guns |
| **AADX** | Army Air Defence Executive Officer |
| **AAM** | Air to Air Missile |
| **AAR** | Air to Air Refueling |
| **AB** | After Burner |
| **ABT** | Abortive / Aborted |
| **ACC** | Area Control Centre |
| **ACM** | Air Combat Maneuvers |
| **ACO** | Airspace Control Order (NATO) |
| **AD** | Air Defence |
| **ADA** | Air Defence Alert |
| **ADC** | Air Defence Command |

| Abbreviations | Description |
|---|---|
| **ADCP** | Air Defence Command Post |
| **ADGE** | Air Defence Ground Environment |
| **ADIZ** | Air Defence Identification Zone |
| **ADOC** | Air Defence Operations Centre |
| **AEF** | Airborne Equipment Failure |
| **AEM** | Advance Employment Manual |
| **AESA** | Active Electronically Scanned Array |
| **AEW** | Airborne Early Warning |
| **AEW&C** | Airborne Early Warning & Control |
| **AEWACS** | Airborne Early Warning Command and Control |
| **AGL** | Above Ground Level |
| **AGM** | Air to Ground Missile |
| **AHQ** | Air Headquarters |
| **AI** | Airborne Intercept Radar |
| **ALR** | Acceptable Level of Risk |
| **ALT** | Altitude |
| **AMPI** | Air Mass Position Indicator |
| **AMR** | Acceptable Merge Ratio |
| **AMSL** | Above Mean Sea Level |
| **AOB** | Air Order of Battle |
| **AOC** | Air Officer Commanding |
| **AOI** | Area of Operational Interest |
| **AOR** | Aircraft Occurrence Report / Area of responsibility |
| **AP** | Anomalous Propagation (of radar energy) |
| **AR** | Air Routes (NATO) |
| **ARH** | Active Radar Homing |
| **ARM** | Anti-Radiation Missile |
| **ARO** | Auxiliary Readout (a part of the data display console) |
| **ASD** | Air Situation Display |

| Abbreviations | Description |
|---|---|
| **ASM** | Air to Surface Missile |
| **ASM** | Airspace Management |
| **ASO** | Air Surveillance Officer |
| **ASR** | Air Staff Requirement / Airport Surveillance Radar |
| **ASV** | Anti-Sub Vessel |
| **ASW** | Anti-Ship Warfare |
| **ATA** | Actual Time of Arrival |
| **ATC** | Air Traffic Control |
| **ATCC** | Air Traffic Control Centre |
| **ATD** | Actual Time of Departure |
| **ATO** | Air Tasking Order |
| **ATO** | Air Tasking Order |
| **AU** | Acquisition Unit |
| **AUTO** | Automatic |
| **AVTR** | Audio Video Tape Recorder |
| **AWACS** | Airborne Warning and Control System |
| **AWCC** | Area Weapon Control Centre |
| **AWR** | After Weapon Release |
| **B** | Bomber - A track classification for friendly bomber a/c |
| **BAI** | Battlefield Air Interdiction |
| **BARCAP** | Barrier CAP |
| **BCN** | Beacon |
| **BEM** | Basic Employment Manual |
| **BITE** | Built-In Test Equipment |
| **BRG** | Bearing |
| **BRH** | Beyond the Radar Horizon |
| **BSA** | Blind Speed Angle / Border Shooting Authority |
| **BSR** | Blip to Scan Ratio |
| **BTY** | Battery (a group of AAD Guns) |

| Abbreviations | Description |
|---|---|
| **BU-SMCC** | Back up Sector Mission Control Center |
| **BVR** | Beyond Visual Range |
| **BWR** | Before Weapon Release |
| **C** | Combined - A track classification to indicate mixed up tracks of similar classification |
| **C/S** | Call Sign |
| **$C^2$** | Command and Control |
| **$C^3$** | Command, Control and Communication |
| **$C^4I$** | Command, Control, Communications, Computer and Intelligence |
| **$C^4I^2SR$** | Command, Control Communications, Computers, Intelligence. Information, Surveillance and Reconnaissance |
| **C4S** | Command, Control, Communications and Computer systems |
| **CAAR** | Civil Aviation Authority Radar |
| **CAF** | Clear Avenue of Fire (for BVR Missile) |
| **CAORP** | Combat Air Operations Rules and Procedures |
| **CAP** | Combat Air Patrol |
| **CAS** | Close Air Support |
| **CAS** | Calibrated Air Speed |
| **CCS** | Combat Commanders School |
| **CDLO** | Civil Defence Liaison Officer |
| **CDX** | Civil Defence Executive Officer |
| **CFAR** | Constant False Alarm Rate |
| **CL** | Contact Lost with aircraft visually or on a radar scope |
| **COC** | Command Operations Centre |
| **COIN** | Counter Insurgency |
| **COMINT** | Communication Intelligence |
| **COMSEC** | Communication Security |
| **CON** | Contact |
| **CONOPS** | Concept of Operations |
| **COP** | Common Operating Picture |

| Abbreviations | Description |
|---|---|
| **CPACS** | Coded Pulse Anti Clutter System |
| **CRC** | Control and Reporting Centre/ Control Radar Coverage |
| **CRCP** | Control and Reporting Command Post |
| **CRT** | Cathode Ray Tube |
| **CSAR** | Combat Search and Rescue |
| **CW** | Continuous Wave |
| **CWI** | Continuous Wave Illumination |
| **CZ** | Combat Zone |
| **D/F** | Direction Finding (Equipment) |
| **D/L** | Data Link |
| **DA** | Defence Attaché |
| **DADCDR** | Deputy Air Defence Commander |
| **DC** | Duty Controller / Detection center |
| **DCA** | Defensive Counter Air |
| **DEAD** | Destruction of Enemy Air Defence |
| **DEZ** | Desired Engagement Zone |
| **DI** | Diversion (of interceptors to another mission) / Direction Indicator (Aircraft) |
| **DIRCM** | Directional Infrared Counter Measures |
| **DK** | Darkness |
| **DLI** | Data Link Interface |
| **DMS** | Dead Man shot |
| **DMTI** | Digital MTI |
| **DR** | Dead Reckoning / Reckoned |
| **DROR** | Desired Roll Out Range |
| **DRS** | Digital Relay System |
| **DSCR** | Digital Search Control and Reporting |
| **DVR** | Digital Video recorder |
| **DWR** | During Weapon Release |
| **EA** | Electronic Attack |

| Abbreviations | Description |
|---|---|
| **EC** | Exercise Controller |
| **ECCM** | Electronic Counter Counter Measures |
| **ECM** | Electronic Counter Measures |
| **ECR** | Expected Contact Range |
| **ECT** | Extended Coverage Telling Area |
| **ELINT** | Electronic Intelligence |
| **ELSEC** | Electronic Security |
| **EMCON** | Emission Control |
| **EOB** | Electronic Order of Battle |
| **EP** | Electronic Protection |
| **EPM** | Electronic Protection Measures |
| **ES** | Electronic Support |
| **ESM** | Electronic Support Measures(Old term for Electronic Support) |
| **ETA** | Estimated Time of Arrival |
| **ETD** | Estimated Time of Departure |
| **ETF** | Expected Time of Fade (on radar) |
| **ETP** | Estimated Time of Pick up (on radar) |
| **EW** | Early Warning / Electronic Warfare |
| **EWO** | Electronic Warfare officer |
| **EWTTR** | Electronic Warfare Test and Training Range |
| **EX** | (Air Defence) Exercise |
| **F** | Fighter - A track classification for interceptor aircraft (Old ID classification) |
| **FA** | Faded from Radar Scope |
| **FAADC** | Forward Army Air Defence Centre |
| **FAC** | Forward Air Controller |
| **FAOR** | Fighter Area of Responsibility |
| **FCC** | Fire Control Center |
| **FCR** | Fire Control Radar |
| **FEBA** | Farthest / Forward Edge of the Battle Area (Army) |

| Abbreviations | Description |
|---|---|
| **FEZ** | Fighter Exclusive Zone |
| **FFS** | Flight Following Service |
| **FIC** | Flight Information Centre |
| **FIR** | Flight Information Region |
| **FITOPS** | Fighter Integrated Transport Operations |
| **FLO** | First Launch OpportUnity |
| **FLOT** | Forward Line of Own Troops ( Army) |
| **FO** | Filtration Officer |
| **FPI** | Faded Prior to Interception |
| **FPN** | Flight Plan |
| **FTM** | Free Text Message (Used in link operations with AEW&C aircraft) |
| **FU** | Firing Unit |
| **FUA** | Flexible Use of Airspace |
| **GBADS** | Ground Based Air Defence Systems |
| **GCA** | Ground Controlled Approach |
| **GCI** | Ground Controlled Interception |
| **GDA** | Gun Defended Area |
| **GEF** | Ground Equipment Failure |
| **GEOREF** | Geographical Reference |
| **GES** | Ground Entry Station |
| **GFR** | Gap Filler Radar |
| **GMCC** | Generic Mission Control Center |
| **GMT** | Greenwich Meridian Time |
| **GPS** | Global Positing System |
| **GS** | Ground Speed |
| **HCA** | Heading Crossing Angle (Also see TCA) |
| **HELO** | Helicopter |
| **HF** | Hold Fire / High Frequency |
| **HFR** | Height finder Radar |

| Abbreviations | Description |
|---|---|
| **HIDACZ** | High Density Airspace Control Zones(NATO) |
| **HIMADS** | High to Medium Level Air Defence Systems |
| **HLDC** | High Level Detection Center |
| **HPD** | Horizontal Polar Diagram |
| **HPR** | High Power Radar |
| **HREZ** | High Risk Engagement Zone |
| **HRI** | Height and Range Indicator |
| **HT** | Height |
| **HUD** | Heads Up Display |
| **HVAA** | High Value Airborne Asset |
| **IADS** | Integrate Air Defence System |
| **IAS** | Indicated Air Speed |
| **IC** | Intercept Controller |
| **ICAO** | International Civil Aviation Organization |
| **ICT** | Intercept Control Technician |
| **ID** | Identification |
| **IFF** | Identification, Friend or Foe |
| **ILS** | Instrument Landing System |
| **IMC** | Instrument Meteorological Conditions |
| **IMINT** | Imagery Intelligence |
| **INAS** | Inertial Navigation and Attack System |
| **INS** | Internal Navigation System |
| **IP** | Identification Point / Initial Point |
| **IPI** | Identified Prior to Interception |
| **IR** | Infra-Red |
| **IRCM** | Infra-Red Counter Measures |
| **IRSTS** | Infra-Red Search and Track System |
| **IS** | Internal security / Information Security |
| **ISR** | Intelligence Surveillance and Reconnaissance |

| Abbreviations | Description |
|---|---|
| **ISTAR** | Intelligence , Surveillance , Target, Acquisition and Reconnaissance |
| **J** | Jammer (Old track ID classification) |
| **JADC** | Joint Air Defence Centre |
| **JDAM** | Joint Direct Attack MUnition |
| **JEZ** | Joint Engagement Zone |
| **JO** | Joint Operations |
| **JOC** | Joint Operations Centre |
| **JSTARS** | Joint Surveillance and Target Attack Radar System |
| **KE-3** | Karakoram Eagle – 3 |
| **KIA** | Killed in Action |
| **KM** | Kilometer |
| **KPH** | Kilometers Per Hour |
| **LAADS** | Low Altitude Air Defence System |
| **LC** | Late Commitment (of weapons) |
| **LFE** | Large Force Employment |
| **LFEZ** | Large Force Employment Zone |
| **LIC** | Low Intensity Conflict |
| **LJF** | Least Jam Frequency |
| **LL** | Low Level |
| **LLR** | Low Looking Radar |
| **LO** | Liaison Officer |
| **LOMADS** | Low to Medium Level Air Defence Systems |
| **LOP** | Line of Position |
| **LOS** | Line of Sight |
| **LPR** | Low Power Radar |
| **LREZ** | Low Risk Engagement Zone |
| **LRR** | Long Range Radar |
| **LRU** | Line Replaceable Unit |
| **LS** | Late Scramble |

| Abbreviations | Description |
|---|---|
| **LUA** | Link Unit Address (AEW&C specific) |
| **M** | 1. Mix up (Mixing up of Interceptors with Hostile or Unknown tracks<br>2. Mach No (Speed of target is M 2.0)<br>3. Indicating Altitude in 1000s of feet (e.g. 5M means 5000ft) |
| **MA** | Mission Accomplished |
| **MA (AWR)** | Mission Accomplished After Weapon Release |
| **MA (BWR)** | Mission Accomplished Before Weapon Release |
| **MASINT** | Measurement and Signature Intelligence |
| **MAWS** | Missile Approach and Warning System |
| **MC** | Mission Commander |
| **MCC** | Mission Control Centre |
| **MCP** | Master Command Post (Army) |
| **MDA** | Missile Defended Area |
| **MEDD** | Manual Entry Data Device |
| **MEZ** | Missile Engagement Zone |
| **MFD** | Multi Functions Display |
| **MFFO** | Mixed Fighter Force Operations |
| **MGC** | Missile Gun Coordinator |
| **MHQ** | Maritime Headquarters |
| **MI** | Missed Interception |
| **MIA** | Missing in Action |
| **MIO** | Movement Identification Officer |
| **ML** | Missile Launcher |
| **MLC** | Movement liaison Clerk |
| **MLD** | Minimum Line of Detection |
| **MLI** | Minimum Line of Interception |
| **MODEM** | Modulation Demodulation |
| **MOU** | Mobile Observer Unit |
| **MR** | Missed Recognition |

| Abbreviations | Description |
|---|---|
| **MREZ** | Medium Risk Engagement Zone |
| **MRR** | Multi Role Radar / Medium Range Radar / Minimum Risk Route (NATO) |
| **MRTT** | Multi Role Transport Tanker |
| **MSL** | Mean Sea Level |
| **MTBCF** | Mean Time Between Critical Failure |
| **MTBF** | Mean Time Between Failure |
| **MTI** | Moving Target Indicator |
| **MTTR** | Mean Time To Repair |
| **N** | North |
| **NAI** | Non Automatic Initiation |
| **NAV** | Navigation |
| **NCTR** | Non Cooperative Target Recognition |
| **NCW** | Net Centric Warfare |
| **NFSL** | No Fighter in Suitable Location |
| **NLT** | Not Later Than |
| **NM** | Nautical Mile |
| **NOE** | Nap Of Earth |
| **NS** | No Scramble |
| **NUDET** | Nuclear Detonation Report |
| **NX** | Naval Executive (officer) |
| **OCA** | Offensive Counter Air |
| **OP** | Observation Post |
| **ORB** | Orbit/Orbiting |
| **ORBAT** | Order of Battle |
| **ORP** | Operational Readiness Platform / Point |
| **OTH** | Over the Horizon |
| **OTR** | On Target Requirement |
| **P** | Transportation or Para dropping |
| **PA** | (Aircraft track) Passed to Adjacent Sector |

| Abbreviations | Description |
|---|---|
| **PDS** | Passive Detection System |
| **PE** | Permanent Echoes / Personal Error (to be used, if applicable) |
| **PESA** | Passive Electronic Steered Array |
| **PGMs** | Precision Guided MUnitions |
| **PK** | Probability of Kill |
| **PPI** | Plan Position Indicator (Radar Scope) |
| **PRF** | Pulse Recurrence Frequency / Pulse Repetition Frequency |
| **PS** | Pilot Simulator |
| **QPQ** | Quid Pro Quo (Term used for some specific pre-defined targets) |
| **R** | Range |
| **R/T** | Radio Telephone |
| **RADEX** | Radar Data Extractor |
| **RAMP** | Recognized Air Maritime Picture |
| **RAP** | Recognized Air Picture |
| **RAREPS** | Radar Weather Reports |
| **RCI** | Radar Coverage Indicator |
| **RCS** | Radar Cross Section |
| **RECCE** | Reconnaissance |
| **RESCORTS** | Rescue Escorts |
| **RESCAP** | Rescue CAP |
| **RGPO** | Range Gate Pull Off |
| **ROA** | Radius of Actions |
| **ROE** | Rules of Engagement |
| **ROZ** | Restricted Operation Zones (NATO) |
| **RR** | Round Robin |
| **RRF** | Receiving and Reporting Flight |
| **RTB** | Return to Base |
| **RV** | Rendezvous Point |
| **RWR** | Radar Warning Receiver |

| Abbreviations | Description |
|---|---|
| **S** | Strike |
| **S/B** | Standby |
| **SAGW** | Surface to Air Guided Weapons |
| **SAM** | Surface to Air Missile |
| **SAM** | Surface to Air Missile |
| **SAR** | Search and Rescue areSynthetic Aperture Radar |
| **SARH** | Semi Active Radar Homing |
| **SC** | Senior Controller |
| **SC&R** | Surveillance Control and Reporting |
| **SCC** | Sub Control Centre |
| **SEAD** | Suppression of Enemy Air Defence |
| **SECOS** | Secure Electronic Protection Measures Communication System |
| **SHORAD** | Short Range Air Defence |
| **SIF** | Selective Identification Feature |
| **SIGINT** | Signal Intelligence |
| **SIGINT** | Signal Intelligence |
| **SIM** | Simulated |
| **SITREP** | Situation Report |
| **SLB** | Side Lobe Blanking |
| **SLC** | Side Lobe Clutter |
| **SLC** | Side Lobe Cancellation |
| **SMCC** | Sector Mission Control Centre |
| **SOA** | Sector Overlap Coverage Area |
| **SODM** | Shot on Dead Man |
| **SOP** | Standard Operating Procedure |
| **SP** | Self-Protection Jammer |
| **SPINS** | Special Instructions |
| **SPJ** | Self-Protection Jammer |
| **SROES** | Special Rules Of Engagement |

| Abbreviations | Description |
|---|---|
| **SRR** | Short Range Radar |
| **SSGW** | Surface to Surface Guided weapons |
| **SSM** | Surface to Surface Missile |
| **SSS** | Saab Surveillance System |
| **STS** | Surveillance and Tactical Station |
| **STZ** | Striker Threat Zone |
| **SUM** | Surface to Underwater Missile |
| **SURV** | Surveillance |
| **SWD** | Senior Weapon Director |
| **TACAN** | Tactical Air Navigation |
| **TAC-D** | Tactical Director |
| **TARCAP** | Target Area CAP |
| **TAS** | True Air Speed |
| **TBM** | Tactical / Theatre Ballistic Missile |
| **TCA** | Track Crossing Angle |
| **TD** | Tabular Display |
| **TDCC** | Terminal Defence Coordination Cell |
| **TDL** | Tactical Data link |
| **TDM** | Missile shot trashed due Maneuver |
| **TEL** | Transportable Erector Launcher |
| **TFC** | Tactical Fluid Control |
| **TFR** | Theater Force Reserve |
| **TH** | Tally Ho |
| **TKR** | Tracker |
| **TOF** | Time of Flight |
| **TOT** | Time Over Target |
| **TPTS** | Tactics, Procedures, Techniques and Solutions |
| **TRANSEC** | Transmission Security |
| **TWCC** | Terminal Weapon Control Centre |

| Abbreviations | Description |
|---|---|
| **TWEZ** | Tail Weapon Engagement Zone |
| **U** | Unidentified - A track classification to denote aircraft which has not yet been positively identified but most likely to be friendly |
| **U/S** | Unserviceable |
| **UAL** | Unit Authorization List |
| **UAV** | Unmanned Aerial Vehicles |
| **UCAV** | Unmanned Combat Aerial Vehicles |
| **UHF** | Ultra High Frequency |
| **UPS / MUPS** | Uninterrupted Power Supply / Mobile Uninterrupted Power Supply |
| **V** | A track classification denoting a track of special interest, like VIP etc. |
| **VA** | Vulnerable Area |
| **VCL** | Voice Communication Links |
| **VCP** | Voice Communication Panel |
| **VDU** | Video Distribution Unit |
| **VFR** | Visual Flight Rules |
| **VGPO** | Velocity Gate Pull Off |
| **VHF** | Very High Frequency |
| **VID** | Visual Identification |
| **VMC** | Visual Meteorological Conditions |
| **VP** | Vulnerable Point |
| **VPD** | Vertical Polar Diagram |
| **V-TWCC** | Vision-Terminal Weapon Control Centre |
| **W** | Airborne Early Warning |
| **WD** | Weapons Director (Used by AWACS Squadron) |
| **W/T** | Wireless Telegraphy |
| **WMD** | Weapons of Mass Destruction |
| **WO** | Wireless Observer |
| **WX** | Weather |
| **X-raid** | A track classification to show that an aircraft is most likely to be unfriendly or hostile |

## CHAPTER - 27

# GLOSSARY OF AIR DEFENCE TERMS

**Introduction**

1. There are certain terms commonly used in Air Defence day-to-day operations, reports and returns. These terms need to be standardized for common understanding by all concerned and smooth conduct of Air Defence operations.

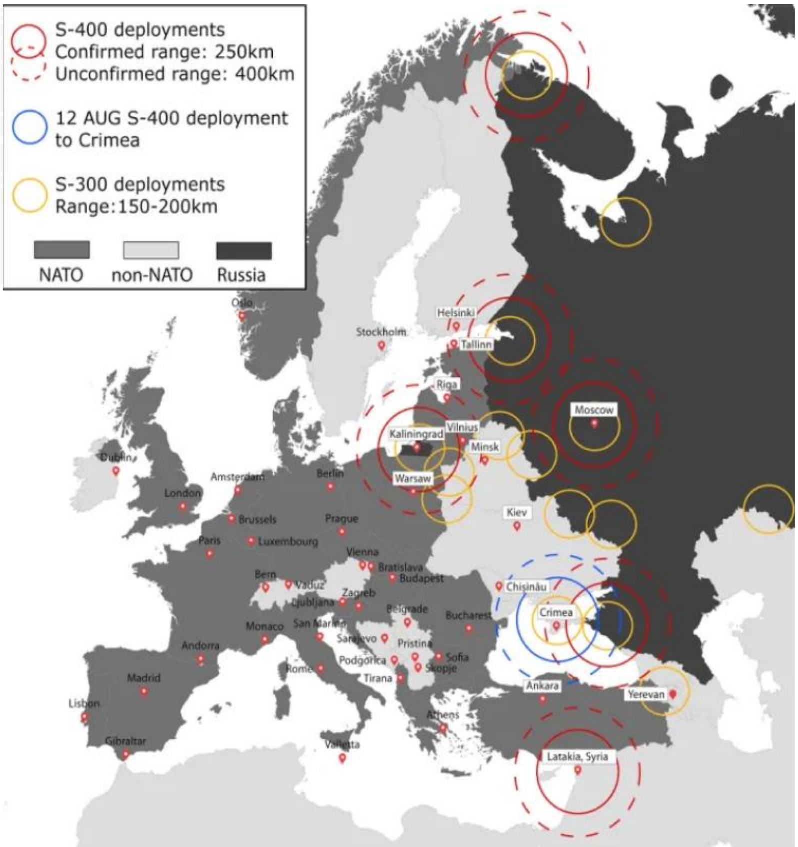
2. Most commonly used terms and their explanation is given as under:

| Term | Explanation |
|---|---|
| **Abeam** | At right angles to the longitudinal axis or to the side of the aircraft. |
| **Abortive (ABT)** | The forced discontinuance of a mission by a weapon for reasons other than enemy action. |
| **Absolute Altitude** | The height of an aircraft directly above the surface or terrain over which it is flying. |
| **Acceleration** | Rate of velocity increase. |
| **ACM** | Air Combat Manoeuvring - The coordinated application of basic fighter manoeuvres by two or more fighters in order to achieve a simulated kill against one or more target aircraft. ACM set-ups are pre-planned and controlled so that fighter flight member may practice and become experienced in all aspects of the fluid attack system. |
| **Advisory Control** | A mode of control in which the controlling agency has communications but no radar capability. |
| **AEW&C** | It is the evolution of AEW having control facilities like intercept aids, C2 consoles, and radio and data link communications. It can conduct interceptions along with providing requisite early warning. |
| **AEWACS** | AEW&C and AWACS when used to describe together is termed as AEWACS. (The abbreviation may not be internationally recognized by contemporary air forces). |

| Term | Explanation |
|---|---|
| **Airspace Violation** | All unauthorized flights by foreign aircraft / helicopters and movements of any unidentified aircraft over Pakistan territory, including the territorial waters, constitute air violation. |
| **Air Battle Management** | The management of operational and tactical activities within the battle space, based on intent, commands, directions, SPINS, TTPs and guidance given by appropriate authority (COC in case of PAF). |
| **Air Burst** | An explosion of a bomb or projectile above the surface as distinguished from an explosion on contact with surface or after penetration. |
| **Air Corridors** | A passage in the air for aircraft, such a passage either leading through anti-aircraft defence or established by international agreement. |
| **Air Defence** | All measures, other than active air defence, taken to (Passive) minimize the effects of hostile air action constitute the Passive Defence. These include the use of cover, concealment, camouflage, dispersal and protective construction. |
| **Air Defence Alert** | An operational status where the Air Defence forces such as fighters, guns SAMs, and radars / AEWACS are required to maintain higher than ordinary preparedness. |
| **Air Defence Committee (ADC)** | Air Defence Committee is an inter-services body under the chairmanship of Air Defence Commander, which decides policy matters of Air Defence. |
| **Air Defence Ground Environment (ADGE)** | It includes the ground radar coverage in a particular area, including low, medium and high-level radars. |
| **Air Defence Identification Zone (ADIZ)** | These are unilaterally declared designated areas of non-territorial airspace where states impose reporting obligations on civil and military aircraft for the purpose of national security. It may be defined as area of airspace over land or beyond territorial waters, extending upward from the surface or water, within which the ready identification, location and control of aircraft are required in the interest of national security. |

| Term | Explanation |
|---|---|
| **Air Mass** | A large body of homogeneous air over a wide area with characteristics which are approximately uniform in the horizontal plane and which change in atypical manner vertically. |
| **Air Sovereignty** | Sovereignty refers to the ownership of airspace. It is the nation's inherent right to exercise absolute control and authority over the airspace above its territory. |
| **Air Speed** | The speed of an aircraft relative to its surrounding air mass. Air speed may be expressed as:<br>(a) **Indicated**. As read from the Air Speed Indicator in the aircraft.<br>(b) **True**. Equivalent air-speed corrected for temperature or atmosphere density based upon ambient temperature. |
| **Air Surveillance** | The systematic observation of air, space, surface or subsurface areas, places, persons or things by visual, aural, electronic, photographic or other means is known as surveillance It is a continuous process, not oriented to a specific target. |
| **Air Tasking Order (ATO)** | A method used to task and disseminate to components, subordinate Units, and command and control agencies projected sorties, capabilities, and / or forces to targets and specific missions. |
| **Air Traffic Control Centre (ATCC)** | A Government agency responsible for providing a safe and expeditious flow of air traffic within an area. |
| **Airborne Equipment Failure** | This term is to be used, when applicable, as a reason for missed interception due to airborne equipment failure. |
| **Airborne Interception Radar (AI)** | A specially designed radar equipment installed in an interceptor aircraft used to assist or affect an air-to-air interception. |
| **Aircraft Scramble** | The process of directing immediate take-off of an aircraft within the shortest possible time from a ground alert condition for an Air Defence mission. |

| Term | Explanation |
|---|---|
| **Airspace Management (ASM)** | Airspace management (ASM) is a planning function with the primary objective of maximizing the utilization of available airspace by dynamic time sharing and at times, the segregation of airspace among various stakeholders based on their needs. |
| **Air-to-Surface Missile** | A missile carried by an aircraft for use against a surface target. |
| **All Weather Fighter** | A fighter aircraft with radar devices and other special equipment which enable it to intercept the target in dark or weather conditions which otherwise do not permit visual interception. |
| **Allied** | Track classification indicating the air-borne object to be of one's own or allied forces. |
| **Alternate Recovery Base** | A base other than the original scramble base, where the following turn around facilities are available. A runway of sufficient length to allow the landing and take-off of interceptors; fuel; starting Unit; minor ordnance; and priority facilities for the servicing of Air Defence interceptors returning from or engaged in Air Defence mission. |
| **Altitude** | The vertical distance of a point or object expressed in number of feet above mean sea level (MSL) 29.92 Hg barometric pressure) Altitude bands are:<br>Low:10K Medium:10-25K<br>High:25-40K Very High: >40K |
| **Anomalous Propagation** | An abnormal pattern formed by radar energy due to unusual stratification of temperature and moisture. |
| **Antenna Train Angle (ATA)** | The angle between the interceptor command heading and a line joining the interceptor and target positions measured from the interceptor heading. |
| **Anti-Aircraft Artillery (AAA)** | A term inclusive of all surface to air weapons with their related equipment, fired from the ground to strike at air borne objects. |

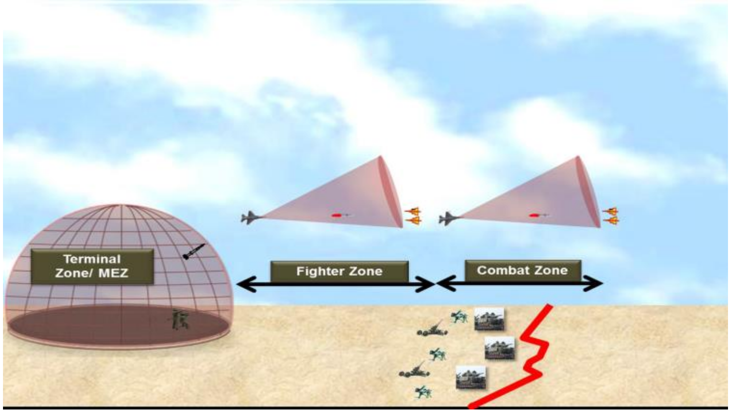
| Term | Explanation |
|---|---|
| **Anti-Access Area-Denial (A2 / AD)** | **Anti-Access (A2) -** Action, activity, or capability, usually long-range, designed to prevent an advancing enemy force from entering an operational area.<br><br>**Area-Denial (AD) -** Action, activity, or capability, usually short-range, designed to limit enemy force's freedom of action within an operational area.<br><br>Developing A2AD capabilities involves building an integrated, multi-layered system that includes forces and means capable of discovering and hitting targets from as longer distances as possible in all environments, including cyber domain. The integrated system could be composed of high-precision, long-range impact systems, advanced air defense systems, anti-ship capabilities in the seaside area, artillery systems and long-range missiles, as well as other capabilities placed on sea, submarine or aerial platforms.<br><br>**Russia's / NATO A2AD Capabilities**<br><br>.<br> |

| Term | Explanation |
|---|---|
| **Area Control Centre** | A Unit established to provide air traffic control service to controlled flights in control areas under its jurisdiction. |
| **Army Air Defence Executive (AADX)** | The Army Air Defence representative at ADOC / SMCC. |
| **Aspect Angle** | The angular difference between the longitudinal axis of the target and line of sight to interceptor, measured from the tail of the target. It is measured from 0˚ to 180° (Right or Left). |
| **Attack Option** | The attack option is defined by its associated attack geometry. Attack options are normally termed as head-on frontal, side beams and astern (tallish). |
| **AWACS** | It is an AEW&C with “Command Function”. An AWACS is capable of generating complete response in isolation, including detection, identification, ordering response and controlling fighters/SAMs. Examples are USAF/NATO E-3C Sentry. |
| **Azimuth** | The angular measurement in a horizontal plane with respect to the longitudinal axis of the aircraft. Also used to designate angle between the magnetic north and the line of sight of an airborne object. |
| **Azimuth Discrimination** | The ability of a radar to distinguish between two targets at the same range but slightly separated in azimuth. |

| Term | Explanation |
|---|---|
| **Balloon Control Order** | (a) **'DE-Camp' (Non Ops Position)**. On receipt of the order 'De-Camp' mobile balloons along with whole equipment and manpower are to move to the 'De-Camp' (Non-Ops) position ensuring that the hoisted balloons move about 300 to 400 feet away on right or left side from the edge of runway or 50 to 100 feet away from the taxi track (as advised by DATCO) in hoisted position. The balloons are then to be lowered immediately to the height of 100 feet AGL at Non-Ops position.<br>(b) **'Re-Trace' (Ops Position).** On receipt of the order 'Re- Trace' balloon trolleys are to be moved to the Ops position and parked on the centre line of the runway taxi track. The balloons are then to be hoisted to the designated heights. |
| **Bandit** | Aircraft identified as enemy according to identification matrix. |
| **Barrier CAP (BARCAP)** | A CAP that protects HVAA against a threat from known direction by putting a screen between the enemy threat and friendly aircraft / position. Examples are protection of tankers / AEWACS or any HVAA. |
| **Barrage Fire** | Anti-Aircraft Guns fire which is designed to fill a volume of space rather than aimed at a given target. Barrage fire is placed as curtain or barrier across the probable course of enemy aircraft. |
| **Barrage Jamming** | Simultaneous jamming of wide range of frequencies. |
| **Battle Damage Assessment (BDA)** | The estimate of damage composed of physical and functional damage assessment, as well as target system assessment, resulting from the application of lethal or nonlethal military force. |
| **Battle Staff** | A group of officers assigned to an Air Defence operation centre to conduct and supervise Air Defence operations. |
| **Beacon** | A moveable or stationary apparatus which transmits light, radio or radar beams for the purpose of orienting aircraft or providing navigational aid to aircraft. |

| Term | Explanation |
|---|---|
| **Beam Width** | The angle between the direction on either side of the axis, at which the intensity of radar beam drops to one half of the transmitter power value, along the axis of max radiation. |
| **Bearing** | The angle from the fighter or interceptor present position to the target position measured from North. North may be true, grid or magnetic, according to the reference system in use. By convention a bearing of North is 360. |
| **Blip** | Display of a received pulse on a Cathode Ray Tube. |
| **Blip to Scan Ratio (BSR)** | The ratio or the number of detections of a specific airborne object to the total number of radar scans (during period of observation). |
| **Bogey** | Unidentified aircraft. |
| **Break Away Heading** | A heading which the interceptor assumes immediately after completing the attack. This is to be accomplished by changing the heading through the shortest arc unless otherwise directed by the intercept controller. |
| **Brevity Code** | A system of code words used in radio or telephone transmissions to expedite and facilitate communications. In the brevity code, words are substituted for phrases to save time and for letters to ensure accuracy. |
| **Broadcast Control** | A type of interception control carried out by transmitting air intelligence data concerning enemy forces to interceptor pilot on R/T. Interceptor pilot uses this data to locate and attack the hostile forces independently. A mode of control that passes target information by referencing a designated location (Bulls eye) or series of locations. |
| **Calibration (CAL)** | Calibration is the process in which radio / radar or any other equipment is tested under operational conditions for assessment of its performance. |
| **CAP Point** | Pre-designated points over which the aircraft is to orbit while on a Combat Air Patrol mission. |

| Term | Explanation |
|---|---|
| **Cardinal Points** | The four main points / directions of the compass card making North, South, East and West. |
| **Cartesian Coordinates** | In space, three planes can be used to locate points by giving their distance from each of the planes along a line parallel to the intersection of the other two. If the planes are mutually perpendicular, these distances are called Cartesian coordinates. The three intersections of these three planes are usually labelled as the X-axis, the Y- axis and the Z-axis. Their common point is called the origin. |
| **CATA** | Collision Antenna Track Angle. The azimuth of the AI radar antenna when tracking (Locked on) a target that is on a collision course with the fighter or interceptor aircraft. |
| **Chaff** | A reflector type of radar countermeasures consisting of metallic strips that, when dropped from an airborne object, create false target returns that can obscure aircraft returns in the same / immediate area. |
| **Chaff Corridor** | Area in which chaff has been dropped. If dropped wide enough and for a period of time, the corridor can hide a large attack force. |
| **Clean** | Configuration of a fighter or interceptor without external fuel tanks. |
| **Clock Code** | Description of position using the aircraft as a reference: the nose is 12 O'clock and the tail is 6 O'clock. |
| **Close Air Support (CAS)** | Close air support missions are air operations against enemy land forces engaged with or in close proximity to friendly forces. CS directly aids friendly land forces in carrying out their assigned tasks. |
| **Close Control** | The control continuously exercised by an Air Defence controller, using radio and radar, over interceptor aircraft in a tactical air situation, into a position from which the target aircraft is within visual range or radar contact. A mode of control varying from providing vectors to providing complete assistance including altitude, speed and heading to the fighter aircraft. |
| **Clutter** | Unwanted signals, echoes, or images displayed, which interfere with the observation of desired signals. |

| Term | Explanation |
|---|---|
| **Collision Course** | A straight-line path that, if maintained will result in a collision of the fighter interceptor and the target at a specific point. |
| **Combat Air Patrol (CAP)** | A mission in which one or more fighter or interceptors are placed on patrol over an area usually for the purpose of intercepting and attacking hostile aircraft before they can reach their object. |
| **Combat Data** | The information which the various Air Defence Units and formations require for proper performance of their operational missions. It consists of air surveillance tactical action and status data. |
| **Combat Mach** | The computer-generated Mach speed at which a fighter or interceptor should fly during the attack phase of an interception. |
| **Combat Zone** | In combat zone both fighters and ground-based weapons operate jointly by following pre-defined procedures. Whenever friendly fighter aircraft are required to operate in this area, the SAM status would be downgraded by SMCC to allow fighter aircraft to operate in combat zone without any restriction.   |
| **Command and Control (C2)** | The exercise of authority and direction by a properly designated commander over assigned and attached forces in the accomplishment of the mission. |
| **Common Operating Picture (COP)** | It is a single identical display of relevant operational information shared by all C2 centres. It facilitates collaborative planning and combined execution and assists all echelons to achieve situational awareness. |

| Term | Explanation |
|---|---|
| **Composite Picture** | The depiction of data from multiple sensors at a single display station. It may or may not be a fused picture. |
| **Compulsory Reporting Track** | Track which requires compulsory reporting. |
| **Control Radar Coverage (CRC)** | An area or areas within Air Defence system where solid radar search, detection and tracking capabilities exist with-in which positive control of interceptors can be maintained throughout. |
| **Corner Velocity** | Minimum speed at which maximum instantaneous placard Gs can be attained. Minimum turn radius and maximum turn rate are achieved by combining corner velocity with maximum available 'Gs'. |
| **Correlation** | The process of associating related items with each other. Associating Air Movement Data with a pending track for identification. |
| **Cross-tell** | Communication of track data to adjacent operations centres. |
| **Darkness (DK)** | Missed interception due to limitations of lighting conditions. Data. A representation of facts or information, used as a basis for reckoning. |
| **Data Link** | Electronic equipment which permits automatic transmission of data in digital form. |
| **Dead Reckoning (DR)** | Determination of position by the application of direction, time and speed data to a previously known position of a track. |
| **Dead Space** | An area within the maximum range of a weapon, radar or observer, which cannot be covered by fire or observation from a particular point / position because of intervening obstacles, nature of ground or limitations of weapons. |
| **Decoy** | Any object or action, a dummy installation, a flying aircraft or a raid used to attract attention and draw enemy action away from real target or military operation. |

| Term | Explanation |
|---|---|
| **Default Value** | Programme setting used when no operator action or message entry has supplied data value required to allow programme operation. |
| **Deployment** | In strategic sense to re-locate forces to desired areas of action or staging bases. To tactically use, send or activate weapons, personnel or Units to combat positions, or area of operations. |
| **Desired Engagement Zone (DEZ)** | Based on ROE, operating environment and available assets it is the geographical area in the FAOR where the DCA aircraft will execute their primary game plan. Academically it can be defined as "Position in AOR where the CAP wants the theoretical merge to occur or enemy's wreckage to fall". |
| **Deviation/Deviate** | The lack of correlation between an aircraft and its flight plan in terms of time or space. |
| **Dew Point** | The temperature at which air, under a given pressure and moisture content becomes saturated with water vapour and at which moisture content condenses. |
| **Digital Data** | Information expressed in numerical values based upon some particular base numbering system. |
| **Direction Finding** | Determining the geographical position or bearing of an airborne or surface object by triangulation of lines of RF energy emitted from the object. |
| **Discontinue** | Discontinue to pass any information (on the track mentioned) till further order. |
| **Drift** | Angle between the heading and the track. |
| **Drop** | An action that causes a track to be dropped from the system, either automatically or as a result of an operator inserted instruction. |
| **Ducting** | The atmospheric phenomenon wherein certain or all rays in a radar beam are channelled through an atmospheric layer that is parallel to the earth's surface, or bending the rays towards the earth's surface. In each case, the rays will detect targets over the horizon. |

| Term | Explanation |
|---|---|
| **Early Warning (EW)** | Early notification about the launch or approach of unknown airborne object obtained by radar or other means. |
| **Echoes** | The reflected radar signals received from the object by a radar receiver. |
| **Effective Range** | Maximum distance at which a weapon is expected to fire accurately to inflict casualties or damage. |
| **Electronic Camouflage** | The use of radar absorbent material to reduce the radar echoing properties of any surface. |
| **Electronic Deception** | The deliberate radiation, re-radiation, alteration of absorption of electromagnetic waves in a manner intended to mislead the enemy in the interpretation of data received by his electronic equipment or to present false indications to his electronic system. |
| **Electronic Warfare** | Military action involving the use of electromagnetic energy to determine, exploit, prevent or reduce an enemy's effective use of radiated electromagnetic energy and action taken to ensure our own effective use of electromagnetic spectrum. |
| **Element** | Two aircraft, when such aircraft are part of a flight of four aircraft. |
| **EMCON** | It is the management of all electromagnetic radiations of a force or Unit to obtain the maximum tactical advantage. |
| **Emergency Recovery Base** | Any landing base where the interceptors can land safely. This base is only used when the interceptors are unable to reach either the original or alternate recovery base due to unforeseen cause. |
| **E-SHORADs** | Extended – Short range AD systems capable to engage airborne targets up to 15 km. |
| **Expedite** | As quickly as possible "Hurry Up". |
| **Extrapolate** | An action taken by the operator to cause the tracking programme to ignore radar data correlation for a period of time and project the track position according to the current velocity and heading. |

| Term | Explanation |
|---|---|
| **Extrapolated Track** | The status of a track that is kept in the system although lacking sufficient data for correlation purposes, or that is being predicted ahead without correlation. |
| **Faded Prior to Interception (FPI)** | This term is to be used as applicable in Missed Interception Reports, when the radar tracking time, after the scramble, is less than the time necessary for the interceptors to proceed from their scramble point to the point of planned interception. |
| **False Return** | Radar echo (return) based on a condition other than actual presence of an aircraft, or shown at a position other than actual aircraft location. |
| **Fighter Area of Responsibility (FAOR)** | An area, bounded by coordinates or geographical features, whose defense is the sole responsibility of the fighters against adversary fighters, cruise missiles, UAVs etc. |
| **Fighter Cover** | The maintenance of a number of fighter aircraft over a specified area or force for the purpose of repelling hostile air activities. |
| **Fighter Exclusive Zone (FEZ)** | Area from MEZ till combat area constitutes Fighter Exclusive Zone (FEZ). In this segment of airspace, fighters are free to operate at all altitudes without any restriction. FEZ operations usually take place in airspace above and beyond the engagement ranges of GBADS of short-range Air Defence systems. |
| **Fighter Sweep** | An offensive mission by fighter aircraft to seek out and destroy enemy aircraft or targets of opportUnity in an allotted area of operations. |
| **Final Attack Heading** | The final heading given to the interceptor pilot to bring him to the position to launch his weapons. |
| **Final Turn Point** | That point in space at which the interceptor is to start his turn to final attack heading so as to roll out in position for a stern attack. |
| **Fire Control System** | A system including radar equipment installed in interceptor aircraft used to assist or affect an air to air interception. |

| Term | Explanation |
|---|---|
| **Fire Power** | The amount of fire which may be delivered by a Unit or a weapon. |
| **Flexible Use of Airspace (FUA)** | This concept is based on the basic principle that airspace should not be designated as either pure civil or military airspace, but rather be considered as one continuum in which all user requirements have to be accommodated to the extent possible. One step towards an effective implementation of the flexible use of airspace (FUA) concept would be to allow civilian users temporary access to military restricted and reserved airspace for optimum use of the airspace. |
| **Flight Following Service** | The process of maintaining contact / reporting surveillance information on a specified aircraft for the purpose of determining en-route progress and / or flight termination. |
| **Flight Information Centre (FIC)** | A Unit established to provide flight information service and alerting service. |
| **Flight Information Region (FIR)** | An airspace of defined dimensions within which flight information service and alerting service are provided. |
| **Flight Information Service** | A service provided for the purpose of giving advice and information useful for the safe and efficient conduct of flight. |
| **Flight Plan** | A statement of the plan and procedure to be followed on a particular flight submitted by the pilot to the ATC. The statement is then forwarded to the movement identification section of the SMCC. |
| **Flight Plan Correlation** | Association of a track with a known flight plan. |
| **Flight Visibility** | The average forward horizontal distance from the cockpit of an aircraft in flight at which prominent unlighted object can be seen or identified by day or prominent lighted object may be seen and identified by night. |

| Term | Explanation |
|---|---|
| **Forward Line of Troops (FLOT)** | The position of own troops passed by forward Army formations in form of a SECRET message at designated timings to Higher formations. |
| **Force Multiplier** | It is the capability that when added to or employed by a combat force, significantly increases the combat potential of that force and thus enhances the probability of successful mission accomplishment. Examples include, but not limited to AEW&Cs, AWACS, AAR and EW Platforms. |
| **Forward Air Controller (FAC)** | An officer (aviator or a qualified officer of the ground forces) who, from a forward ground or air position, controls aircraft engaged in close air support of ground troops. |
| **Forward Airfield** | An airfield used to support tactical operations without establishing full support facilities. |
| **Forward Area** | An area in proximity to combat. |
| **Fratricide** | The accidental destruction of own, allied or friendly forces. |
| **Frequency** | The number of cycles occurring in a given period of time:<br>(a) High Frequency (HF) 3-30 MHz<br>(b) Very High Frequency (VHF) 30-300 MHz |
| **Functional check** | The battle staff will declare Ops equipment on “functional check status” in the following two conditions.<br>(a) After completion of scheduled maintenance<br>(b) On new deployment / redeployment<br>The term indicates that the equipment is currently being assessed for its operational performance / status. After functional check, the equipment will be declared S/A along with its status or U/S.<br>The length of time period for functional check is at the discretion of battle staff, unless specified by higher HQs. The time should be kept as short as possible. |

| Term | Explanation |
|---|---|
| **Fused Picture** | A composite picture with unified / singular depiction of data. When different tracks / plots of a composite picture are merged, it forms a fused picture. |
| **Gap Filler Radar** | Radar which is used to supplement the coverage of long-range radar in areas where coverage is inadequate. |
| **GEOREF** | Refers to a point represented in the World Geographic Reference Co-ordinate System in which the world is divided into major and minor geographic areas represented by alphanumeric designations. |
| **Grid** | Two sets of parallel lines intersecting at right angles and forming squares. The grid is super imposed on maps, charts and other similar representations of the earth surface in an accurate and consistent manner to permit identification of ground locations. |
| **Ground Controlled Approach (GCA)** | The techniques or procedures for bringing down, through use of both surveillance and precision approach radars, an aircraft during its approach so as to place it in a position for landing. |
| **Ground Controlled Interceptions (GCI)** | A system employing radar techniques which will permit. Control of friendly aircraft or guided missiles for the purpose of effecting interception. |
| **Ground Environment** | A network of ground radars, associated communications and control facilities which provide for the detection, identification, interception and engagement of airborne vehicles. |
| **Ground Equipment Failure (GEF)** | This term is used, when applicable, with Missed Interception Report (MI GEF). |
| **Guidance System (Missile)** | A system which evaluates flight information, and correlates it with target data, determines the desired flight path of the missile and communicates the necessary commands to the missile flight control system. |

| Term | Explanation |
|---|---|
| **Guided Missile** | A missile that is directed to its target while in flight either by a pre-set or self-reacting device within the missile or by radio command out-side the missile. |
| **Hand over** | The process of transferring the air surveillance and control responsibilities for interceptions, between adjacent Units of the same operational level. |
| **Hard Turn** | A turn, the intensity of which is normally governed by the position of a bandit, but generally performed at a briefed 'G' / AOA. |
| **Height Range Indicator** | A special radar scope which indicates the altitude as well as the range of aircraft. |
| **High Value Airborne Asset (HVAA)** | Airborne assets that are so important to a fighting force, that the loss of even one could prove extremely detrimental to friendly forces and their war-fighting capabilities, or provide an enemy with significant propaganda and morale boosting value. Examples are AEW&C, EA aircraft, special ops platforms etc. |
| **HIMADs** | High to Medium AD systems capable of engaging targets 100 kms and beyond generally up to 400 kms. |
| **Horizontal Polar Diagram** | A graphic representation of radar detection ranges at various azimuth is called Horizontal Polar Diagram (HPD). |
| **Identification** | The determination of an aircraft's classification by any means or combination of means including visual recognition, flight plan correlation, electronic interrogation, and track behaviour etc. |
| **Identification Friend or Foe (IFF)** | An electronic identification system consisting of a ground interrogator and an airborne transponder, used to identify friendly aircraft Identified Friendly This term is to be used when reporting results of scramble Prior to action. It is applicable when the reported track is identified Interception (IPI) friendly, by means available to the Air Defence System, prior to the completion of interception. When no weapons have been committed against the track this term is not to be reported. |
| **Initial Point (IP)** | The first point or beginning of the active flight plan or exercise flight route. |

| Term | Explanation |
|---|---|
| **Integrated Air Defence (IAD)** | An integrated Air Defence system is the one in which multiple sensors as well as anti-aircraft weapons (SAMs, air superiority fighters and interceptors) are under a common system of command, control, communications and intelligence (C3I). With modern long range / strategic SAMs such as S-400/300 and Chinese HQ-9, employed under IAD, they can threaten to restrict freedom of maneuver well outside their own land borders. Both Russia and China rely heavily on their respective IADS to provide the core of their A2/AD challenge to Western military freedom of action, by contesting the latter's ability to gain and maintain air superiority near their borders. |
| **Intercept Controller (IC)** | An officer in the operations centre responsible to the Assistant Chief Controller or Weapon Assignment Officer for directing or monitoring assigned fighters or interceptors on identification or attack intercept missions. In some systems the Intercept Controller is referred to as either a Fighter Controller an Intercept Director a Weapon Director. |
| **Interception** | To effect visual or radar contact with an airborne object by an interceptor aircraft for the purpose of interrogation, intervention or engagement. |
| **Interceptor** | An aircraft primarily designed or used for interception. Intercept Point. The point in space towards which an interceptor is vectored to complete an interception. |
| **Interface** | The common boundary between two or more systems or parts of single system and the functional intersystem relationships that influence mission accomplishments. |
| **International Airspace** | The airspace over contiguous zones, exclusive economic zones, high seas and territory not subject to sovereignty of any state. |
| **International Civil Aviation Organization (ICAO)** | A civilian organization with headquarters at Montreal, Canada that is responsible for establishing all flight rules and standards for international air traffic. Most countries apply ICAO rules to their own airspace and airfield. |
| **Interrogation** | Interrogation is an action aimed at determining the identity of the intercepted aircraft. |

| Term | Explanation |
|---|---|
| **In-Trail Turn** | A turn used while directing a flight in a combat trail formation. An in-trail turn is used to change the direction of a flight while maintaining an in-trail position. |
| **Joint Engagement Zone (JEZ)** | Volume of airspace in the theatre with specific dimensions (latitude / longitude along with min / max altitudes) in which multiple Air Defence weapons systems of one or more services are employed simultaneously and in concert with each other to engage enemy airpower. |
| **Kill Box** | A segment of sanitized airspace which is free of friendly aircraft. The airspace can be defined for a period of time from the surface level to infinite altitude. |
| **Late Scramble (LS)** | Missed interception as a result of being airborne late (MI LS) ie, when the difference between the scramble time and the airborne time is greater than the state of readiness of the interceptors. |
| **Lateral Displacement** | The perpendicular distance between the interceptor's Displacement position and the target's projected flight path. |
| **Layered Integrated Air Defence System** | IADS entails that all sensors and shooters are under single and unified C2; however, its efficacy can be enhanced if layered IADS is exercised. Layered IADS provides the requisite flexibility to engage hostile airborne object even if an enemy attempt to exploit weakness of one system; still other system can successfully engage the target. |
| **Line of Control (LoC)** | The line of Control separates Pakistan and India in the disputed territory of Jammu and Kashmir, running from the confluence of the river Tawi and Chenab in the south and thence in the north to point NJ 9842. Detailed description of the Line of Control is given in “Agreement of 11 December 1972 on delineation of Line of Control in Jammu and Kashmir resulting from cease fire on 17 December 1971 in accordance with Shimla Agreement of 2nd July 1972” and is marked on official maps duly signed by the representatives of Pakistan and India. |

| Term | Explanation |
|---|---|
| **Line of Position (LOP)** | A line of constant bearing with its origin (target) moving at a constant rate and direction, on which an object (Interceptor) having the correct speed and heading will remain. |
| **Line replaceable Unit (LRU)** | An essential support item which can be removed and replaced at field level in a relatively short time to restore the U/S item. LRUs improve maintenance operations because they can be stocked and replaced quickly from inventory. |
| **LOMADs** | Low to Medium AD systems capable of engaging airborne targets up to 50 to 75 km. |
| **Long Range Radar (LRR)** | A radar that possesses a detection range in excess of approximately 150 NM (278 Km). |
| **Lost** | A track status indicating that insufficient radar data is available to sustain continuous correlation. |
| **Mach Number** | The speed of a moving body as measured against the speed of sound at the same height. |
| **Magnetic North** | The direction along the magnetic isogonal. Most compasses based on magnetic attraction principles are oriented to magnetic North. The angular difference between true and magnetic north is expressed in degrees of magnetic variation. |
| **Manual Entry Data Device (MEDD)** | A computer operator station that includes display and a key board. The operator is capable of transmitting manually inserted data into the computer programme. |
| **Mass Raid (MRD)** | This term is to be used to indicate a number of individual tracks in close proximity, heading for the same area/VP. |
| **Maximum Turning** | Turn performance at which the maximum obtainable turn rate in number of degrees/second is achieved. It is generally associated with comer velocity. |

| Term | Explanation |
|---|---|
| **Mean Time Between Critical Failure (MTBCF)** | It is also called as mission reliability and it is a subset of MTBF because it only counts those failures that results in a mission abort or mission failure. To calculate MTBCF, “critical failure” for that equipment /mission should first be defined. |
| **Mean Time Between Failure(MTBF)** | It is a maintenance metric indicating the duration that an equipment operates without disturbances. It is measured in hours. It is important to note that MTBF is only applicable to reparable items. |
| **Mean Time to Failure (MTTF)** | It is maintenance metric indicating the duration that equipment operates before it totally breaks down. While MTBF is used for repairable items, MTTF is used for items that are not repairable, thus it is assumed that once the MTTF is reached, the item has reached its maximum hours of service. |
| **Mean Time To Repair (MTTR)** | It represents the average time required to repair a failed component or device. |
| **Medium Range Radar (MRR)** | A radar that possesses a detection range from approximately 60 NM (111 Km) to approximately 150 NM (278 Km). |
| **Minimum Radius of Turn** | Turn radius generated at any altitude when maximum of obtainable 'G' is attained at the lowest possible airspeed. |
| **Minimum Line of Detection (MLD)** | The minimum distance from a VP at which an "Enemy" aircraft is to be detected, so as to provide enough time for tactical action and for disposing it off at safe distance from VP, is known as the MLD. |
| **Minimum Line of Interception (MLI)** | This is the minimum distance from a VP / VA where the enemy aircraft is to be intercepted / destroyed. This is so, because a loaded bomber if crashes over a target after interception, it will achieve its purpose despite successful interception. |

| Term | Explanation |
|---|---|
| **Missed Interception (MI)** | This term is to be used to indicate that the attempted interception did not result in visual or electronic contact with the target or the target aircraft was beyond the maximum effective range of the interceptor's weapons or did not meet the criteria for MA. The reason for failing to intercept is to be indicated e.g. MIAEF (Missed Interception due to Airborne Equipment Failures etc). |
| **Missed Recognition** | This term is to be used to report that although the interception of an unknown track was successful, weather or darkness precluded positive identification. Example, MRDK or MRWX. |
| **Missile Engagement Zone (MEZ)** | Terminal weapon zone is Missile Exclusive are (MEZ), where only SAM would engage the targets while own fighter would stay outside this zone. MEZ are ideal for point defence of critical assets, protection of maneuvering Units in the forward area, and area coverage of rear operations. |
| **Mission** | The task assigned to a fighter or interceptor or flight. |
| **Mission Accomplished (MA)** | This term is used to denote successful accomplishment of a mission. It is used to specify the result of interception by SAM, AAD guns or Interceptors:<br>(a) MA 1 MIG-21.<br>(b) MA 1 Jaguar (BWR)<br>(c) MA 1 MIG-27 (AWR)<br>(d) MA 2 MIG-27 (DWR)<br>(e) MA (No Claim)*<br>*It applies to interceptors only where after sighting the interceptors in threatening posture, enemy aircraft abandoned the mission or when interceptors were suitably placed but shooting could not take place due to enemy's timely evasive action and exit. |
| **Mixed Fighter Force Operations** | The employment of AD fighters of varying capability (mix of active missile shooters and less capable aircraft), coordinating tactics, procedures and communications with an aim to increase the combat effectiveness of entire AD force. |

| Term | Explanation |
|---|---|
| **Mix-up** | This is a track classification to indicate mixing up of interceptor and hostile or unknown aircraft in an engagement. This is not to be confused with the merged tracks where only the tracks of aircraft get mixed up on the radar scope even when the aircraft were flying at different altitudes at some position. |
| **Mode** | The method of operation used to conduct Air Defence of a specific area. (Auto-mode or Manual Mode). |
| **Nap of Earth Flying** | Very low altitude flight course used by military aircraft to avoid enemy detection. The geographical features / terrain are used as cover. |
| **Noise** | Unwanted sound or electronic disturbances introduced in the communication or radar system. |
| **Non Scheduled Maintenance (NSMC)** | Maintenance of any Ops equipment which does not take place as per pre-defined schedule, rather it is a preventive maintenance measure, to prevent any future major un-serviceability of the equipment. |
| **Off-Center** | The process by which a selected portion of a display is presented at the centre of the situation Information or Plan Position Indicator display. |
| **Operational Control** | Those functions of command involving the assignment of tasks, the designation of objectives and the authoritative direction necessary to accomplish mission. This control shall be exercised by Air Headquarters unless specified otherwise. Operational Control over Bases and Sector HQs is exercised by the Air Headquarters through RACs and ADC respectively. |
| **Operational Status** | The ability of a Unit to carry out its assigned mission at a particular time. The Ops status of the Unit can be Status I, II or III. |
| **Ops Clearance** | When any Ops equipment of PAF AD network is giving live AD operations, situation may arise where its operations may be hampered due to any test / trial activity. In all such cases, "Ops clearance" is to be sought by the battle staff from higher echelons. |
| **Orbit Point** | A geographically defined reference point over land or water, used for stationing airborne aircraft. |

| Term | Explanation |
|---|---|
| **Out of Action** | This term is to be used for indicating Ack Ack or Missile Units that are non-operational due to enemy action. |
| **Out of Range** | This term is to be used, when applicable, as a reason for **'No Scramble'** when detection line is such that the target is at a position and heading which does not permit a successful interception. |
| **Outbound Tracks** | Tracks with a movement away from the reporting radar complex are to be classified as outbound. |
| **Overlap Radar** | A radar located in one sector whose area of useful radar coverage includes a portion of another radar coverage within the same sector or of another sector. |
| **Permanent Echo** | A radar echo or return from a fixed object. |
| **Plan Position Indicator (PPI)** | The cathode ray indicator with a circular screen used by the Air Defence system for surveillance and interceptions. This type of radar scope gives a true picture of the position of a target in relation to the radar. |
| **Plot** | The visual display of the geographical location of an airborne object. |
| **Point Defence** | Its purpose is the defence of specified individual geographical area, city, or vital installation. Its distinguishing feature is, its limited area of defence. |
| **Point of No Return** | The farthest or critical point along an aircraft track beyond which its endurance will not permit return to its own or some other associated base on its own fuel supply. |
| **Polar Grid** | A type of circular grid wherein range and azimuth are represented from a central reference point. |
| **Positioning Aid** | A device such as an azimuth protector, an inter scan line, cursor or an overlay that is used in conjunction with a radar scope to conduct an interception. |

| Term | Explanation |
|---|---|
| **Positive Control** | The operation of air traffic within a radar ground control environment in which positive identification, tracking and direction of aircraft within airspace is conducted safely, by an agency having the capability, authority and responsibility therein. |
| **Pulse Length** | The time in microseconds during which a single pulse is transmitted in a pulse modulated radar system. Also called as pulse width. |
| **Radar Coverage** | The effective coverage of an area or airspace provided by a radar set or network, within which objects can be detected. |
| **Radar Cross Section (RCS)** | Measure of the targets ability to reflect back radar signals in the direction of radar receiver, measured in meter Units. |
| **Radar Shadow** | Area of reduced pick-up or no pick-up due to irregular terrain, cloud or other obstructions. |
| **Radar Frequency Bands** | **The frequency range of bands is** :<br>VHF 30 - 300 MHz<br>UHF 300 - 1000 -“-<br>P 230 - 1000 -“-<br>L 1000 - 2000 -“-<br>S 2000 - 4000 -“-<br>C 4000 - 8000 -“-<br>X 8000 - 12500 -“-<br>Ku 12.5 - 18 GHz<br>K 18 - 26.5 -“-<br>KA 26.5 - 40 -“-<br>Millimetre More than 40 GHz |

| Term | Explanation |
|---|---|
| **Radius of Actions** | The maximum distance a ship, aircraft, or vehicle can travel away from its base along a given course with normal combat load and return without refueling, allowing for all safety and operating factors. |
| **Radome** | Ruberoid sphere (dome) for protecting radar antennas from weather / Climate effects. |
| **Raid** | A consolidation of a number of unknown or enemy tracks, having the same classification, into a single display. |
| **Range** | Distance from a given reference point, usually a radar site, fighter, or the operations centre. |
| **Range Resolution** | The ability of radar set to distinguish between two targets at the same azimuth but slightly separated in range. |
| **Rate of Closure** | The speed with which two moving objects close the distance between them. |
| **Rate of Overtake** | The speed advantage with which one aircraft approach another, directly from behind or within 70º of that position. |
| **Raw Radar Data** | Radar data before being digitalized for transmission or display. In digital radars it is no longer possible for an operator to display radar data before it is digitalized. In these cases, raw radar data is referred to that data before being processed by a digital target extractor. |
| **Reclassified (RC)** | This term is to be used to indicate change in the classification of a track |
| **Recognition** | The visual Determination of type, nationality, ownership and number of an intercepted target aircraft |

| Term | Explanation |
|---|---|
| **Reaction Time** | Surveillance Delay + System Delay + Tactical Action Time, as explained below:<br><br>(a) **Surveillance Delay**. Time taken for detection / observation, confirmation of the track and its upward telling to SMCC.<br><br>(b) **System Delay**. Time taken for plotting filtering and identification of the track at SMCC.<br><br>(c) **Tactical Action Time**. Time taken from ordering scramble till the interceptors take off.<br><br>OR<br><br>Time taken from ordering the diversion of interceptors till they are made suitably available to the intercept controller.<br><br>OR<br><br>Time taken from ordering the Air Defence Artillery till it is activated. |
| **Recognized Air Picture (RAP)** | An electronically produced display complied from a variety of sources (active and passive sensors). It covers three-dimensional volume of interest in which all detected air contacts have been evaluated against specific threat parameters and recognition criteria and then assigned an identification category and track number. |
| **Reconnaissance** | Mission designator for tactical aircraft on a mission of observation of an area either visually or with the aid of photography or electronic devices. |
| **Restricted Area** | A specified area of airspace designated for other than air traffic control purposes, over which the flight is restricted in accordance with certain specific conditions / rules |
| **Retrograde** | The act of withdrawing from a present anchor position in response to a threat, to accomplish a quick landing, while preserving threat bubble. AEW&C mission support would no longer be available and AEW&C survival gets priority over mission support. |
| **Rules of Engagement** | The Air Defence rules of engagement or directives that describe the circumstances under which weapons can fire at an aircraft. |

| Term | Explanation |
|---|---|
| **Safety Zone** | A zone through which aircraft movement is safe. This may be indicated with height band or area-wise. |
| **Salvo Fire** | Capability of a SAM system to launch multiple numbers of missiles (two or more) on a single target in one go. |
| **Scheduled Maintenance** | Maintenance of any ops equipment which takes places as pre-defined schedule (including time period / date). |
| **Scope** | Face of the cathode ray tube on which radar echoes are displayed. |
| **Screening Angle** | This is the angle formed by the inter-section of the radar line of sight and the horizontal plane at the radar antenna. |
| **SEAD / DEAD** | SEAD operations aim to destroy, degrade or neutralize components of enemy surface-to-Air Defence systems which may include AD sensors, SAMs and AAAs. Surface-to-air elements may be neutralized by hard or soft kill methods. |
| **Search and Rescue (SAR)** | Mission designation for operations essential to the recovery of downed airmen. This mission is also referred to as a Rescue Mission. |
| **Second Time Around Echoes** | When the transmitted pulse is emitted by the radar, a sufficient length of time must elapse to allow any echo signals to return and be detected before the next pulse may be transmitted. Echoes that arrive after the transmission of the next pulse are called "Second Time Around Echoes". |
| **Selective Identification Feature (SIF)** | A modification of the basic IFF system. Two basic components, the coder and the decoder have been added. |
| **SHORADs** | Short Range AD systems capable to engage airborne targets up to 8km. |
| **Simulation** | A method of presenting an artificially produced air picture of set of conditions representative of a live (real) situation for training or programme test purpose. |
| **Simultaneous Engagement** | The engagement of hostile targets by a combination of Air Defence weapons at the same time. |

| Term | Explanation |
|---|---|
| **Slant Range** | The line of sight distance between two points not at the same elevation. |
| **Standard Operating Procedure** | An approved and standing method or procedure for accomplishing a task. |
| **Status Data** | Information that includes the number and availability of Air Defence weapons, sensors and weather reports in air defence. |
| **Strike** | Aircraft that are directed to destroy enemy ground or sea targets. |
| **Sub-Refraction** | A less than normal bending downward or refraction of the radar energy (i.e. upward bending of RF energy from the calculated path). |
| **Strategic Orbit Point** | A geographical location of continuous radar coverage to which interceptors are ordered so as to place them in an advantageous position for subsequently attacking hostile targets. As opposed to placing aircraft on Combat Air Patrol. Interceptors are ordered to the strategic orbit point based upon the latest available intelligence, so that the interceptors are quickly directed on an attack vector when a hostile track is established or re-established. |
| **Strobe** | Report of numerous radar data along a single azimuth at a given time, usually caused by electronic signals emitted by a jammer aircraft or a radiation source such as the sun. If the strobe was created by ECM it is referred to as an ECM strobe. If the strobe was created by the sun it is referred to as a sun strobe. |
| **Surface-to-Air Missile (SAM)** | Missiles capable of striking aerial targets from the ground. |
| **Surveillance and Tactical station (STS)** | Term used to describe the working station of SMCC (includes screens and computer). |
| **Symbology** | The characters or symbols (letters, numerals, pictures, etc.) used to present information on displays. |

| Term | Explanation |
|---|---|
| **Essentially required Tabular Display (TD)** | This display is used for displaying secondary Air Defence data that is not for displaying on the Situation information Display (e.g. detailed weapon status, weather status, amplified track status, radar status, etc). It is displayed in a tabular type format. This display is sometimes referred to as Auxiliary Readout Display (ARD). |
| **Tactical Action** | The action initiated or decision made on an unknown or hostile surveillance track, so as to achieve the objective. |
| **Tactical Action Data** | Information that includes a report of action taken or decision made on air surveillance data. |
| **Tactical Control** | A mode of Control that provides information in bearing, range, altitude and aspect (BRAA) references. |
| **Tactical Decision (TD)** | The term is to be used to indicate the decision taken in committing the air-defence weapons against unknown or hostile airborne weapons. |
| **Tactical Fluid Control (TFC)** | Using different levels of control w.r.t the situation wherever applicable; controlling agency is required to have capability of exercising whole continuum of control. |
| **Tactical Report** | A report which gives the tactical action taken against a track requiring tactical action. |
| **Tactical Reposition (by AEWACs)** | The act of adjusting anchor position/parameters to maintain separation from threat, or to optimize senor coverage. Mission support is more important than tactical maneuvering. |
| **Tactical Retreat (by AEWACs)** | The act on continuing present mission support while extending range from a perceived threat. In simple it is range preservation with mission support. Mission support is as important as tactical maneuvering. |
| **Tally Ho** | This term is to be used by a fighter interceptor crew to indicate that the aircraft which has been sighted is presumably the one that is intended to be intercepted, that the interceptor pilot is going in for the attack and does not need any more instructions from the GCI controller. Controller is to listen out and monitor the engagement closely for repositioning, if required. |

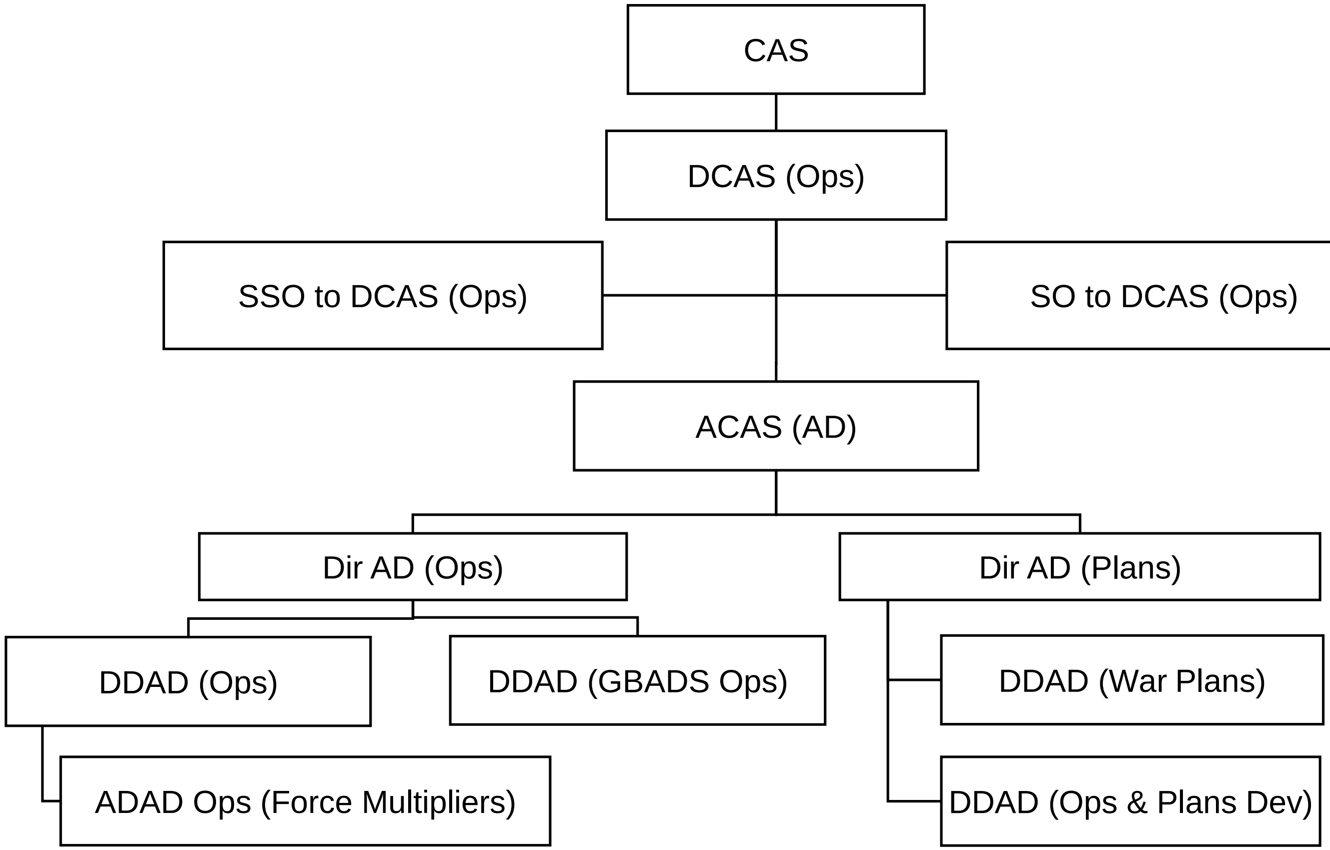
| Term | Explanation |
|---|---|
| **TARCAP** | An offensive tactic employed by the Sweep / Escorts of OCA package after achieving Air superiority in an enemy territory to ensure survival of the friendly strikes from enemy aerial activity becoming, or likely to become a threat during and after their attack on the target. |
| **Target** | A friendly aircraft simulating as hostile during an Air Defence exercise in order to exercise a part or whole of the Air Defence system. |
| **National Airspace** | Airspace above land territory and internal, archipelagic, and territorial waters. |
| **Territorial Waters** | Waters beyond land and internal waters to the adjacent belt of sea. Its territorial waters limit is up to 12 NMs measured from the base line of the coast. |
| **Telling** | The process of communicating air surveillance information between Air Defence Units:<br>(a) **Downward Telling**. Telling from a higher to a lower level of command.<br>(b) **Lateral Telling**. Telling between adjacent Air Defence Units at the same operational level.<br>(c) **Upward Telling**. Telling from lower to higher level of command. |
| **Threat Bubble** | Pre-defined range between own aircraft and threat at which own aircraft must undertake defensive measure to ensure its survival e.g. Minimum range between an AEWACs and a threat, at which an AEWACs, following a defensive profile, would be able to land at a retrograde airfield while being chased. |
| **Threat Warning Information (TWI)** | Information, other than routine surveillance reports, which gives location, magnitude and progress of any threat. These warnings are passed by the most expeditious means to higher Air Defence Units. |
| **Track** | A display representing the position of an airborne object alongwith its strength, height, speed and heading. |
| **Track Classification** | The classification of tracks that requires identification under current directives, tracks are to be classified as Hostile, Unknown, and Allied etc as given in the operating procedures of this manual. |

| Term | Explanation |
|---|---|
| **Track Data** | Information pertinent to a track. |
| **Transponder** | A transmitter-receiver capable of accepting the electronic challenge of an interrogator and automatically giving an appropriate reply. |
| **Triangulation** | The method of determining the position of aircraft by obtaining bearings of the moving object with reference to two or more fixed stations at a known distance apart. |
| **True North** | The direction along the global meridians of longitude. |
| **Turn around** | The re-fuelling or rearming of an aircraft following a mission. (Unclassified Track. The term applied to a track during the period between establishment of track and its classification). |
| **Unauthorized Flight** | When foreign aircraft, permitted to fly over Pakistan, deviate from their flight plan, their flight will be considered unauthorized. Such deviations will occur when these aircraft fly: -<br>(a) On dates other than those permitted.<br>(b) Three hours before or after the scheduled time.<br>(c) In areas outside designated air corridors.<br>(d) In excess of the number allowed.<br>(e) Make other unexpected departures from the given flight plan. |
| **Under check** | The battle staff will declare an Ops equipment on “under check status” in the following two conditions:<br>(a) After completion of non-scheduled maintenance.<br>(b) After completion of maintenance, when an equipment was declared unserviceable.<br>(c) The term indicates that the equipment is currently being assessed for its operational performance / status. After under check the equipment will be declared S/A from “back time” the time its under check started or U/S from “back time” the time it got U/S initially.<br>(d) The length of time period for under check is at the discretion of battle staff, unless specified by higher HQs. However, the time should be kept as short as possible. |

| Term | Explanation |
|---|---|
| **Un-identified (U)** | Classification of a track on which complete information for positive identification is not available and based upon established criteria; it is most likely to be a friendly aircraft. |
| **Unknown (X)** | Classification of track on which no information is available, and based upon established criteria; it is most likely to be an enemy aircraft. |
| **Vertical Polar Diagram** | The graphic representation on radar detection ranges, at various altitudes, is called a VPD (Vertical Polar Diagram). There are theoretical and practical VPDs. |
| **Wireless Observer Squadron** | A corps of specially trained ground observers deployed at suitable points to provide information of all aircraft movement within their area of responsibility. |
| **Working Border / Boundary** | It is border commencing south of Taku Chak to Abial Dogran between Pakistan and India. Any violation along the working border is dealt as violation of International border between Pakistan and India. |
| **Zone Time** | The time kept in an area of a 15º zone of longitude, the central meridian of each zone being 15º or a multiple of 15º from the Greenwich Meridian. The times of successive zones differ by one hour. |
| **Zulu Time** | An alphabetic expression indicating Greenwich Mean Time. |

APPENDIX 'A' TO
AFM NO 355–2
DATED 29 JAN 21

## ORGANOGRAM OF DCAS (OPS) SECTT (AD DOMAIN)

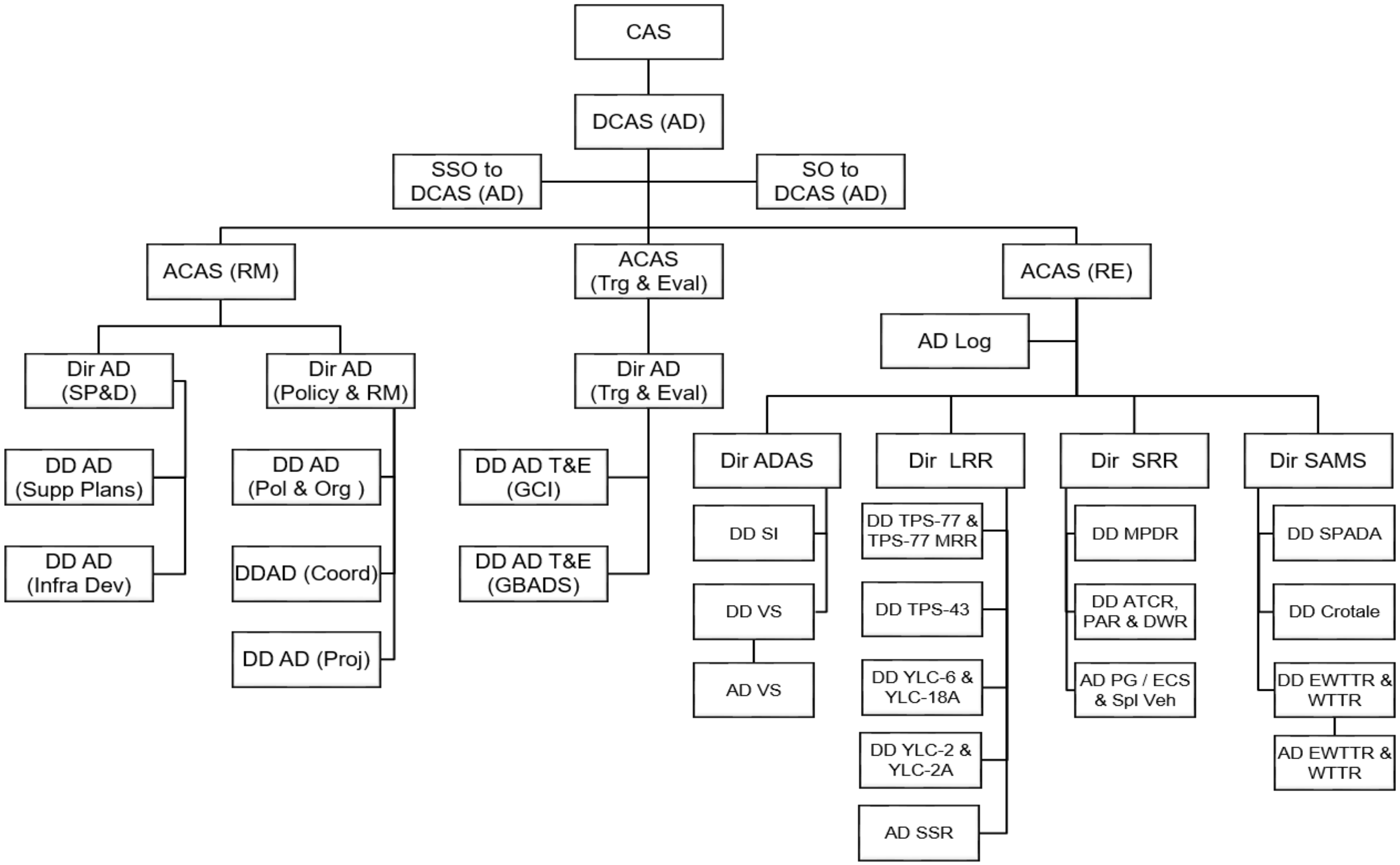





Revised on 29 Jan 21

APPENDIX 'B' TO
AFM NO 355–2
DATED 29 JAN 21

# ORGANOGRAM OF DCAS (AD) SECTT

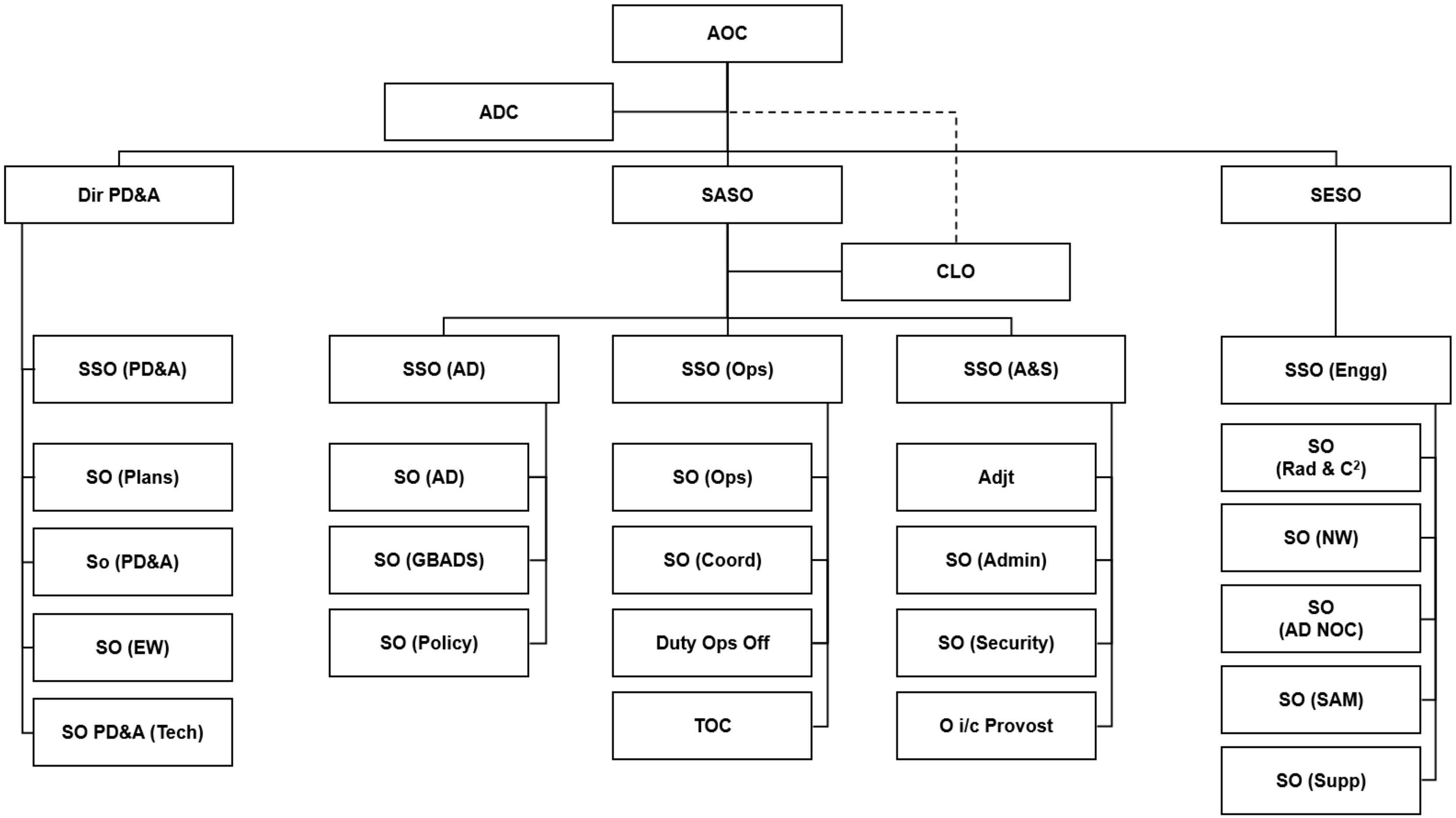





Revised on 29 Jan 21

APPENDIX 'C' TO
AFM NO 355–2
DATED 29 JAN 21

# ORGANOGRAM OF HEADQUARTERS AIR DEFENCE COMMAND

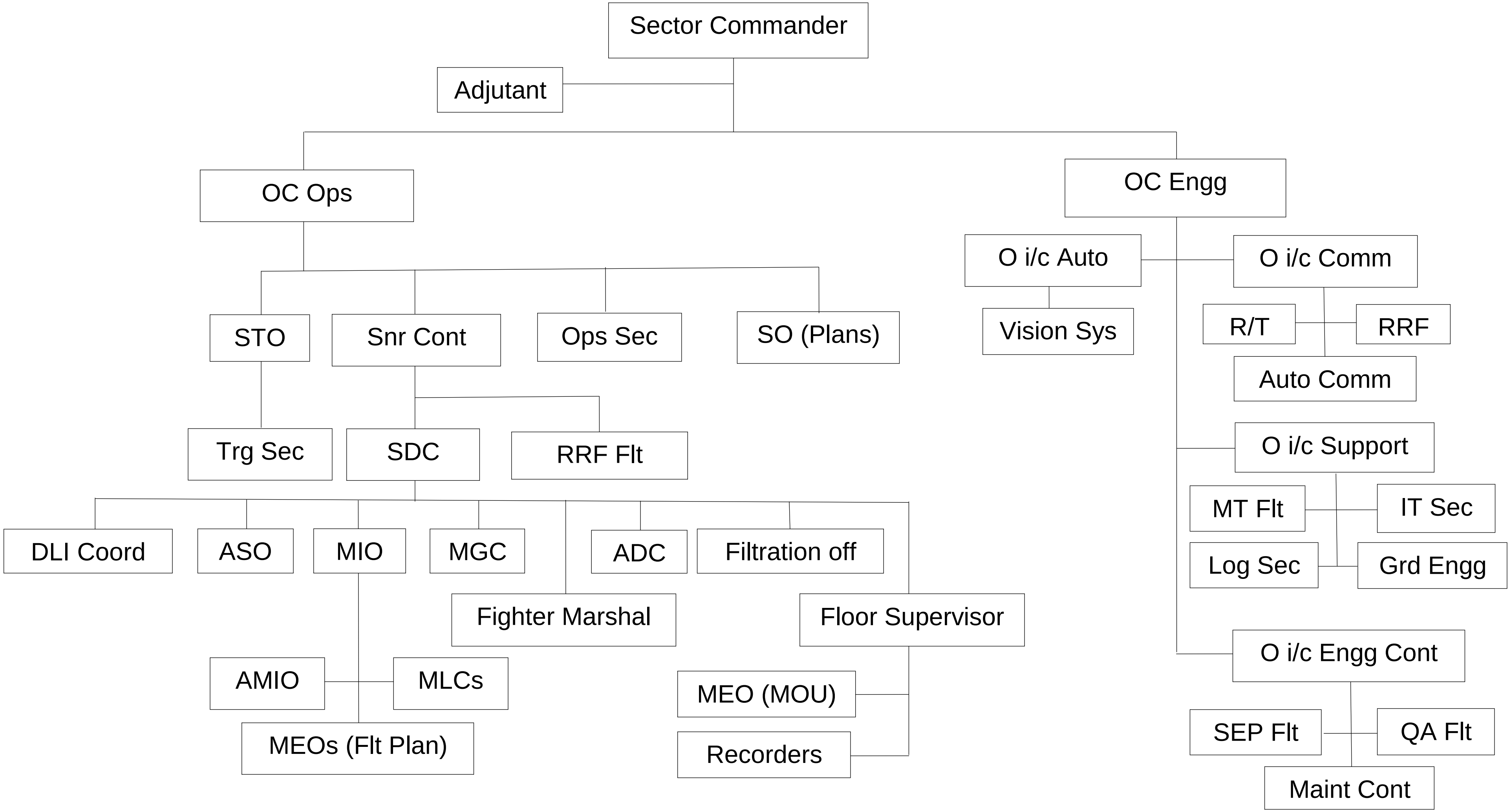





Revised on 29 Jan 21

APPENDIX 'D' TO
AFM NO 355 – 2
DATED 29 JAN 21

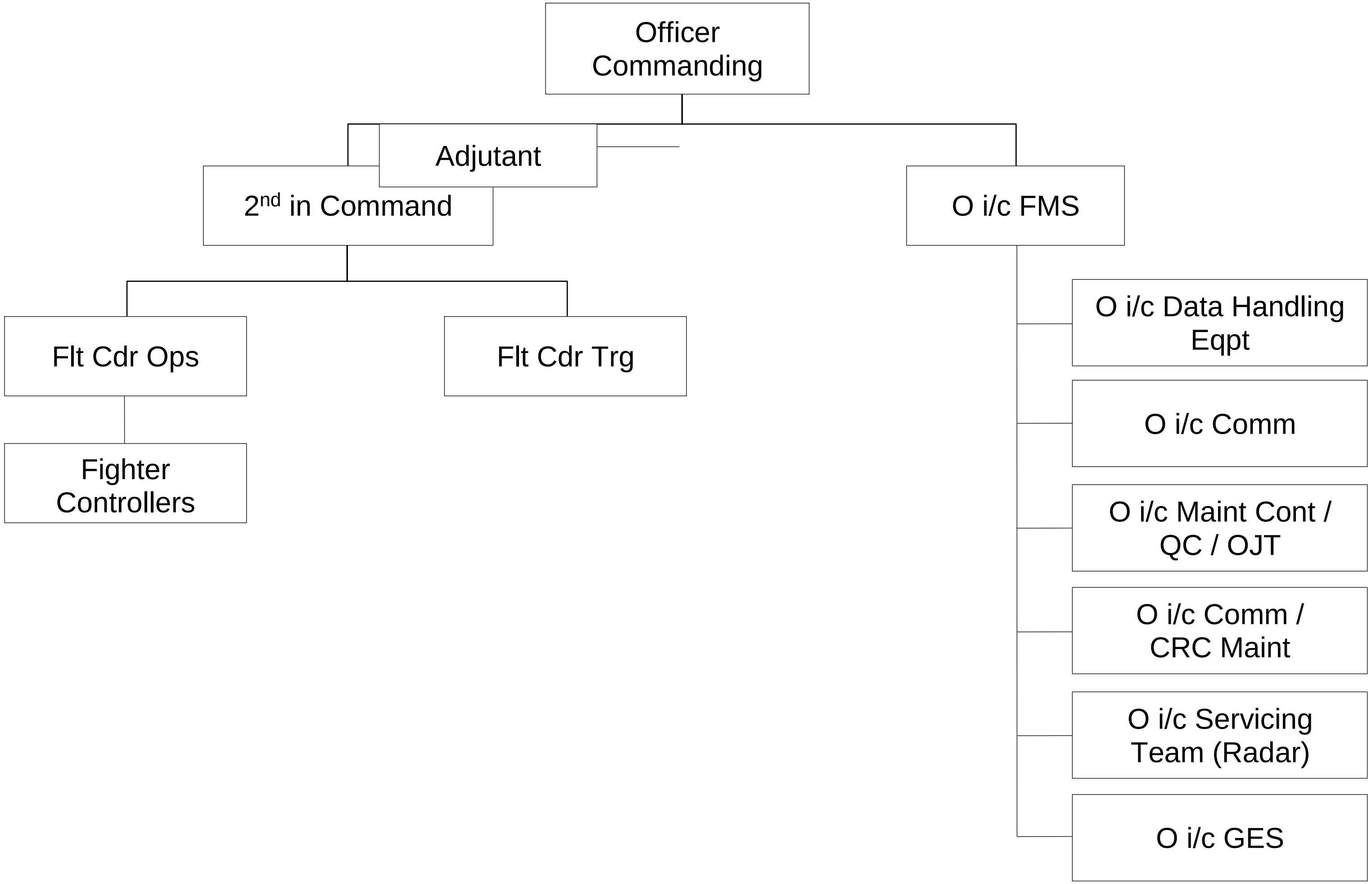
# ORGANOGRAM OF SECTOR HEADQUARTERS





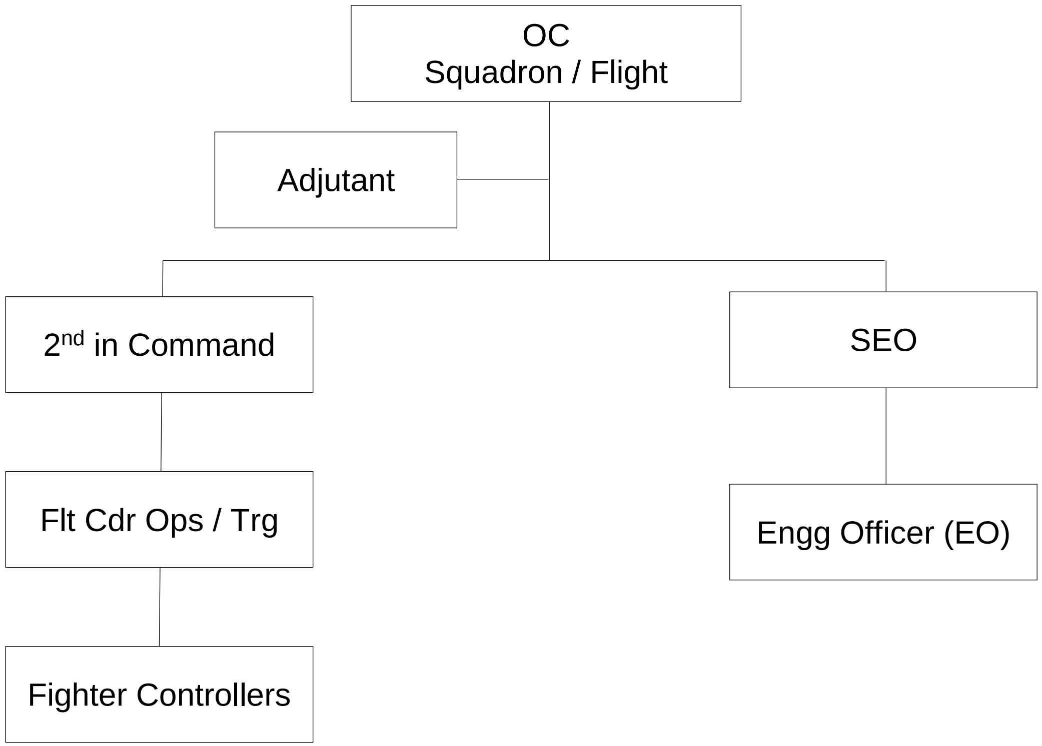

Revised on 29 Jan 21

## ORGANOGRAM OF GMCC HEADQUARTERS



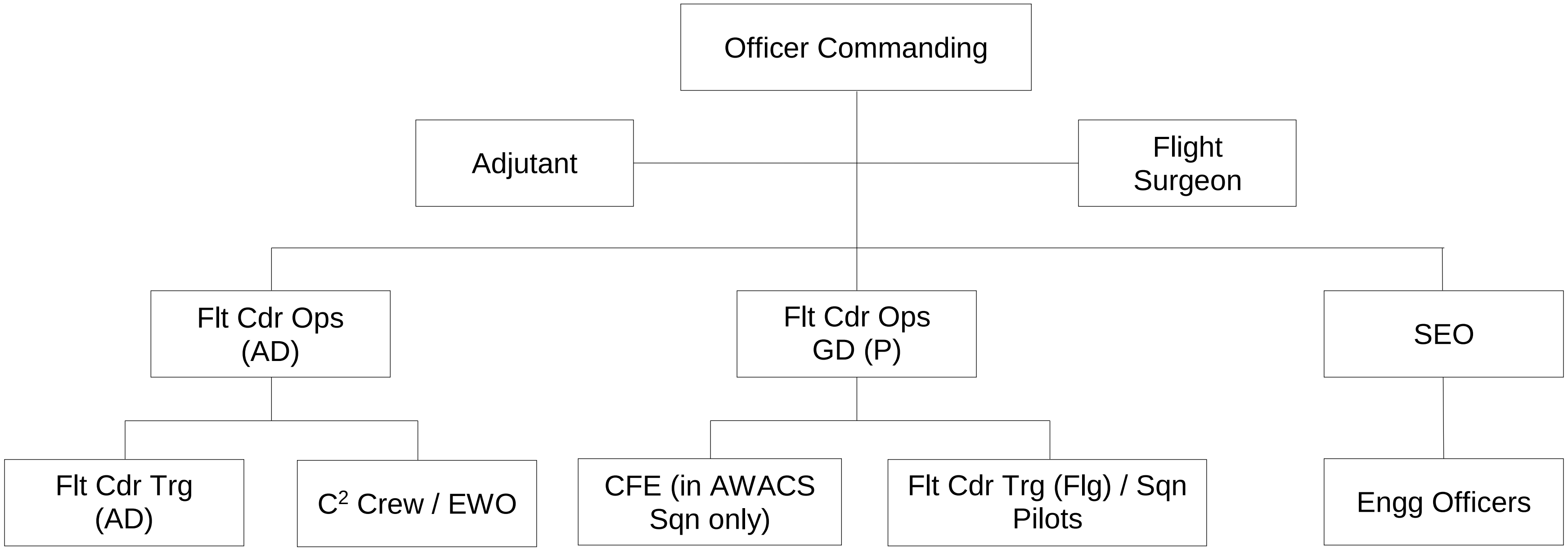
APPENDIX 'F' TO
AFM NO 355 – 2
DATED 29 JAN 21

# ORGANOGRAM OF HIGH LEVEL RADAR SQUARDON



APPENDIX 'G' TO
AFM NO 355 – 2
DATED 29 JAN 21

## ORGANOGRAM OF AEWACS SQUADRON HEADQUARTERS

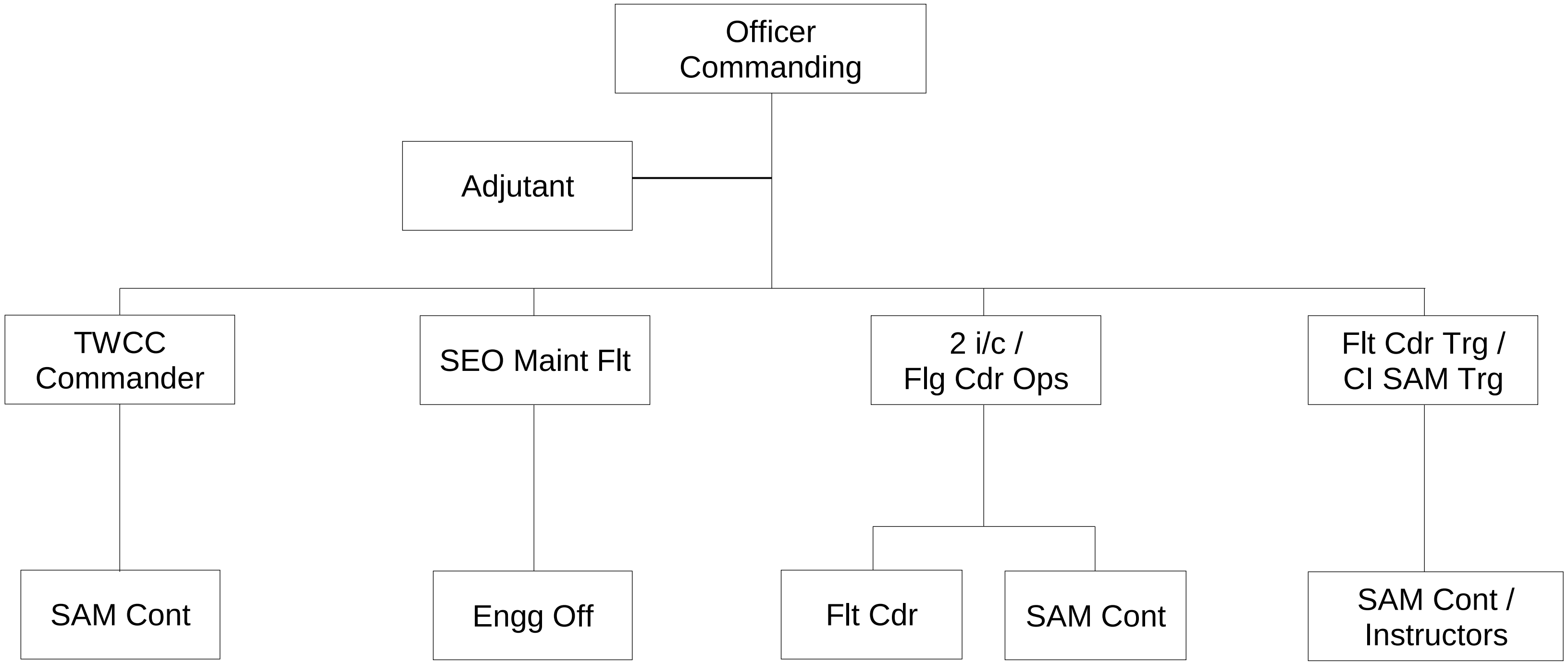





Revised on 29 Jan 21

APPENDIX ‘H’ TO
AFM NO 355 – 2
DATED 29 JAN 21

# ORGANOGRAM OF COMPOSITE SAM SQUADRON (CSS)

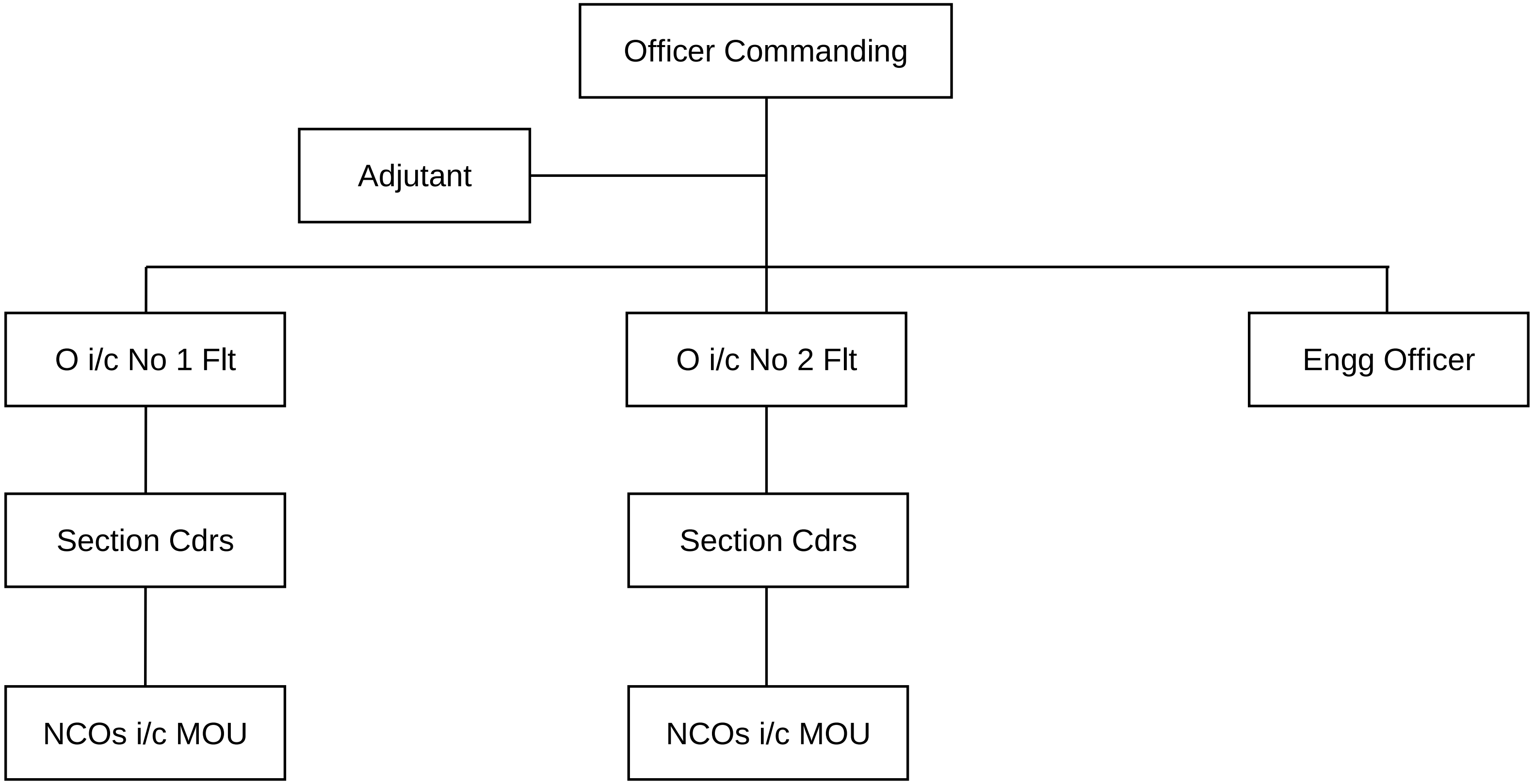





Revised on 29 Jan 21

APPENDIX 'J' TO
AFM NO 355 – 2
DATED 29 JAN 21

## MOBILE OBSERVER UNIT (MOU)






Revised on 29 Jan 21

APPENDIX 'K' TO
AFM NO 355 – 2
DATED 29 JAN 21

**SECRET**
(When filled in)

**TACTICAL ACTION REPORT**
PART – I
(To be filled in at SMCC)

**Date**: ___________________

1. **Target**

(a) Track No _______________ (b) Initial detection position GEOREF ____________________________________

(c) Strength _______________ (d) Height _______________________

(e) Type (if any) ____________ (f) Time ________________________

(g) Reported by ________________________________________________

2. **Interceptors**

(a) Mission No ____________ (b) Time, Type & No of interception place on S/ By ______________________

(c) Time, Type & No of interception (scrambled / diverted) (d) Initial Vector and Channel ______________________________________________

(e) Time Interceptor Airborne: ____________________________ (f) Interceptor's Callsign: ____________________________________

(g) Time R/T contact with interceptor (SMCC) ________________________

(h) Time handed over the GCI control (give callsign)

(i) ______________________ (ii) ___________________________

(iii) ______________________ (iv) ___________________________

(j) Air Defence Officers GCI callsign

(i) ______________________ (ii) ___________________________

(iii) ______________________ (iv) ___________________________

(k) Time off GCI / Off Ch / SMCC / GCI ______________________________

1 of 2
(When filled in)
**SECRET**



Revised on 29 Jan 21

APPENDIX 'K' TO
AFM NO 35 – 2(CONTD)
DATED 29 JAN 21

**SECRET**
(When filled in)

3. **Result (GCI)**

(a) Intercept Position

(i) GEOREF ______________ (ii) Range/ Brg from final GCU _____

(iii) Height ______________ (iv) Time ____________________

(b) Time off GCI / Off Ch / SMCC / GCI

4. **Surface to Air Missile**

(a) Surface to Air Missile (Type and Location):

(i) Time and Position of engagement (with Ref to VP) _____________

______________________________________________________________

(ii) Mission result with No and Type of aircraft (if known) ___________

______________________________________________________________

(b) AAD Weapons:

(i) No of type of aircraft (if known) shot down ____________________

______________________________________________________________

(ii) Time ____________________________________________________

5. **Remarks by**

(a) OC Ops (SMCC)

(b) Sector Commander

2 of 2
(When filled in)
**SECRET**

APPENDIX 'L' TO
AFM NO 355 – 2
DATED 29 JAN 21

**SECRET**
(When filled in)

**PART – II**
(to be completed by GCI Unit)

**INTERCEPTION TRACING**

**Date**: ___________________

**Track No**: _______________

**LEGEND**:

Splash / Tally Ho position in
Green Circle

Target / Hostile track initial
Plot in red circle

Interceptors track in green
Target / Hostile track in red

**Remarks**

1 of 1

**SECRET**
(When filled in)

APPENDIX 'M' TO
AFM NO 355 – 2
DATED 29 JAN 21

# INTERCEPTOR STATE REPORT

| Bases | Type | Available | --------Mts Readiness | --------Mts Readiness | Cockpit Standby | Committed | Recovery | Turn Around |
|---|---|---|---|---|---|---|---|---|
| | | | | | | | | |
| | | | | | | | | |
| | | | | | | | | |
| | | | | | | | | |
| | | | | | | | | |



Revised on 29 Jan 21

**SPECIMEN**

APPENDIX 'N' TO
AFM NO 355 – 2
DATED 29 JAN 21

**SECRET**
(When filled in)

# RADAR QUALITY CONTROL PERFORMA
# (Long Range Radars / KE-03)

**Sqn** ____________ **Site** ____________________ **Type** ________________________ **Date** _____________________ **Time** ____________

| Track No | Aircraft Details | | Flight Details | | | Azimuth (In / Out) or GCI Area | Ref Range | PSR | | RPC (Max / Ref)x100% | SSR M-3 Code_____ | | Height Finding Capability | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | Type | DB Rating | Call Sign | FPN | Flight Level | | | Max | Min | | Max | Min | Actual Height | Radar Height | Difference | HF Status |
| | | | | | | | | | | | | | | | | |

**RESULTANT BLIP TO SCAN RATIO**

| Range Segment No | Total Scans | Blips Picked | BSR | Standard BSR as per Range Segment | Resultant BSR 1 of 1 | Average RBSR (Radar) |
|---|---|---|---|---|---|---|
| | | | | | | |
| | | | | | | |
| | | | | | | **Average RBSR (GMCC)** |
| | | | | | | |
| | | | | | | |
| | | | | | | |
| | | | | | | **Data Loss** |
| | | | | | | |
| | | | | | | |
| | | | | | | |

**STATUS**

| | |
|---|---|
| **PSR** | |
| **SSR** | |
| **Height Finding** | |
| **Data Link** | |

**Remarks**:

**(Duty Controller)**

1 of 1
(When filled in)
**SECRET**



Revised on 29 Jan 21

APPENDIX 'P' TO
AFM NO 355 – 2
DATED 29 JAN 21

**SPECIMEN**

**SECRET**

(When filled in)

## RADAR QUALITY CONTROL PERFORMA

## (Medium / Short Range Radars)

**Unit** ______________________ **Date** ____________________

| S No | Site | Radar Type | Tilt | Track No. & Time | Flight Details | | | REF Range | Quality Carried out from | PSR | | SSR | | RPC | Total Scans | Blips Picked | BSR | BSR Probability from TOs | RBSR (if Applicable) | Status |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | A/C Type | Height | Azimuth / Area | | | MAX | MIN | MAX | MIN | | | | | | | |
| | | | | | | | | | Radar | | | | | | | | | | | PSR : |
| | | | | | | | | | GMCC | | | | | | | | | | | SSR : |
| | | | | | | | | | Data Loss between Radar & GMCC | | | | | | | | | | | HT Finding :<br>Data Link : |
| | | | | | | | | | Radar | | | | | | | | | | | PSR : |
| | | | | | | | | | GMCC | | | | | | | | | | | SSR : |
| | | | | | | | | | Data Loss between Radar & GMCC | | | | | | | | | | | HT Finding :<br>Data Link : |
| | | | | | | | | | Radar | | | | | | | | | | | PSR : |
| | | | | | | | | | GMCC | | | | | | | | | | | SSR : |
| | | | | | | | | | Data Loss between Radar & GMCC | | | | | | | | | | | HT Finding :<br>Data Link : |
| | | | | | | | | | Radar | | | | | | | | | | | PSR : |
| | | | | | | | | | GMCC | | | | | | | | | | | SSR : |
| | | | | | | | | | Data Loss between Radar & GMCC | | | | | | | | | | | HT Finding :<br>Data Link : |
| | | | | | | | | | Radar | | | | | | | | | | | PSR : |
| | | | | | | | | | GMCC | | | | | | | | | | | SSR : |
| | | | | | | | | | Data Loss between Radar & GMCC | | | | | | | | | | | HT Finding :<br>Data Link : |

1 of 1

(When filled in)

**SECRET**

**(Duty Controller)**



Revised on 29 Jan 21